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EET307 – SYNCHRONOUS & INDUCTION MACHINES

Module 1

Principle of Operation of three phase alternators, Constructional features, Types of Armature Windings (detailed winding diagram not required), EMF equation, Numerical Problems.

Harmonics-causes, suppression, Rating of alternators, Parameters of armature winding, Armature reaction, Equivalent Circuit, Phasor Diagram, Load characteristics, Power Flow Equations.

Module 2

Voltage regulation of three phase Alternators-Direct loading, EMF Method, MMF Method, Potier Method, ASA Method -Numerical Problems.

Blondel's two reaction theory, Phasor Diagram under lagging power factor, Determination of X_d and X_q by slip test, Power developed by a Salient pole machine, Numerical Problems.

Parallel Operation of Alternators- Necessary Conditions, Synchronisation- Synchronising current, Power and Torque, Effect of reactance, Numerical Problems, Methods of Synchronisation.

Module 3

Principle of Synchronous Motor, Equivalent circuit, Phasor diagrams, Power flow diagram and equations, Losses and efficiency -Numerical Problems, Power-angle Characteristics, V Curve and Inverted V Curves.

Three phase Induction motor – Constructional features, Expressions for Power and Torque - Torque-Slip characteristics, Phasor diagram, Equivalent Circuit of Induction motor- Tests on Induction motors for determination of equivalent circuit-Numerical Problems.

Module 4

Performance of three phase Induction motors using Circle diagram, Numerical Problems. Cogging and Crawling in cage motors, Double cage Induction motor-Torque-Slip Characteristics.

Starting of Induction motors – Types of Starters – DOL starter, Autotransformer Starter, Star Delta starter, Rotor Resistance Starter-Numerical Problems.

Braking of Induction motors – Plugging, Dynamic braking, Regenerative braking

Speed control – Stator Voltage control, V/f control, Rotor Resistance Control.

Module 5

Induction generator – Principle of operation, Grid Connected and Self Excited Operation of Induction Generators, Torque-Slip Characteristics of an Induction machine.

Single phase Induction motors - Double field revolving theory, Equivalent Circuit, Torque Slip Characteristics, Types of Single-phase induction motor, Applications.

Selection of AC motors for different applications.

References:

S.No.	Author	Title of book	Publisher
1	P.S. Bhimbra	Electrical Machinery	Khanna
2	M.G. Say	Performance & Design of AC Machines	CBS
3	Kothari & Nagrath	Electric Machines	Tata McGraw-Hill
4	K Murugesh Kumar	Induction & Synchronous Machines	Vikas
5	J.B. Gupta	Theory & Performance of Electrical Machines	S.K. Kataria

COURSE OUTCOMES

CO1	Analyse the performance of different types of alternators
CO2	Analyse the performance of a synchronous motor
CO3	Analyse the performance of different types of induction motors.
CO4	Describe the operating principle of induction generator
CO5	Explain the types of single-phase induction motors and their working principle

Course Contents and Lecture Schedule:

Sl. No.	Topic	No. of Hours
1	Basics of Alternators (10 hours)	
1.1	Principle of operation and classification of alternators, Synchronous speed.	2
1.2	Construction of synchronous machines. Salient and Cylindrical types, Turbogenerators. Stationary and Rotating armature types	1
1.3	Armature windings-Types.: Single layer, Double layer, Full pitched winding, Short pitched winding, Concentrated and Distributed winding	1
1.4	EMF Equation, Pitch factor and Distribution factor, Numerical problems	3
1.5	Harmonics in Alternators: Space and slot harmonics, Suppression, Effect of pitch factor on harmonics.	1
1.6	Armature Reaction, Equivalent Circuit and Phasor Diagrams, Power Flow Equations	2
2	Voltage Regulation and Synchronisation of Alternators (10 hours)	
2.1	Voltage Regulation of Alternators: EMF, MMF, Potier and ASA Method.	4
2.2	Blondel's Two Reaction Theory, Phasor Diagram under lagging power factor based on two reaction theory, Slip Test	3
2.3	Parallel Operation of Alternators, Necessity of Parallel Operation. Advantages.	1
2.4	Synchronisation of Alternators: Dark Lamp and Bright Lamp Method.	2
3	Three Phase Synchronous and Induction Motors (10 hours)	
3.1	Synchronous Motors-Principle, Equivalent Circuit, Phasor Diagrams, Power Flow Diagram, Power and Torque Equations, Numerical Problems	3
3.2	Effects of excitation on armature current and power factor- V and Inverted V Curves, advantages, disadvantages and applications of Synchronous motors.	1
3.3	Three phase Induction Motors-Principle, Constructional details, Slip ring and Cage types.	1
3.4	Slip, frequency and rotor current, Expression for torque and Power Starting torque, Full load and Pull out torque, Torque- Slip characteristics, Phasor diagram	3
3.5	Tests on Induction motors for determination of Equivalent circuit, Equivalent Circuit of Induction motor-Numerical Problems	2
4	Three Phase Induction Motors Contd. (8 hours)	
4.1	Circle Diagram, Numerical Problems.	3
4.2	Cogging, Crawling—remedial measures, Double Cage Induction Motor-Principle.	1
4.3	Starters for three phase Induction Motors: DOL, Autotransformer, Star Delta and Rotor Resistance Starters.	2
4.4	Speed Control in Induction Motors	1
4.5	Braking in Induction Motors	1
5	Induction Generators and Single-phase Induction Motors (7 hours)	
5.1	Induction Generators: Grid Connected and Self Excited types.	1
5.2	Single phase induction motors-principle, Double field revolving theory, Torque-Slip characteristics, Applications	2
5.3	Types-Split phase, Capacitor Start, Capacitor Start and Run types, Shaded pole motor, Shaded Pole Motor-Principle of operation and applications.	3
5.4	Selection of AC motors for different Applications.	1

Module 1

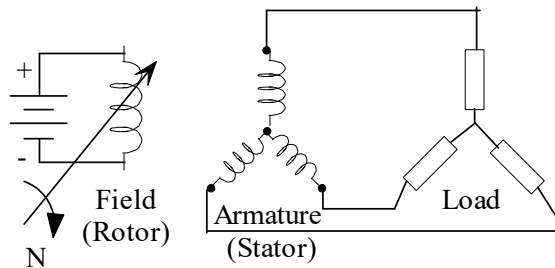
Principle of Operation of three phase alternators, Constructional features, Types of Armature Windings (detailed winding diagram not required), EMF equation, Numerical Problems.

Harmonics-causes, suppression, Rating of alternators, Parameters of armature winding, Armature reaction, Equivalent Circuit, Phasor Diagram, Load characteristics, Power Flow Equations.

SYNCHRONOUS GENERATOR (ALTERNATOR)

Working principle of alternator is same as that of a dc generator (“when the flux linked with a coil changes, an emf is induced in it”), but in a synchronous generator there is no need of commutator.

In fact, all dc generators are alternators in which alternating voltages are set up and it is the commutator which converts this ac into dc.



The construction of dc machines are always rotating armature type for commutation (for getting unidirectional voltage). However, alternator may be rotating armature type with field stationary or rotating field type with armature stationary.

Usually, armature is kept stationary and the field rotates under it.

Advantages of keeping the armature stationary :-

- The load can be directly fed without any sliprings or brushes.
- It is easier to insulate stationary armature winding from the core for high voltages upto 33kV (Stationary armature is not subjected to mechanical stresses due to centrifugal action and more space is available on the stator for providing more insulation).
- Only 2 slip rings are required to supply dc to the field winding (rotor). Since the exciting current is small (low voltage dc supply), slip rings and brushes may be of light construction. (Field system requires about 2% of the power capacity of the machine).

The field winding is excited by direct current to produce flux in the air gap. When the rotor rotates, voltage is induced in the armature winding placed on the stator. The armature current produces a revolving flux in the air gap whose speed is the same as the speed of the rotor – hence the name synchronous machine.

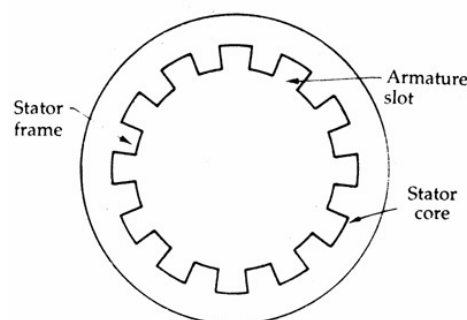
Construction

A three-phase synchronous machine consists of two parts

- Stator
- Rotor

Stator (Armature)

The core is hollow cylindrical in shape and is made of sheet steel laminations (0.35mm to 0.65mm thick) insulated from each other. (Stator of synchronous machines and induction machines are similar)



It consists of number of slots in its inner periphery to accommodate armature conductors (winding). A three phase distributed, full pitched/short pitched windings are placed in these slots. Windings are suitably connected to form a balanced three phase star or delta connected circuit. It is wound for a definite number of poles as per requirement of speed.

Star connection is preferred for stator winding: -

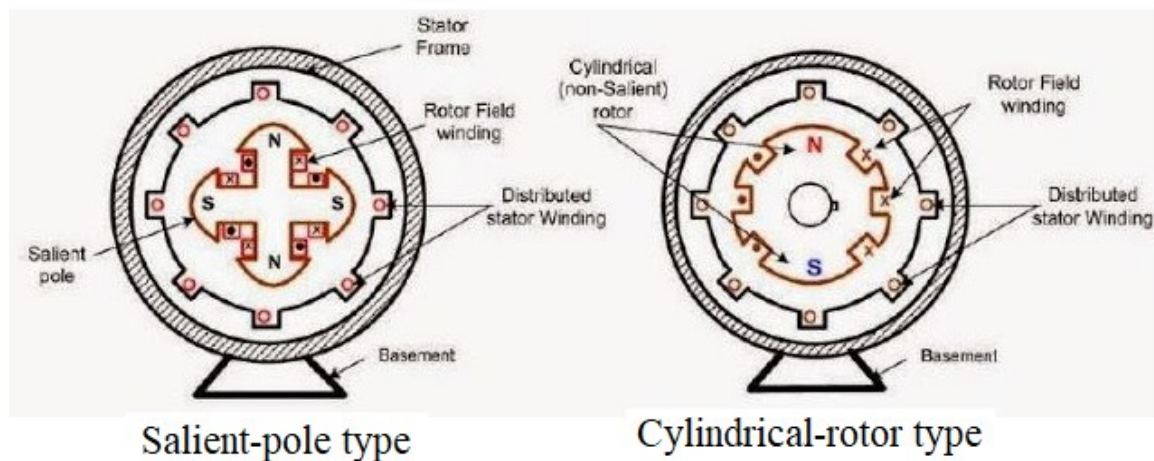
- 1) Since the voltage per phase is only $\frac{1}{\sqrt{3}}$ or 58% of the voltage between the lines, amount of insulation in the slots can be reduced, conductors of larger cross section can be used, hence more power output possible.
 - 2) When a synchronous generator is under load, the voltage induced in each phase becomes distorted, and the waveform is no longer sinusoidal. The distortion is mainly due to an undesired 3rd harmonic voltage. With star connection, third harmonic voltages will not be present in line voltages & hence line voltage remains sinusoidal under all load conditions. But, when a delta connection is used, the harmonic voltages do not cancel, but add up. Because, the delta is closed, they produce third harmonic circulating current which increases the I²R losses.
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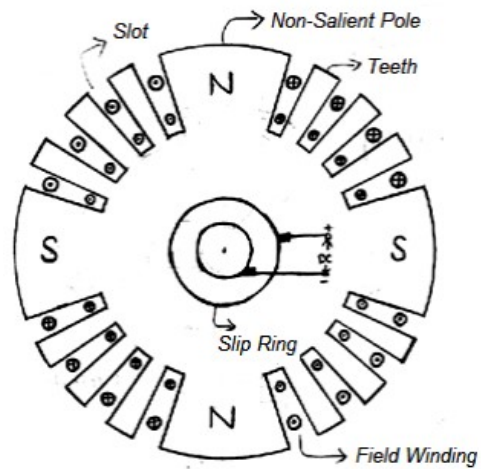
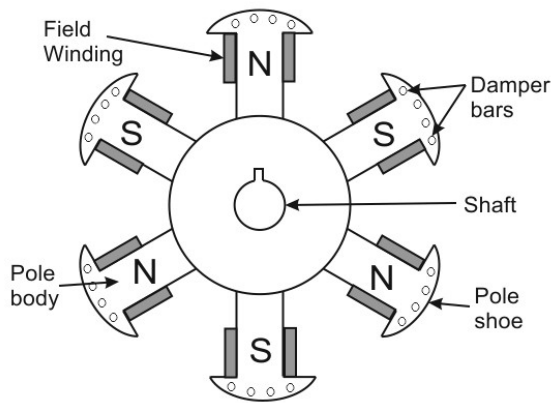
Rotor

Rotor carries field winding which is supplied with dc through 2 slip rings.

Two types

- a) Salient (or projecting) pole type
- b) Cylindrical rotor type





Salient Pole

1. It consists of projected poles, laminated, made of cast iron or cast steel
2. Poles carry concentrated field windings
3. Air gap is not uniform
4. It has large diameter and short axial length
5. It is used for low and medium speed machines (water wheel drive or diesel engine drive)

Smooth cylindrical

1. It is built from solid steel forging (usually chromium – nickel steel)
2. Poles consists of radial slots in which field windings are placed
3. Air gap is nearly uniform
4. It is of small diameter and of very long axial length.
5. It is used for high speeds (Steam driven)
6. Noiseless operation
7. Less windage loss
8. Better in dynamic balancing

Principle of operation

When the rotor (excited by dc source) is rotated by the prime mover (diesel engine, steam turbine etc), the stator conductors are cut by the magnetic flux of the rotor poles. Hence an emf is induced in the stator conductors. This emf is alternating in nature. The frequency of this induced emf is given by $f = \frac{PN}{120}$ and the direction of induced emf is found by Fleming's right hand rule.

Frequency

Number of revolutions/minute = N

Number of revolutions/second = $\frac{N}{60}$

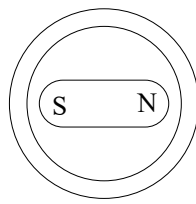
Number of cycles/revolution = $\frac{P}{2}$

Hence, number of cycles/second $f = \frac{P}{2} \times \frac{N}{60} = \frac{PN}{120}$

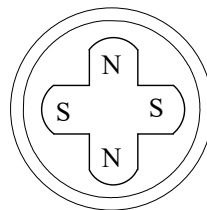
Relation between electrical & mechanical degrees

1 revolution = 360 mechanical degrees

1 cycle = 360 electrical degrees



1 revolution → 1 cycle



1 revolution → 2 cycles

Electrical degree = $P/2 \times$ Mechanical degree

$$\theta_e = \frac{P}{2} \times \theta_m$$

3 phase armature winding

CLASSIFICATION:-

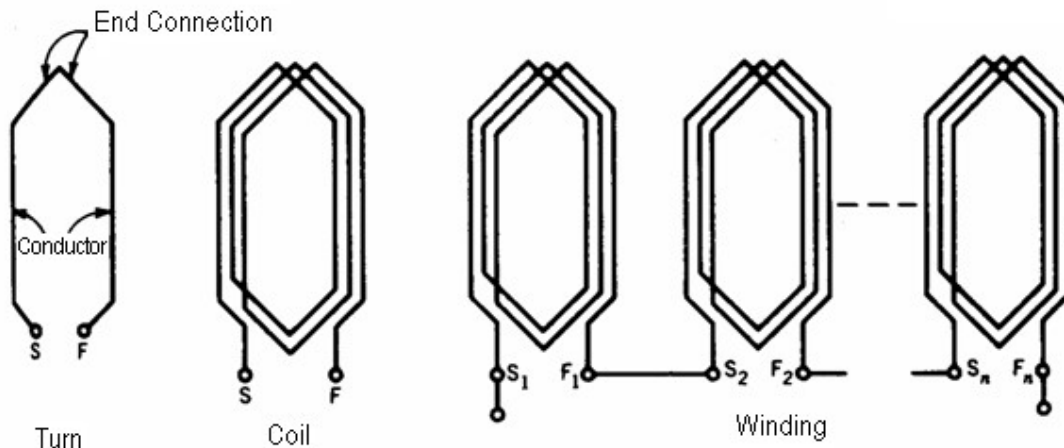
- Type of coil: Concentric (for single layer) & lap (for double layer)
- Layers: Single layer (for small machine) and double layer
- Overhang: Diamond, multi-plane, mush, involute
- phase spread: 60° , 120°
- slotting: Integral or fractional
- coil span: Full-pitched or short-pitched
- Coils: Single turn, multi-turn
- slots: closed, semi-enclosed, open i) connection: Star or Delta

Turn – consists of two conductors connected one end by an end connector

Coil – is formed by connecting several turns in series

Winding – is formed by connecting several coils in series

Turn, coil and winding are shown in figure below.



Pole pitch – distance between the centers of two adjacent poles (expressed in terms of number of slots, conductors or in electrical degrees)

Pole pitch = 180° electrical

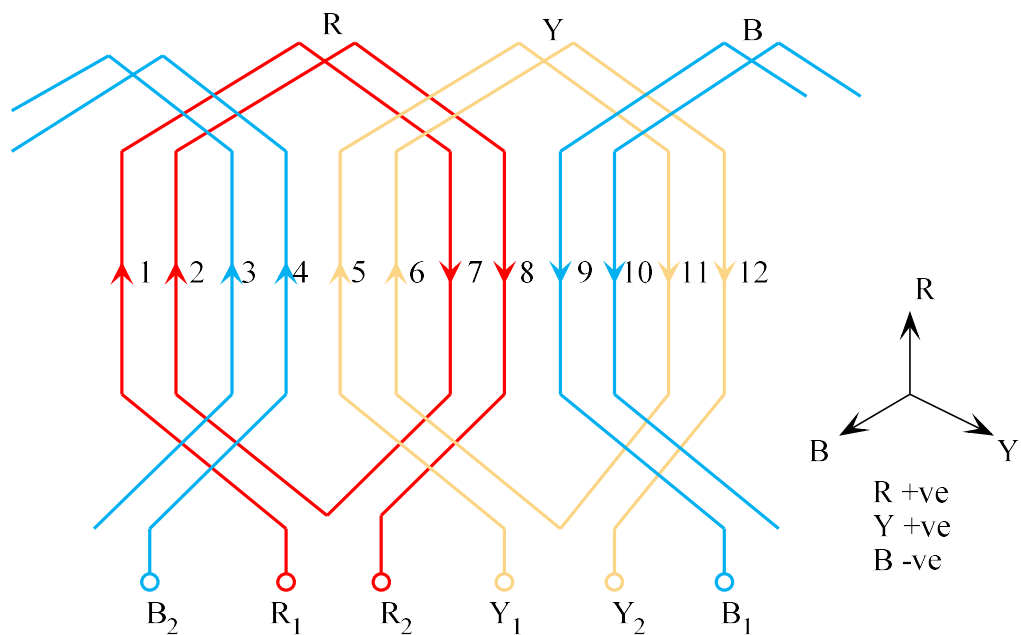
Coil span - distance between the two sides of a coil (expressed in terms of number of slots, conductors or in electrical degrees)

If the coil span is one pole pitch, it is called full-pitched coil.

If the coil span is less than one pole pitch, the coil is known as short-pitched/short-chorded coil.

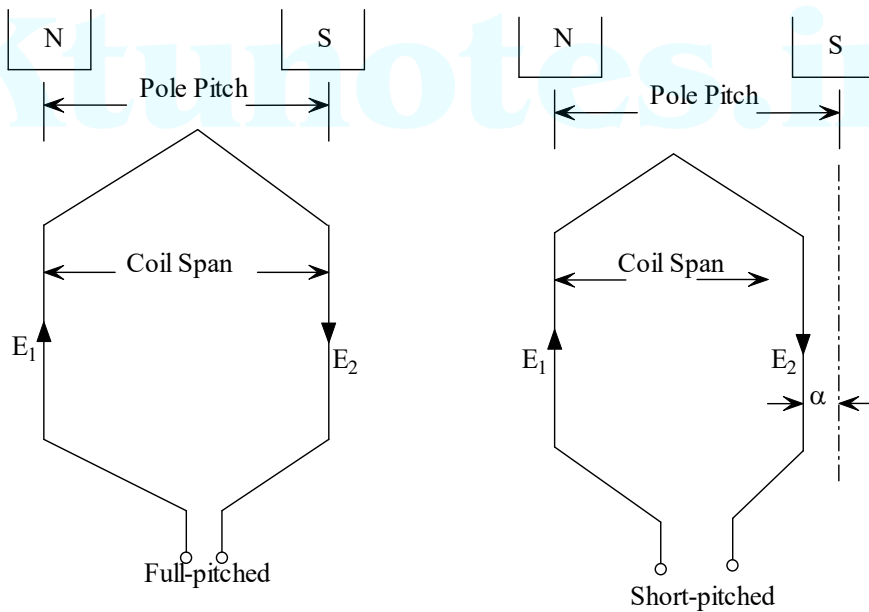
Short-pitched coils are desirable in ac machines to eliminate some harmonics.

Winding diagram of 3 phase, 12 slots, 2 pole, single layer, full pitched winding

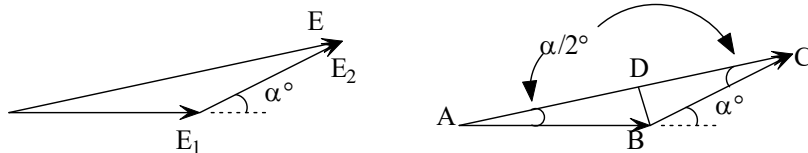


Coil Span factor (Pitch factor) K_c

In full pitched (coil span = pole pitch) winding, the induced emfs in the two sides of the coil will be in phase and the resultant voltage between the two terminals of the coil will be the arithmetic sum.



However, in short chorded (coil span < pole pitch) winding, the induced emfs in the two sides of the coil will not be in phase, hence their resultant given by the phasor sum is always less than their arithmetic sum.



$$\text{Arithmetic sum} = AB + BC = 2E_1 \quad (E_1 = E_2 \text{ in magnitude})$$

$$\text{Phasor sum} = AD + DC = 2E_1 \cos \frac{\alpha}{2}$$

$$K_c = \frac{\text{Phasor Sum of coilside emfs}}{\text{Arithmetic Sum of coilside emfs}} \\ = \frac{\text{Emf induced in shortpitched coil}}{\text{Emf induced in fullpitched coil}}$$

$$K_c = \cos \frac{\alpha}{2} \text{ where } \alpha = \text{chording angle}$$

For example, consider a 3-phase, 4-pole, 36 slots machine. Here, the pole pitch = $\frac{\text{slots}}{\text{poles}} = 9 = 180^\circ$.

Here, slot pitch, $\beta = \frac{180^\circ}{\text{slots/pole}} = \frac{180^\circ}{9} = 20^\circ$. If the coil span is 9 slots (full-pitched), $\alpha = 0^\circ$. If the coil

span is 8 slots, $\alpha = 20^\circ$. If the coil span is 7 slots, $\alpha = 40^\circ$.

Advantages of short pitched winding :-

Due to chording, the resultant induced emf is reduced by a factor $K_c = \cos \frac{\alpha}{2}$ (always less than unity) however, some harmonics are eliminated from the emf wave and hence the waveform is more sinusoidal. Also, there is saving of copper required in the end connections of coils.

Coil span factor for n^{th} harmonic is $K_{cn} = \cos \frac{n\alpha}{2}$.

If $\alpha = 60^\circ$, for 3rd harmonic, $K_{c3} = \cos \frac{3 \times 60^\circ}{2} = 0$ i.e. 3rd harmonics are eliminated.

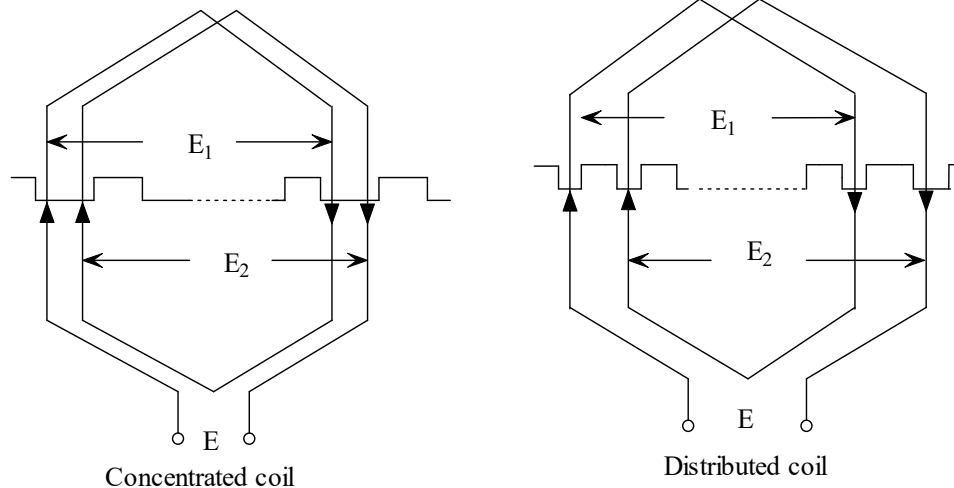
Distribution factor K_d (Breadth factor)

The coils belonging to one phase of armature winding under one pole pair are not concentrated in one slot, instead they are distributed in several adjacent slots.

eg 1:- 12 slots, 2 pole, 3 phase

$$m = \frac{\text{slots}}{\text{pole} \times \text{phase}} = \frac{12}{2 \times 3} = 2;$$

there are 2 coils belonging to each phase under each pole pair, these 2 coils are distributed in 2 adjacent slots



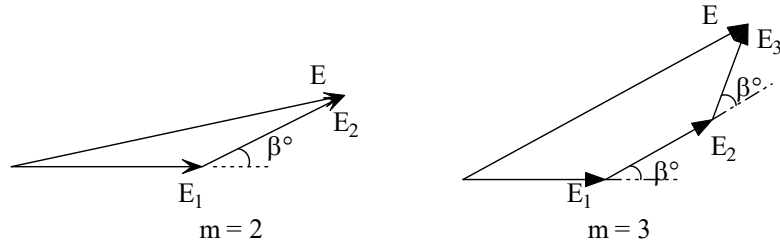
eg 2:- 36 slots, 4 pole, 3 phase

$$m = \frac{\text{slots}}{\text{pole} \times \text{phase}} = \frac{36}{4 \times 3} = 3;$$

there are 3 coils belonging to each phase under each pole pair, these 3 coils are distributed in 3 adjacent slots; number of pole pairs = 2

(m = integer → integral slot winding; m = fraction → fractional slot winding)

The emf induced in each coil are equal in magnitude; but due to distribution, these coil emfs will have a phase displacement of β° (electrical) (β° = slot pitch/angle) between them. Therefore, the phasor sum of coil emfs will be less than the arithmetic sum of coil emfs.



Distribution factor K_d is given by

$$K_d = \frac{\text{Emf induced in distributed winding}}{\text{Emf induced in concentrated winding}} \\ = \frac{\text{Phasor Sum of coil emfs}}{\text{Arithmetic Sum of coil emfs}}$$

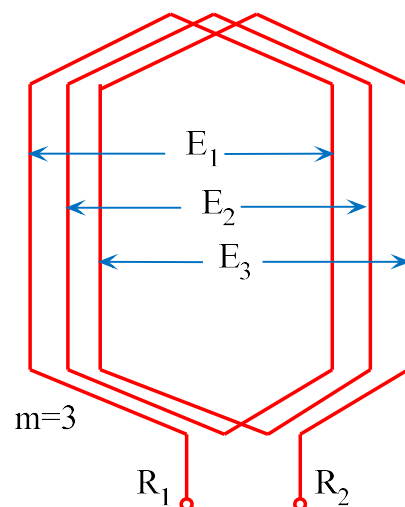
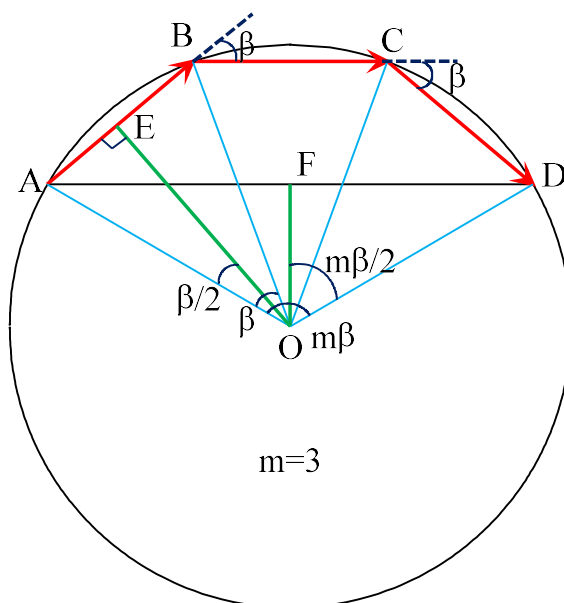
Expression of distribution factor K_d :

m = Number of slots/pole/phase

$$\beta = \text{slot pitch} = \frac{180^\circ}{(\text{slots} / \text{pole})}$$

Draw bisectors to AB, BC and CD. All this bisectors will meet at a common point O. O is the centre of a circle with AB, BC and CD as chords.

AB = BC = CD



$$|E_1| = |E_2| = |E_3| = AB = AE + BE = 2 \times OA \times \sin \frac{\beta}{2}$$

$$|E_1| + |E_2| + |E_3| = 2 \times m \times OA \times \sin \frac{\beta}{2}$$

$$|E_R| = AD = AF + DF = 2 \times OA \times \sin \frac{m\beta}{2}$$

$$K_d = \frac{2 \times OA \times \sin \frac{m\beta}{2}}{2 \times m \times OA \times \sin \frac{\beta}{2}} = \frac{\sin \frac{m\beta}{2}}{m \sin \frac{\beta}{2}}$$

$$\text{Hence, } K_d = \frac{\sin(m\beta/2)}{m \sin(\beta/2)}$$

K_d is always less than unity.

Due to distribution, the resultant emf is reduced by a factor K_d , however some harmonics are reduced and hence the emf waveform becomes more sinusoidal.

Distribution factor for n th harmonic is given by

$$K_{dn} = \frac{\sin(mn\beta/2)}{m \sin(n\beta/2)}$$

EMF EQUATION

Let Φ = flux per pole in Wb

P = Number of poles

Z = Number of armature conductors in series/phase

T = Number of turns in series / phase = $Z/2$

N = revolutions / minute

In one revolution of the rotor, each armature conductor is cut by a flux of ΦP wb.

$$\text{Number of revolutions/second} = \frac{N}{60}$$

$$\text{Time taken to complete one revolution} = \frac{60}{N} \text{ sec}$$

$$\text{Average emf induced/conductor} = \frac{d\phi}{dt} = \frac{\phi P}{(60/N)} = \frac{\phi P N}{60} = 2\phi \frac{PN}{120} = 2f\phi$$

Number of conductors in series/phase, $Z = 2T$

$$\text{Average value of emf induced/phase} = 2f\phi \times 2T = 4f\phi T$$

$$\text{RMS value of induced emf/phase} = 4K_f f\phi T$$

$$K_f = \text{form factor} = \frac{\text{RMS value}}{\text{Average value}}$$

= 1.11 for sine wave

$$E_{ph} = 4K_f f\phi T \text{ volts}$$

If K_c = coil span factor and K_d = distribution factor,

$$E_{ph} = 4K_f K_c K_d f\phi T \text{ volts}$$

$K_c K_d = K_w$ = winding factor

$$Z = \frac{\text{No. of conductors/slot} \times \text{No. of slots}}{\text{No. of phases}}$$

Note:-

For n^{th} harmonic, $E_{phn} = 4K_f K_{cn} K_{dn} n f \phi_n T$

$$\frac{E_{phn}}{E_{ph1}} = \frac{K_{wn}}{K_{w1}} \frac{n\phi_n}{\phi_1}$$

$$\frac{\phi_n}{\phi_1} = \frac{1}{n} \frac{B_n}{B_1} \quad \left(\text{since } \phi_1 = \frac{2DL}{P} B_1 \text{ and } \phi_n = \frac{2DL}{nP} B_n \right)$$

$$\frac{E_{phn}}{E_{ph1}} = \frac{K_{wn}}{K_{w1}} \frac{B_n}{B_1}$$

$$E_{ph} = \sqrt{E_{ph1}^2 + E_{ph3}^2 + \dots E_{phn}^2}$$

HARMONICS IN EMF WAVEFORM – CAUSES AND SUPPRESSION

In alternators, the primary cause of harmonics in the emf waveform is due to the non-sinusoidal field flux waveform. In case the field flux waveform is a sine wave, there would be no harmonics in the alternator generated emf. Thus first an attempt is made to have almost sinusoidal field flux wave and then the means to reduce the harmonics from the emf waveform should be adopted.

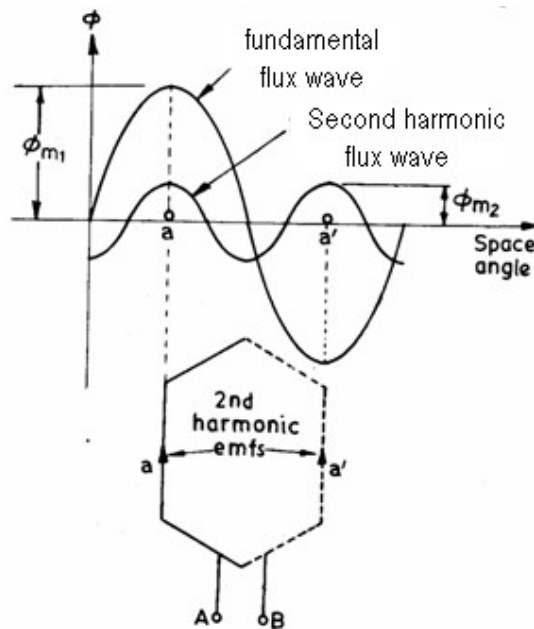
Field flux waveform can be made as much sinusoidal as possible by the following methods:

- i) Small air gap at the pole centre and large airgap towards the pole tips in salient-pole synchronous machines tends to make the field flux sinusoidal.
- ii) Skew the pole faces if possible
- iii) In turbo-alternators, the air gap is uniform. So field winding is distributed in the slots in such a manner as to make the field flux waveform almost a sine wave.

In spite of all these attempts, field flux waveform along the air-gap periphery is not sinusoidal. As a result, harmonic emfs are always generated in an alternator. These can, however, be eliminated or suppressed as follows:

- iv) Distribution: The distribution of the armature winding along the air-gap periphery tends to make the emf waveform sinusoidal
- v) Chording: With coil-span less than pole pitch, the harmonics can be eliminated
- vi) Skewing: By skewing the armature slots, only tooth harmonics or slot harmonics can be eliminated
- vii) Fractional slot winding – if the number of slots / pole / phase is a fraction, there will be considerable reduction in harmonic emf.
- viii) Alternator connections: Star or delt connections of alternators suppress triplen harmonics from appearing across the lines.

Even harmonics are absent in the generated emf (phase or line) :- The induced emf in the two sides, a and a' of a coil due to second harmonic flux wave will be in the same direction (upward). Hence, resultant emf between the two terminals of the coil will be zero (the two coil side emfs cancel each other).



Elimination of even and triplen harmonics:-

Fundamental components of generated emf are

$$e_{R1} = E_{Rm} \sin(\omega t)$$

$$e_{Y1} = E_{Ym} \sin(\omega t - 120^\circ)$$

$$e_{B1} = E_{Bm} \sin(\omega t + 120^\circ)$$

Third harmonic voltages are

$$e_{R3} = E_{R3m} \sin(3\omega t)$$

$$e_{Y3} = E_{Y3m} \sin 3(\omega t - 120^\circ) = E_{Y3m} \sin(3\omega t)$$

$$e_{B3} = E_{B3m} \sin 3(\omega t + 120^\circ) = E_{B3m} \sin(3\omega t)$$

If the three phases are connected in star, line voltage, $e_{RY3} = e_{R3} - e_{Y3} = 0$. Hence 3rd harmonics are absent in line voltage.

Hence, third harmonic voltages E_{R3} , E_{Y3} , E_{B3} are in phase.

Qn.1. A 3 phase 8 pole 750 rpm star-connected alternator has 72 slots on the armature. Each slot has 12 conductor and winding is short chorded by 2 slots. Find the induced emf between lines, given the flux per pole is 0.06 wb. (Ans: $K_d=0.956$, $K_c=0.94$, $E_{line}=2998V$)

Qn.2. A 4 pole, 50Hz star connected alternator has a flux per pole of 0.12wb. It has 4 slots per pole per phase, conductors/slot being 4. If the winding coil span is 150° , find the emf. (Ans: $K_d=0.958$, $K_c=0.966$, $E_{line}=1366V$)

Qn3 A three phase star connected synchronous generator driven at 750 rpm is required to generate a line-to-line voltage of 440V at 50Hz on open circuit. The stator is wound with 2 slots per pole per phase and each coil has 4 turns. Calculate the useful flux per pole.

(Ans: $K_d=0.966$, $K_c=1$, $\Phi=36.9 \times 10^{-3} \text{Wb}$)

$$P = 8; \beta = \frac{180^\circ}{6} = 30^\circ$$

$$K_d = \frac{\sin(\frac{m\beta}{2})}{m \sin(\frac{\beta}{2})} = \frac{\sin 30^\circ}{2 \sin 15^\circ} = 0.966$$

For a double-layer winding, No. of coils = No. of slots = $2 \times 8 \times 3 = 48$

Since number of turns/coil is 4, no. of turns per phase $T = \frac{48 \times 4}{3} = 64$

(Note : No. of conductors / slot = $4 \times 2 = 8$)

$$E_{ph} = 4K_f K_c K_d f \phi T = 4 \times 1.11 \times 0.966 \times 1 \times 50 \times \phi \times 64 = 254$$

$$\phi = 18.5 \text{ mWb}$$

For a single-layer winding, No. of coils = No. of slots / 2 = 24

Since number of turns/coil is 4, no. of turns per phase $T = \frac{24 \times 4}{3} = 32$

(Note : No. of conductors / slot = 4)

$$E_{ph} = 4K_f K_c K_d f \phi T = 4 \times 1.11 \times 0.966 \times 1 \times 50 \times \phi \times 32 = 254$$

$$\phi = 37 \text{ mWb}$$

Qn4 For a 3-phase winding with 3 slots/pole/phase and coil span of 8 slots, compute the coil span factor and distribution factors. The flux density wave in the air gap of this machine is found to have 20% third harmonic. Calculate the percentage increase in per phase r.m.s. emf due to harmonic.

Solution: $m = 3$, pole pitch = $3 \times 3 = 9$ slots

$$\beta = \frac{180^\circ}{9} = 20^\circ$$

$$K_{d1} = \frac{\sin(m\beta/2)}{m \sin(\beta/2)} = \frac{\sin 30^\circ}{3 \sin 10^\circ} = 0.9598$$

$$\alpha = 20^\circ$$

$$K_{c1} = \cos \frac{\alpha}{2} = \cos \frac{20^\circ}{2} = 0.985$$

$$K_{w1} = K_{c1} K_{d1} = 0.9494$$

$$K_{d3} = \frac{\sin(mn\beta/2)}{m \sin(n\beta/2)} = \frac{\sin 90^\circ}{3 \sin 30^\circ} = 0.667$$

$$K_{c3} = \cos \frac{n\alpha}{2} = \cos \frac{60^\circ}{2} = 0.866$$

$$K_{w3} = K_{c3} K_{d3} = 0.5776$$

$$\frac{E_{ph3}}{E_{ph1}} = \frac{K_{w3} B_n}{K_{w1} B_1} = \frac{0.5776}{0.9454} \times 0.2 = 0.1222$$

$$E_{ph} = \sqrt{E_{ph1}^2 + E_{ph3}^2} = 1.00744 E_{ph1}$$

$$\% \text{ increase in generated emf} = \frac{1.00744 - 1}{1} \times 100 = 0.744\%$$

Qn5 A 3-phase star-connected alternator has 81 slots, 6 poles and double-layer short-pitched winding with coil-span equal to 13 slots. The flux density distribution in the air gap is given by $B(\theta) = \sin \theta + 0.4 \sin 3\theta + 0.25 \sin 5\theta$. Determine the rms value of third and fifth harmonic phase emfs in terms of fundamental phase emf. Also, calculate the ratio of resultant line emf to the resultant phase emf.

$$m = \frac{81}{3 \times 6} = \frac{9}{2}$$

$$\text{Slot pitch} = \beta = \frac{180}{(81/3)} = \frac{40^\circ}{3}$$

$$\text{Coil span} = 13 \times \beta = \frac{13 \times 40}{3} = \frac{520^\circ}{3}$$

$$\text{Chording angle } \alpha = 180 - \frac{520^\circ}{3} = \frac{20^\circ}{3}$$

$$K_{d1} = \frac{\sin(60/2)}{9 \sin(60/2 \times 9)} = 0.9555$$

$$K_{d3} = \frac{\sin(90)}{9 \sin(60/6)} = 0.6399$$

$$K_{d5} = \frac{\sin(150)}{9 \sin(300/18)} = 0.1937$$

$$K_{c1} = \cos \frac{\alpha}{2} = \cos \frac{20^\circ}{3 \times 2} = 0.9988$$

$$K_{c3} = \cos 3 \times \frac{10^\circ}{3} = 0.9848$$

$$K_{c5} = \cos 5 \times \frac{10^\circ}{3} = 0.958$$

$$K_{w1} = K_{c1} K_{d1} = 0.9539$$

$$K_{w3} = K_{c3} K_{d3} = 0.6302$$

$$K_{w5} = K_{c5} K_{d5} = 0.1856$$

$$E_{ph3} = E_{ph1} \frac{K_{w3}}{K_{w1}} \frac{B_n}{B_1} = \frac{0.6302}{0.9539} \times 0.4 = 0.2643 E_{ph1}$$

$$E_{ph5} = E_{ph1} \frac{K_{w5}}{K_{w1}} \frac{B_n}{B_1} = \frac{0.1856}{0.9539} \times 0.25 = 0.0486 E_{ph1}$$

$$E_{ph} = \sqrt{E_{ph1}^2 + E_{ph3}^2 + E_{ph5}^2} = 1.0355 E_{ph1}$$

$$E_L = \sqrt{3} \times \sqrt{E_{ph1}^2 + E_{ph5}^2} = 1.734 E_{ph1}$$

$$\frac{E_L}{E_{ph}} = \frac{1.734}{1.0355} = 1.675$$

Previous University Questions :

01. Find the distribution and pitch factor of a 3-phase, 4-pole, 24 slots alternator having its armature coils short pitched by one slot. (APJAKTU July 2017)
02. Two alternators run on the same shaft (ie, runs at same speed). One generates voltage at 50Hz and the other at 40Hz. Find the maximum possible speed. Also find another possible speed. (APJAKTU April 2018) (Hint: $N = \frac{120 \times 50}{P_1} = \frac{120 \times 40}{P_2}$; $\frac{P_1}{P_2} = \frac{5}{4} = \frac{10}{8} = \frac{20}{16}$; $N = 600\text{rpm}$ or 300rpm)
03. The armature of 60Hz 11kV 450 rpm star connected alternator has 8 conductors per slot. The winding is short chorded by 3 slots to eliminate 5th harmonics completely. Find the flux per pole to generate rated voltage on open circuit. (APJAKTU April 2018) (Hint : $\cos \frac{5 \times \alpha}{2} = 0$ or $\alpha = 36^\circ$; $3\beta = 36^\circ$; $\beta = 12^\circ$; $S = 36$ slots; $Z = \frac{8 \times 36}{3} = 96$; $T = 48$)
04. A 3-phase, 10-pole alternator has 2 slots/pole/phase on its stator with 10 conductors per slot. The air gap flux is sinusoidally distributed and equals 0.05Wb. The stator has a double layer winding with a coil span of $150^\circ E$. If the alternator is running at 600rpm, calculate the emf generated/phase at no load. (APJAKTU December 2018)

Synchronous generator on no load

When a synchronous generator is driven at constant speed, the terminal voltage on open circuit is,

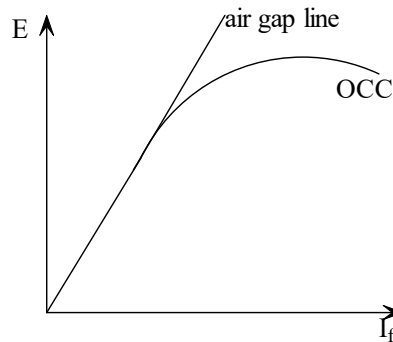
$$E = K \cdot \Phi \quad \text{where} \quad K = 4K_f K_c K_d f T$$

$$\phi \propto I_f$$

Therefore, $E \propto I_f$

If the reluctance of the magnetic flux path through the iron is neglected and only the reluctance of air-gap is considered, the open circuit characteristic (OCC) (E Vs I_f) will be a straight line.

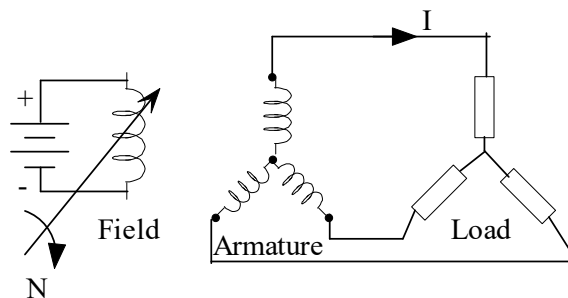
However, if the reluctance of both air-gap and iron path are considered, the OCC will be as shown in figure.



The deviation of the OCC from a straight line is due to the saturation effect of the iron, i.e. at higher values of I_f , the rate of rise of induced emf gets reduced.

Synchronous generator on load

When a synchronous generator is running on no-load, there will be no current flowing through the armature windings. The flux produced in the air-gap will be due to the field ampere-turns only.



When load is connected across the armature terminals, current will flow through the armature windings. This three phase currents will produce a rotating magnetic field in the air-gap. The effect of the armature flux on the flux produced by the field ampere-turns is called armature reaction. The armature flux will distort, oppose or aid the field flux causing reduction or increase in the air-gap flux depending upon the power factor of the load.

ARMATURE REACTION

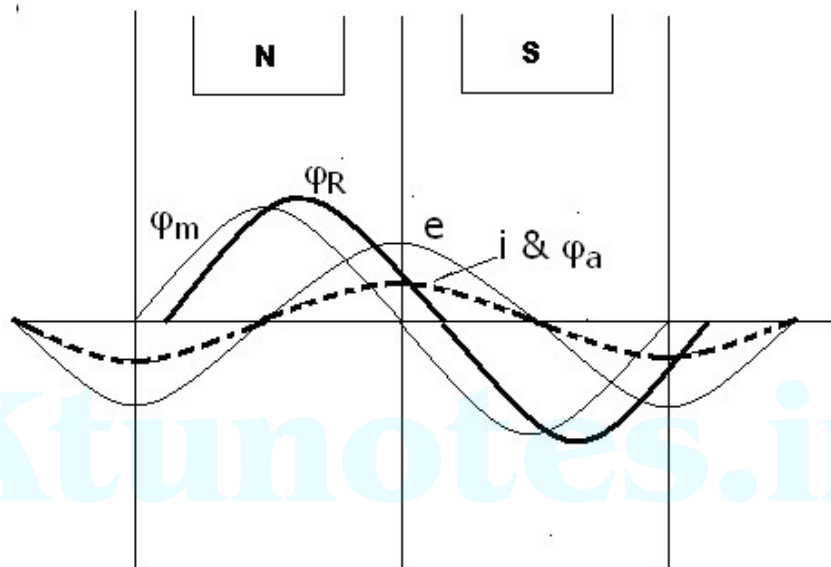
The effect of the armature flux (produced by the armature current) on the main flux (produced by the field winding) is called armature reaction. The effect of the armature flux depends on the magnitude of the armature current and load power factor.

Main flux ϕ_m available in the air-gap is sinusoidal. Sinusoidal main flux produces a sinusoidal emf and hence a sinusoidal current. Sinusoidal armature current causes the armature flux also to be sinusoidal. Armature flux ϕ_a and main flux ϕ_m are rotating at the same speed in same direction. Hence the relative speed between ϕ_a and ϕ_m is zero or they are stationary in space.

Note : In ac machines, the induced emfs in the coils will be sinusoidal only if the flux density wave in the air gap is sinusoidal. This is achieved in the salient-pole construction (with concentrated field coils) by providing non-uniform air gap above pole-shoes; minimum air gap in the middle of the pole-shoe progressively increasing towards outer edges. Another method of obtaining sinusoidal flux-density wave in the air gap is the use of non-salient pole construction (cylindrical rotor with uniform air gap but with a suitable distributed field winding along the rotor periphery).

Armature reaction at unity power factor

Consider a purely resistive load connected to the alternator having unity power factor.

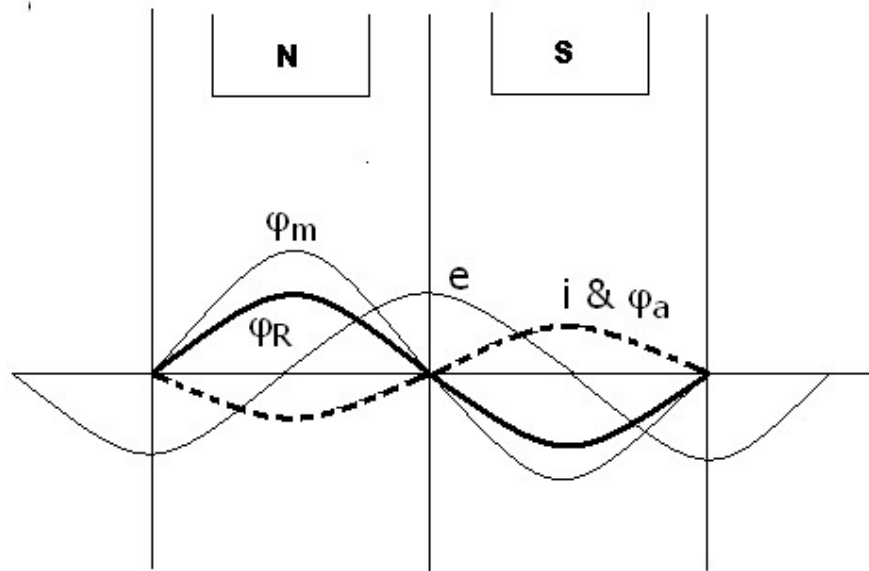


Induced emf e lags behind ϕ_m by 90° ($e \propto -d\phi/dt$). Since power factor is unity, e and i will be in phase. i sets up a flux ϕ_a in phase with i . Resultant flux in the air-gap ϕ_R will be the sum of ϕ_a and ϕ_m . Here, ϕ_a lags behind ϕ_m by 90° since power factor is unity.

Due to armature reaction, flux in the first quarter portion is decreased and in the second quarter portion flux is increased. If saturation is neglected, amount of flux increased in the first quarter and amount of flux decreased in the second quarter will be equal. Hence, the effect of armature reaction is only distortional or cross-magnetizing at unity power factor. If saturation is accounted (i.e. if the armature current is large), amount of flux increased in the second quarter will be less than the amount of flux decreased in the first quarter and hence the resultant flux ϕ_R will be less than ϕ_m in magnitude. Hence, in a saturated machine, the effect of armature reaction is both distortional and demagnetizing.

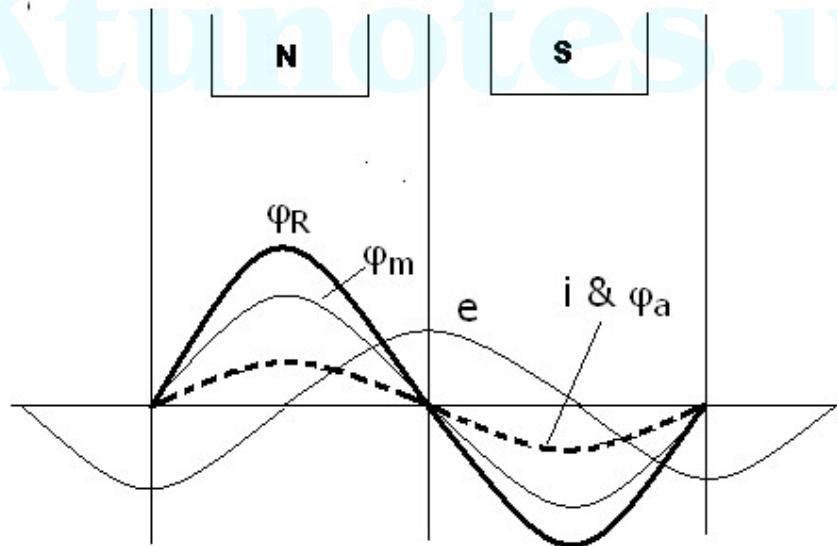
Armature reaction at zero power factor lagging

Induced emf e lags behind ϕ_m by 90° . Since power factor is zero lagging, i lags behind e by an angle 90° . i sets up a flux ϕ_a in phase with i . ϕ_a lags behind ϕ_m by 180° or ϕ_a is in phase opposition to ϕ_m . Resultant flux in the air-gap ϕ_R will be the sum of ϕ_a and ϕ_m . Resultant flux ϕ_R is decreased. Effect of armature reaction is purely demagnetizing for a purely inductive load (Zero power factor lagging). Since the resultant flux is decreased, generated emf will be decreased.



Armature reaction at zero power factor leading

Induced emf e lags behind ϕ_m by 90° . Since power factor is zero leading, i leads from e by an angle 90° . i sets up a flux ϕ_a in phase with i . ϕ_a is in phase with ϕ_m . Resultant flux in the air-gap ϕ_R is increased. Effect of armature reaction is purely magnetizing for a purely capacitive load (Zero power factor leading). Since the resultant flux is increased, generated emf will be increased.



For an R-L load ($0 < \text{p.f. lag} < 1$), effect of armature reaction is partly distortional and partly demagnetizing. For an R-C load ($0 < \text{p.f. lead} < 1$), effect of armature reaction is partly distortional and partly magnetizing.

The change in terminal voltage due to effect of armature reaction can be accounted by assuming the presence of a fictitious reactance X_{ar} in series with the armature.

Armature leakage reactance

When current flows in the armature (i.e. when the machine is loaded), it sets up armature flux. Interaction of the armature flux with the main flux causes energy conversion. The armature flux which links only with the armature winding not with the field winding is called leakage flux. The various leakage fluxes may be i) slot leakage flux, ii) overhang leakage flux and iii) zig-zag leakage flux. Slot leakage includes the fluxes that link the coils of a winding by crossing the slots in which the coils are

placed. Overhang leakage includes all the leakage fluxes around the coil ends protruding from the slots at each end of the core. Zig-zag leakage is due to fluxes produced by a given winding that cross the air gap but do not link the other winding.

The effect of this leakage flux is accounted by connecting a leakage reactance X_{al} in series with the armature.

Synchronous impedance

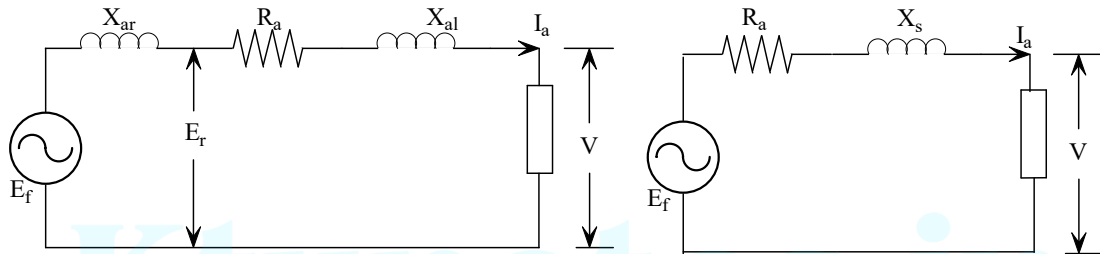
When a synchronous generator is loaded, there will be a change in the terminal voltage from that under no-load condition, due to :

- armature resistance R_a
- armature leakage reactance X_{al}
- effect of armature reaction X_{ar}

$$V = E_f - I_a R_a - jI_a X_{al} - jI_a X_{ar} = E_f - I_a R_a - jI_a (X_{al} + X_{ar})$$

$$\text{or } E_f = V + I_a R_a + jI_a (X_{al} + X_{ar}) = V + I_a R_a + jI_a X_s$$

where $X_s = X_{al} + X_{ar} = \text{Synchronous reactance}$



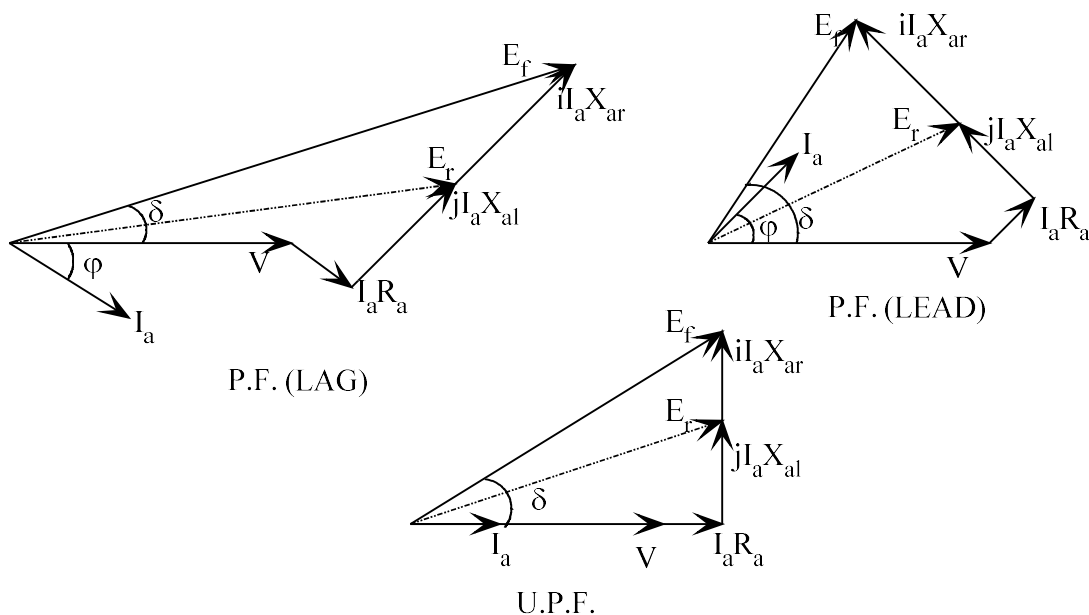
$$Z_s = R_a + jX_s = \text{synchronous impedance}$$

$$\bar{\phi}_r = \bar{\phi}_f + \bar{\phi}_{ar}$$

$$\bar{\phi}_r = \text{air gap flux (resultant flux)} \quad \bar{\phi}_f = \text{main flux} \quad \bar{\phi}_{ar} = \text{armature flux}$$

$$E_r = \text{air gap emf} \quad E_f = \text{excitation emf} \quad V = \text{terminal voltage}$$

PHASOR DIAGRAMS



δ = load angle = angle between E_f and V

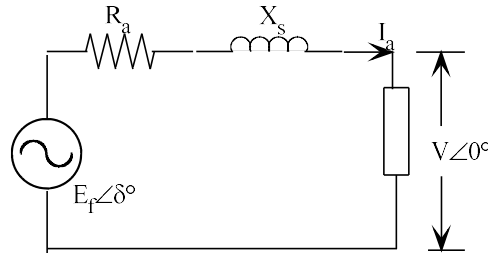
ϕ = power factor angle = angle between V and I_a

θ = synchronous impedance angle = $\tan^{-1} \frac{X_s}{R_a}$

Range of synchronous impedance

Practically, armature resistance R_a is usually of the order of 0.01 p.u. (i.e. the voltage drop in the armature resistance at rated armature current is about 1% of rated voltage). Leakage reactance value ranges from 0.1 to 0.2 p.u. and the synchronous reactance is of the order of 1 to 1.5 p.u. Armature resistance of a synchronous machine is so low that it can be neglected for all practical purposes except in the computation of losses, temperature rise and efficiency.

POWER FLOW EQUATIONS (Synchronous Generator)



V = Terminal voltage/phase

E_f = Excitation voltage/phase

I_a = Armature current

δ = Load or power angle (angle between E_f and V)

θ = Synchronous impedance angle = $\tan^{-1}(\frac{X_s}{R_a})$ (if R_a is neglected, $\theta = 90^\circ$)

ϕ = Power factor angle = angle between V and I_a .

$\bar{V} = V \angle 0^\circ$, $\bar{E}_f = E_f \angle \delta^\circ$

Synchronous impedance = $R_a + jX_s = Z_s \angle \theta$

Complex Power Output per phase

$$S_o = P_o + jQ_o = \bar{V} \times \bar{I}_a^*$$

$$= V \angle 0^\circ \times \left(\frac{E_f \angle \delta - V \angle 0}{Z_s \angle \theta} \right)^*$$

$$= V \angle 0^\circ \times \left(\frac{E_f}{Z_s} \angle (\delta - \theta) - \frac{V}{Z_s} \angle -\theta \right)^*$$

$$= \frac{VE_f}{Z_s} \angle (\theta - \delta) - \frac{V^2}{Z_s} \angle \theta$$

$$\text{Real power output per phase, } P_o = \frac{VE_f}{Z_s} \cos(\theta - \delta) - \frac{V^2}{Z_s} \cos \theta = \frac{VE_f}{Z_s} \cos(\theta - \delta) - \frac{V^2 R_a}{Z_s^2}$$

$$= \frac{VE_f}{Z_s} \cos(\theta - \delta) - \frac{V^2 R_a}{Z_s^2} \text{ since } \cos \theta = \frac{R_a}{Z_s}$$

= Electrical power developed by the generator

$$\text{Reactive power output per phase } Q_o = \frac{VE_f}{Z_s} \sin(\theta - \delta) - \frac{V^2}{Z_s} \sin \theta$$

Complex Power Input per phase

$$\begin{aligned} S_i &= P_i + jQ_i = \bar{E}_f \times \bar{I}_a^* \\ &= E_f \angle \delta \times \left(\frac{E_f \angle \delta - V \angle 0}{Z_s \angle \theta} \right)^* \\ &= E_f \angle \delta \times \left(\frac{E_f}{Z_s} \angle (\delta - \theta) - \frac{V}{Z_s} \angle -\theta \right)^* \\ &= \frac{E_f^2}{Z_s} \angle \theta - \frac{VE_f}{Z_s} \angle (\theta + \delta) \end{aligned}$$

$$\text{Real power input per phase, } P_i = \frac{E_f^2}{Z_s} \cos \theta - \frac{VE_f}{Z_s} \cos(\theta + \delta)$$

$$\text{Reactive power input per phase } Q_i = \frac{E_f^2}{Z_s} \sin \theta - \frac{VE_f}{Z_s} \sin(\theta + \delta)$$

Mechanical power input to generator = P_i + rotational losses

Rotational losses = Core loss + Mechanical loss

Core loss = Eddy current loss + Hysteresis loss

Mechanical loss = Friction loss + Windage loss

Maximum Power Output per phase

For maximum power output, $\frac{dP_o}{d\delta} = 0$

$$\frac{d}{d\delta} \left(\frac{VE_f}{Z_s} \cos(\theta - \delta) - \frac{V^2 R_a}{Z_s^2} \right) = 0$$

Since V , E_f , Z_s and R_a are constants,

$$\sin(\theta - \delta) = 0 \text{ or } \delta = \theta$$

Maximum power output occurs if load angle = impedance angle

$$\text{Maximum power output, } P_{o\max} = \frac{VE_f}{Z_s} - \frac{V^2 R_a}{Z_s^2}$$

If R_a is neglected, for maximum output, $\delta = 90^\circ$ and $P_{o\max} = \frac{VE_f}{X_s}$

Maximum Power Input per phase

For maximum power input, $\frac{dP_i}{d\delta} = 0$

$$\frac{d}{d\delta} \left(\frac{E_f^2}{Z_s} \cos \theta - \frac{VE_f}{Z_s} \cos(\theta + \delta) \right) = 0$$

Since V , E_f , Z_s and θ are constants,

$$\sin(\theta + \delta) = 0 \text{ or } \theta + \delta = 180^\circ$$

For maximum power input, $\delta = 180 - \theta$.

$$\text{Maximum power input} = P_{i\max} = \frac{E_f^2}{Z_s} \cos \theta + \frac{VE_f}{Z_s}$$

If R_a is neglected, $\delta = 90^\circ$, $P_{i\max} = \frac{VE_f}{X_s}$.

Q1 : A 3-phase, 11kV, 10MW, star connected synchronous generator has a synchronous impedance of $(0.8+j8)\Omega/\text{phase}$. If the excitation is such that the open-circuit voltage is 14kV, determine i) the maximum output of the generator and ii) the current and power factor at the maximum output.

$$\text{Ans: } V = \frac{11000}{\sqrt{3}} = 6350.85V$$

$$E_f = \frac{14000}{\sqrt{3}} = 8082.9V$$

$$Z_s = 0.8 + j8 = 8.04 \angle 84.3^\circ$$

$$\begin{aligned} \text{Maximum power output/phase} &= P_{o\max} = \frac{VE_f}{Z_s} - \frac{V^2}{Z_s} \cos \theta \\ &= \frac{6350.85 \times 8082.9}{8.04} - \frac{6350.85^2}{8.04} \cos 84.3^\circ \\ &= 5.886 \text{ MW} \end{aligned}$$

$$\text{Maximum power output} = 3 \times P_{o\max} = 17.66 \text{ MW}$$

For maximum output, $\delta = \theta = 84.3^\circ$

$$I_a = \frac{E_f \angle \delta - V \angle 0}{Z_s \angle \theta} = \frac{8082.9 \angle 84.3 - 6350.85 \angle 0}{8.04 \angle 84.3} = 1215.28 \angle 40.3^\circ A$$

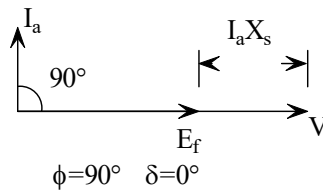
$$\text{Power factor} = \cos 40.3^\circ = 0.763 \text{ lead}$$

Q2 : A given 3 MVA, 50Hz, 11kV, 3-phase, star connected alternator, when supplying 100A at zero p.f. leading has line to line voltage of 12370V; when load is removed the terminal voltage falls down to 11000V. Predict the regulation of the alternator when supplying full-load at 0.8 pf lag. Neglect armature resistance.

$$\text{Ans: } V = \frac{12370}{\sqrt{3}} = 7142V$$

$$E_f = \frac{11000}{\sqrt{3}} = 6351V$$

$$I_a = \frac{3000}{\sqrt{3} \times 11} = 157.46A$$



$$X_s = \frac{V - E_f}{I_a} = \frac{7142 - 6351}{100} = 7.91 \Omega$$

At 0.8 p.f. lag,

$$E_f = 6351 + 157.46 \angle -36.87^\circ \times j7.91 = 7167.9 \angle 8^\circ V$$

$$\text{V.R.} = \frac{7167.9 - 6351}{6351} \times 100 = 12.86\%$$

Q3 : A 3-phase star-connected alternator has a synchronous impedance of $1 + j10 \Omega$ per phase. It is operating at a constant voltage of 6.6kV and its field current is adjusted to give an excitation voltage of 6.4kV. Find i) the maximum power output ii) armature current and power factor at maximum power output iii) maximum power input and iv) armature current and power factor at maximum power input.

$$\text{Ans: } V = \frac{6600}{\sqrt{3}} = 3810.5V$$

$$E_f = \frac{6400}{\sqrt{3}} = 3695V$$

$$Z_s = 1 + j10 = 10.05 \angle 84.3^\circ$$

$$\begin{aligned} \text{i) Maximum power output/phase} &= P_{o\max} = \frac{VE_f}{Z_s} - \frac{V^2}{Z_s} \cos \theta \\ &= \frac{3810.5 \times 3695}{10.05} - \frac{3810.5^2}{10.05} \cos 84.3^\circ \\ &= 1257.48 \text{ KW} \end{aligned}$$

$$\text{Maximum power output} = 3 \times P_{i\max} = 3772.4 \text{ KW}$$

For maximum output, $\delta = \theta = 84.3^\circ$

$$I_a = \frac{E_f \angle \delta - V \angle 0}{Z_s \angle \theta} = \frac{3695 \angle 84.3 - 3810.5 \angle 0}{10.05 \angle 84.3} = 501.2 \angle 48.8^\circ \text{ A}$$

Power factor = $\cos 48.8^\circ = 0.659$ lead

$$\begin{aligned} \text{ii) Maximum power input/phase} &= P_{i\max} = \frac{E_f^2}{Z_s} \cos \theta + \frac{VE_f}{Z_s} \\ &= \frac{3695^2}{10.05} \cos 84.3^\circ + \frac{3695 \times 3810.5}{10.05} \\ &= 1535.9 \text{ KW} \end{aligned}$$

$$\text{Maximum power input} = 3 \times P_{i\max} = 4607.7 \text{ KW}$$

For maximum input, $\delta = 180 - \theta = 95.7^\circ$

$$I_a = \frac{E_f \angle \delta - V \angle 0}{Z_s \angle \theta} = \frac{3695 \angle 95.7 - 3810.5 \angle 0}{10.05 \angle 84.3} = 553.74 \angle 54.35^\circ \text{ A}$$

Power factor = $\cos 54.35^\circ = 0.583$ lead

(Note: Maximum Power input = $3E_f I_a \cos(\delta - \phi)$)

$$= 3 \times 3695 \times 553.74 \cos(95.7 - 54.35) = 4607.8 \text{ kW}$$

Q4 : A 3-phase star-connected alternator with synchronous impedance of $0 + j5\Omega$ per phase is connected to a 11 kV system. The alternator power output is found to be 10MW and reactive power output is 3MVA. Compute the magnitude of the excitation voltage, load angle, line current and power factor.

$$S_o = P_o + jQ_o = \bar{V} \times \bar{I}_a^*$$

$$S_o = 10 + j3 = 10.44 \angle 16.7^\circ \text{ MVA}$$

$$S_o = \bar{V} \times \bar{I}_a^* = 10.44 \angle 16.7^\circ$$

$$\bar{I}_a^* = \frac{S_o}{V \angle 0^\circ} = \frac{10.44 \times 10^6 \angle 16.7^\circ}{\sqrt{3} \times 11 \times 10^3} = 547.96 \angle 16.7^\circ$$

$$\bar{I}_a = 547.96 \angle -16.7^\circ$$

$$E_f = 6351 + 547.96 \angle -16.7^\circ \times (j5) = 7605.4 \angle 20.18^\circ \text{ V}$$

$$E_f(\text{line}) = 13172.9 \text{ V}$$

Load angle $\delta = 20.18^\circ$

Power factor $\cos \phi = \cos(-16.7) = 0.958 \text{ lag}$

LOSSES & EFFICIENCY

The various losses in synchronous machines are

- i) No-load rotational losses (friction and windage loss, open-circuit core loss)
- ii) Field circuit loss (I^2R loss in the field winding)
- iii) Direct load loss (I^2R loss in the armature winding)

- iv) Stray load losses (core loss in the teeth, core end-plates etc due to armature leakage flux, eddy current loss in the armature conductors due to armature leakage flux)

Direct load loss + stray load loss = short circuit load loss

Qn. 1 : A 100kVA, 440V, 3-phase, star-connected alternator has the following data: Friction & windage loss = 340W, open-circuit core loss = 480W, field winding resistance at 75°C = 180Ω, effective armature resistance per phase = 0.02Ω. The voltage applied to field winding is 220V. Calculate the alternator efficiency at 0.8 pf and at i) half full load and ii) full load.

$$\text{Ans: } I_a = \frac{100000}{\sqrt{3} \times 440} = 131.22A$$

$$\text{Field circuit loss} = \frac{V_f^2}{R_f} = \frac{220^2}{180} = 268.89W$$

a) At half full load,

$$\text{Direct load loss (Armature Cu loss)} = 3 \times \left(\frac{I_a}{2}\right)^2 \times R_a = 3 \times \left(\frac{131.22}{2}\right)^2 \times 0.02 = 258.28W$$

$$\text{Total losses at half full load} = 340 + 480 + 258.28 + 268.89 = 1347.17W$$

$$\text{Output at half full load} = \frac{1}{2} \times 100000 \times 0.8 = 40000W$$

$$\text{Input at half full load} = 40000 + 1347.17 = 41347.17W$$

$$\text{Efficiency at half full load} = \frac{\text{output}}{\text{input}} \times 100 = \frac{40000}{41347.17} \times 100 = 96.742\%$$

b) At full load,

$$\text{Direct load loss (Armature Cu loss)} = 3 \times I_a^2 \times R_a = 3 \times 131.22^2 \times 0.02 = 1033.12W$$

$$\text{Total losses at full load} = 340 + 480 + 1033.12 + 268.89 = 2122W$$

$$\text{Output at full load} = 100000 \times 0.8 = 80000W$$

$$\text{Input at full load} = 80000 + 2122 = 82122W$$

$$\text{Efficiency at full load} = \frac{\text{output}}{\text{input}} \times 100 = \frac{80000}{82122} \times 100 = 97.416\%$$

Module 2

Voltage regulation of three phase Alternators-Direct loading, EMF Method, MMF Method, Potier Method, ASA Method -Numerical Problems.

Blondel's two reaction theory, Phasor Diagram under lagging power factor, Determination of X_d and X_q by slip test, Power developed by a Salient pole machine, Numerical Problems.

Parallel Operation of Alternators- Necessary Conditions, Synchronisation- Synchronising current, Power and Torque, Effect of reactance, Numerical Problems, Methods of Synchronisation.

VOLTAGE REGULATION OF ALTERNATOR

It is defined as the change in terminal voltage, expressed as a percentage of the rated voltage, when the load at a given power factor is removed, with speed and field current remaining unchanged.

$$\text{Voltage regulation} = \frac{E_f - V}{V} \times 100\%$$

E_f = no-load excitation voltage

V = full-load terminal voltage at the same speed and field excitation

For a lagging power factor load, E_f always increases (voltage regulation always positive) and for a leading power factor load, E_f may decrease (voltage regulation may be positive or negative).

In case of small alternators, the voltage regulation can be obtained by actually loading it. In large machines, it may not be possible to obtain the voltage regulation by actual loading because of huge wastage of power.

Simple tests (open circuit & short circuit tests) can be conducted to predetermine the voltage regulation without conducting load test.

Q1 : A 1200 kVA, 6600V, 3-phase star-connected alternator has $R_a = 0.25\Omega/\text{phase}$ and $X_s = 5\Omega/\text{phase}$. Calculate the voltage regulation if it delivers full load current at i) 0.8 p.f. lag ii) 0.8 p.f. lead and iii) unity p.f.

Ans: $\sqrt{3}V_L I_L \times 10^{-3} = \text{kVA}$

$$I_a = \frac{1200}{\sqrt{3} \times 6600} \times 1000 = 104.97 \text{ A}$$

$$V = \frac{V_L}{\sqrt{3}} = \frac{6600}{\sqrt{3}} = 3810.5 \text{ V}$$

a) 0.8 p.f. lag

$$\Phi = 36.87^\circ$$

$$E_f = V + I_a(R_a + jX_s) = 3810\angle 0^\circ + 104.97\angle -36.87^\circ \times (0.25 + j5) = 4165.56\angle 5.57^\circ \text{ V}$$

$$\text{V.R.} = \frac{4165.56 - 3810.512}{3810.512} \times 100 = 9.32\%$$

b) 0.8 p.f. lead

$$\Phi = 36.87^\circ$$

$$E_f = V + I_a(R_a + jX_s) = 3810\angle 0^\circ + 104.97\angle +36.87^\circ \times (0.25 + j5) = 3542.97\angle 7.06^\circ \text{ V}$$

$$\text{V.R.} = \frac{3542.97 - 3810.512}{3810.512} \times 100 = -7.02\%$$

c) Unity p.f.

$$\Phi = 0^\circ$$

$$E_f = V + I_a(R_a + jX_s) = 3810\angle 0^\circ + 104.97\angle 0^\circ \times (0.25 + j5) = 3871.98\angle 7.79^\circ \text{ V}$$

$$V.R. = \frac{3871.97 - 3810.512}{3810.512} \times 100 = 1.61\%$$

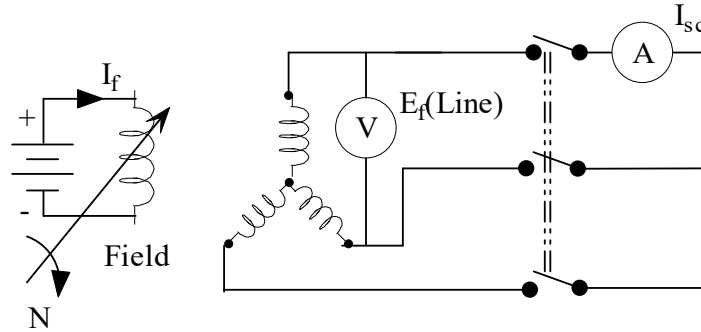
Q2 : Find the voltage regulation at full-load 0.8 p.f. lagging for a 3-phase, 1500 kVA, 6600V, 3-phase star-connected alternator having $R_a = 0.093 \Omega/\text{phase}$ and $X_s = 8.5 \Omega/\text{phase}$.

(Ans: $V=3810.5V$, FL current = 131.22A, $E_f = 4575.9V$, V.R. = 20.09%)

PREDETERMINATION OF VOLTAGE REGULATION

A. EMF method or SYNCHRONOUS IMPEDANCE method (Pessimistic Method)

The synchronous impedance can be found from the results of open circuit (OC) and short circuit (SC) tests on an alternator.

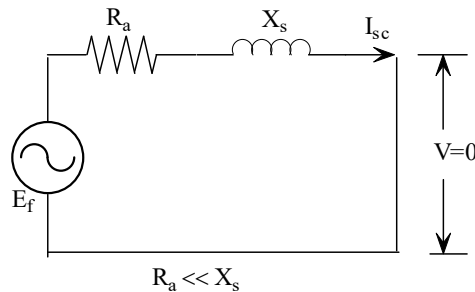


OC TEST

The alternator is driven at synchronous speed, and the open circuit terminal voltage E_f (per phase) is measured as the field current I_f is varied. The curve showing the variation of E_f with I_f is known as the open-circuit characteristic (OCC). For low values of field current, this characteristic is a straight line. However, for higher values of field current, the characteristic departs from the straight line due to the saturation of magnetic circuit.

SC TEST

Short circuit test is performed by running the alternator at synchronous speed. Keeping the output terminals short circuited through an ammeter, reduced excitation current is allowed to flow through the field winding. The relationship between short circuit armature current I_{sc} and the field current I_f gives the short circuit characteristics (SCC). (This characteristic is a straight line, because the air gap flux, under short circuit conditions, is only about 15% of that under rated voltage conditions. This is due to two reasons. Firstly the field current is very small. Secondly the power factor of the machine is about zero lagging, so that armature reaction is demagnetizing in nature. Therefore, there is no saturation of the magnetic circuit and the characteristic is a straight line.)



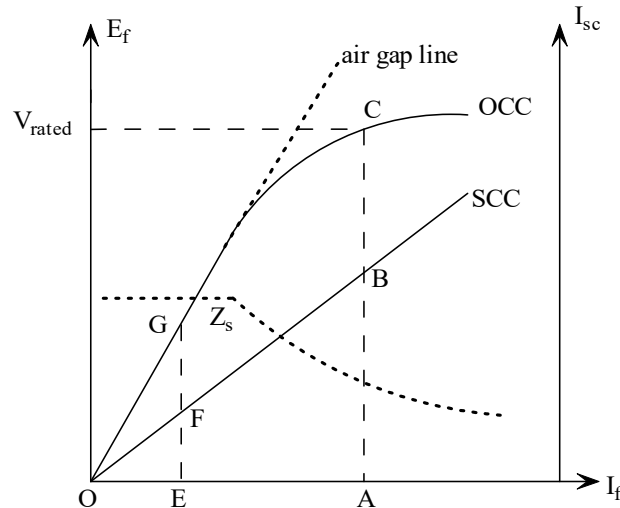
When the machine is short circuited, the emf generated is consumed as a voltage drop in the synchronous impedance (because the terminal voltage is zero).

For any particular value of I_f , the ratio of open circuit voltage and short circuit armature current gives the synchronous impedance Z_s .

$$X_s = \sqrt{Z_s^2 - R_a^2}$$

The armature resistance R_a is usually measured by the voltmeter-ammeter method. If OCC and SCC were both straight lines, synchronous impedance Z_s would be constant. SCC is always a straight line, but OCC includes a partially saturated region. Therefore Z_s decreases with increasing excitation.

If the value of open circuit voltage is read from the air gap line, the resulting value of Z_s is known as unsaturated synchronous impedance.



The saturated synchronous impedance at the rated voltage is obtained as follows:

$$Z_{s(sat)} = \frac{V_{rated}}{AB}$$

The unsaturated synchronous impedance is obtained as follows:

$$Z_{s(unsat)} = \frac{GE}{EF}$$

Since a synchronous machine under normal operating conditions works in a somewhat saturated region of the magnetization characteristic, the performance of the machine as calculated from $X_{s(unsat)}$, will differ considerably from the actual value during its normal operation.

Q1: The open circuit and short circuit test is conducted on a 3-phase, star connected, 220V, 15.25kVA alternator. The OC test results are,

I_f amp	0.2	0.4	0.6	0.8	1	1.2	1.4	1.8	2.2	2.6
E_f line volts	29	58	87	116	146	172	194	232	262	284

Field current of 1.2A produces a short circuit current of 40A. $R_a = 0.06\Omega/\text{phase}$. Calculate its full load regulation at 0.8 p.f. lag using emf method.

Ans:

$$\text{Full load current, } I_a = \frac{15250}{\sqrt{3} \times 220} = 40A$$

$$V = \frac{220}{\sqrt{3}} = 127V$$

From OCC, corresponding to rated current ($=40A$), $I_f = 1.2A$, $E_f = 99.3V$;

$$Z_s = \frac{99.3}{40} = 2.48\Omega$$

$$X_s = \sqrt{Z_s^2 - R_a^2} = \sqrt{2.48^2 - 0.06^2} = 2.48\Omega$$

$$\Phi = 36.87^\circ$$

$$E_f = V + I_a(R_a + jX_s) = 127 \angle 0^\circ + 40 \angle -36.87^\circ \times (0.06 + j2.48) = 203.9 \angle 22.46^\circ V$$

$$\text{V.R.} = \frac{203.9 - 127}{127} \times 100 = 60.55\%$$

From OCC, corresponding to rated voltage (=127V), $I_f = 1.7\text{A}$, $I_{sc} = 55\text{A}$;

$$Z_s = \frac{127}{55} = 2.31\Omega$$

$$X_s = \sqrt{Z_s^2 - R_a^2} = \sqrt{2.31^2 - 0.06^2} = 2.3\Omega$$

$$\Phi = 36.87^\circ$$

$$E_f = V + I_a(R_a + jX_s) = 127 \angle 0^\circ + 40 \angle -36.87^\circ \times (0.06 + j2.3) = 197.76 \angle 21.4^\circ V$$

$$\text{V.R.} = \frac{197.76 - 127}{127} \times 100 = 55.71\%$$

Note :

Actual value of voltage regulation can be determined by conducting load test on synchronous generator. But it causes excessive wastage of power. The synchronous reactance determined by emf method is large and it leads to high value of voltage regulation compared to actual value. Hence emf method is called pessimistic method.

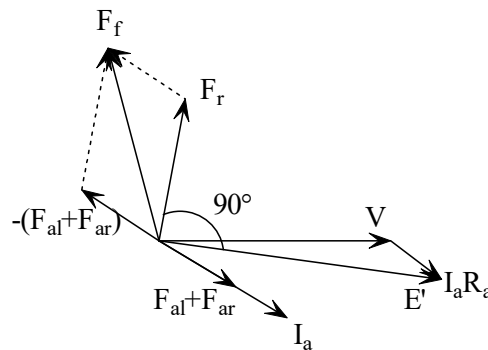
Minimum value of Z_s , corresponding to maximum short circuit current (allowed to flow through armature winding without excessive heating) should be used as far as possible. Z_s calculated corresponding to rated armature current will be higher than Z_s calculated corresponding to rated voltage. Since emf method is a pessimistic method (voltage regulation computed by emf method is much higher than the actual value), Z_s corresponding to rated voltage is preferred.

B. MMF method

$I_a R_a$ and $I_a X_{al}$ are emf quantities and armature reaction is an mmf quantity. In emf method, all are treated as emf quantities (armature reaction mmf is replaced by $I_a X_{ar}$) while in mmf method, all of them are treated as mmf quantities.

Armature reaction mmf F_{ar} is in phase with I_a . Armature leakage reactance drop $I_a X_{al}$ is transformed into equivalent mmf F_{al} . As F_{ar} is in phase with I_a , similarly F_{al} is also in phase with I_a .

Field mmf $\bar{F}_f = \bar{F}_r - (\bar{F}_{al} + \bar{F}_{ar})$ where $\bar{F}_r$ is the mmf corresponding to $\bar{E}' = \bar{V} + \bar{I}_a \times R_a$



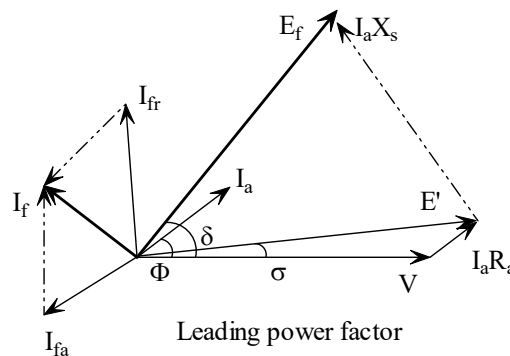
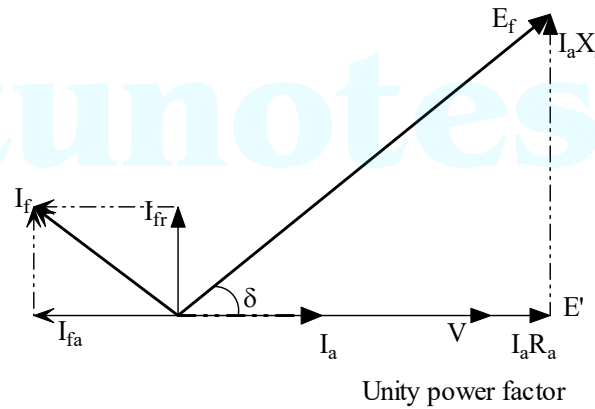
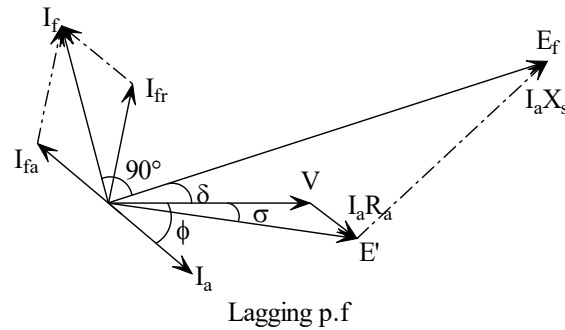
F_r (obtained from OCC corresponding to E') leads from E' by 90° .

$\bar{F}_{al} + \bar{F}_{ar}$ is the mmf (field current) required to circulate full-load armature current under short circuit test. $\bar{F}_{al} + \bar{F}_{ar}$ is in phase with the armature current I_a .

Steps to obtain voltage regulation by mmf method:

- 1) Plot O.C.C. & S.C.C.

- 2) Find $\bar{E}' = V \angle 0^\circ + I_a \angle^\pm \Phi \times R_a = |E'| \angle^\pm \sigma^\circ$ and obtain I_{fr} corresponding to E' from OCC.
 $\bar{I}_{fr} = I_{fr} \angle^\pm \sigma + 90^\circ$ (Note: $-\Phi$ for lag & $+\Phi$ for lead)
- 3) Find I_{fa} from SCC. I_{fa} is the field current to circulate full-load armature current on short circuit (from SC test). $\bar{I}_{fa} = I_{fa} \angle^\pm \Phi + 180^\circ$ for lag and $\bar{I}_{fa} = I_{fa} \angle^\pm \Phi + 180^\circ$ for lead
- 4) Find $\bar{I}_f = \bar{I}_{fr} + \bar{I}_{fa} = I_f \angle 90^\circ + \delta^\circ$
- 5) Corresponding to I_f , obtain E_f from OCC.
- 6) Voltage regulation = $\frac{E_f - V}{V} \times 100\%$



Note : MMF methods gives low value of voltage regulation compared to actual value. Hence mmf method is also called optimistic method.

Q1: The open circuit and short circuit test is conducted on a 3-phase, star connected, 220V, 15.25kVA alternator. The OC test results are,

I_f amp	0.2	0.4	0.6	0.8	1	1.2	1.4	1.8	2.2	2.6
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E_f line volts 29 58 87 116 146 172 194 232 262 284
 Field current of 1.2A produces a short circuit current of 40A. $R_a = 0.06\Omega/\text{phase}$. Calculate its full load regulation at 0.8 p.f. lag using mmf method.

Ans:

$$\text{Full load current, } I_a = \frac{15250}{\sqrt{3} \times 220} = 40A \quad V = \frac{220}{\sqrt{3}} = 127V$$

$$\bar{E}' = V \angle 0^\circ + I_a \angle -\Phi \times R_a = 127 \angle 0^\circ + 40 \angle -36.87^\circ \times 0.06 = 128.9 \angle -0.64^\circ V$$

$$I_{fr} = 1.7A \text{ (from OCC corresponding to } E_f = 128.9V)$$

$$\bar{I}_{fr} = I_{fr} \angle -\sigma + 90^\circ = 1.7 \angle 89.36^\circ$$

$$I_{fa} = 1.2A \text{ (from SCC corresponding to } I_{sc} = 40A)$$

$$\bar{I}_{fa} = I_{fa} \angle -\Phi + 180^\circ = 1.2 \angle 143.13^\circ A$$

$$\bar{I}_f = \bar{I}_{fr} + \bar{I}_{fa} = 1.7 \angle 89.36^\circ + 1.2 \angle 143.13^\circ = 2.6 \angle 111.25^\circ A$$

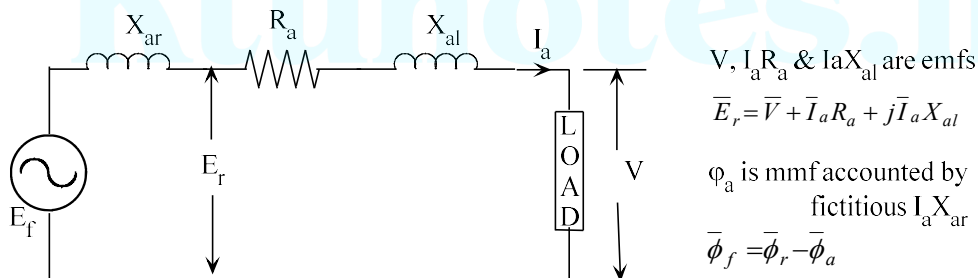
$$E_f = 165V \text{ (from OCC corresponding to } I_f = 2.6A)$$

$$\text{Voltage regulation} = \frac{165 - 125}{125} \times 100\% = 29.92\%$$

C. POTIER method

In Potier method, $I_a R_a$ and $I_a X_{al}$ which are emf quantities are treated as emfs and armature reaction which is an mmf quantity as mmf itself.

Potier method is based on the separation of armature leakage reactance and armature reaction mmf.

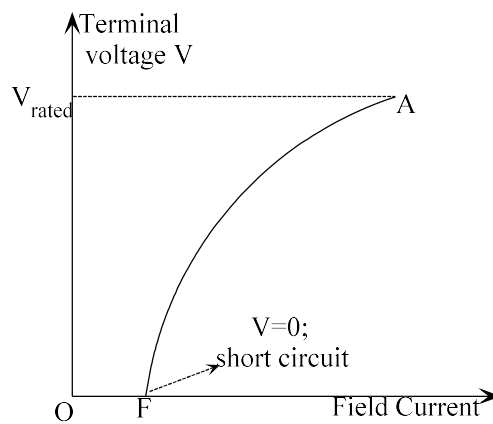


To determine the armature leakage reactance and armature reaction mmf separately, three tests are performed on the given alternator.

1. Open Circuit test
2. Short Circuit test &
3. Zero power factor test

Zero power factor characteristics:

Keeping the speed constant, the relationship between the voltage across pure inductive load and field current at same armature current (full load current) is called zero power factor characteristics. Only two points A and F are sufficient to complete the ZPF characteristics. Point A corresponds to rated voltage and point F corresponds to short circuit (ie. $V=0$). Zero power factor characteristics can be plotted by conducting i) zero power factor test at rated voltage and rated armature current and ii) short circuit test.

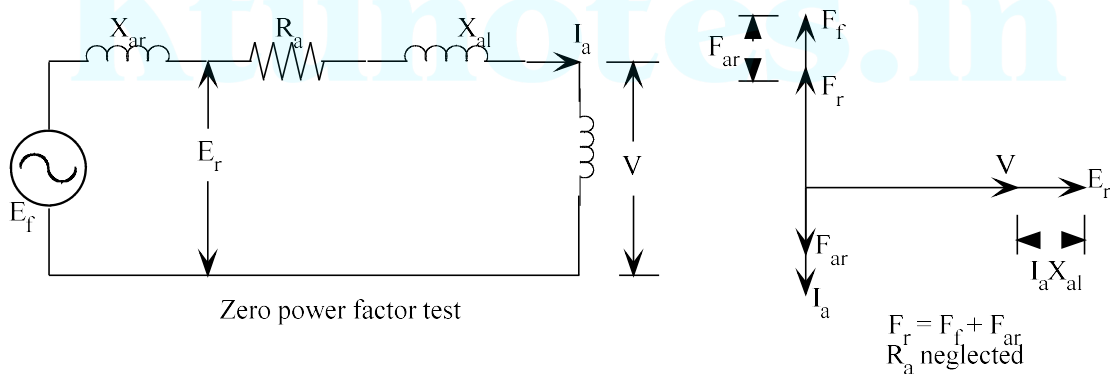


Zero Power Factor test:

Method 1) Alternator is run at synchronous speed by the prime mover. A purely inductive load is connected across the armature terminals. Vary the excitation I_f till the voltage becomes rated value. Vary the load inductance till the armature current becomes rated value. Note the field current. Point A on zero power factor characteristics corresponds to this field current and rated voltage.

Method 2) Alternator is connected in parallel with infinite bus bar. After synchronization, increase the excitation (Note: overexcited synchronous generator operates at lagging p.f.) till the armature current becomes rated value. Note the field current. Point A on zero power factor characteristics corresponds to this field current and rated voltage.

Point F on ZPF characteristics corresponds to field current to circulate full-load short circuit current during SC test.



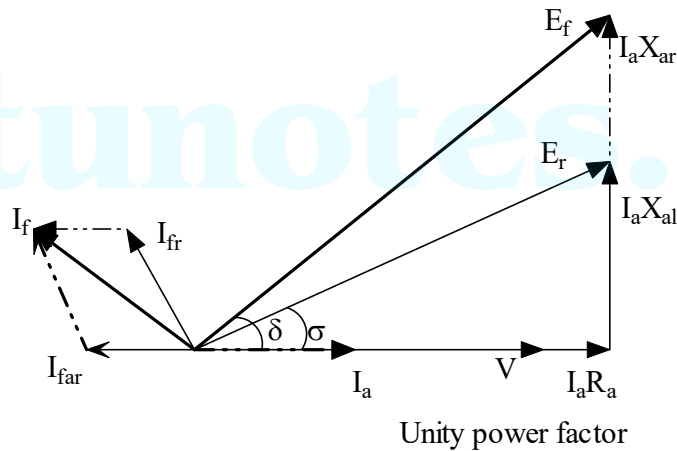
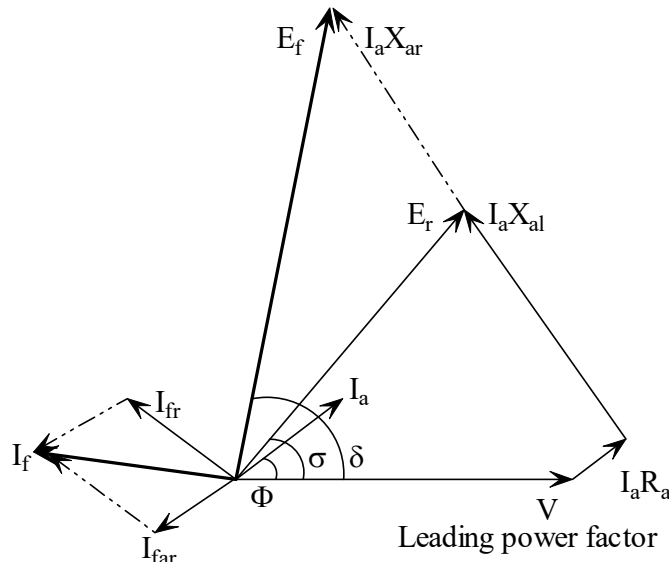
OF = AD = Field current required to deliver full load short circuit armature current (from SC test) = Field current required to overcome the demagnetizing effect of armature reaction and to balance the leakage reactance drop.

Draw line DC from D parallel to air gap line meeting OCC at C.

- 1) Plot O.C.C.
- 2) To plot Z.P.F.C., obtain the points F & A. Point F corresponds to full load armature current during short circuit. Point A corresponds to rated terminal voltage and rated armature current when the load is zero power factor lagging. Draw Potier triangle. Find X_{al} .
- 3) Find $\bar{E}_r = V \angle 0 + I_a \angle \pm \Phi \times (R_a + jX_{al}) = E_r \angle \sigma^\circ$. Refer OCC and find I_{fr} corresponding to E_r . $\bar{I}_{fr} = I_{fr} \angle 90 + \sigma^\circ$
- 4) Find I_{far} . $\bar{I}_{far} = I_{far} \angle 180 - \Phi^\circ$ for lag $\bar{I}_{far} = I_{far} \angle 180 + \Phi^\circ$ for lead.
- 5) Find $\bar{I}_f = \bar{I}_{fr} + \bar{I}_{far} = I_f \angle 90 + \delta^\circ$

6) Refer OCC and find E_f corresponding to I_f . $\bar{E}_f = E_f \angle \delta^\circ$

7) Voltage Regulation = $\frac{E_f - V}{V} \times 100\%$



Q1: The following data were obtained for the OCC of a 10MVA, 13kV, 3-phase, 50 Hz, star-connected synchronous generator:

I_f (A) :	50	75	100	125	150	162.5	200	250	300
$V_{oc}(\text{line})(\text{kV})$	6.2	8.7	10.5	11.8	12.8	13.2	14.2	15.2	15.9

An excitation of 100A causes the full-load current to flow during the short-circuit test. The excitation required to give the rated current at zero p.f. and rated voltage is 290A. Find the voltage regulation at full load and 0.8 p.f lag by a) emf method b) mmf method and c) potier method. Neglect armature resistance.

a) emf method

$$\text{Full load current, } I_a = \frac{10 \times 10^6}{\sqrt{3} \times 13000} = 444.1 \text{ A}$$

$$V = \frac{13000}{\sqrt{3}} = 7505.6 \text{ V}$$

From OCC, corresponding to rated voltage (=7505.6V), $I_f = 155 \text{ A}$, $I_{sc} = 700 \text{ A}$;

$$Z_s = \frac{7505.6}{700} = 10.72 \Omega$$

$$X_s = 10.72\Omega$$

$$\Phi = 36.87^\circ$$

$$E_f = V + I_a(R_a + jX_s) = 7505.6\angle 0^\circ + 444.1\angle -36.87^\circ \times j10.72 = 11039.8\angle 20.2^\circ V$$

$$V.R. = \frac{11039.8 - 7505.6}{7505.6} \times 100 = 47.1\%$$

b) mmf method

$$\bar{E}' = V\angle 0 + I_a\angle -\Phi \times R_a = 7505.6\angle 0 V$$

$$I_{fr} = 155 A \text{ (from OCC corresponding to } E' = 7505.6V)$$

$$\bar{I}_{fr} = I_{fr}\angle -\sigma + 90^\circ = 155\angle 90^\circ$$

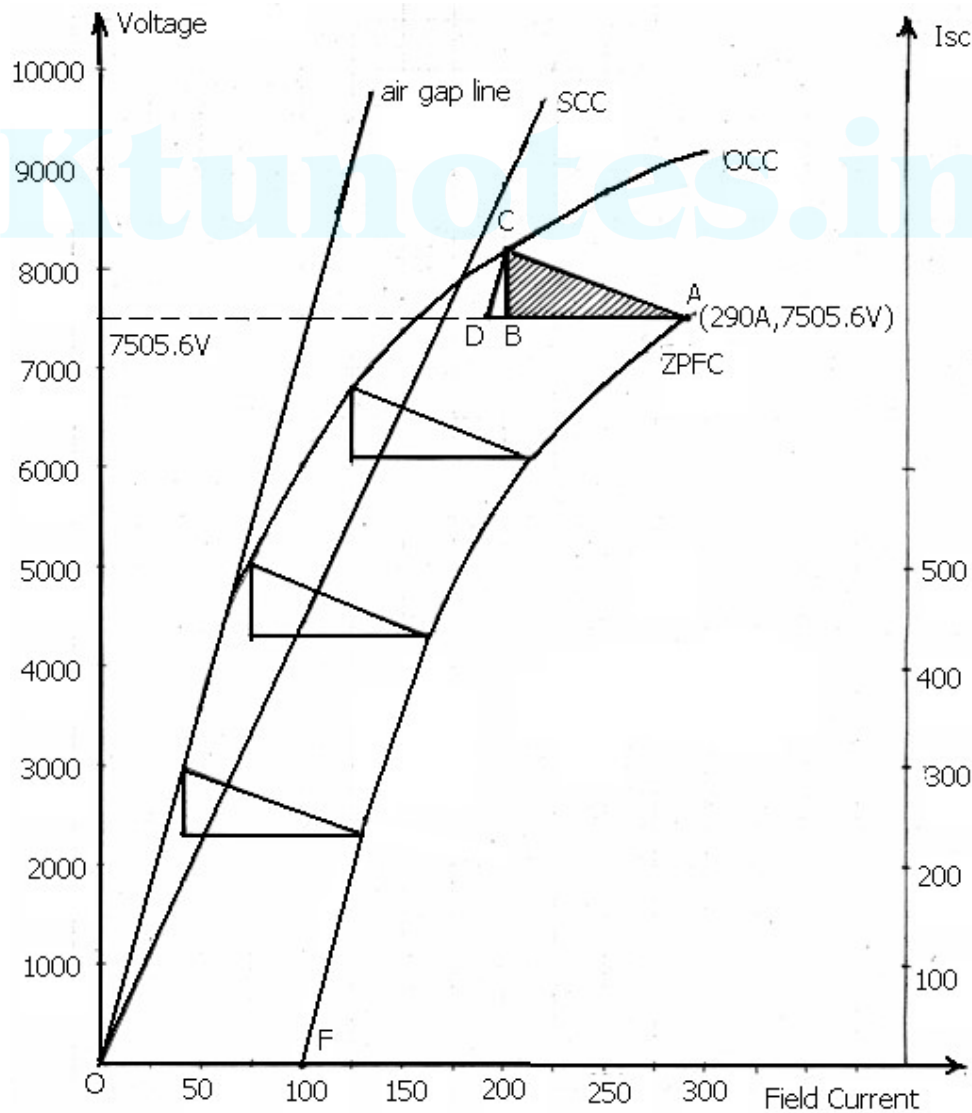
$$I_{fa} = 100 A \text{ (from SCC corresponding to } I_{sc} = 444.1 A)$$

$$\bar{I}_{fa} = I_{fa}\angle -\Phi + 180^\circ = 100\angle 143.13^\circ A$$

$$\bar{I}_f = \bar{I}_{fr} + \bar{I}_{fa} = 155\angle 90^\circ + 100\angle 143.13^\circ = 229.4\angle 110.4^\circ A$$

$$E_f = 8500 V \text{ (from OCC corresponding to } I_f = 229.4 A)$$

$$\text{Voltage regulation} = \frac{8500 - 7505.6}{7505.6} \times 100 = 13.25\%$$



c) POTIER method

$$X_{al} = \frac{BC}{I_a} = \frac{1.4 \times 500}{444.1} = 1.58 \Omega$$

$$\begin{aligned}\bar{E}_r &= V \angle 0^\circ + I_a \angle -\Phi \times (R_a + jX_{al}) = 7505.6 \angle 0^\circ + 444.1 \angle -36.87^\circ \times j1.58 \\ &= 7505.6 \angle 0^\circ + 444.1 \angle -36.87^\circ \times j1.58 = 7946.46 \angle 4.05^\circ = E_r \angle \sigma^\circ\end{aligned}$$

$I_{fr} = 185 \text{ A}$ (from OCC corresponding to 7946.46V)

$$\bar{I}_{fr} = I_{fr} \angle 90^\circ + \sigma^\circ = 185 \angle 94.05^\circ \text{ A}$$

$$I_{far} = AB = 3.6 \times 25 = 90 \text{ A}$$

$$\bar{I}_{far} = I_{far} \angle 180^\circ - \Phi^\circ = 90 \angle 143.13^\circ \text{ A}$$

$$\bar{I}_f = \bar{I}_{fr} + \bar{I}_{far} = 185 \angle 94.05^\circ + 90 \angle 143.13^\circ = 253.25 \angle 109.6^\circ = I_f \angle 90^\circ + \delta^\circ$$

$E_f = 8800 \text{ V}$ (corresponding to $I_f = 253.25 \text{ A}$)

$$\bar{E}_f = 8800 \angle 19.6^\circ$$

$$\text{Voltage regulation} = \frac{8800 - 7505.6}{7505.6} \times 100\% = 17.25\%$$

D) ASA Modified MMF method

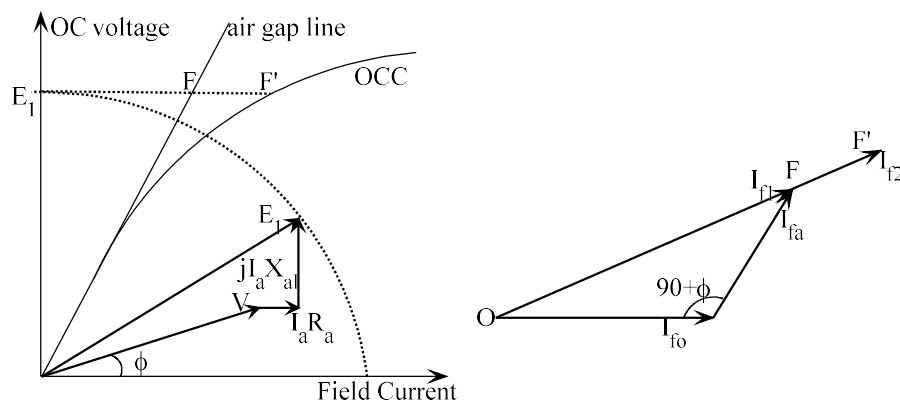
Neither of the two methods, MMF method and EMF method is capable of giving the reliable values of the voltage regulation due to the following reasons :

- In these methods, the magnetic circuit is assumed to be unsaturated. This assumption is unrealistic as in practice. It is not possible to have completely unsaturated magnetic circuit.
- In salient pole alternators, it is not correct to combine field ampere turns and armature ampere turns. This is because the field winding is always concentrated on a pole core while the armature winding is always distributed. Similarly, the field and armature mmfs act on magnetic circuits having different reluctances in case of salient pole machine. Hence phasor combination field and armature mmf is not fully justified.

In ASA (American Standards Association) modified MMF method, additional excitation FF' required to compensate for the partial saturation of the machine is found.

I_{f0} is the excitation required to produce rated terminal voltage on open circuit and I_{fa} is the mmf required to produce rated armature current on short circuit (from SC test). ϕ is the power factor angle. Here, resultant OF ($= I_{f1}$) is based on the assumption of unsaturated magnetic circuit which is not true in practice. Additional mmf FF' is required to take into account the effect of partially saturated magnetic circuit.

From the Potier triangle, determine the leakage reactance X_{al} and find $\bar{E}_1 = V \angle 0^\circ + I_a \angle -\Phi \times (R_a + jX_{al})$. From E_1 , draw a horizontal line intersecting air gap line at F and OCC at F'. FF' gives the additional excitation required to take into account the effect of partially saturated magnetic circuit. Adding FF' to I_{f1} , we get the total excitation I_{f2} . Corresponding to I_{f2} , we can determine the open circuit voltage E_f from OCC.



The results obtained by ASA method are reliable for both salient as well as non salient pole machines.

Q2: The following data were obtained for the OCC of a 10MVA, 13kV, 3-phase, 50 Hz, star-connected synchronous generator:

I _f (A) :	50	75	100	125	150	162.5	200	250	300
V _{oc} (line)(kV)	6.2	8.7	10.5	11.8	12.8	13.2	14.2	15.2	15.9

An excitation of 100A causes the full-load current to flow during the short-circuit test. The excitation required to give the rated current at zero p.f. and rated voltage is 290A. Find the voltage regulation at full load and 0.8 p.f lag by ASA method. Neglect armature resistance.

$$\text{Full load current, } I_a = \frac{10 \times 10^6}{\sqrt{3} \times 13000} = 444.1 \text{ A}$$

$$V = \frac{13000}{\sqrt{3}} = 7505.6 \text{ V}$$

$$X_{al} = \frac{BC}{I_a} = \frac{1.4 \times 500}{444.1} = 1.58 \Omega$$

$$\bar{E}_1 = V \angle 0^\circ + I_a \angle -\Phi \times (R_a + jX_{al}) = 7505.6 \angle 0^\circ + 444.1 \angle -36.87^\circ \times j1.58 = 7946.46 \angle 4.05^\circ$$

Corresponding to E₁, distance between airgap line and OCC, FF'=70A

$$I_{fo} = 155 \text{ A (from OCC corresponding to } V = 7505.6 \text{ V)}$$

$$I_{fa} = 100 \text{ A}$$

$$\bar{I}_{f1} = \bar{I}_{fo} + \bar{I}_{far} = 155 \angle 0^\circ + 100 \angle (90 - 36.87)^\circ = 229.4 \angle 20.4^\circ \text{ A}$$

$$I_{f2} = 229.4 + 70 = 299.4 \text{ A}$$

$$E_f = 9180 \text{ V (corresponding to } 299.4 \text{ A)}$$

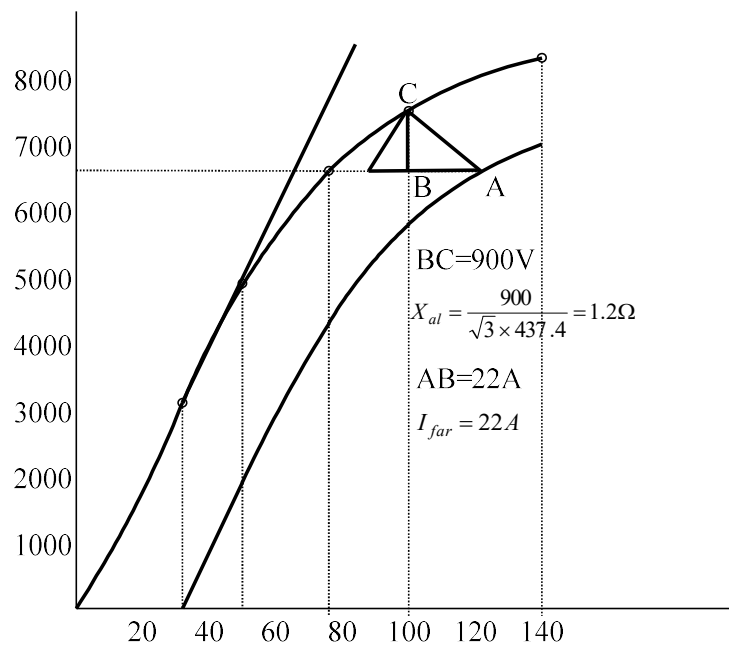
$$\text{Voltage regulation} = \frac{9180 - 7505.6}{7505.6} \times 100 = 22.3\%$$

Note : voltage regulation by mmf method is 13.25% only.

The following data pertains to a 5000kVA, 6600V, 3-phase, 50 Hz star connected alternator:-

Field current	32	50	75	100	140
OC Voltage	3100	4900	6600	7500	8300
ZPF voltage (line) (full load)	0	1850	4250	5800	7000

Determine the regulation by ZPF method at full load unity power factor. Neglect armature resistance. Draw its phasor diagram also.



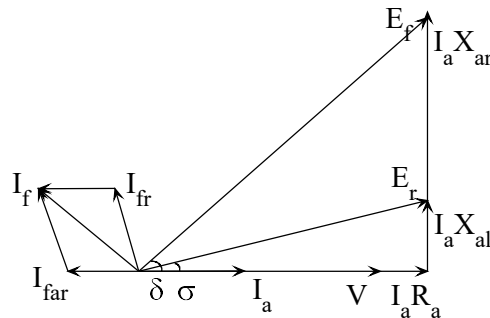
Full load current, $I_a = \frac{5000 \times 10^3}{\sqrt{3} \times 6600} = 437.4 A$

$V = \frac{6600}{\sqrt{3}} = 3810.5V$

Plot of OCC & ZPFC

$X_{al}=1.2\Omega$, $I_{far}=22A$ from Potier triangle (3 marks)

Phasor diagram



$$\bar{E}_r = V \angle 0^\circ + I_a \angle 0^\circ \times (R_a + jX_{al}) = 3810.5 \angle 0^\circ + 437.4 \angle 0^\circ \times j1.2 = 3846.5 \angle 7.8^\circ V = E_r \angle \sigma^\circ$$

$$E_r(\text{line}) = 6662V$$

$I_{fr} = 80A$ (from OCC corresponding to 6662V)

$$\bar{I}_{fr} = I_{fr} \angle 90^\circ + \sigma^\circ = 80 \angle 97.8^\circ A$$

$I_{far} = AB = 22A$

$$\bar{I}_{far} = I_{far} \angle 180^\circ - \Phi^\circ = 22 \angle 180^\circ A$$

$$\bar{I}_f = \bar{I}_{fr} + \bar{I}_{far} = 80 \angle 97.8^\circ + 22 \angle 180^\circ = 85.8 \angle 112.5^\circ = I_f \angle 90^\circ + \delta^\circ$$

$E_f(\text{line}) = 7100V$ (corresponding to $I_f = 85.8A$)

$$\bar{E}_f = 4099 \angle 22.5^\circ V$$

Voltage regulation = $\frac{4099 - 3810.5}{3810.5} \times 100\% = 7.6\%$

PARALLEL OPERATION OF ALTERNATORS

Need for parallel operation is

- i) The total load requirement can not be met by a single alternator
- ii) During periods of light load, one or more alternators may be shutdown, and the remaining operates at or near full load so that efficiency is maximum.
- iii) If there is a breakdown of a generator, there is no total interruption of the power supply.
- iv) When one machine is taken out of service for its maintenance and inspection, the remaining machines maintain the continuity of supply.
- v) In order to meet the increasing future demand of load, more machines can be added without disturbing the original installation.

SYNCHRONOUS GENERATORS ON INFINITE BUSBARS

A supply system with a large number of synchronous generators connected in parallel is referred to as infinite bus-bar. Any additional synchronous machine, whether to work as a generator or as a motor is connected in parallel with the system. *The characteristics of an infinite busbar system are constant terminal voltage (zero synchronous impedance – since large number of generators are in parallel), constant bus-bar frequency (infinite rotational inertia – speed remains constant).*

Before a synchronous generator can be put to share a load, it should be properly connected in parallel with the common bus-bar. *Interconnection of the terminals of a generator with the terminals of another alternator or infinite bus-bar is called synchronizing.*

CONDITIONS OF PARALLEL CONNECTION (SYNCHRONIZATION)

Before the alternator can be connected to the infinite bus, the incoming alternator and the infinite bus must have the same

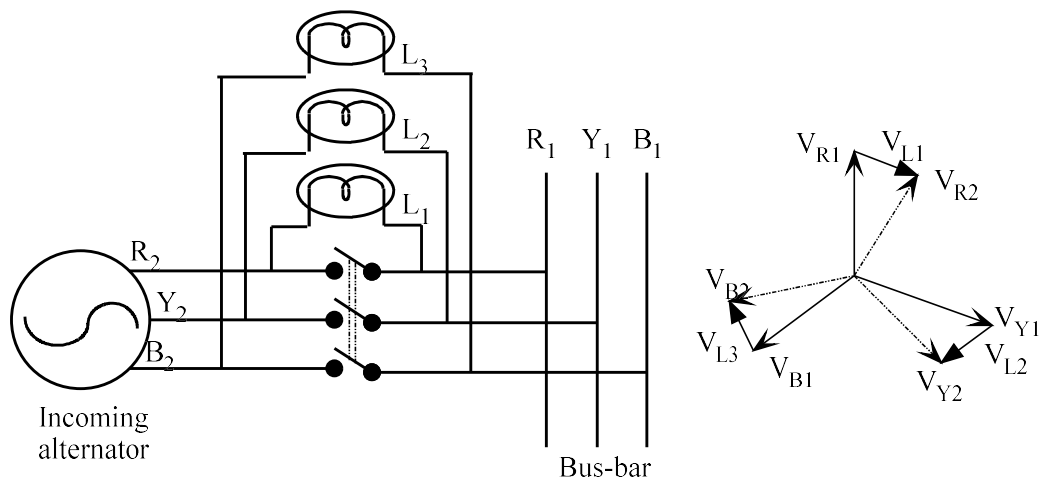
- i) Voltage
- ii) Frequency
- iii) Phase sequence
- iv) Phase

Synchronization or parallel connection of alternators can be achieved by any one of the following two methods:

- a) By 3-lamp (one dark & two bright lamp/three dark lamp) method
- b) By using a synchroscope

SYNCHRONIZATION BY DARK LAMP METHOD

In this method, three lamps are connected as shown in figure. By running the alternator at synchronous speed and by adjusting the field excitation, the armature voltage is increased near to rated value. If all the three lamps become bright and dim at the same time, the phase sequences of both the incoming generator and the bus-bar are the same. If they become bright and dim in sequence, the phase sequence of the incoming alternator should be reversed (by interchanging any two leads of incoming alternator).



Now, the field excitation is adjusted such that voltages of the incoming alternator and the bus-bar are equal. The speed of the prime-mover of the incoming machine is further adjusted slowly until the lamps flicker at a very low rate. The parallel switch is closed at the instant all the three lamps are dark (condition 4 : in phase). The incoming alternator thus gets connected in parallel with the bus-bar.

Note:

Voltage across lamp 1, $\vec{V}_{L1} = \vec{V}_{R1} - \vec{V}_{R2}$

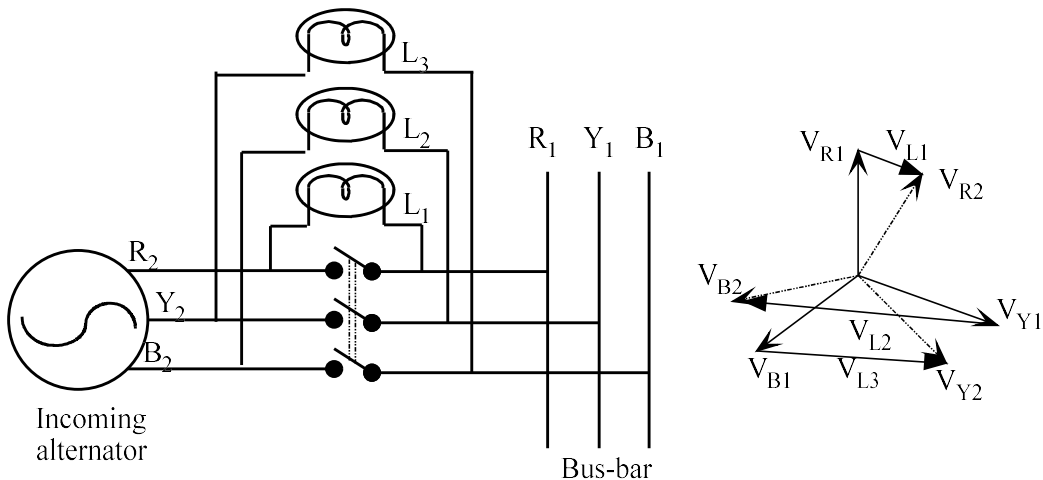
Voltage across lamp 2, $\vec{V}_{L2} = \vec{V}_{Y1} - \vec{V}_{Y2}$

Voltage across lamp 3, $\vec{V}_{L3} = \vec{V}_{B1} - \vec{V}_{B2}$

If the phase sequence is correct, voltages are equal in magnitude and the voltages are in phase, voltages across the lamps will be zero and hence all the lamps will be dark.

SYNCHRONIZATION BY BRIGHT LAMP METHOD (ONE DARK & TWO BRIGHT)

In this method, three lamps are connected as shown in figure. By running the alternator at synchronous speed and by adjusting the field excitation, the armature voltage is increased near to rated value. If all the three lamps become bright and dim in sequence, the phase sequences of both the incoming generator and bus-bar are the same. If all the three lamps become bright or dim at the same time, the phase sequence of the incoming alternator should be reversed (by interchanging any two leads of incoming alternator).



Now, the field excitation is adjusted such that voltages of the incoming alternator and the bus-bar are equal. The speed of the prime-mover of the incoming machine is further adjusted slowly until the lamps flicker at a very low rate. The parallel switch is closed at the instant when the lamp L_1 is dark and lamps L_2 and L_3 are equally bright. The incoming alternator thus gets connected in parallel with the bus-bar.

Note:

Voltage across lamp 1, $\vec{V}_{L1} = \vec{V}_{R1} - \vec{V}_{R2}$

Voltage across lamp 2, $\vec{V}_{L2} = \vec{V}_{Y1} - \vec{V}_{B2}$

Voltage across lamp 3, $\vec{V}_{L3} = \vec{V}_{B1} - \vec{V}_{Y2}$

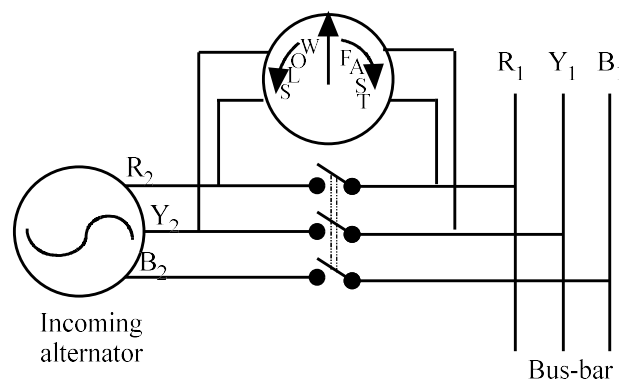
If the phase sequence is correct, voltages are equal in magnitude and the voltages are in phase, $\vec{V}_{L1} = \vec{V}_{R1} - \vec{V}_{R2} = 0$, $\vec{V}_{L2} = \vec{V}_{Y1} - \vec{V}_{B2} = \sqrt{3}V$, $\vec{V}_{L3} = \vec{V}_{B1} - \vec{V}_{Y2} = \sqrt{3}V$. L_1 will be dark, L_2 & L_3 will be equally bright.

For high voltage alternators, it is not possible to connect the lamps directly. In such cases lamps are connected through potential transformers.

SYNCHRONIZATION BY SYNCHROSCOPE

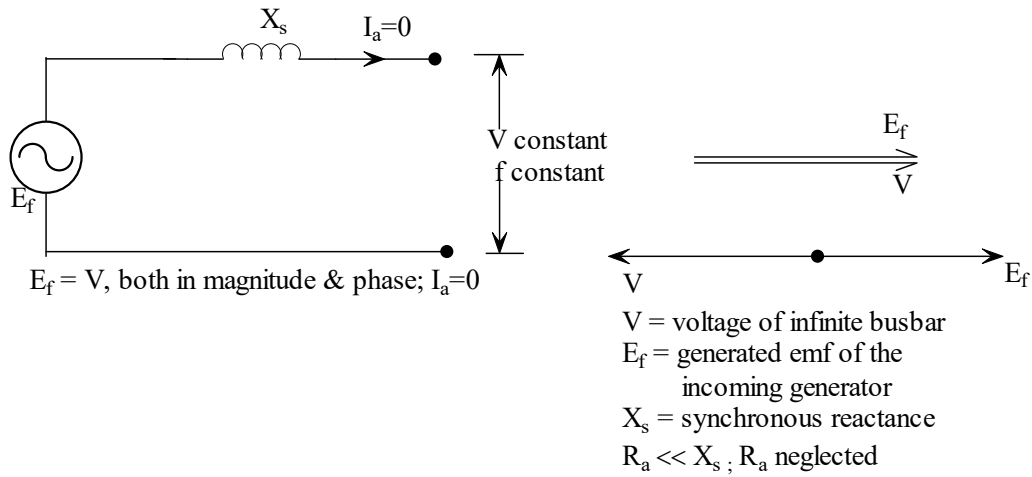
Lamp method is not accurate since it requires correct sense of judgement of the operator.

A synchroscope is an instrument that indicates by means of a revolving pointer the phase difference and the frequency difference between the voltages of the incoming alternator and the busbars. It is essentially a small motor, the field being supplied from the busbars through a potential transformer and the rotor from the incoming alternator. A pointer is attached to the rotor. When the incoming alternator is running fast (ie, frequency of the incoming alternator is higher than that of the busbars), the rotor and hence the pointer moves in the clockwise direction. When the incoming alternator is running slow (ie, frequency of the incoming alternator is lower than that of the busbars), the pointer moves in anticlockwise direction. When the frequency of the incoming alternator is equal to that of the busbars, no torque acts on the rotor and the pointer points vertically upwards. When the pointer moves very slowly (ie. frequencies are almost the same) and passes through the zero phase point (vertical up position), the synchronizing switch is closed and the alternator is connected to the infinite bus.



Machine floats on the bus-bar

Just after synchronization, the generated emf of the incoming machine is equal to the bus-bar voltage. The incoming machine is then said to be floating on busbars. At this instant, it will neither deliver nor receive any power.



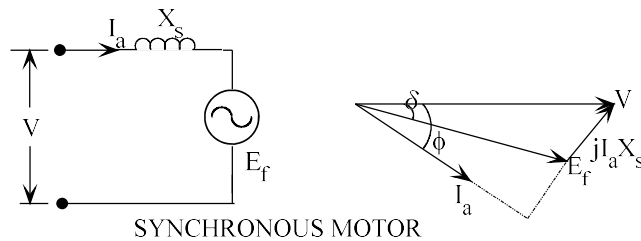
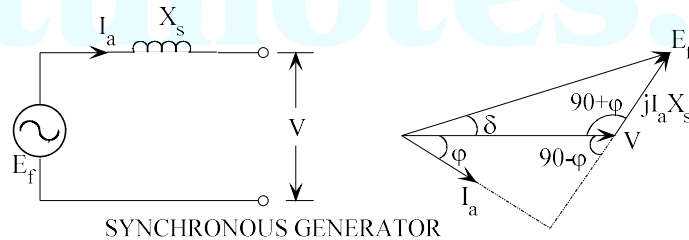
When the generator is connected in parallel, it should share a portion of the total load depending upon its kVA rating.

Power Angle Characteristic

Consider a synchronous machine connected to infinite bus bar. In generating mode, the generated emf (excitation emf) E_f leads V by angle δ . (In motoring mode, E_f lags behind V by δ).

In generating mode, $\vec{E}_f = \vec{V} + j\vec{I}_a X_s$ (R_a is neglected)

(In motoring mode, $\vec{E}_f = \vec{V} - j\vec{I}_a X_s$)



From the phasor diagram,

$$\frac{E_f}{\sin(90 + \phi)} = \frac{I_a X_s}{\sin \delta} \text{ for generator}$$

$$\frac{E_f}{\sin(90 - \phi)} = \frac{I_a X_s}{\sin \delta} \text{ for motor}$$

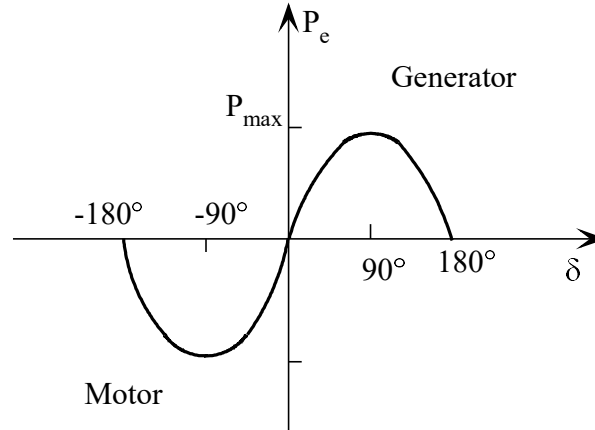
$$I_a \cos \phi = \frac{E_f}{X_s} \sin \delta$$

Multiplying both sides by V ,

$$VI_a \cos \phi = \frac{E_f V}{X_s} \sin \delta$$

$$P_e = \frac{E_f V}{X_s} \sin \delta = \text{Electrical power exchanged with the busbar}$$

Power angle characteristic is shown below.



Maximum power occurs at $\delta = 90^\circ$ beyond which the machine falls out of step (loses synchronism). The machine is normally operated at δ much less than 90° .

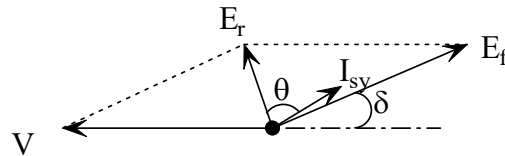
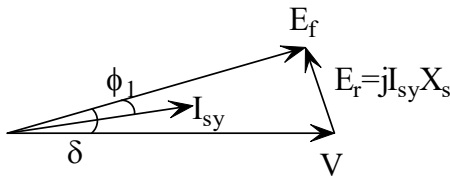
Effect of change in mechanical input (change in fuel supply)

Just after synchronization, the synchronous generator is floating. Let the governor set points be raised to increase the mechanical power input keeping the excitation constant.

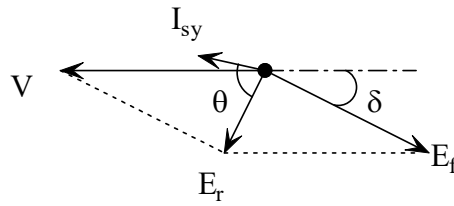
The increase in power input would tend to accelerate the rotor and E_f will move ahead of V . Load angle δ is increased. It will cause a phase difference between V and E_f , the magnitude of E_f remaining unchanged. The resultant of these two voltages E_r will cause a synchronizing current I_{sy} . Thus the power delivered by the generator, whose input power has been increased will increase by an amount equal to $E_f I_{sy} \cos \phi_1$ where ϕ_1 is the angle between E_f and I_{sy} . Thus with increase in input, the generator delivers more power to infinite bus. The frequency and terminal voltage of the generator remains same as it is connected to infinite bus bar.

$$(\text{Power output} = \frac{E_f V}{X_s} \sin \delta)$$

Since E_f , V , X_s are constants, power output increases as δ increases.)



(Note : After synchronization, if the prime mover input is removed, E_f shifts and lags behind V by angle δ , i.e. δ is reversed and the synchronous machine will act as motor; $E_f I_{sy} \cos \phi_1$ will be negative where ϕ_1 is the angle between E_f and I_{sy})

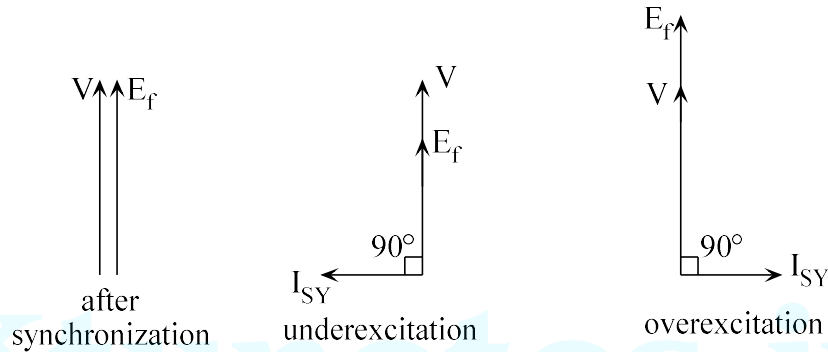


Effect of change of excitation

Alternator on no-load

Just after synchronization, the synchronous generator is floating. E_f and V are equal. Now, if the excitation of alternator is reduced, E_f will be less than V . Then the resultant voltage is $E_r = E_f - V$.

This will cause a circulating current I_{SY} to flow. I_{SY} will lead E_f by 90° . Due to magnetizing effect of armature reaction, leading current in the alternator will try to increase the flux and hence the excitation emf equal to busbar voltage. Hence synchronism is maintained.



If just after synchronization, if the excitation is increased, E_f will be greater than V and it causes a circulating current I_{SY} to flow. I_{SY} will lag E_f by 90° . Due to demagnetizing effect of armature reaction, lagging current in the alternator will try to decrease the flux and hence the excitation emf equal to busbar voltage. Again synchronism is maintained.

Here, no active power is delivered since I_{SY} is in quadrature with V . But alternator takes reactive power from bus since $E_f < V$ and delivers it to bus if $E_f > V$.

Alternator on load

Consider a synchronous generator is supplying power to the infinite bus with induced emf E_f , power angle δ and working at unity power factor with current I_a .

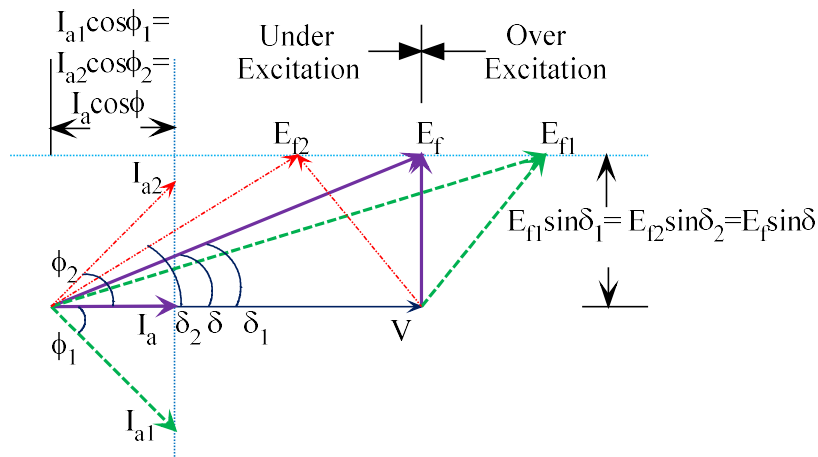
Now, assume that excitation is increased so that E_f is increased to E_{f1} . Since mechanical input remains

same, $\frac{E_f V}{X_s} \sin \delta$ remains constant. Hence, load angle δ is decreased to δ_1 so that

$$E_{f1} \sin \delta_1 = E_f \sin \delta$$

$$P_e = VI \cos \phi = \text{constant}$$

Armature current increases from I_a to I_{a1} . But, $I_a \cos \phi$ remains same. Power factor decreases to $\cos \phi_1$ (lag) so that $I_{a1} \cos \phi_1 = I_a \cos \phi$



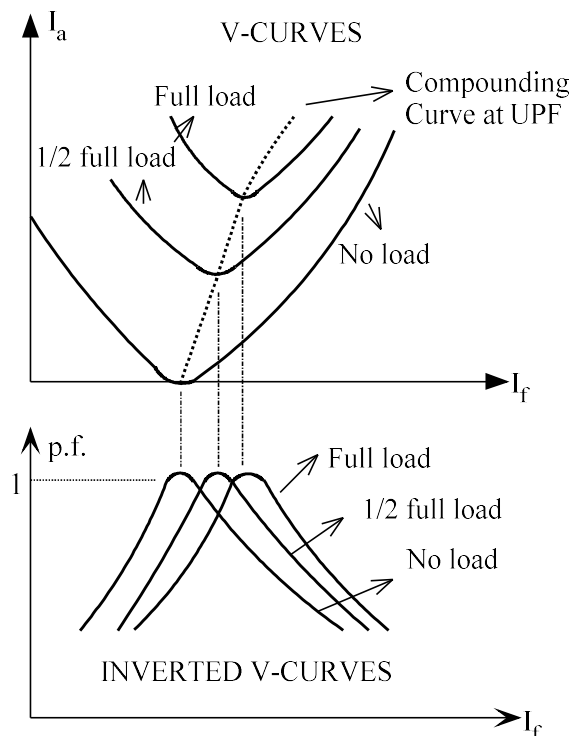
Now, assume that excitation is decreased so that E_f is decreased to E_{f2} . But, $\frac{E_f V}{X_s} \sin \delta$ remains constant. Hence, load angle δ is increased to δ_2 so that $E_{f2} \sin \delta_2 = E_f \sin \delta$. But, $P_e = VI \cos \phi =$ constant. Armature current increases from I_a to I_{a2} . But, $I_a \cos \phi$ remains same. Power factor decreases to $\cos \phi_2$ (lead) so that $I_{a2} \cos \phi_2 = I_a \cos \phi$.

The excitation corresponding to unity power factor is called normal excitation, while the excitation larger than this is called over-excitation and less than this is called under-excitation.

Note: Synchronous generator supplies a lagging power factor current when over-excited and supplies a leading power factor current when under-excited.

I_a Vs I_f for various values of load (real power) is called V-curves and power factor Vs I_f is called inverted V-curve.

Line joining all the minimum armature current points of the V-curves is called unity power factor compounding curve.



Hence, the change in prime-mover input to the synchronous generator connected to an infinite bus bar causes change in real (active) power delivered, but the frequency remains same. Change in excitation causes a change in reactive power delivered (power factor changes) but the terminal voltage remains same.

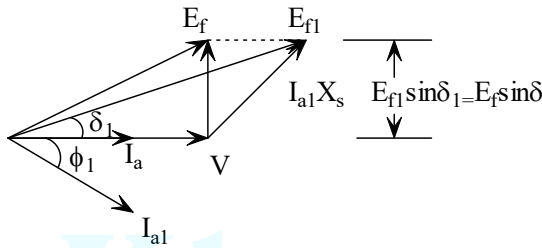
Qn 1: A 3-phase, 11kV, star-connected synchronous generator having $X_s=6\Omega/\text{phase}$ and negligible R_a has an armature current of 200A at unity power factor when operating on infinite bus bar. If the prime mover input remains the same and the emf is raised by 25%, determine the new values of armature current and power factor.

$$\text{Ans: } V = \frac{11000}{\sqrt{3}} = 6350.85V \quad I_a = 200A \quad \cos\phi = 1$$

$$E_f = V + jI_a X_s = 6350.85\angle 0^\circ + 200\angle 0^\circ \times j6 = 6463.2\angle 10.7^\circ$$

If excitation is increased by 25%,

$$E_{f1} = 1.25 \times E_f = 8079\angle \delta_1$$



$$E_{f1} \sin \delta_1 = E_f \sin \delta = 8079 \times \sin 10.7^\circ$$

$$\sin \delta_1 = \frac{6463.2 \times \sin 10.7^\circ}{8079} = 0.1485$$

$$\delta_1 = 8.54^\circ$$

$$jI_{a1} X_s = E_{f1} \angle \delta_1 - V \angle 0^\circ = 8079 \angle 8.54^\circ - 6350.85 \angle 0^\circ = 2030.83 \angle 36.21^\circ V$$

$$I_{a1} = \frac{2030.83 \angle 36.21^\circ}{j6} = 338.47 \angle -53.79^\circ$$

$$\text{p.f. } \cos \phi_1 = \cos 53.79^\circ = 0.59 \text{ lag}$$

Qn 2: A 3-phase, 11kV, star-connected synchronous generator having $Z_s=1+j10\Omega/\text{phase}$ delivers an armature current of 100A at unity power factor when operating on infinite bus bar. If the generator output remains constant and the excitation is increased by 15%, determine the new values of armature current and power factor.

$$\text{Ans: } V = \frac{11000}{\sqrt{3}} = 6350.85V \quad I_a = 100A \quad \cos\phi = 1 \quad Z_s = 1 + j10 = 10.05 \angle 84.29^\circ \Omega \quad \theta = 84.29^\circ$$

$$\bar{E}_f = \bar{V} + \bar{I}_a \bar{Z}_s = 6350.85\angle 0^\circ + 100\angle 0^\circ \times (1 + j10) = 6527.9\angle 8.81^\circ V$$

$$\delta = 8.81^\circ$$

$$\text{Real power output per phase, } P_o = \frac{VE_f}{Z_s} \cos(\theta - \delta) - \frac{V^2}{Z_s} \cos \theta$$

$$= \frac{6350.85 \times 6527.9}{10.05} \cos(84.29 - 8.81) - \frac{6350.85^2}{10.05} \cos 84.29$$

$$= 634954.7W$$

If excitation is increased by 15%,

$$E_{f1} = 1.15 \times E_f = 7507 \angle \delta_1$$

Since real power output remains same,

$$\frac{6350.85 \times 7507}{10.05} \cos(84.29 - \delta_1) - \frac{6350.85^2}{10.05} \cos 84.29 = 634954.7W$$

$$\cos(84.29 - \delta_1) = 0.218$$

$$\delta_1 = 6.883^\circ$$

$$I_{a1} Z_s = E_{f1} \angle \delta_1 - V \angle 0^\circ = 7507 \angle 6.883^\circ - 6350.85 \angle 0^\circ = 1422.63 \angle 39.23^\circ V$$

$$I_{a1} = \frac{1422.63 \angle 39.23^\circ}{1 + j10} = 141.56 \angle -45.06^\circ$$

$$\text{p.f. } \cos \phi_1 = \cos 45.06^\circ = 0.706 \text{ lag}$$

Qn 3: A 3-phase, star connected alternator with $R_a = 0.4 \Omega/\text{ph}$ and $X_s = 6 \Omega/\text{ph}$ delivers 300A at a power factor of 0.8 lag to constant voltage, constant frequency 10kV bus bar. If the steam supply is unchanged, find the percentage change in the induced emf necessary to raise the power factor to unity. Ignore changes in losses.

Ans:

$$E_{f1} = V \angle 0^\circ + I_{a1} Z_s = 5773.5 \angle 0^\circ + 300 \angle -36.87^\circ \times (0.4 + j6) = 7082.87 \angle 11.14^\circ V$$

Since steam supply is unchanged, output is constant. Hence, $I_{a1} \cos \phi_1 = I_{a2} \cos \phi_2$

$$300 \times 0.8 = I_{a2} \times 1$$

$$I_{a2} = 240 \angle 0^\circ A$$

$$E_{f2} = V \angle 0^\circ + I_{a2} Z_s = 5773.5 \angle 0^\circ + 240 \angle 0^\circ \times (0.4 + j6) = 6043.56 \angle 13.78^\circ V$$

$$\% \text{ change in excitation} = I_{a1} = \frac{E_{f1} - E_{f2}}{E_{f1}} \times 100 = \frac{7082.87 - 6043.56}{7082.87} \times 100 = 14.67\%$$

Qn 4: A 3-phase star-connected alternator, rated at 11kV, 1600kVA, has negligible armature resistance and a synchronous reactance of 30Ω per phase. When delivering full-load current at a certain power factor, the voltage regulation of the machine is zero. Estimate the power factor of the load and the power delivered by the machine. (Ans: 0.9801 leading, 1568.24kW)

Expression for synchronizing power P_{SYN} and torque T_{SYN}

A synchronous machine, whether a generator or a motor, when synchronized to infinite busbars has a tendency to remain in synchronism.

Consider a synchronous generator transferring a steady power P_o at a steady load angle δ_o . Suppose that, due to a transient disturbance, the rotor of the generator accelerates, so that the load angle increases by an angle $d\delta$. The load on the machine increases to $P_o + \Delta P$. Since the steady power input remains unchanged, this additional load decreases the speed of the machine and brings it back to synchronism.

Similarly, if due to a transient disturbance, the rotor of the machine retards so that the load angle decreases. Load on the machine decreases to $P_o - \Delta P$. Since the steady power input remains unchanged, the reduction in the load accelerates the rotor. Consequently, the machine again comes in synchronism.

The rate at which synchronous power P varies with δ is called the synchronizing power coefficient P_{sy} .

$$\text{Power output, } P = \frac{3V}{Z_s} [E_f \cos(\theta - \delta) - V \cos \theta]$$

$$\text{Synchronizing power per electrical radian, } P_{\text{syn}} = \frac{dP}{d\delta} = \frac{3VE_f}{Z_s} \sin(\theta - \delta)$$

If R_a is neglected,

$$P_{syn} = \frac{3VE_f}{X_s} \sin(90 - \delta) = \frac{3VE_f}{X_s} \cos \delta$$

Unit is watts/electrical radian.

Synchronizing power is transient in nature. In case of a generator, a sudden disturbance in prime-mover torque causes the synchronizing power to flow from, or to, the bus in order to restore the rotor to its previous position. A sudden change in the generator or motor field current, also causes the synchronizing power to come into play, so as to maintain the synchronism. In case of motor, a sudden disturbance in shaft-load torque causes the synchronizing power to flow from or to the motor, so that rotor is brought back to its initial steady position. In short, the synchronizing power comes into play only when the steady-state operating conditions are disturbed. Once, the steady state condition is reached after the disturbance, the synchronizing power reduces to zero. Thus the synchronizing power is transient in nature and exists only for the time during which the disturbance persists. The synchronizing power flows from, or to, the bus, in order to maintain the relative velocity between interacting stator and rotor fields zero.

Significance of synchronizing power coefficient: (Effect of synchronous reactance)

Synchronizing power coefficient is a measure of the stiffness of electromagnetic coupling between stator and rotor fields. Large value of P_{syn} indicates that the coupling is stiff or rigid.

Too large stiffness of coupling means that the motor tends to follow closely, the variation of speed caused by the disturbance in electric power supply. In case there is no power supply disturbance, then too much stiffness of coupling would cause the motor speed to remain practically constant, regardless of the mechanical load fluctuations. Too rigid electromagnetic coupling causes undue mechanical shocks, whenever there are fluctuations in the mechanical load or the supply.

Too rigid a coupling means that the machine will be subjected to shocks with change of load or supply. These shocks may damage the rotor or the windings.

$$P_{syn} = \frac{3VE_f}{X_s} \cos \delta$$

P_{syn} is inversely proportional to the synchronous reactance. Machines with large air gaps have relatively small reactances. Therefore, a synchronous machine with a larger air gap is more stiff than a machine with smaller air gap. Since P_{syn} is directly proportional to E_f , an overexcited machine is more stiff than an underexcited machine.

$$1 \text{ radian} = \frac{180}{\pi} \text{ degrees}$$

$$\text{Watts/electrical radian} = \text{watts}/\left(\frac{180}{\pi} \text{ elec}^\circ\right)$$

$$P_{syn} = \frac{3VE_f}{X_s} \cos \delta \times \frac{\pi}{180} \text{ W/elec}^\circ$$

$$\text{Elec}^\circ = \frac{P}{2} \times \text{mech}^\circ \quad (\text{or} \quad \text{Elec}^\circ = \frac{2}{P} \text{ mech}^\circ)$$

$$P_{syn} = \frac{3VE_f}{X_s} \cos \delta \times \frac{\pi}{180} \times \frac{P}{2} \text{ W/mech}^\circ$$

$$\text{Synchronizing torque} = T_{syn} = \frac{60}{2\pi N_s} \times P_{sy} \text{ Nm/ mech}^\circ$$

$$T_{syn} = \frac{60}{2\pi N_s} \frac{3VE_f}{X_s} \cos \delta \times \frac{\pi}{180} \times \frac{P}{2} \text{ Nm/mech}^\circ$$

Qn 1: A 2 MVA, 3-phase, 8-pole alternator is connected to 6000V, 50 Hz busbars and has a synchronous reactance of 4Ω /phase. Calculate the synchronizing power and the synchronizing torque per mechanical degree of rotor displacement at no load. Assume normal excitation.

Ans:

$$V = \frac{6000}{\sqrt{3}} = 3464.1V$$

On no-load, $E_f = V$ and $\delta = 0^\circ$

$$E_f = 3464.1V$$

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{8} = 750 \text{ rpm}$$

$$P_{syn} = \frac{3VE_f}{X_s} \cos \delta \times \frac{\pi}{180} \times \frac{P}{2} \text{ W/mech}^\circ$$

$$P_{syn} = \frac{3 \times 3464.1 \times 3464.1}{4} \cos 0^\circ \times \frac{\pi}{180} \times \frac{8}{2} = 628317.9 \text{ W/mech}^\circ$$

$$T_{syn} = \frac{60}{2\pi N_s} \times P_{sy} = \frac{60}{2\pi \times 750} \times 628317.9 = 8000 \text{ Nm/mech}^\circ$$

Qn 2: A 6.6kV, 3MVA, 3 phase, 50Hz, 8 pole alternator has synchronous reactance of 2.9Ω and negligible resistance. When it is operating on infinite bus, calculate the synchronizing power and the corresponding torque per mechanical degree of phase displacement at i) no load and ii) full load 0.85 pf lag.

Ans:

$$V = \frac{6600}{\sqrt{3}} = 3810.5V \text{ bm}$$

$$I_a = \frac{3 \times 10^6}{\sqrt{3} \times 6600} = 262.43A$$

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{8} = 750 \text{ rpm}$$

i) On no-load, $E_f = V$ and $\delta = 0^\circ$

$$E_f = 3810.5V$$

$$P_{syn} = \frac{3VE_f}{X_s} \cos \delta \times \frac{\pi}{180} \times \frac{P}{2} \text{ W/mech}^\circ$$

$$P_{syn} = \frac{3 \times 3810.5 \times 3810.5}{2.9} \times \cos 0^\circ \times \frac{\pi}{180} \times \frac{8}{2} = 1048635 \text{ W/mech}^\circ$$

$$T_{syn} = \frac{60}{2\pi N_s} \times P_{sy} = \frac{60}{2\pi \times 750} \times 1048635 = 13351.64 \text{ Nm/mech}^\circ$$

ii) at full load & 0.85 pf lag

$$\bar{E}_f = V \angle 0^\circ + I_a \angle -\phi^\circ \times jX_s = 3810.5 \angle 0^\circ + 262.43 \angle -31.79^\circ \times j2.9 = 4260.8 \angle 8.73^\circ$$

$E_f = 4260.8V$ and $\delta = 8.73^\circ$

$$P_{syn} = \frac{3 \times 3810.5 \times 4260.8}{2.9} \times \cos 8.73^\circ \times \frac{\pi}{180} \times \frac{8}{2} = 1158972 \text{ W/mech}^\circ$$

$$T_{syn} = \frac{60}{2\pi N_s} \times P_{sy} = \frac{60}{2\pi \times 750} \times 1158972 = 14756.49 \text{ Nm/mech}^\circ$$

Qn 3: A 1500kVA, 3-phase, star-connected 6.6kV, 8 pole, 50Hz synchronous generator has a reactance of 0.6p.u. and negligible resistance. Calculate the synchronizing power per mechanical degree at full load and 0.8 pf lagging.

Ans:

$$V = \frac{6600}{\sqrt{3}} = 3810.5V$$

$$I_a = \frac{1500 \times 10^3}{\sqrt{3} \times 6600} = 131.2A$$

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{8} = 750rpm$$

$$X_s(p.u.) = \frac{I_{rated} \times X_s}{V_{rated}}$$

$$X_s = \frac{V_{rated} \times X_s(p.u.)}{I_{rated}} = \frac{3810.5 \times 0.6}{131.2} = 17.42\Omega$$

At full load & 0.8 pf lag

$$\bar{E}_f = V \angle 0^\circ + I_a \angle -\phi^\circ \times jX_s = 3810.5 \angle 0^\circ + 131.2 \angle -36.87^\circ \times j17.42 = 5494.92 \angle 19.44^\circ V$$

$$E_f = 5494.92V \text{ and } \delta = 19.44^\circ$$

$$P_{syn} = \frac{3 \times 3810.5 \times 5494.2}{17.42} \times \cos 19.44^\circ \times \frac{\pi}{180} \times \frac{8}{2} = 237358W / \text{mech}^\circ$$

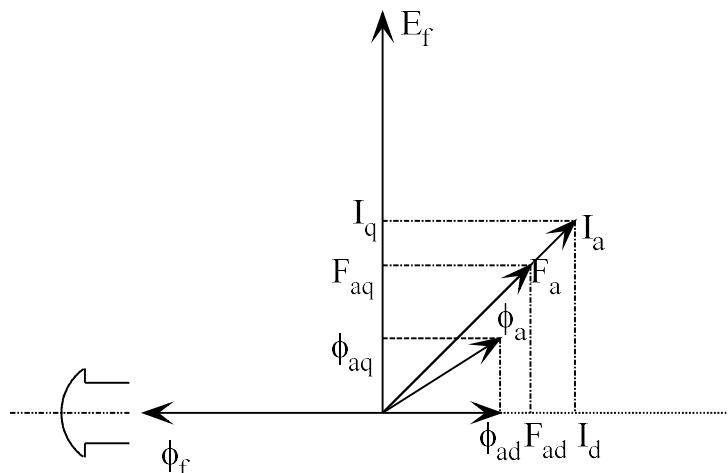
BLONDEL'S TWO REACTION THEORY

In a cylindrical rotor type machine, air gap is uniform so that the reluctance of the magnetic path is the same in all directions. Hence, effect of armature reaction can be accounted by one reactance X_s . X_s is constant for all directions of armature flux relative to rotor.

However, in a salient pole type machine, the radial length of the air gap varies so that the reluctance of the magnetic circuit along polar axis (direct axis) is much less than the reluctance along the interpolar axis (quadrature axis). Because of the lower reluctance along the polar axis, more flux is produced along d-axis than along q-axis. Hence, reactance due to armature reaction will be different along d-axis and q-axis.

According to two reaction theory, the armature mmf F_a can be divided into two components as i) component acting along direct axis F_{ad} and ii) component acting along quadrature axis F_{aq} .

Φ_f is the main flux and is along d-axis. Φ_f produces excitation emf E_f lagging by 90° .



When generator is loaded, I_a flows. Assume that the power factor is lagging. Hence, I_a lags behind E_f . I_a creates armature mmf F_a . F_a can be decomposed into 2 components F_{ad} and F_{aq} . F_{ad} produces flux Φ_{ad} and F_{aq} produces flux Φ_{aq} .

If S_d and S_q are the reluctances along direct and quadrature axes,

$$S_q > S_d$$

$$\frac{F_{aq}}{\Phi_{aq}} > \frac{F_{ad}}{\Phi_{ad}}$$

$$\frac{F_{aq}}{F_{ad}} > \frac{\Phi_{aq}}{\Phi_{ad}}$$

Resultant mmf, $F_a = F_{ad} + jF_{aq}$ and resultant flux, $\Phi_a = \Phi_{ad} + j\Phi_{aq}$

Slope of F_a will be more than slope of Φ_a .

Φ_a is not in phase with armature mmf F_a but lags behind F_a .

Φ_{ad} will oppose the field flux Φ_f if the power factor is lagging.

Φ_{ad} will be in the same direction of the field flux Φ_f if the power factor is leading.

Φ_{aq} is cross magnetizing the main flux.

F_a is proportional to current I_a . Decomposing F_a into F_{ad} and F_{aq} in effect means decomposition of I_a into I_d and I_q .

Φ_{ad} and Φ_{aq} are rotating at synchronous speed. They induce armature reaction emfs in each phase of stator winding. These emfs are E_{ad} and E_{aq} respectively.

Resultant emf, $\bar{E}_r = \bar{E}_f + \bar{E}_{ad} + \bar{E}_{aq}$

Armature reaction emf can be considered as a voltage drop across a fictitious reactance.

Hence,

$$\bar{E}_{ad} = -j\bar{I}_d X_{ad}$$

$$\bar{E}_{aq} = -j\bar{I}_q X_{aq}$$

$$\bar{E}_r = \bar{E}_f - j\bar{I}_d X_{ad} - j\bar{I}_q X_{aq}$$

Terminal voltage, $\bar{V} = \bar{E}_f - j\bar{I}_d X_{ad} - j\bar{I}_q X_{aq} - \bar{I}_a R_a - j\bar{I}_a X_{al}$

But, $\bar{I}_a = \bar{I}_d + \bar{I}_q$

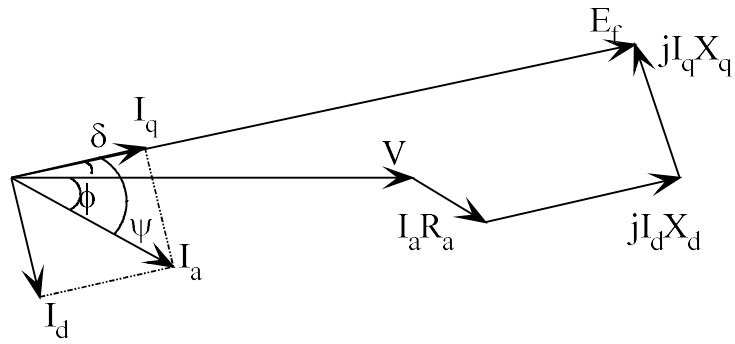
$$\begin{aligned} \bar{V} &= \bar{E}_f - j\bar{I}_d X_{ad} - j\bar{I}_q X_{aq} - \bar{I}_a R_a - j(\bar{I}_d + \bar{I}_q) X_{al} \\ &= \bar{E}_f - \bar{I}_a R_a - j\bar{I}_d (X_{ad} + X_{al}) - j\bar{I}_q (X_{aq} + X_{al}) \\ &= \bar{E}_f - \bar{I}_a R_a - j\bar{I}_d X_d - j\bar{I}_q X_q \end{aligned}$$

$X_d = X_{ad} + X_{al}$ = d-axis synchronous reactance

$X_q = X_{aq} + X_{al}$ = q-axis synchronous reactance

Note: X_{al} is independent of rotor position

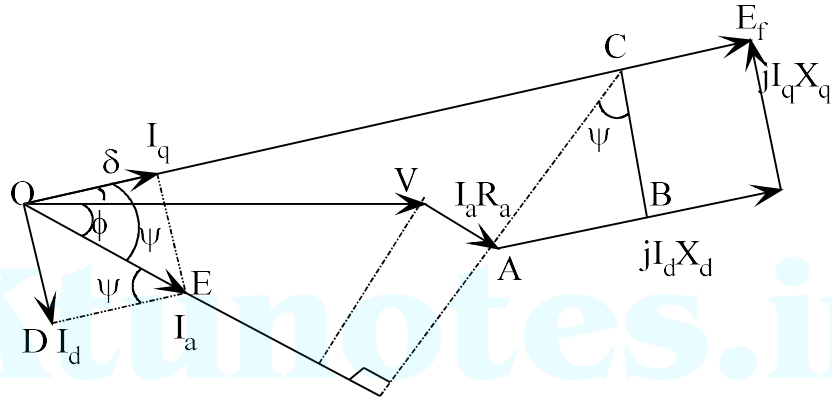
Note: I_q is along E_f .



$$I_q = I_a \cos \Psi$$

$$I_d = I_a \sin \Psi$$

Ψ = angle between I_a and E_f



$\triangle ODE$ and $\triangle ABC$ are similar. Hence, $\frac{AC}{OE} = \frac{BC}{DE}$

$$AC = \frac{BC \times OE}{DE} = \frac{jI_q X_q \times I_a}{I_q} = jI_a X_q$$

$$\vec{OC} = \vec{V} + \vec{I}_a R_a + j\vec{I}_a X_q = \vec{V} + \vec{I}_a (R_a + jX_q)$$

Angle of $\vec{OC}$ will give δ . (Taking V as reference)

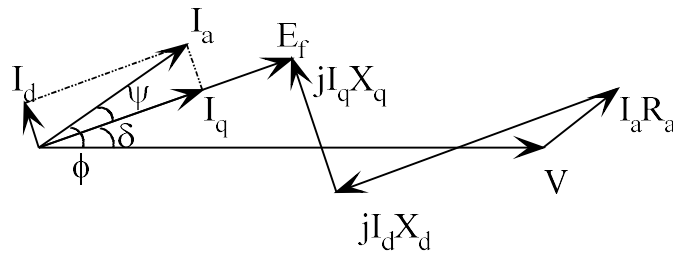
$\psi = \delta + \phi$ for lag

$\psi = \delta - \phi$ for lead

(or use the following formula for finding ψ ,

$$\tan \Psi = \frac{V \sin \phi + I_a X_q}{V \cos \phi + I_a R_a}$$

$$E_f = V \cos \delta + I_a R_a \cos \Psi + I_a X_d \sin \Psi$$



Phasor diagram at leading p.f.

Qn.1 : A 50Hz, 3-phase, 480V delta connected salient pole synchronous generator has $X_d=0.1\Omega$ and $X_q=0.075\Omega$. The generator is supplying 1200A at a) 0.8 p.f. lag and b) 0.8 p.f. lead. Find the excitation emf. Neglect armature resistance.

$$\text{Ans: } I_a = \frac{1200}{\sqrt{3}} = 692.82 \text{ A}$$

a) 0.8 p.f. lag

$$\vec{OC} = \vec{V} + \vec{I}_a (R_a + jX_q) = 480 \angle 0^\circ + 692.82 \angle -36.86^\circ \times j0.075 = 512.85 \angle 4.65^\circ$$

$$\delta = 4.65^\circ$$

$$\psi = \delta + \phi = 4.65 + 36.86 = 41.51^\circ$$

$$E_f = V \cos \delta + I_a R_a \cos \Psi + I_a X_d \sin \Psi$$

$$= 480 \times \cos 4.65^\circ + 692.82 \times 0.1 \times \sin 41.51^\circ = 524.33 \text{ V}$$

$$\bar{E}_f = 524.33 \angle 4.63^\circ \text{ V}$$

b) 0.8 p.f. lead

$$\vec{OC} = \vec{V} + \vec{I}_a (R_a + jX_q) = 480 \angle 0^\circ + 692.82 \angle +36.86^\circ \times j0.075 = 450.75 \angle 5.29^\circ$$

$$\delta = 5.29^\circ$$

$$\psi = \delta - \phi = 5.29 - 36.86 = -32.57^\circ$$

$$E_f = V \cos \delta + I_a R_a \cos \Psi + I_a X_d \sin \Psi$$

$$= 480 \times \cos 5.29^\circ + 692.82 \times 0.1 \times \sin(-32.57^\circ) = 440.66 \text{ V}$$

$$\bar{E}_f = 440.66 \angle 5.29^\circ \text{ V}$$

Qn 2: A salient pole synchronous generator has the following per-unit parameters: $X_d=1.2$, $X_q=0.8$, $R_a=0.025$. Compute the excitation emf on a per unit basis, when the generator is delivering rated kVA at rated voltage and at power factor of a) 0.8 lagging and b) 0.8 leading.

a) 0.8 p.f. lag

$$\vec{OC} = \vec{V} + \vec{I}_a (R_a + jX_q) = 1 \angle 0^\circ + 1 \angle -36.86^\circ \times (0.025 + j0.8) = 1.62 \angle 22.62^\circ$$

$$\delta = 22.62^\circ$$

$$\psi = \delta + \phi = 22.62 + 36.86 = 59.48^\circ$$

$$E_f = V \cos \delta + I_a R_a \cos \Psi + I_a X_d \sin \Psi$$

$$= 1 \times \cos 22.62^\circ + 1 \times 0.025 \times \cos(59.48) + 1 \times 1.2 \times \sin(59.48)^\circ = 1.957 \text{ p.u.}$$

$$\bar{E}_f = 1.957 \angle 22.62^\circ \text{ V}$$

b) 0.8 p.f. lead

$$\vec{OC} = \vec{V} + \vec{I}_a (R_a + jX_q) = 1 \angle 0^\circ + 1 \angle +36.86^\circ \times (0.025 + j0.8) = 0.849 \angle 50.5^\circ$$

$$\delta = 50.5^\circ$$

$$\psi = \delta - \phi = 50.5 - 36.86 = 13.64^\circ$$

$$E_f = V \cos \delta + I_a R_a \cos \Psi + I_a X_d \sin \Psi$$

$$= 1 \times \cos 50.5^\circ + 1 \times 0.025 \times \cos(13.64) + 1 \times 1.2 \times \sin(13.64)^\circ = 0.943 \text{ p.u.}$$

Qn.3 : A 20kVA, 220V, 50Hz, star connected 3-phase salient pole synchronous generator supplies full load at a lagging power factor angle of 45° . Given $X_d=4\Omega$, $X_q=2\Omega$ and $R_a=0.5\Omega$. Calculate i) the power angle and ii) voltage regulation under the given load conditions.

$$\text{Ans: } V = \frac{220}{\sqrt{3}} = 127V$$

$$I_a = \frac{20000}{\sqrt{3} \times 220} = 52.49A$$

$$\vec{OC} = \vec{V} + \vec{I}_a(R_a + jX_q) = 127 \angle 0^\circ + 52.49 \angle -45^\circ \times (0.5 + j2) = 226.73 \angle 14.21^\circ$$

$$\delta = 14.21^\circ$$

$$\psi = \delta + \phi = 14.21 + 45 = 59.21^\circ$$

$$E_f = V \cos \delta + I_a R_a \cos \Psi + I_a X_d \sin \Psi$$

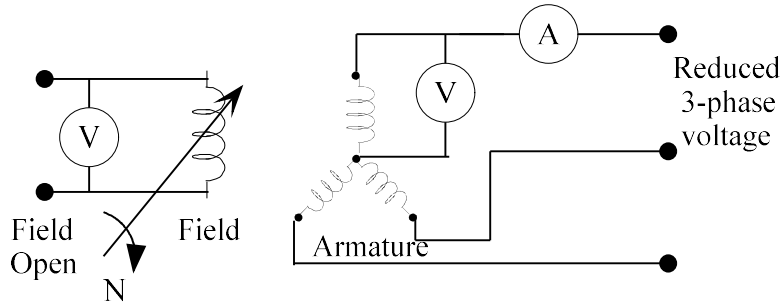
$$= 127 \times \cos 14.21^\circ + 52.49 \times 0.5 \times \cos(59.21) + 52.49 \times 4 \times \sin(59.21)^\circ = 316.91V$$

$$\bar{E}_f = 316.91 \angle 14.21^\circ V$$

$$V.R. = \frac{E_f - V}{V} \times 100 = \frac{316.91 - 127}{127} \times 100 = 149.53\%$$

SLIP TEST (To determine X_d and X_q)

In this test, a small voltage at rated frequency, and not more than about 25% of the rated voltage is applied to the 3-phase stator winding. The field winding is unexcited and left open circuited. Three phase currents in armature produce a rotating magnetic field. The armature mmf wave will be rotating at synchronous speed.

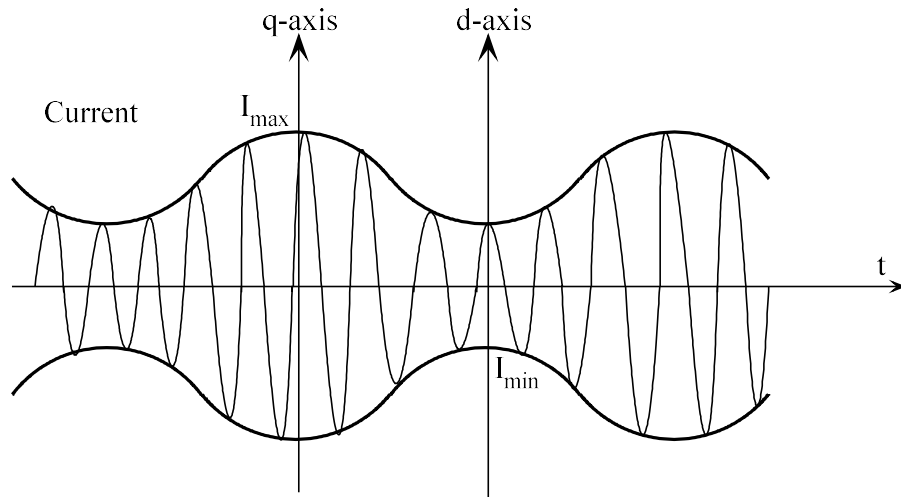


The rotor is driven by an auxiliary motor at a speed slightly less or slightly more than synchronous speed. The direction of rotation should be the same as that of the rotating field produced by the stator. A small voltage reading indicated by the voltmeter across the open field winding terminals shows that the direction of rotation of rotor is proper.

Since the rotor is running at a speed, N , close to synchronous speed N_s , there will be a small slip between the rotating magnetic field produced by the armature and the actual salient field poles. The relative speed between the armature mmf and the field poles is equal to the slip speed ($N_s - N$). Since the stator mmf moves slowly past the actual field poles, there will be an instant when the peak of the armature mmf wave is in line with the axis of the actual salient field poles.

When the peak of the armature mmf wave is in line with the axis of the salient field poles (d-axis), reluctance offered by the small air gap is minimum. This results in minimum magnetizing current ($X_d > X_q$). (In this position, the armature flux linkage with the field winding is maximum and the rate of change of flux linkage is zero. Therefore, the induced voltage across the field winding is zero).

$$X_d = \frac{\text{maximum voltage}}{\text{minimum current}}.$$



After one-quarter of slip cycle, the peak armature mmf wave is in line with the q-axis. In this position, the reluctance offered by long air gap is maximum. A large magnetizing current is needed to establish same air gap flux. In this position, the armature flux linking the field winding is zero, and the rate of change of flux linkage is maximum. Consequently, induced voltage across the field winding is maximum.

$$X_q = \frac{\text{minimum voltage}}{\text{maximum current}}.$$

Slip test is generally used to find the ratio $\frac{X_d}{X_q}$. The d-axis synchronous reactance X_d is

determined from the OC and SC tests as in the case of cylindrical rotor machine. Knowing X_d from the OC and SC tests and the ratio $\frac{X_d}{X_q}$ from the slip test, we can calculate X_q .

Note: Normally, X_q is between 50% and 80% of X_d .

Qn 1: Slip test was performed on a 3-phase, 415V, star-connected synchronous machine. The ammeter fluctuates between 4.7A and 7A and the fluctuation in the voltmeter connected across the lines is between 87V and 98V. Estimate the direct and quadrature axis reactances. The effective resistance per phase is 0.8Ω .

Ans:-

$$V_{\min} = \frac{87}{\sqrt{3}} = 50.23V ; I_{\min} = 4.7A$$

$$V_{\max} = \frac{98}{\sqrt{3}} = 56.58V ; I_{\max} = 7A$$

$$Z_d = \frac{V_{\max}}{I_{\min}} = \frac{56.58}{4.7} = 12.04\Omega$$

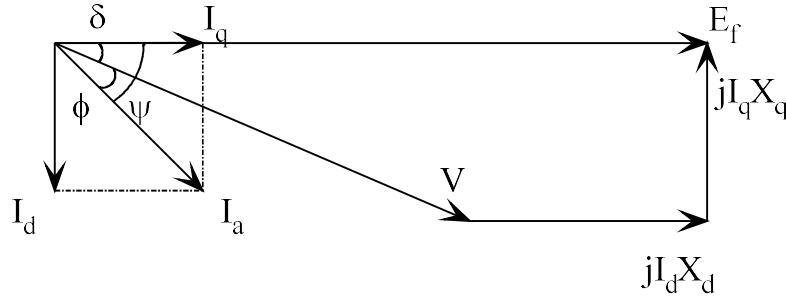
$$Z_q = \frac{V_{\min}}{I_{\max}} = \frac{50.23}{7} = 7.176\Omega$$

$$X_d = \sqrt{Z_d^2 - R_a^2} = \sqrt{12.04^2 - 0.8^2} = 12.01\Omega$$

$$X_q = \sqrt{Z_q^2 - R_a^2} = \sqrt{7.176^2 - 0.8^2} = 7.13\Omega$$

POWER ANGLE CHARACTERISTICS OF SALIENT POLE SYNCHRONOUS GENERATOR

The phasor diagram of a salient pole synchronous generator at lagging power factor (neglecting R_a) is shown below.



Complex Power Output per phase

$$S_o = P_o + jQ_o = \bar{V} \times \bar{I}_a^*$$

Take E_f as reference.

$$V \angle -\delta = V \cos \delta - jV \sin \delta$$

$$I_a = I_q - jI_d$$

$$I_a^* = I_q + jI_d$$

$$S_o = P_o + jQ_o = \bar{V} \times \bar{I}_a^* \\ = (V \cos \delta - jV \sin \delta)(I_q + jI_d)$$

$$X_q I_q = V \sin \delta$$

$$I_q = \frac{V \sin \delta}{X_q}$$

$$X_d I_d = E_f - V \cos \delta$$

$$I_d = \frac{E_f - V \cos \delta}{X_d}$$

Substituting,

$$S_o = (V \cos \delta - jV \sin \delta) \left(\frac{V \sin \delta}{X_q} + j \frac{E_f - V \cos \delta}{X_d} \right) \\ = \left(\frac{V^2 \sin \delta \cos \delta}{X_q} + \frac{VE_f \sin \delta}{X_d} - \frac{V^2 \sin \delta \cos \delta}{X_d} \right) + j \left(\frac{VE_f \cos \delta}{X_d} - \frac{V^2 \cos^2 \delta}{X_d} - \frac{V^2 \sin^2 \delta}{X_q} \right)$$

$$\text{Real power output per phase} = \frac{VE_f \sin \delta}{X_d} + \frac{V^2}{2} \left(\frac{1}{X_q} - \frac{1}{X_d} \right) \sin 2\delta$$

$$\text{Total real power output, } P = \frac{3VE_f \sin \delta}{X_d} + \frac{3V^2}{2} \left(\frac{1}{X_q} - \frac{1}{X_d} \right) \sin 2\delta$$

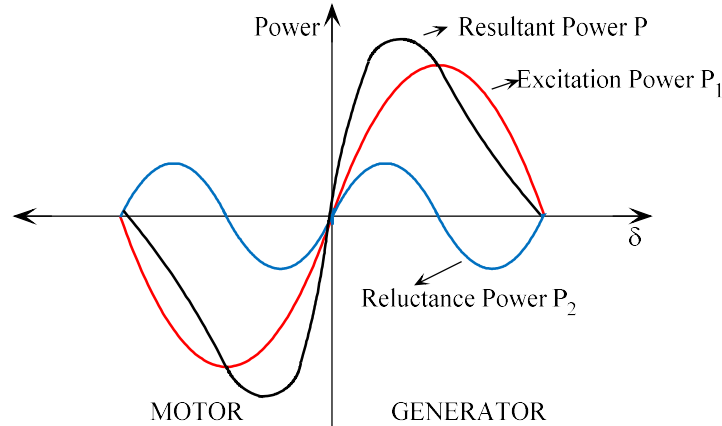
First term $\frac{3VE_f \sin \delta}{X_d} = P_1$ is called excitation power (power due to field excitation)

Second term $\frac{3V^2}{2} \left(\frac{1}{X_q} - \frac{1}{X_d} \right) \sin 2\delta = P_2$ is called reluctance power (power due to saliency). P_2 does

not depend on field excitation and is present even when field current is zero. Reluctance power exists only if the machine is operating in parallel with other machines or infinite bus bar.

The machine can continue to operate on infinite-bus at synchronous speed provided the load on the machine is less than the maximum value of reluctance power, the value of which is around 30% of the rated power.

For a cylindrical rotor machine, $X_d = X_q$ and hence reluctance power is zero.



Reluctance power is maximum, when $\delta = 45^\circ$.

Maximum power occurs when $\delta \leq 90^\circ$.

Synchronizing power in a salient pole synchronous generator

$$\text{Real power output, } P = \frac{3VE_f \sin \delta}{X_d} + \frac{3V^2}{2} \left(\frac{1}{X_q} - \frac{1}{X_d} \right) \sin 2\delta$$

$$\text{Synchronizing power, } P_{sy} = \frac{3VE_f \cos \delta}{X_d} + 3V^2 \left(\frac{1}{X_q} - \frac{1}{X_d} \right) \cos 2\delta \text{ W/elec.radian}$$

$$P_{sy} = \left\{ \frac{3VE_f \cos \delta}{X_d} + 3V^2 \left(\frac{1}{X_q} - \frac{1}{X_d} \right) \cos 2\delta \right\} \times \frac{\pi}{180} \times \frac{P}{2} \text{ W/mech}^\circ$$

Qn 1: A 4kVA, 3-phase, 110V, 50Hz, star connected alternator has $X_d = 3\Omega$ and $X_q = 2\Omega$. The machine is delivering full-load current at 0.8 p.f. lagging at rated voltage. Find the induced emf, load angle and maximum power output of the alternator.

$$\text{Ans: } V = \frac{110}{\sqrt{3}} = 63.51V$$

$$I_a = \frac{4 \times 1000}{\sqrt{3} \times 110} = 21A$$

a) 0.8 p.f. lag

$$\vec{OC} = \vec{V} + \vec{I}_a (R_a + jX_q) = 63.51 \angle 0^\circ + 21 \angle -36.86^\circ \times j2 = 94.86 \angle 20.75^\circ$$

$$\delta = 20.75^\circ$$

$$\psi = \delta + \phi = 20.75 + 36.86 = 57.61^\circ$$

$$\begin{aligned} E_f &= V \cos \delta + I_a R_a \cos \Psi + I_a X_d \sin \Psi \\ &= 63.51 \times \cos 20.75^\circ + 21 \times 3 \times \sin 57.61^\circ = 112.59V \end{aligned}$$

$$\bar{E}_f = 112.59 \angle 20.75^\circ V$$

$$\begin{aligned} \text{Power output, } P &= \frac{3VE_f \sin \delta}{X_d} + \frac{3V^2}{2} \left(\frac{1}{X_q} - \frac{1}{X_d} \right) \sin 2\delta \\ &= \frac{3 \times 63.51 \times 112.59 \times \sin \delta}{3} + \frac{3 \times 63.51^2}{2} \left(\frac{1}{2} - \frac{1}{3} \right) \sin 2\delta \\ &= 7150.6 \sin \delta + 1008.3 \sin 2\delta \end{aligned}$$

For maximum power output,

$$\frac{dP}{d\delta} = 0$$

$$7150.6 \cos \delta + 2016.76 \cos 2\delta = 0$$

$$7150.6 \cos \delta + 2016.76(2 \cos^2 \delta - 1) = 0$$

$$4033.52 \cos^2 \delta + 7150.6 \cos \delta - 2016.76 = 0$$

$$\cos \delta = 0.247 \text{ or } \delta = 75.7^\circ$$

$$P_{\max} = 7150.6 \sin \delta + 1008.3 \sin 2\delta = 7150.6 \sin 75.7^\circ + 1008.3 \sin 2 \times 75.7^\circ = 7411.7W$$

Qn.2 : A 3.7MVA, 3-phase, star-connected synchronous generator rated at 6.6kV has 24 poles. Given $X_d=9.8\Omega$, $X_q=6.1\Omega$ and $R_a=0\Omega$. Calculate the excitation emf and the voltage regulation when supplying a real power of 2.8MW at 0.8 p.f lagging to the 6.6kV bus-bars. What maximum power can the generator supply, if the field circuit becomes open circuited?

$$\text{Ans: } V = \frac{6600}{\sqrt{3}} = 3810.5V$$

$$I_a = \frac{2.8 \times 10^6}{\sqrt{3} \times 6600 \times 0.8} = 306.17A$$

$$\vec{OC} = \vec{V} + \vec{I}_a(R_a + jX_q) = 3810.5 \angle 0^\circ + 306.17 \angle -36.86^\circ \times j6.1 = 5152.3 \angle 16.86^\circ$$

$$\delta = 16.86^\circ$$

$$\psi = \delta + \phi = 16.86 + 36.86 = 53.72^\circ$$

$$E_f = V \cos \delta + I_a R_a \cos \Psi + I_a X_d \sin \Psi$$

$$= 3810.5 \times \cos 16.86^\circ + 306.17 \times 9.8 \times \sin(53.72^\circ) = 6065.5V$$

$$\bar{E}_f = 6065.5 \angle 16.86^\circ V$$

$$V.R. = \frac{E_f - V}{V} \times 100 = \frac{6065.5 - 3810.5}{3810.5} \times 100 = 59.17\%$$

$$\text{If excitation is zero, power output} = \frac{3V^2}{2} \left(\frac{1}{X_q} - \frac{1}{X_d} \right) \sin 2\delta = \text{reluctance power}$$

For maximum reluctance power, $\delta = 45^\circ$

$$\begin{aligned} \text{Maximum reluctance power} &= \frac{3V^2}{2} \left(\frac{1}{X_q} - \frac{1}{X_d} \right) \\ &= \frac{3 \times 3810.5^2}{2} \left(\frac{1}{6.1} - \frac{1}{9.8} \right) = 1348034.5W \end{aligned}$$

SHORT CIRCUIT RATIO (SCR)

Short circuit ratio (SCR) of a synchronous machine is defined as the ratio of the field current required to generate rated voltage on open circuit to the field current required to circulate rated armature current on short circuit.

SCR can be calculated from the OCC and SCC.

$$SCR = \frac{I_f \text{ for rated O.C. voltage}}{I_f \text{ for rated S.C. current}} = \frac{1}{X_{sp.u.}} = \frac{1}{X_{dp.u.}}$$

Effect of SCR on machine performance:

- 1) Voltage regulation – a low value of SCR means that the synchronous reactance has a large value. With a low value of SCR, the synchronous generator has a large variation in terminal voltage with a change in load (Machine is very sensitive to load variations). In order to keep the terminal voltage constant, large variation in field current is required.
- 2) Stability – A machine with a low value of SCR (high value of X_s) has a lower stability limit as the maximum power output of machine is inversely proportional to X_s .
- 3) Parallel operation – Machines with a low value of SCR are difficult to operate in parallel because a high value of X_s gives a small synchronizing power. This power is responsible for keeping the machines in synchronism.
- 4) Short circuit current – A small value of SCR indicates a smaller value of current under short circuit conditions owing to large value of synchronous reactance.
- 5) A large SCR will increase the size, weight and cost of the machine.

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Module 3

Principle of Synchronous Motor, Equivalent circuit, Phasor diagrams, Power flow diagram and equations, Losses and efficiency -Numerical Problems, Power-angle Characteristics, V Curve and Inverted V Curves.

SYNCHRONOUS MOTOR

The synchronous motor construction is similar to rotating field type alternator. It consists of two parts:

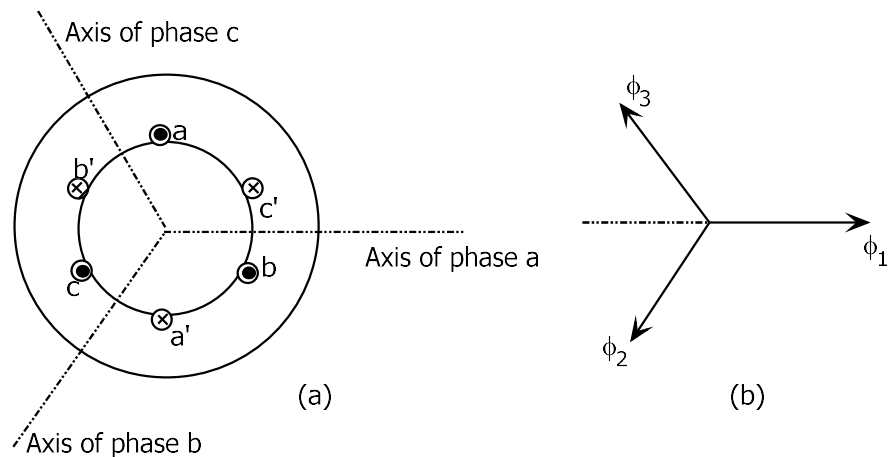
1. Stator – Consisting of 3-phase star or delta connected winding. This is excited by a 3-phase supply.
2. Rotor – Rotor is a field winding, the construction of which can be salient or cylindrical rotor type. Practically most of the synchronous motors use salient pole type construction. The field winding is excited by a separate dc supply through slip rings.

Salient features of synchronous motor are

- i) A synchronous motor runs at synchronous speed or not at all. Its speed is constant ($N_s = \frac{120f}{P}$) at all loads. The only way to change its speed is to alter the supply frequency.
- ii) It can be made to operate over a wide range of power factors (lag, unity or lead) by adjustment of its field excitation. Hence, a synchronous motor can be made to carry the mechanical load at constant speed and at the same time improve the power factor of the system.
- iii) Synchronous motors are generally of the salient pole type.
- iv) A synchronous motor is not self-starting and an auxiliary means has to be used for starting it.

PRODUCTION OF ROTATING MAGNETIC FIELD

When 3-phase windings displaced in space by 120° are supplied by 3-phase balanced currents, a rotating magnetic field is produced.

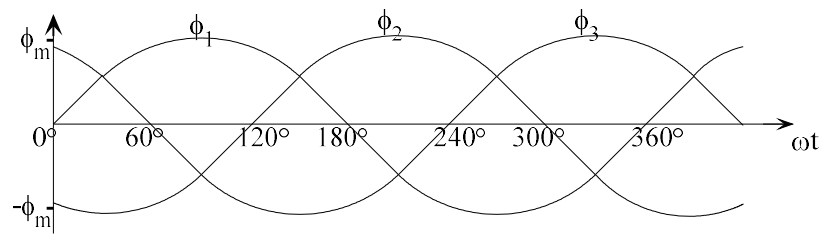


Consider three identical coils located on axes physically at 120° in space as shown in figure (a). Let each coil be supplied from one phase of a balanced 3-phase supply. Each coil will produce an alternating flux along its own axis. Let the instantaneous fluxes be given by

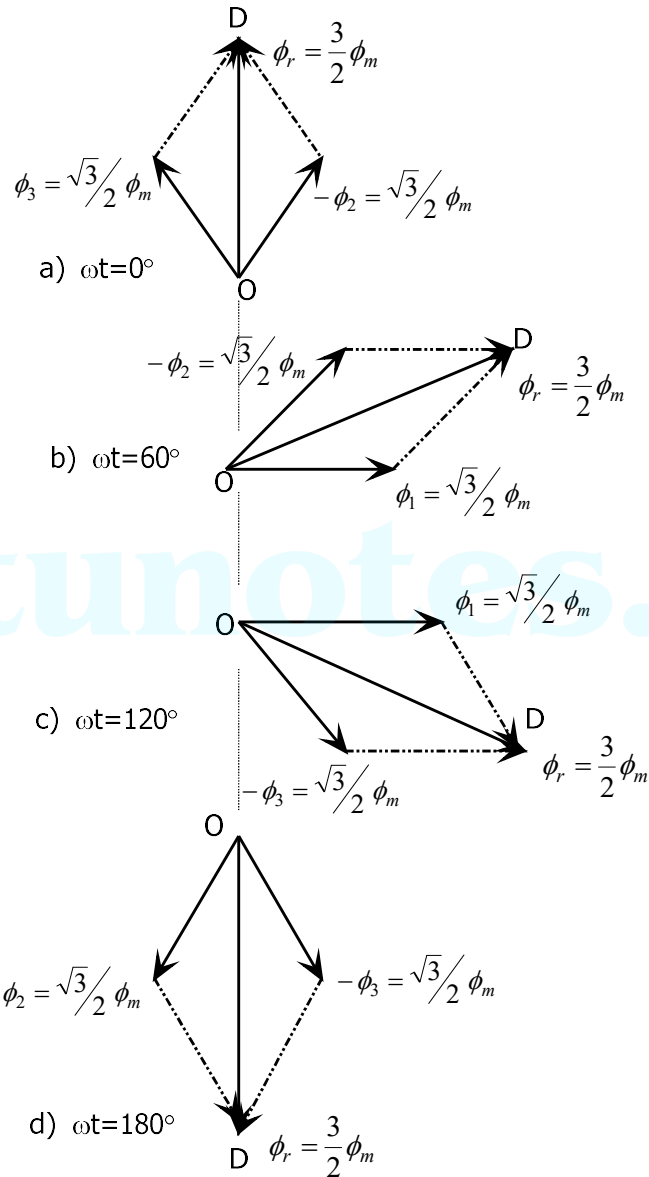
$$\phi_1 = \phi_m \sin \omega t$$

$$\phi_2 = \phi_m \sin(\omega t - 120^\circ)$$

$$\phi_3 = \phi_m \sin(\omega t + 120^\circ)$$



The positive directions of flux phasors for each phase are shown in figure (b).



The resultant flux ϕ_r , at any instant, is equal to the phasor sum of the fluxes due to three phases.

i) When $\omega t = 0^\circ$,

$$\phi_1 = \phi_m \sin \omega t = \phi_m \sin 0^\circ = 0$$

$$\phi_2 = \phi_m \sin(\omega t - 120^\circ) = \phi_m \sin(-120^\circ) = -\frac{\sqrt{3}}{2}\phi_m \text{ (acting along the direction opposite to positive direction of } \phi_2 \text{)}$$

$$\phi_3 = \phi_m \sin(\omega t + 120) = \phi_m \sin(120^\circ) = \frac{\sqrt{3}}{2} \phi_m \text{ (acting along positive direction of } \phi_3 \text{)}$$

$$\text{Resultant flux, } \phi_r = OD = 2 \times \frac{\sqrt{3}}{2} \phi_m \times \cos 30^\circ = 2 \times \frac{\sqrt{3}}{2} \phi_m \times \frac{\sqrt{3}}{2} = \frac{3}{2} \phi_m$$

ii) When $\omega t = 60^\circ$,

$$\phi_1 = \phi_m \sin \omega t = \phi_m \sin 60 = \frac{\sqrt{3}}{2} \phi_m \text{ (acting along positive direction of } \phi_1 \text{)}$$

$$\phi_2 = \phi_m \sin(\omega t - 120) = \phi_m \sin(-60^\circ) = -\frac{\sqrt{3}}{2} \phi_m \text{ (acting along the direction opposite to positive direction of } \phi_2 \text{)}$$

$$\phi_3 = \phi_m \sin(\omega t + 120) = \phi_m \sin(180) = 0$$

$$\text{Resultant flux, } \phi_r = OD = 2 \times \frac{\sqrt{3}}{2} \phi_m \times \cos 30^\circ = 2 \times \frac{\sqrt{3}}{2} \phi_m \times \frac{\sqrt{3}}{2} = \frac{3}{2} \phi_m$$

Magnitude of the resultant flux remains same, but it has rotated clockwise through an angle of 60° from its position at instant $\omega t = 0^\circ$.

iii) When $\omega t = 120^\circ$,

$$\phi_1 = \phi_m \sin \omega t = \phi_m \sin 120^\circ = \frac{\sqrt{3}}{2} \phi_m \text{ (acting along positive direction of } \phi_1 \text{)}$$

$$\phi_2 = \phi_m \sin(\omega t - 120) = \phi_m \sin 0^\circ = 0$$

$$\phi_3 = \phi_m \sin(\omega t + 120) = \phi_m \sin(240^\circ) = -\frac{\sqrt{3}}{2} \phi_m \text{ (acting along the direction opposite to positive direction of } \phi_3 \text{)}$$

$$\text{Resultant flux, } \phi_r = OD = 2 \times \frac{\sqrt{3}}{2} \phi_m \times \cos 30^\circ = 2 \times \frac{\sqrt{3}}{2} \phi_m \times \frac{\sqrt{3}}{2} = \frac{3}{2} \phi_m$$

Magnitude of the resultant flux remains same, but it has further rotated clockwise through an angle of 60° from its position at instant $\omega t = 60^\circ$.

iv) When $\omega t = 180^\circ$,

$$\phi_1 = \phi_m \sin \omega t = \phi_m \sin 180^\circ = 0$$

$$\phi_2 = \phi_m \sin(\omega t - 120) = \phi_m \sin 60^\circ = \frac{\sqrt{3}}{2} \phi_m \text{ (acting along positive direction of } \phi_1 \text{)}$$

$$\phi_3 = \phi_m \sin(\omega t + 120) = \phi_m \sin(300^\circ) = -\frac{\sqrt{3}}{2} \phi_m \text{ (acting along the direction opposite to positive direction of } \phi_3 \text{)}$$

$$\text{Resultant flux, } \phi_r = OD = 2 \times \frac{\sqrt{3}}{2} \phi_m \times \cos 30^\circ = 2 \times \frac{\sqrt{3}}{2} \phi_m \times \frac{\sqrt{3}}{2} = \frac{3}{2} \phi_m$$

Magnitude of the resultant flux remains same, but it has further rotated clockwise through an angle of 60° from its position at instant $\omega t = 120^\circ$.

Here, the speed of the rotating field is $\frac{f}{p}$ revolutions/second (corresponding to f cycles/second where p = pole- pair)

The direction of rotation of resultant flux in the air gap depends upon the phase sequence. By changing the phase sequence, we can change the direction of rotation.

PRINCIPLE OF OPERATION

A synchronous motor is not self starting. If the rotor field poles are excited by the field current and the stator terminals are connected to the ac supply, the motor will not start; instead, it vibrates. This can be explained as follows.

Let us consider a two-pole synchronous machine. If it is connected to a 3-phase, 50Hz ac supply, stator currents will produce a rotating magnetic field that will rotate at 3000rpm in the air gap. Let us represent this rotating field by two stator poles rotating at 3000rpm as shown in figure.

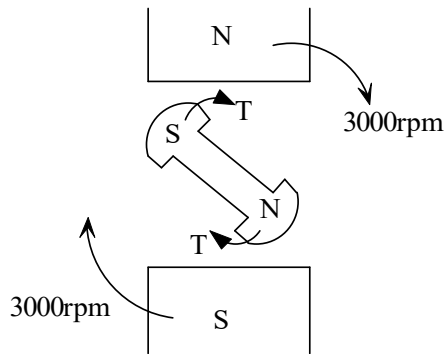


Figure (a) at $t = 0$

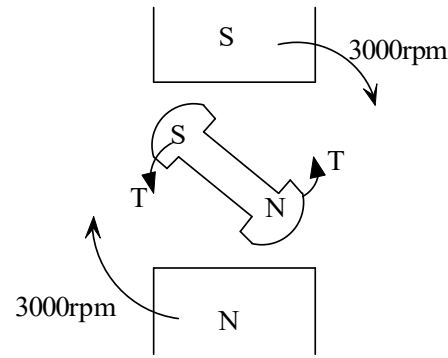


Figure (b) at $t = t_1$

At start ($t=0$), let the rotor poles be at the position shown in figure (a). The rotor will therefore experience a clockwise torque, making it rotate in the direction of the stator rotating poles. At $t=t_1$ (nearly 0.01 sec since $T=1/f=0.02\text{sec}$), let the stator poles move by half a revolution, shown in figure (b). The rotor poles have hardly moved, because of the high inertia of the rotor. Therefore, at this instant the rotor experiences a counterclockwise torque tending to make it rotate in the direction opposite to that of the stator poles. The net torque on the rotor in one revolution will be zero, and therefore the motor will not develop any starting torque. The stator field is rotating so fast that the rotor poles cannot catch up or lock onto it. The motor will not speed up but will vibrate.

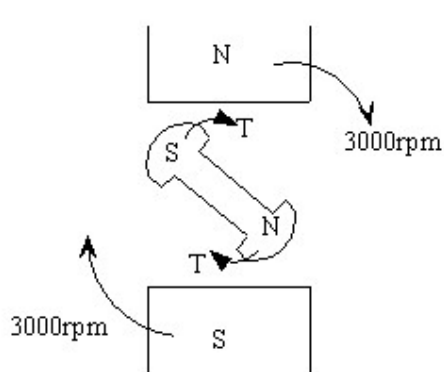


Figure (c) at $t = 0$

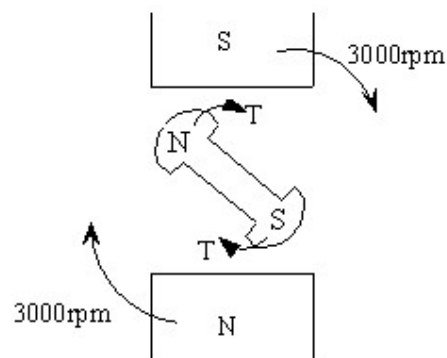


Figure (d) at $t = t_1$

However, if the rotor is rotated by some means in the same direction of the rotating field at a speed near the synchronous speed, rotor will overcome its moment of inertia (Figure (c) and (d)). If the rotor field is established at a particular instant when unlike poles come closer, a magnetic locking takes place between the stator and rotor poles. Hence, the rotor continues to rotate due to this magnetic locking even if the external force is removed.

General procedure to start a synchronous motor is

1. Give a 3-phase supply to 3-phase stator winding. This will produce a rotating field revolving at synchronous speed.
2. Drive the rotor by some external means like dc motor in the direction of rotating field at a speed close to synchronous speed.
3. Switch on the dc supply given to the rotor circuit which will produce rotor poles.
4. At a particular instant, both the fields get magnetically locked. The stator field pulls rotor field into synchronism. Now, the external devices used to rotate the rotor can be removed. But, the rotor will continue to rotate at the same speed as that of the rotating field due to magnetic locking.

Note: A synchronous motor has no starting torque. It has torque only when running at synchronous speed. Either a squirrel-cage winding is added to the rotor to cause it to start, or a DC motor is used to bring the rotor to near synchronous speed, at which time the AC is then applied. Once up to speed, the rotor snaps into step with the rotating magnetic field and will continue to rotate at synchronous speed.

The torque developed when the motor snaps into synchronous speed is called *pull-in torque*. If the load on the motor is increased to the point where the rotor is pulled out of synchronism (excessive torque), the motor will stop. The maximum torque a motor can develop before synchronization is lost is called *pull-out torque*. The pull-out torque is generally 1.5 times the continuously rated torque.

Methods of starting synchronous motor

1. By using an external motor

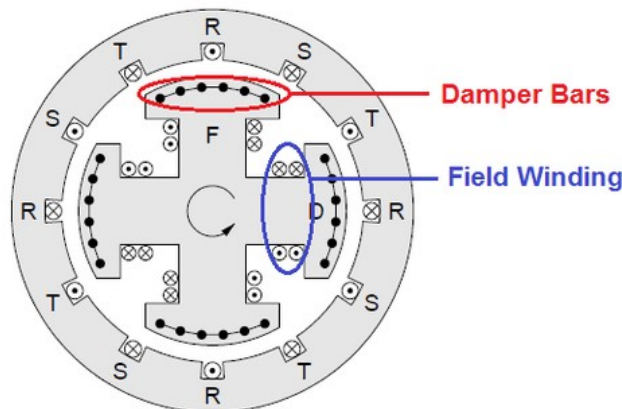
The rotor is brought to synchronous speed by using an external motor. Now, dc excitation is given to the rotor winding. The synchronous machine is then synchronized with the bus bar as a synchronous generator. The starting motor is then disconnected. Once in parallel, the synchronous machine will work as a motor. Now, the load can be connected to the synchronous motor. (Here, synchronization is needed as direction of rotation of the rotor as well as that of the rotating magnetic field of the stator should be the same).

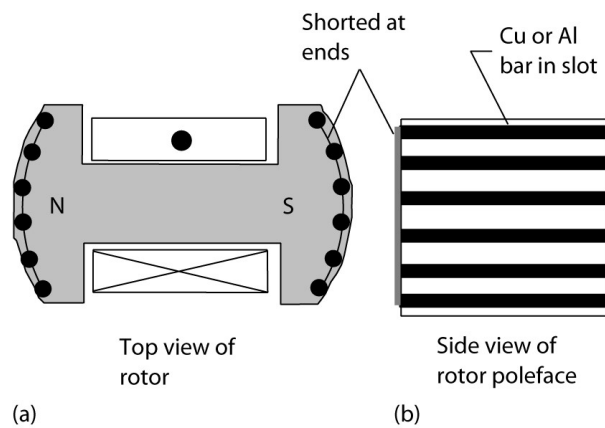
Note: The starting motor has to overcome the inertia of the synchronous motor at no load. Hence the rating of the starting motor can be much smaller than the rating of the synchronous motor.

2. By using damper (amortisseur) winding

In synchronous motor, in addition to normal field winding, an additional winding consisting of copper bars is placed in the slots in the pole faces. The bars are short circuited by end rings at both ends. These short circuited bars form a squirrel cage winding.

When a 3-phase supply is connected to the stator, the synchronous motor with damper winding will start as a 3-phase induction motor. As the motor approaches synchronous speed, the dc excitation is applied to the field windings. The rotor will then pull into step with the stator magnetic field.



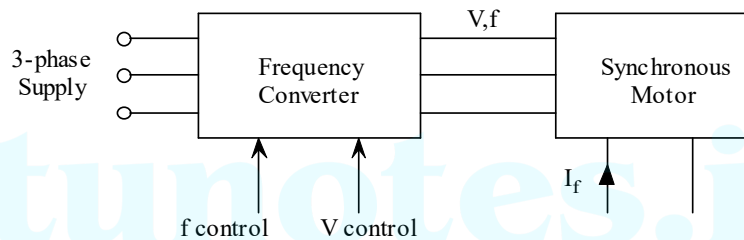


When the machine is running at synchronous speed, no current will be induced in the damper winding.

Instead of direct-on-line starting, the synchronous motor is sometimes started by star-delta starting, reactor starting or auto-transformer starting in order to limit the starting current.

3. Start with variable supply frequency

By using a frequency converter, a synchronous motor can be brought from standstill to its desired speed.



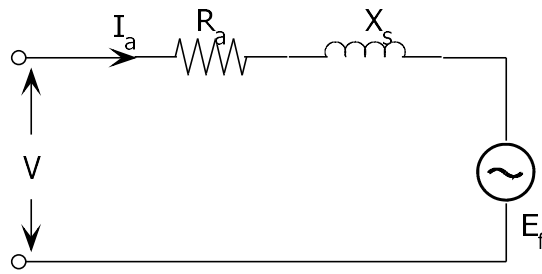
Initially, the field winding is excited with dc supply. The motor is started with a low voltage, low-frequency supply. This will make the stator field rotate slowly so that the rotor poles can follow the stator poles. Afterward, the frequency is gradually increased and the motor brought to its desired speed.

This method is expensive. However, if the synchronous motor has to run at variable speeds, this method may be used.

NOTE: Field winding has usually a large number of turns as compared to the stator turns. At the time of starting, very high voltage may be induced in the field winding. On starting, rotating magnetic field cuts the field winding at synchronous speed and hence high value of emfs are induced in it. This high value of emf may cause damage to the field winding insulation. The voltage induced in the field winding can be limited by short-circuiting the field winding or by connecting it to a resistance is about 7 to 10 times field winding resistance. As the speed increases, induced emf decreases and the resistance is gradually reduced. When the rotor reaches normal speed, external resistance is reduced to zero and the field winding is opened. After this, field winding is connected to the dc source. By this method, additional torque is produced due to interaction between stator field and rotor field due to current in field circuit. This additional torque adds to the induction motor torque developed by damper bars and the starting torque is increased.

EQUIVALENT CIRCUIT & PHASOR DIAGRAMS

A synchronous motor is same as a synchronous generator except that the direction of power flow is reversed. Equivalent circuit of a synchronous motor is given below,



V = Supply voltage/phase

E_f = Excitation voltage/phase

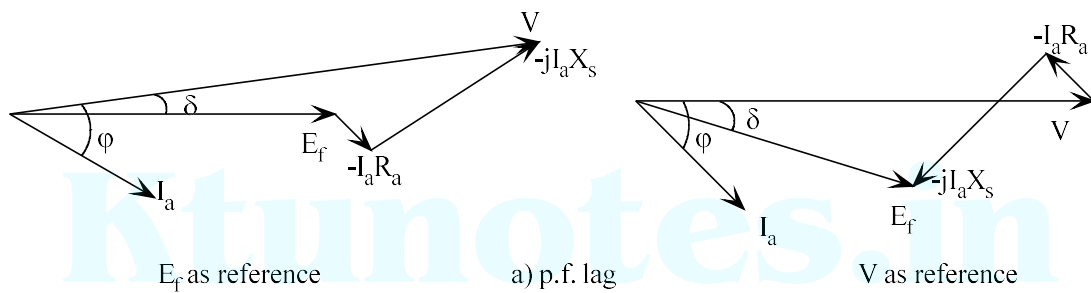
I_a = Armature current

δ = Power/Torque angle (angle between E_f and V)

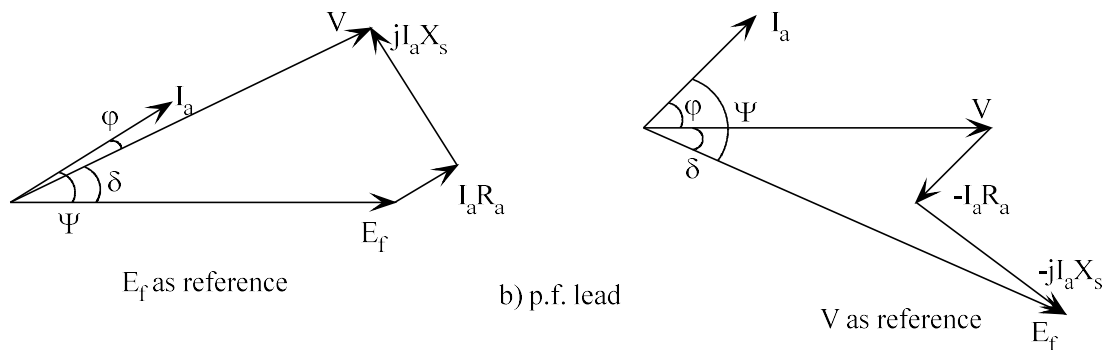
θ = Synchronous impedance angle = $\tan^{-1}(\frac{X_s}{R_a})$ (if R_a is neglected, $\theta = 90^\circ$)

ϕ = Power factor angle = angle between V and I_a .

a) Lagging power factor



b) Leading power factor



$$\bar{E}_f = \bar{V} - \bar{I}_a Z_s$$

$$E_f \angle -\delta^\circ = V \angle 0^\circ - I_a \angle -\phi \times (R_a + jX_s)$$

Qn 1: A 1000kVA, 11000V, 3-phase star connected synchronous motor has an armature resistance and reactance per phase of 3.5Ω and 40Ω respectively. Determine the induced emf and angular retardation of the rotor when fully loaded at a) unity power factor b) 0.8 p.f. lag and c) 0.8 p.f. lead.

Ans: $V = \frac{11000}{\sqrt{3}} = 6351V$

$$I_a = \frac{1000 \times 10^3}{\sqrt{3} \times 11000} = 52.49A$$

a) unity p.f.

$$\bar{E}_f = \bar{V} - \bar{I}_a(R_a + jX_s) = 6350.85\angle 0^\circ - 52.49\angle 0^\circ \times (3.5 + j40) = 6514.74\angle -18.8^\circ V$$

b) 0.8 p.f.lag

$$\bar{E}_f = \bar{V} - \bar{I}_a(R_a + jX_s) = 6350.85\angle 0^\circ - 52.49\angle -36.86^\circ \times (3.5 + j40) = 5187.57\angle -17.6^\circ V$$

c) 0.8 p.f.lead

$$\bar{E}_f = \bar{V} - \bar{I}_a(R_a + jX_s) = 6350.85\angle 0^\circ - 52.49\angle +36.86^\circ \times (3.5 + j40) = 7675\angle -13.49^\circ V$$

Qn 2: A 3-phase, 400V, star connected synchronous motor takes a power input of 5472W at rated voltage. $Z_s = 0 + j10\Omega/\text{phase}$. If its excitation voltage is adjusted equal to the rated voltage of 400V, calculate the power factor, load angle and armature current.

Ans:-

$$E_f = V = \frac{400}{\sqrt{3}} = 231V$$

$$\text{Power input} = \sqrt{3} \times V_L \times I_a \times \cos \phi = \sqrt{3} \times 400 \times I_a \times \cos \phi = 5472W$$

$$\text{Hence, } I_a \times \cos \phi = 7.898 \text{ -----(1)}$$

$$\sqrt{3} \times 400 \times I_a \times \cos \phi = 5472W$$

$$E_f \angle -\delta = V \angle 0 - I_a \angle \phi \times jX_s$$

$$E_f (\cos \delta - j \sin \delta) = V - I_a (\cos \phi + j \sin \phi) \times j10$$

Equating real and imaginary parts,

$$E_f \cos \delta = V + 10I_a \sin \phi$$

$$E_f \sin \delta = -10I_a \cos \phi = -10 \times 7.898 = -78.98$$

$$\sin \delta = -0.3419 \text{ or } \delta = -19.99^\circ$$

$$I_a \sin \phi = \frac{E_f \cos \delta - V}{X_s} = -1.3917 \text{ -----(2)}$$

From (1) and (2),

$$\tan \phi = -0.1762 \text{ or } \phi = -9.994 \text{ (hence current is lagging)}$$

$$I_a = \frac{7.898}{\cos \phi} = 8.02A$$

Qn 3: A 3980V, 50Hz, 4-pole, star connected synchronous motor generates back emf of 1790V per phase. $Z_s = 2.2 + j22\Omega/\text{phase}$. The torque angle is 30° electrical. Calculate armature current/phase ii) power factor of the motor iii) gross torque developed by the motor.

Ans:-

$$V = \frac{3980}{\sqrt{3}} = 2297.85V \quad E_f \angle -\delta = V \angle 0 - I_a \angle \phi \times Z_s$$

$$I_a \angle -\phi = \frac{V \angle 0 - E_f \angle -\delta}{Z_s} = \frac{2297.85\angle 0 - 1790\angle -30}{2.2 + j22} = 52.746\angle -34.16^\circ A$$

Armature current per phase = 52.746A

Power factor = $\cos(-34.16) = 0.827$ lag

$$\text{Power input} = \sqrt{3} \times V_L \times I_a \times \cos \phi = \sqrt{3} \times 3980 \times 52.746 \times \cos(-34.16) = 300875.6W$$

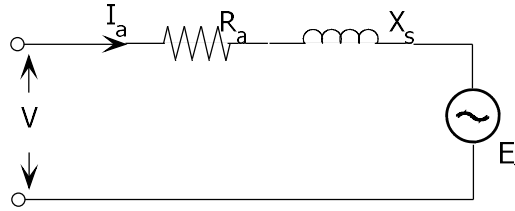
$$\text{Armature Cu loss} = 3 \times I_a^2 \times R_a = 3 \times 52.746^2 \times 2.2 = 18362.1W$$

$$\text{Mechanical power developed} = P_m = 300875.6 - 18362.1 = 282513.5W$$

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{4} = 1500rpm$$

$$T_g = \frac{60}{2\pi N_s} \times P_m = \frac{60}{2\pi \times 1500} \times 282513.5 = 1798.5 \text{ N-m}$$

POWER FLOW EQUATIONS (Synchronous Motor)



V = Supply voltage/phase

E_f = Excitation voltage/phase

I_a = Armature current

δ = Torque angle (angle between E_f and V)

θ = Synchronous impedance angle = $\tan^{-1}\left(\frac{X_s}{R_a}\right)$ (if R_a is neglected, θ = 90°)

φ = Power factor angle = angle between V and I_a.

$$\vec{V} = V \angle 0^\circ, \vec{E}_f = E_f \angle -\delta^\circ$$

$$\text{Synchronous impedance} = R_a + jX_s = Z_s \angle \theta$$

Complex Power input per phase

$$\begin{aligned} S_i &= P_i + jQ_i = \vec{V} \times \vec{I}_a^* \\ &= V \angle 0^\circ \times \left(\frac{V \angle 0^\circ - E_f \angle -\delta^\circ}{Z_s \angle \theta} \right)^* \\ &= V \angle 0^\circ \times \left[\frac{V}{Z_s} \angle -\theta - \frac{E_f}{Z_s} \angle (-\delta - \theta) \right]^* \\ &= \frac{V^2}{Z_s} \angle \theta - \frac{VE_f}{Z_s} \angle (\theta + \delta) \end{aligned}$$

$$\text{Real power input per phase, } P_i = \frac{V^2}{Z_s} \cos \theta - \frac{VE_f}{Z_s} \cos(\theta + \delta)$$

$$= \frac{V^2 R_a}{Z_s^2} - \frac{VE_f}{Z_s} \cos(\theta + \delta) \quad \text{since } \cos \theta = \frac{R_a}{Z_s}$$

$$\text{Reactive power input per phase } Q_i = \frac{V^2}{Z_s} \sin \theta - \frac{VE_f}{Z_s} \sin(\theta + \delta)$$

Maximum Power Input per phase

For maximum power input, $\frac{dP_i}{d\delta} = 0$

$$\frac{d}{d\delta} \left[\frac{V^2}{Z_s} \cos \theta - \frac{VE_f}{Z_s} \cos(\theta + \delta) \right] = 0$$

Since V, E_f, Z_s and θ are constants,

$$\sin(\theta + \delta) = 0 \quad \text{or } \theta + \delta = 180^\circ$$

For maximum power input, $\delta = 180 - \theta$.

$$\text{Maximum power input} = P_{i\max} = \frac{V^2}{Z_s} \cos \theta + \frac{VE_f}{Z_s}$$

Complex Power developed per phase

$$\begin{aligned} S_o &= P_o + jQ_o = \bar{E}_f \times \bar{I}_a^* \\ &= E_f \angle^{-\delta} \times \left(\frac{V \angle 0 - E_f \angle^{-\delta}}{Z_s \angle \theta} \right)^* \\ &= E_f \angle^{-\delta} \times \left[\frac{V}{Z_s} \angle(-\theta) - \frac{E_f}{Z_s} \angle(-\delta - \theta) \right]^* \\ &= \frac{VE_f}{Z_s} \angle(\theta - \delta) - \frac{E_f^2}{Z_s} \angle \theta \end{aligned}$$

$$\text{Mechanical power developed, } P_m = \frac{VE_f}{Z_s} \cos(\theta - \delta) - \frac{E_f^2}{Z_s} \cos \theta$$

$$\text{If } R_a \text{ neglected, } P_m = \frac{VE_f}{X_s} \sin \delta$$

Power Output (shaft power) = P_m - Rotational losses

Maximum mechanical power developed $P_{m\max}$

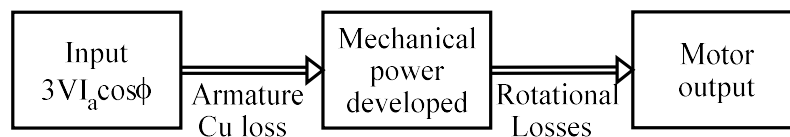
$$\begin{aligned} \frac{dP_m}{d\delta} &= 0 \\ \frac{d}{d\delta} \left(\frac{VE_f}{Z_s} \cos(\theta - \delta) - \frac{E_f^2}{Z_s} \cos \theta \right) &= 0 \end{aligned}$$

$$\sin(\theta - \delta) = 0$$

$$\delta = \theta$$

$$P_{m\max} = \frac{VE_f}{Z_s} - \frac{E_f^2}{Z_s} \cos \theta$$

$$\text{If } R_a \text{ is neglected, for maximum power developed, } \delta = 90^\circ \text{ and } P_{m\max} = \frac{VE_f}{X_s}$$



Qn1 : The excitation of a 415V, 3-phase, delta connected synchronous motor is such that the induced emf is 520V. The impedance per phase is $0.5 + j4\Omega$. If the friction and iron losses are constant at 1000W, calculate the power output, line current, power factor and efficiency for maximum power output.

Ans:

$$Z_s = 0.5 + j4 = 4.031 \angle 82.875^\circ$$

For maximum power output, $\delta = \theta = 82.875^\circ$

$$P_{m\max} = 3 \times \left(\frac{VE_f}{Z_s} - \frac{E_f^2}{Z_s} \cos \theta \right)$$

$$= 3 \times \left(\frac{415 \times 520}{4.031} - \frac{520^2}{4.031} \cos 82.875^\circ \right) = 135644.6 \text{ W}$$

Maximum power output, $P_{\text{out}} = P_m - \text{Rotational losses} = 135644.6 - 1000 = 134644.6 \text{ W}$

$$\text{Armature current, } I_a = \frac{V \angle 0^\circ - E_f \angle -\delta}{Z_s \angle \theta} = \frac{415 \angle 0^\circ - 520 \angle -82.875^\circ}{4.031 \angle 82.875^\circ} = 154.74 \angle -27.06^\circ \text{ A}$$

$$\text{Line current} = \sqrt{3} \times 154.74 = 268 \text{ A}$$

$$\text{Armature Cu loss} = 3 I_a^2 R_a = 3 \times 154.74^2 \times 0.5 = 35916.7 \text{ W}$$

$$\text{Power input} = \text{Power developed} + \text{Cu losses} = 135644.6 + 35916.7 = 171561.3 \text{ W}$$

$$\text{Power input} = \sqrt{3} \times V_L I_L \cos \phi = 171561.3 \text{ W}$$

$$\cos \phi = \frac{171561.3}{\sqrt{3} \times 415 \times 268} = 0.89 \text{ lag}$$

$$\text{Efficiency, } \eta = \frac{134644.6}{171561.3} = 78.48\%$$

Qn 2: A 3-phase, 500V, star connected synchronous motor gives net output of 17kW on full load operating at 0.9 lagging power factor. Its armature resistance is $0.8\Omega/\text{phase}$. The rotational losses are 1300W. Estimate the current drawn by the motor and full load efficiency.

$$\text{Power output} = 17000 \text{ W}$$

$$\text{Mechanical power developed, } P_m = P_{\text{out}} + \text{rotational losses} = 17000 + 1300 = 18300 \text{ W}$$

$$\text{Power input} = \sqrt{3} \times V_L \times I_a \times \cos \phi = \sqrt{3} \times 500 \times I_a \times 0.9 = 779.4 \times I_a$$

$$\text{Armature Cu loss} = 3 I_a^2 R_a = 3 \times I_a^2 \times 0.8 = 2.4 I_a^2$$

$$\text{Power input} = \text{Mechanical power developed} + \text{Armature Cu loss}$$

$$779.4 \times I_a = 2.4 I_a^2 + 18300$$

$$2.4 I_a^2 - 779.4 \times I_a + 18300 = 0$$

$$I_a = 299.27 \text{ or } 25.48 \text{ A}$$

$$I_a = 25.48 \text{ A (Since output is 17kW and } V = 500 \text{ V; } I_a = 299.57 \text{ A not possible)}$$

$$\text{Power input} = 779.4 \times 25.48 = 19859.1 \text{ W}$$

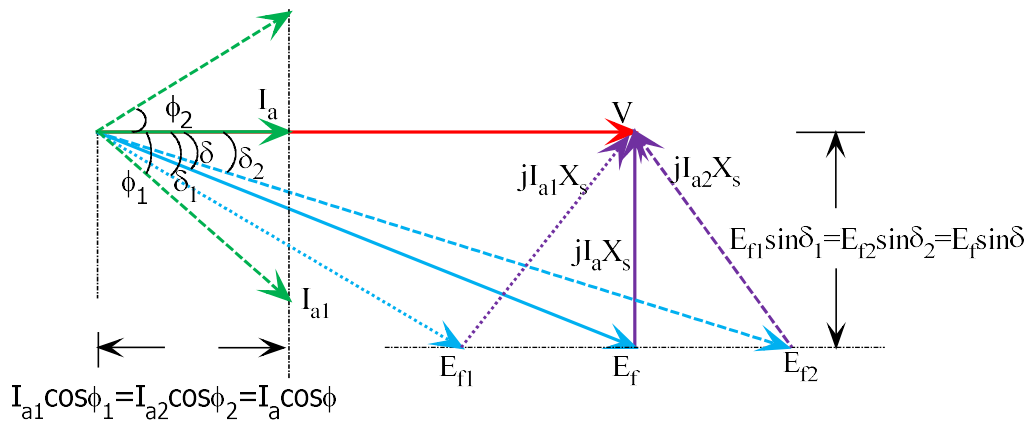
$$\text{Efficiency, } \eta = \frac{17000}{19859.1} \times 100 = 85.6\%$$

Qn3 : A 500V, 3-phase, mesh connected synchronous motor has an excitation emf of 600V. The synchronous impedance per phase is $0.4 + j5\Omega$ while the windage, friction and iron losses are 1200W. What maximum power output can it deliver? What is the corresponding line current, power factor and motor efficiency?

Ans:- 160.8kW, 258.5A, 0.842 lag, 85.5%

Effect of varying field current – V CURVES & INVERED V CURVES

Consider a synchronous motor connected to the infinite bus and operating at constant load with induced emf E_f , torque angle δ and working at unity power factor with current I_a .



Now, assume that excitation is increased so that E_f is decreased to E_{f1} . Since the synchronous motor is operating at constant load, $P_m = \frac{E_f V}{X_s} \sin \delta$ remains constant. Hence, torque angle δ is increased to

δ_1 so that $E_{f1} \sin \delta_1 = E_f \sin \delta$.

Also, $V I_a \cos \phi = \text{constant}$ (since $R_a = 0$)

Hence, $I_a \cos \phi = \text{constant}$.

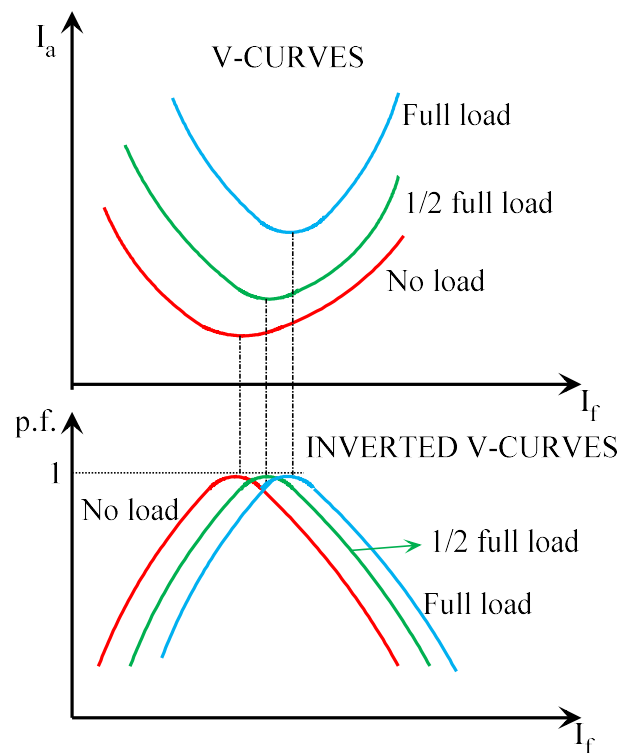
Armature current increases from I_a to I_{a1} . But, $I_a \cos \phi$ remains same. Power factor decreases to $\cos \phi_1$ (lag) so that $I_{a1} \cos \phi_1 = I_a \cos \phi$

Now, assume that excitation is increased so that E_f is decreased to E_{f2} . But, $\frac{E_f V}{X_s} \sin \delta$ remains constant. Hence, torque angle δ is decreased to δ_2 so that $E_{f2} \sin \delta_2 = E_f \sin \delta$. But, $V I_a \cos \phi = \text{constant}$. Armature current increases from I_a to I_{a2} . But, $I_a \cos \phi$ remains same. Power factor decreases to $\cos \phi_2$ (lead) so that $I_{a2} \cos \phi_2 = I_a \cos \phi$.

The excitation corresponding to unity power factor is called normal excitation, while the excitation larger than this is called over-excitation and less than this is called under-excitation.
Note: Synchronous motor draws a leading power factor current when over-excited and draws a lagging power factor current when under-excited.

I_a Vs I_f for various values of load is called V-curves and power factor Vs I_f is called inverted V-curve.

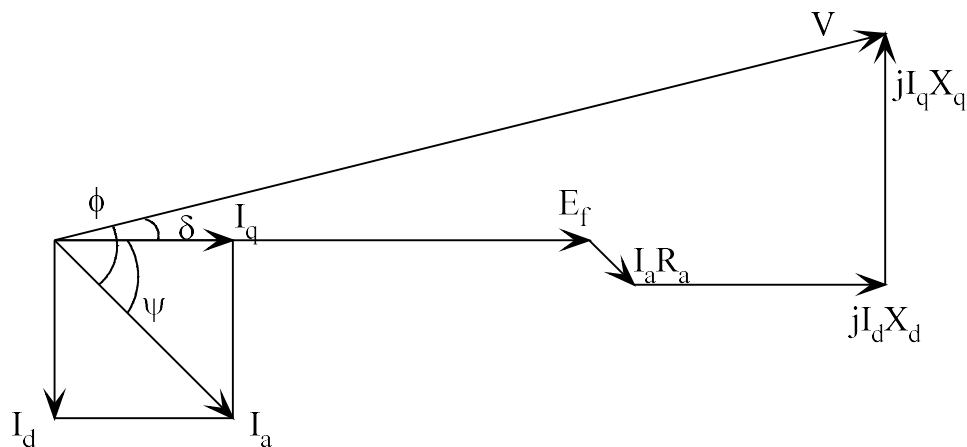
Line joining all the minimum armature current points of the V-curves is called unity power factor compounding curve.

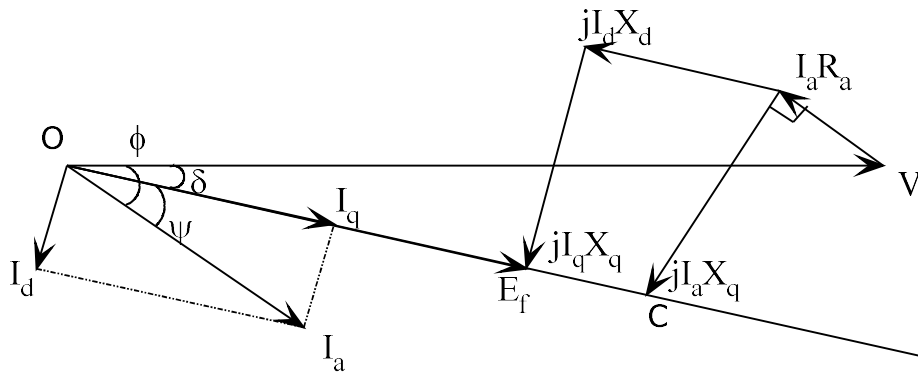


Minimum excitation : Since $E_f \sin \delta$ remains constant, as E_f is reduced, δ is increased. Maximum possible δ is 90° . Hence minimum permissible excitation is $E_{f \min} = \frac{P_{m \max} X_s}{V}$.

PHASOR DIAGRAMS OF A SALIENT POLE SYNCHRONOUS MOTOR

a) at lagging power factor





$$\vec{OC} = \vec{V} - \vec{I}_a (R_a + jX_q)$$

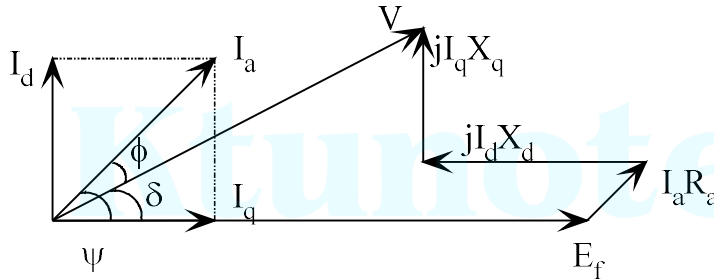
Angle of $\vec{OC}$ will give δ . (Taking V as reference)

$$\psi = \phi - \delta \text{ for lag}$$

$$\psi = -\phi - \delta \text{ for lead}$$

$$E_f = V \cos \delta - I_a R_a \cos \Psi - I_a X_d \sin \Psi$$

b) at leading power factor



Qn 1: A 12 pole 50Hz, 11kV, 3-phase, star connected, synchronous motor has $R_a=0.2\Omega$; $X_d=5\Omega$ and $X_q=3\Omega$ per phase. Find the excitation voltage when the motor draws an input of 20MW at 0.6 p.f. leading.

Ans:-

$$V = \frac{11000}{\sqrt{3}} = 6350.85V$$

$$\text{Power input} = \sqrt{3} \times V_L \times I_a \times \cos \phi = \sqrt{3} \times 11000 \times I_a \times 0.6 = 20000000W$$

$$I_a = 1749.55A$$

$$\vec{OC} = \vec{V} - \vec{I}_a (R_a + jX_q) = 6350.85 \angle 0^\circ - 1749.55 \angle 53.13^\circ \times (0.2 + j3) = 10893.6 \angle -18.35^\circ$$

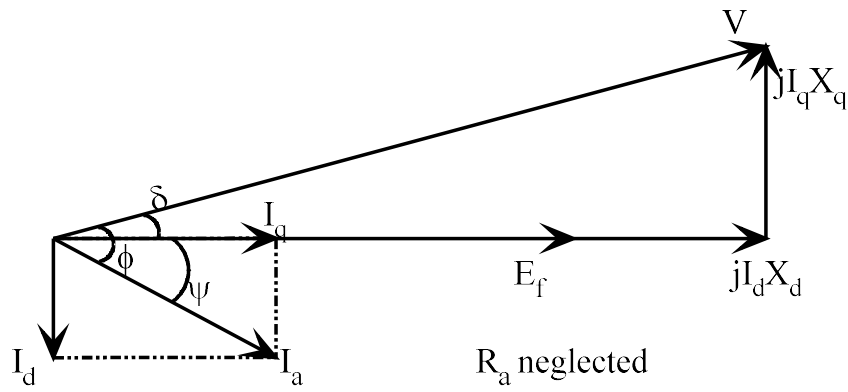
$$\delta = 18.35^\circ$$

$$\psi = -\phi - \delta = -53.13 - 18.35 = -71.48^\circ$$

$$\begin{aligned} E_f &= V \cos \delta - I_a R_a \cos \Psi - I_a X_d \sin \Psi \\ &= 6350.85 \times \cos 18.35^\circ - 1749.55 \times 0.2 \times \cos 71.48^\circ - 1749.55 \times 5 \times \sin(-71.48)^\circ \\ &= 14211.5V \end{aligned}$$

TORQUE ANGLE CHARACTERISTICS OF SALIENT POLE SYNCHRONOUS MOTOR

The phasor diagram of a salient pole synchronous generator at lagging power factor (R_a neglected) is shown below.



Complex Power developed per phase

$$S = P + jQ = \bar{V} \times \bar{I}_a^* \text{ (if } R_a \text{ is neglected)}$$

Take E_f as reference vector.

$$V \angle \delta = V \cos \delta + jV \sin \delta$$

$$I_a = I_q - jI_d$$

$$I_a^* = I_q + jI_d$$

$$\begin{aligned} S = P + jQ &= \bar{V} \times \bar{I}_a^* \\ &= (V \cos \delta + jV \sin \delta)(I_q + jI_d) \end{aligned}$$

$$X_q I_q = V \sin \delta$$

$$I_q = \frac{V \sin \delta}{X_q}$$

$$X_d I_d = V \cos \delta - E_f$$

$$I_d = \frac{V \cos \delta - E_f}{X_d}$$

Substituting,

$$S = (V \cos \delta + jV \sin \delta) \left(\frac{V \sin \delta}{X_q} + j \frac{V \cos \delta - E_f}{X_d} \right)$$

$$\begin{aligned} \text{Real power developed per phase} &= \frac{V^2 \sin \delta \cos \delta}{X_q} + \frac{VE_f \sin \delta}{X_d} - \frac{V^2 \sin \delta \cos \delta}{X_d} \\ &= \frac{VE_f \sin \delta}{X_d} + \frac{V^2}{2} \left(\frac{1}{X_q} - \frac{1}{X_d} \right) \sin 2\delta \end{aligned}$$

$$\text{Total real power, } P = \frac{3VE_f \sin \delta}{X_d} + \frac{3V^2}{2} \left(\frac{1}{X_q} - \frac{1}{X_d} \right) \sin 2\delta$$

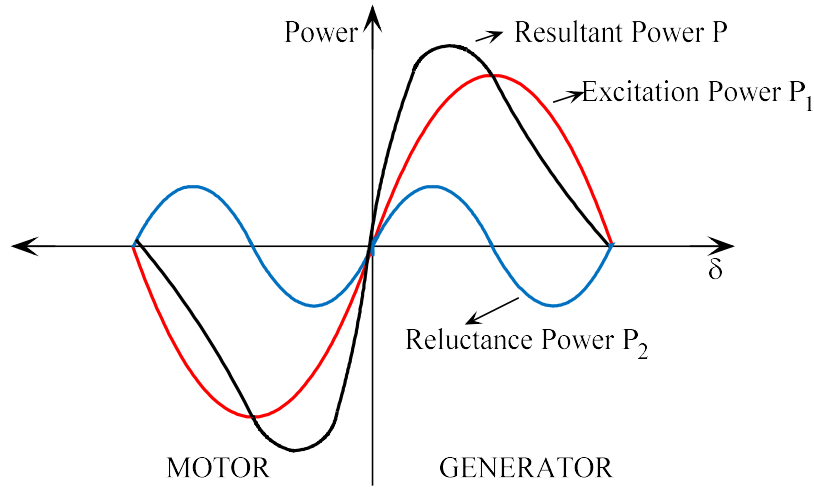
First term $\frac{3VE_f \sin \delta}{X_d} = P_1$ is called excitation power (power due to field excitation)

Second term $\frac{3V^2}{2} \left(\frac{1}{X_q} - \frac{1}{X_d} \right) \sin 2\delta = P_2$ is called reluctance power (power due to saliency). (power

developed in salient pole synchronous motor due to the variable reluctance offered by the salient pole).

P_2 does not depend on field excitation and is present even when field current is zero.

For a cylindrical rotor machine, $X_d=X_q$ and hence reluctance power is zero.



Reluctance power is maximum, when $\delta = 45^\circ$.

Maximum power occurs when $\delta \leq 90^\circ$.

Qn 1: A 1500kVA, star-connected, 6.6kV salient-pole synchronous motor has $X_d=23.2\Omega$ and $X_q=14.5\Omega$ /phase respectively; armature resistance being negligible. Calculate the excitation emf when the motor is supplying rated load at 0.8 leading power factor. What maximum load the motor can supply without loss of synchronism, if the excitation is cut off? What will be the value of torque angle under this condition?

Ans:-

$$V = \frac{6600}{\sqrt{3}} = 3810.5V$$

$$I_a = \frac{1500000}{\sqrt{3} \times 6600} = 131.2A$$

$$\vec{OC} = \vec{V} - \vec{I}_a(R_a + jX_q) = 3810.5\angle 0^\circ - 131.2\angle 36.86^\circ \times j14.5 = 5180.3\angle -17.09^\circ$$

$$\delta = 17.09^\circ$$

$$\psi = -\phi - \delta = -36.86 - 17.09 = -53.95^\circ$$

$$\begin{aligned} E_f &= V \cos \delta - I_a R_a \cos \Psi - I_a X_d \sin \Psi \\ &= 3810.5 \times \cos 17.05^\circ - 131.2 \times 23.2 \times \sin(-53.95^\circ) \\ &= 6103.98V \end{aligned}$$

With excitation cut off, output is only reluctance power,

$$P = \frac{3V^2}{2} \left(\frac{1}{X_q} - \frac{1}{X_d} \right) \sin 2\delta$$

For maximum reluctance power,

$$\sin 2\delta = 0 \text{ or } \delta = 45^\circ$$

$$P_{\max} = \frac{3V^2}{2} \left(\frac{1}{X_q} - \frac{1}{X_d} \right) = \frac{3 \times 3810.5^2}{2} \left(\frac{1}{14.5} - \frac{1}{23.2} \right) = 563272.38W$$

Qn 2: A 6.6kV, 3-phase, star-connected synchronous motor is running in parallel with an infinite bus. Its direct and quadrature axis synchronous reactances are 10Ω and 5Ω respectively. If the field current is reduced to zero, find the maximum load that can be put on the synchronous motor. Also calculate the armature current and the maximum power. Neglect armature resistance.

Ans:-

$$V = \frac{6600}{\sqrt{3}} = 3810.5V$$

$$P = \frac{3VE_f \sin \delta}{X_d} + \frac{3V^2}{2} \left(\frac{1}{X_q} - \frac{1}{X_d} \right) \sin 2\delta$$

When the field current is zero, $E_f=0$.

$$P = \frac{3V^2}{2} \left(\frac{1}{X_q} - \frac{1}{X_d} \right) \sin 2\delta$$

For maximum reluctance power,

$$\sin 2\delta = 0 \text{ or } \delta = 45^\circ$$

$$P_{\max} = \frac{3V^2}{2} \left(\frac{1}{X_q} - \frac{1}{X_d} \right) = \frac{3 \times 3810.5^2}{2} \left(\frac{1}{5} - \frac{1}{10} \right) = 2177986.5W$$

$$I_q = \frac{V \sin \delta}{X_q} = \frac{3810.5 \times \sin 45^\circ}{5} = 538.89A$$

$$I_d = \frac{V \cos \delta}{X_d} = \frac{3810.5 \times \cos 45^\circ}{10} = 269.44A$$

$$\text{Armature current at maximum power, } I_a = \sqrt{I_d^2 + I_q^2} = \sqrt{269.44^2 + 538.89^2} = 602.5A$$

HUNTING

When synchronous motor is on no-load, the stator and rotor pole axes almost coincide with each other.

When motor is loaded, the rotor pole axis falls back with respect to stator. The angle by which rotor retards is called load angle δ . Under steady state operation of synchronous motor, the electromagnetic torque T_e is equal and opposite to the load torque T_L . In steady state, the rotor runs at synchronous speed, thereby maintaining constant value of the torque angle δ .

If there is a sudden change in the load torque, the equilibrium is disturbed and there is a resultant torque which changes the speed of the motor.

$$T_e - T_L = J \frac{d\omega_m}{dt} \text{ where } J = \text{moment of inertia and } \omega_m = \text{angular velocity of rotor}$$

When there is sudden change in load torque, the motor slows down temporarily and δ is sufficiently increased to restore the torque equilibrium and the synchronous speed.

$$T_e = \frac{3}{\omega_s} \frac{VE_f}{X_s} \sin \delta \quad (\text{Note: Power developed } P = \omega_s T_e = \frac{3VE_f}{X_s} \sin \delta)$$

As δ increases, T_e increases. Motor is accelerated. By the time rotor reaches synchronous speed, δ increases beyond δ_1 which is required for new equilibrium state. Hence, speed continues to increase beyond synchronous speed. When rotor accelerates beyond synchronous speed, δ decreases. T_e decreases. At the point when $T_e = T_L$, equilibrium is not restored. Rotor continues to swing backwards. δ goes on decreasing beyond δ_1 . When $\delta < \delta_1$, $T_L > T_e$, motor slows down. δ increases again. The rotor oscillates around δ_1 and synchronous speed before reaching the new steady state.

This phenomenon of oscillation of the rotor about its final equilibrium position is called hunting.

Causes of hunting :-

- 1) Sudden changes of load
- 2) Sudden changes in excitation

Effects of hunting:-

- 1) It can lead to loss of synchronism
- 2) During hunting, current drawn by the motor fluctuates; causes problem to other appliances connected to same supply.
- 3) Large mechanical stress may develop in the rotor shaft
- 4) Machine losses increase and temperature of the machine rises

Reduction of hunting:-

- a) Use of damper windings
- b) Use of flywheels
- c) By designing synchronous machines with suitable synchronizing power coefficients

Use of damper winding to prevent hunting

In the slots provided in the pole faces, a short circuited winding is placed. When rotor oscillates, emf is induced in damper winding and hence current flows. According to Lenz's law, the direction of induced emf is always so as to oppose the cause producing it. The cause is oscillations. Hence induced currents oppose the hunting. The oscillations are damped out quickly.

Synchronous condenser

When synchronous motor is over excited it takes leading power factor current. An overexcited synchronous motor on no load can be operated at nearly zero power factor leading. Then, it will act as a normal capacitor for improving the power factor. Hence, an overexcited synchronous motor on no load is called synchronous condenser.

MULTIPLE CHOICE QUESTIONS:

01	When the excitation of an unbalanced salient pole synchronous motor gets disconnected	
	A. The motor will burn	B. The motor will stop
	C. The motor will run as a reluctance motor at the same speed	D. The motor will run as a reluctance motor at lower speed
02	The back emf set up in the stator of a synchronous motor will depend on	
	A. Rotor speed only	B. Rotor excitation only
	C. Rotor excitation and rotor speed	D. Coupling angle, rotor speed and excitation
03	The armature current of a synchronous motor has large values for	
	A. Low excitation only	B. High excitation only
	C. Both low and high excitation	D. Depends on other factors
04	What happens if field winding of the synchronous motor is short circuited?	
	A. First starts as induction motor and then run as synchronous motor	B. Not start
	C. Motor will burn out	D. Run as induction motor
05	The speed regulation of a synchronous motor is	
	A. 100%	B. 50%
	C. 25%	D. 0%
06	In a 3-phase synchronous motor on no-load, if one of the phases is short circuited, the motor will	
	A. Run as before	B. Overheated and eventually burn
	C. Will stop	D. None of the above
07	In a 3-phase synchronous motor on rated load, if one of the phases is short circuited, the motor will	
	A. Run as before	B. Overheated and eventually burn
	C. Will stop	D. None of the above

08	A 3-phase salient pole synchronous motor with damper winding is fed from an infinite bus and is running at no-load. Now, if the field current is reduced to zero, then the	
	A. Motor will run at synchronous speed	B. Motor will stop
	C. Motor will run at low speed	D. Motor will burn
09	What is the ratio of no-load speed to full-load speed of a 200kVA, 12 poles, 2200V, 3-phase, 50Hz synchronous motor?	
	A. 1	B. Less than 1
	C. Greater than 1	D. Infinite
10	The direction of rotation of a synchronous motor can be reversed by reversing	
	A. Field winding	B. Polarity of rotor poles
	C. Supply phase sequence	D. None of the above

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Module 3

Three phase Induction motor – Constructional features, Expressions for Power and Torque - Torque-Slip characteristics, Phasor diagram, Equivalent Circuit of Induction motor- Tests on Induction motors for determination of equivalent circuit-Numerical Problems.

THREE PHASE INDUCTION MOTOR

The induction motor is the most rugged and the most widely used machine in industry. The induction machine can operate both as a motor and as a generator. However, it is seldom used as a generator supplying electrical power to a load. The performance characteristics as a generator are not satisfactory for most applications. The induction machine is extensively used as a motor in many applications.

Induction motor was invented by Nikola Tesla in 1888.

Construction

A three phase induction motor consists of two parts

- c) Stator
- d) Rotor

Stator

The core is hollow cylindrical in shape and is made of high grade sheet steel laminations (0.35mm to 0.65mm thick) insulated from each other. It consists of number of slots in its inner periphery to accommodate armature conductors (winding). A three phase distributed, full pitched/short pitched windings are placed in these slots. Windings are suitably connected to form a balanced three phase star or delta connected circuit. It is wound for a definite number of poles as per requirement of speed.

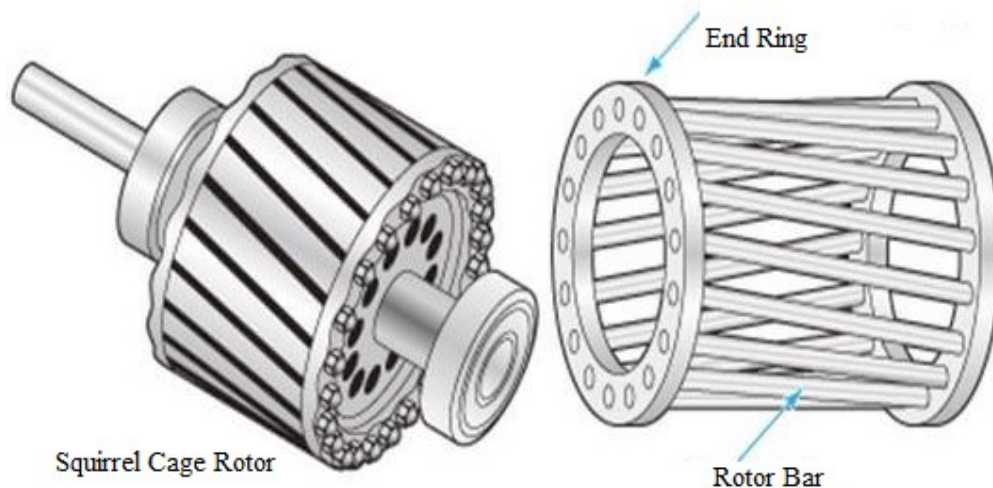
Rotor

Two types

- c) Squirrel cage
- d) Wound rotor (slip ring type)

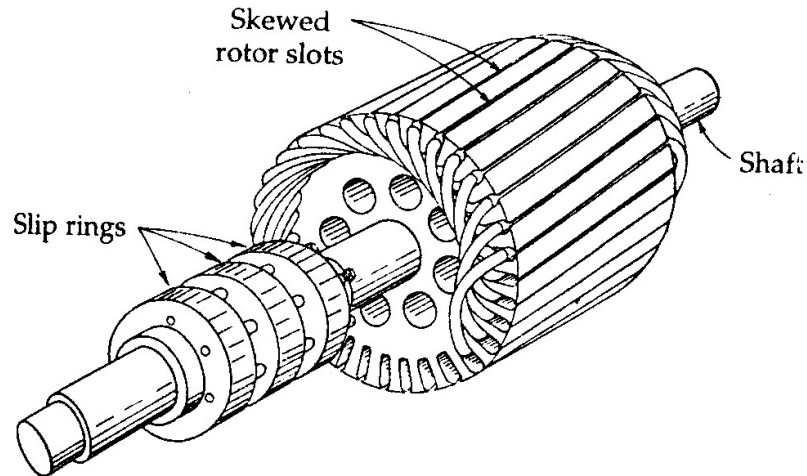
Squirrel cage

It is cylindrical (solid) in shape and built up of number of laminations having slots on its outer surface. Solid copper or aluminium bars short circuited at both ends by copper rings are inserted in these slots. There is no insulation between the iron core and the conductors. The current induced in the rotor is contained within the circuit formed by the conductors and end rings. Skewing the rotor slots helps avoid crawling (locking in at sub synchronous speed) and reduces vibration.

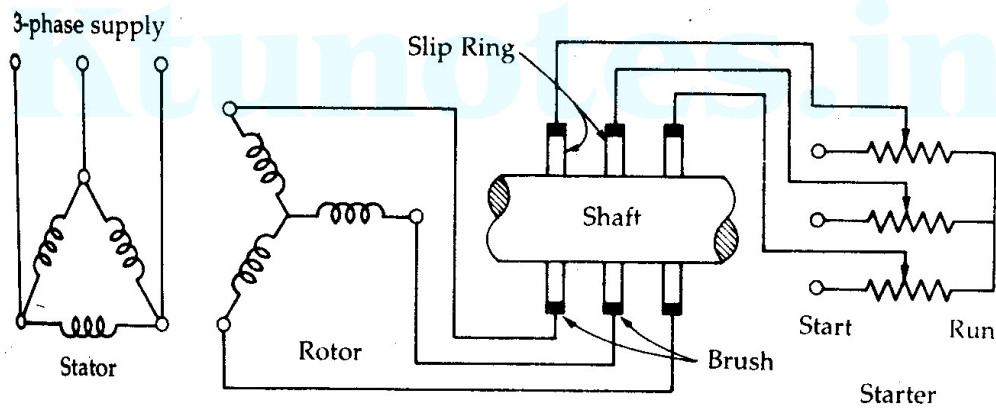


Wound rotor (Slip ring type)

It consists of a laminated cylindrical core having slots on its outer surface. These slots carry a 3 phase winding similar to the one in the stator. The rotor winding is uniformly distributed in the slots and is usually star connected.



The rotor is wound for as many poles as the number of stator poles. The open ends of the rotor winding are brought out and joined to 3 insulated slip rings mounted on the rotor shaft with one brush resting on each slip ring. The 3 brushes are connected to a 3 phase star connected rheostat. At starting the external resistance is included in the rotor circuit to give a large starting torque and to reduce the starting current. As the motor speeds up, the resistance is gradually reduced up to zero.



Comparison of squirrel cage (SCIM) and slip ring induction motors (SRIM)

SCIM	SRIM
1. Its rotor consists of copper bars permanently short circuited at both ends	1. Its rotor consists of 3 phase windings which is then connected to external resistance through slip rings and brushes
2. Starting torque is poor	2. Better starting torque can be achieved by inserting external resistance in the rotor circuit.
3. Separate starting methods are needed in order to reduce the starting current	3. Rotor resistance starting can be used
4. Its rotor can adjust to any number of stator poles. So different speeds can be obtained by different arrangements of stator winding.	4. Rotor and stator are wound for same number of poles
5. It has less components and hence less labor since it has no rotor winding. So it is cheaper	5. Slip rings, brushes, starting resistance etc. increases the cost
6. Better efficiency	6. Low efficiency

7. Better cooling (larger space to provide fan blades on rotor)	7. Cooling not efficient
8. Less maintenance	8. More maintenance

The transfer of energy from the stator to the rotor, whether squirrel cage or wound rotor, is by means of electromagnetic induction and occurs in a manner similar to that in a transformer.

Principle of operation of 3 phase induction motor

When the stator winding is connected to three phase ac supply, a rotating magnetic field is established in the air gap which rotates at synchronous speed. Initially, rotor is stationary. Due to relative speed between the rotating magnetic field and stationary rotor conductors, an emf is induced in the rotor. As the rotor circuit is closed, currents will circulate through them. According to Lenz's law, these induced currents will flow in such a direction so as to oppose the cause producing it. Here the cause is relative speed. In order to reduce the relative speed, the currents in the rotor produce a torque tending to rotate the rotor in the same direction of rotating field.

At synchronous speed of the rotor, the relative speed is zero, no emf and no torque developed, rotor tends to stop, hence rotor cannot attain synchronous speed. Motor runs at a speed slightly less than synchronous speed.

Speed of the rotating magnetic field is called synchronous speed N_s . ($N_s = \frac{120f}{P}$)

Slip

Difference between the speed of the rotating magnetic field N_s and the actual speed of the rotor N is called slip speed.

Slip, $s = \frac{N_s - N}{N_s} \times 100\%$ where N = Rotor speed

At starting, $N=0$, hence $s=1$ (standstill)

At synchronous speed, $N=N_s$ and $s=0$

At starting, rotor is stationary and the frequency of rotor emf or current is same as the supply frequency.

When the rotor starts revolving, then the frequency of the rotor currents depends on the relative speed or slip.

Let f_2 be the rotor frequency at any slip s

$$f_2 = \frac{P(N_s - N)}{120} = \frac{(N_s - N)}{N_s} \frac{PN_s}{120} = sf$$

Rotor frequency = slip x supply frequency

Note: The normal full-load slip of the induction motor is of the order of 2%-8%, so the frequency of the rotor currents is as low as 1-4Hz.

Let E_1 and E_2 be the stator and rotor induced voltages at standstill

$$E_1 = 4.44K_{w1}f\Phi T_1$$

$$E_2 = 4.44K_{w2}f\Phi T_2$$

where

T_1 =No. of stator turns/phase

T_2 =No. of rotor turns/phase

K_{w1} =Stator winding factor

K_{w2} =Rotor winding factor

$$\frac{E_1}{E_2} = \frac{K_{w1}T_1}{K_{w2}T_2} = \text{Effective turns ratio}$$

or

$$E_1 \propto E_2$$

Under running condition, $f_2 = sf$

$$\text{Rotor emf/phase} = 4.44K_{w2}(sf)\Phi T_2 = sE_2$$

Slip depends on the load on the motor. As the load increases, slip also increases (or speed decreases).

Let R_1 = stator winding resistance/phase

X_1 = stator leakage reactance/phase

R_2 = rotor winding resistance/phase

X_2 = rotor leakage reactance/phase at standstill

sX_2 = rotor leakage reactance/phase during running condition

(Note: At standstill, $X_2 = 2\pi f_2 L_2 = 2\pi f L_2$)

During running condition, $f_2 = sf$; rotor leakage reactance/phase = sX_2)

A 3 phase induction motor is similar to a three phase transformer (principle of operation is mutual induction) with secondary short circuited and rotating.

Note: Induced currents in the 3-phase rotor windings also produce a rotating field. Its speed with respect to rotor is $N_{s2} = \frac{120f_2}{P} = \frac{120sf}{P} = sN_s$. Because the rotor itself is rotating at N rpm, speed of the rotor field with respect to stator is $N + sN_s = N_s(1-s) + sN_s = N_s$. Hence, both the stator field and rotor field are rotating in the air gap at the same synchronous speed N_s .

Note: The stator magnetic field and rotor magnetic field are stationary with respect to each other. The interaction between these two fields can be considered to produce the torque.

Note: Assume that 3-phase supply is given to the rotor windings of a slip ring induction motor and stator windings are shorted. Initially, rotor is stationary. When 3-phase currents flow through the 3-phase rotor winding, a rotating magnetic field is produced in the rotor. Assume that the direction of rotating field is clockwise. This rotating field induces emfs and hence currents in the stator winding. This current will flow such that it opposes the cause (relative speed). Hence, the rotor will rotate in anticlockwise direction.

Qn.1: A 3 phase induction motor has 2 poles and is connected to 400V, 50Hz supply. Calculate the actual rotor speed and rotor frequency when the slip is 4%.

Ans:-

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{2} = 3000 \text{ rpm}$$

$$\text{Motor speed, } N = (1-s)N_s = (1-0.04) \times 3000 = 2880 \text{ rpm}$$

$$\text{Rotor speed, } f_2 = sf = 0.04 \times 50 = 2 \text{ Hz}$$

Qn.2: A 8 pole alternator runs at 750rpm and supplies power to a 6 pole induction motor which has at full load a slip of 3%. Find the full load speed of the induction motor and the frequency of its rotor emf.

Ans:

$$\text{Supply frequency } f = \frac{PN}{120} = \frac{8 \times 750}{120} = 50 \text{ Hz}$$

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{6} = 1000 \text{ rpm}$$

$$N = (1-s)N_s = (1-0.03) \times 1000 = 970 \text{ rpm}$$

Rotor frequency, $f_2 = sf = 0.03 \times 50 = 1.5\text{Hz}$

Qn.3: In a 6 pole, 3 phase, 50Hz motor with star-connected rotor, the rotor resistance per phase is 0.3Ω , the reactance at standstill is 1.5Ω per phase and an emf between the slip rings on open circuit is 175V. Calculate i) slip at a speed of 950 rpm ii) rotor emf per phase iii) rotor frequency and reactance at a speed of 950 rpm.

Ans:-

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{6} = 1000\text{rpm}$$

$$\text{i) Slip, } s = \frac{N_s - N}{N_s} \times 100\% = \frac{1000 - 950}{1000} \times 100 = 5\%$$

$$\text{ii) Rotor emf per phase at standstill, } E_2 = \frac{175}{\sqrt{3}} = 101\text{V}$$

$$\text{iii) Rotor frequency, } f_2 = sf = 0.05 \times 50 = 2.5\text{Hz}$$

$$\text{iv) Rotor reactance, } X_2' = sX_2 = 0.05 \times 1.5 = 0.075\Omega$$

Qn.4: The stator of a 3-phase, 4-pole, slip ring induction motor is connected to 50Hz supply. At the rotor terminals, a frequency of 30Hz is required. Find the possible speeds at which the rotor must be driven. What is the ratio of slip-ring emfs at these speeds at no load?

Ans:-

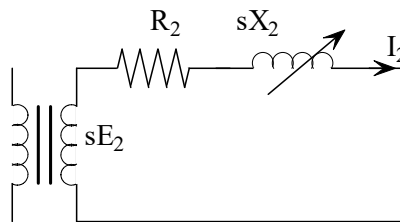
$$\text{slip, } s = \frac{f_2}{f} = \frac{30}{50} = 0.6$$

$$N = (1 \pm s)N_s = (1 \pm 0.6) \times 1500 = 600 \text{ or } 2400\text{rpm}$$

Rotor emfs per phase at both speeds are equal ($=sE_2$). Hence, ratio = 1.

Torque Equation

Equivalent rotor circuit is



Rotor equivalent circuit

Torque, $T \propto \Phi I_2 \cos \Phi_2$

where Φ = flux per pole

I_2 = rotor current

Φ_2 = angle between rotor emf and rotor current

$$\text{Rotor current, } I_2 = \frac{sE_2}{\sqrt{R_2^2 + (sX_2)^2}}$$

$$\text{Power factor, } \cos \Phi_2 = \frac{R_2}{\sqrt{R_2^2 + (sX_2)^2}}$$

But,

$$\text{Rotor induced emf at standstill, } E_2 = 4.44 K_{w2} f \Phi T_2$$

$$\text{or } \Phi \propto E_2$$

Hence,

$$T \propto E_2 \cdot \frac{sE_2}{\sqrt{R_2^2 + (sX_2)^2}} \cdot \frac{R_2}{\sqrt{R_2^2 + (sX_2)^2}}$$

$$T \propto \frac{sE_2^2 R_2}{R_2^2 + (sX_2)^2}$$

Here, the proportionality constant is $\frac{3}{2\pi} \frac{N_s}{60}$

$$T = \frac{3}{2\pi} \frac{N_s}{60} \cdot \frac{sE_2^2 R_2}{R_2^2 + (sX_2)^2} \text{ N-m}$$

$$T = \frac{sE_2^2 R_2}{R_2^2 + (sX_2)^2} \text{ synchronous watts}$$

Synchronous watts is the unit of torque. It is the torque which develops a power of 1 watt across the air gap while running at synchronous speed of the motor.

Qn.1: A 200HP, 1000V, 25Hz, star connected induction motor has a star connected, slip ring rotor with a transformation ratio of 3.6. The rotor resistance per phase is 0.01Ω and inductance 0.64mH . The stator losses may be neglected. Find a) the rotor starting current per phase on normal voltage with slip rings shorted; b) the rotor power factor at starting; c) the rotor current at 3% slip; d) the rotor power factor at 3% slip.

Ans:-

$$\frac{\text{Stator voltage per phase}}{\text{rotor voltage per phase}} = 3.6$$

Rotor voltage per phase $E_2 = 160.375\text{V}$

At starting,

$$\text{Rotor current, } I_2 = \frac{E_2}{\sqrt{R_2^2 + X_2^2}} = \frac{160.375}{\sqrt{0.01^2 + 0.1^2}} = 1595.79\text{A}$$

$$\text{Power factor, } \cos \Phi_2 = \frac{R_2}{\sqrt{R_2^2 + X_2^2}} = \frac{0.01}{\sqrt{0.01^2 + 0.1^2}} = 0.099 \text{ lag}$$

At slip 3%,

$$\text{Rotor current, } I_2 = \frac{sE_2}{\sqrt{R_2^2 + (sX_2)^2}} = \frac{0.03 \times 160.375}{\sqrt{0.01^2 + (0.03 \times 0.1)^2}} = 460.8\text{A}$$

$$\text{Power factor, } \cos \phi_2 = \frac{R_2}{\sqrt{R_2^2 + (sX_2)^2}} = \frac{0.01}{\sqrt{0.01^2 + (0.03 \times 0.1)^2}} = 0.808 \text{ lag}$$

Qn.2: A 1100V, 50Hz delta connected induction motor has a star connected slip ring rotor with phase transformation ratio 3.8. Rotor resistance and leakage reactance are 0.012Ω and 0.25Ω per phase respectively. Neglecting stator impedance and magnetizing currents, determine : i) rotor current at start with slip rings shorted, ii) rotor power factor at start with slip rings shorted, iii) rotor current at 4% slip with slip rings shorted and iii) rotor power factor at 4% slip with slip rings shorted. (Calicut University, June 2009)

Ans:-

$$\frac{\text{Stator voltage per phase}}{\text{rotor voltage per phase}} = 3.8$$

Rotor voltage per phase $E_2 = 289.47\text{V}$

At starting,

$$\text{Rotor current, } I_2 = \frac{E_2}{\sqrt{R_2^2 + X_2^2}} = \frac{289.47}{\sqrt{0.012^2 + 0.25^2}} = 1156.55 A$$

$$\text{Power factor, } \cos \Phi_2 = \frac{R_2}{\sqrt{R_2^2 + X_2^2}} = \frac{0.012}{\sqrt{0.012^2 + 0.25^2}} = 0.048 \text{ lag}$$

At slip 4%,

$$\text{Rotor current, } I_2 = \frac{sE_2}{\sqrt{R_2^2 + (sX_2)^2}} = \frac{0.04 \times 289.47}{\sqrt{0.012^2 + (0.04 \times 0.25)^2}} = 741.26 A$$

$$\text{Power factor, } \cos \phi_2 = \frac{R_2}{\sqrt{R_2^2 + (sX_2)^2}} = \frac{0.012}{\sqrt{0.012^2 + (0.04 \times 0.25)^2}} = 0.768 \text{ lag}$$

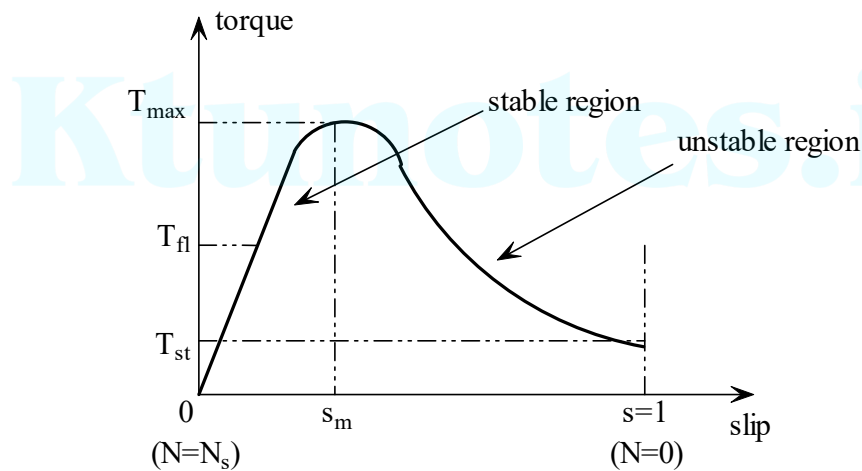
Torque – Slip (T-s) characteristics

$$T = \frac{sE_2^2 R_2}{R_2^2 + (sX_2)^2} \text{ synchronous watts}$$

For small values of s (speed near to N_s), $(sX_2)^2$ is negligible compared to R_2^2 .

$$T \propto \frac{sE_2^2}{R_2}$$

$$T \propto s \quad (\text{Equation of straight line})$$



For low speeds, $(sX_2)^2$ is large and R_2^2 is negligible compared to $(sX_2)^2$.

$$T = \frac{sE_2^2 R_2}{(sX_2)^2} = \frac{E_2^2 R_2}{sX_2^2}$$

$$T \propto \frac{1}{s} \quad (\text{Equation of rectangular hyperbola})$$

When $N=N_s$, $s=0$ and torque T is zero. For small values of s , torque increases linearly with s .

As the load increases, s increases and the torque reaches its maximum value when $s = \frac{R_2}{X_2}$.

Further increase in load beyond the point of maximum torque (Breakdown torque or pull out torque), slip increases further, but the torque begins to decrease. The result is that the motor slows down and eventually stops.

Note: The normal full-load slip of the induction motor is of the order of 2%-8%.

CONDITION FOR MAXIMUM TORQUE

$$T = \frac{sE_2^2 R_2}{R_2^2 + (sX_2)^2} \text{ sync. Watts}$$

For maximum torque with variation of slip s (as load changes),

$$\frac{dT}{ds} = 0$$

$$\frac{dT}{dR_2} = \frac{E_2^2 R_2 [(R_2^2 + s^2 X_2^2) - s \times 2s X_2^2]}{[R_2^2 + s^2 X_2^2]^2} = 0$$

$$R_2^2 + s^2 X_2^2 - 2s^2 X_2^2 = 0$$

$$R_2^2 = s^2 X_2^2 \quad \text{OR} \quad s = \frac{R_2}{X_2}$$

$$\text{Slip at maximum torque, } s_m = \frac{R_2}{X_2}$$

Maximum torque,

$$T_{\max} = \frac{sE_2^2 R_2}{R_2^2 + (sX_2)^2} = \frac{\frac{R_2}{X_2} E_2^2 R_2}{R_2^2 + \left(\frac{R_2}{X_2} \times X_2\right)^2}$$

$$T_{\max} = \frac{E_2^2}{2X_2} \text{ sync. watts}$$

Qn 1: A 3-phase induction motor has a 4-pole star connected stator winding. Motor runs at 50Hz supply with 200V between lines. Rotor resistance and standstill reactance are 0.1Ω and 0.9Ω per phase respectively. Calculate a) total torque at 4% slip b) maximum torque and c) speed at maximum torque. Ratio of rotor turns to stator turns = 0.67. (Calicut University June 2009).

Ans:-

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{4} = 1500 \text{ rpm}$$

$$E_2 = \frac{200}{\sqrt{3}} \times 0.67 = 77.365 \text{ V}$$

$$\text{Torque at 4\% slip, } T = \frac{3 \times 60}{2\pi N_s} \cdot \frac{sE_2^2 R_2}{R_2^2 + (sX_2)^2} = \frac{3 \times 60}{2\pi \times 1500} \times \frac{0.04 \times 77.365^2 \times 0.1}{0.1^2 + (0.04 \times 0.9)^2} = 40.479 \text{ N-m}$$

$$\text{Slip at maximum torque, } s_m = \frac{R_2}{X_2} = \frac{0.1}{0.9} = 0.111$$

$$\text{Maximum torque, } T_{\max} = \frac{3 \times 60}{2\pi N_s} \cdot \frac{E_2^2}{2X_2} = \frac{3 \times 60}{2\pi \times 1500} \times \frac{77.365^2}{2 \times 0.9} = 63.5 \text{ N-m}$$

$$\text{Speed at maximum torque, } N = (1 - s_m) N_s = (1 - 0.111) \times 1500 = 1333.5 \text{ rpm}$$

Effect of variation of rotor resistance for a slip ring induction motor

$$T = \frac{sE_2^2 R_2}{R_2^2 + (sX_2)^2} \text{ sync. watts}$$

For maximum torque with variation of R_2 ,

$$\frac{dT}{dR_2} = 0$$

$$\frac{dT}{dR_2} = \frac{sE_2[(R_2^2 + s^2 X_2^2) - R_2 \cdot 2R_2]}{[R_2^2 + s^2 X_2^2]^2} = 0$$

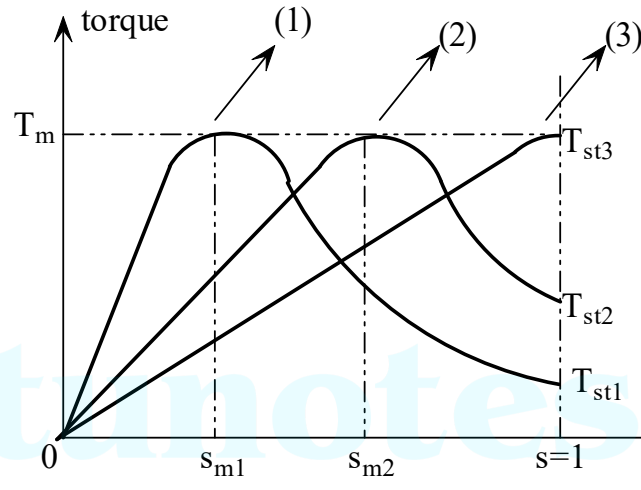
$$R_2^2 + s^2 X_2^2 - 2R_2^2 = 0$$

$$s^2 X_2^2 = R_2^2 \text{ OR } \underline{R_2 = sX_2}$$

$$\text{Maximum torque, } T_{\max} = \frac{E_2^2 \cdot s \cdot sX_2}{(sX_2)^2 + (sX_2)^2} \text{ since } R_2 = sX_2$$

$$T_{\max} = \frac{E_2^2}{2X_2} \text{ sync. watts}$$

$T_{\max}$ is independent of rotor resistance.



Curve (1) shows the T-s characteristics with no external resistance in the rotor circuit.

Curve (2) and (3) shows the T-s characteristics with external resistance added to the rotor circuit. If the total rotor resistance is equal to X_2 , maximum torque occurs at starting.

Addition of rotor resistance does not affect the maximum torque T_m but it controls the slip at which maximum torque occurs.

During normal working region, slip is small, T-s characteristics is a straight line.

When $R_2 = 0.2X_2$, $T_{\max}$ occurs when $s = 0.2$

When $R_2 = 0.5X_2$, $T_{\max}$ occurs when $s = 0.5$

When $R_2 = X_2$, $T_{\max}$ occurs when $s = 1$ (at starting)

Effect of variation of rotor resistance is

- i) Starting torque increases with increase in value of rotor resistance

$$T_{st} = \frac{E_2^2 R_2}{R_2^2 + X_2^2} \approx \frac{E_2^2 R_2}{X_2^2}$$

$$T_{st} \propto R_2 \text{ since } R_2^2 \ll X_2^2$$

- ii) Maximum torque remains constant and is independent of the value of rotor resistance.

$$T_{\max} = \frac{E_2^2}{2X_2} \text{ sync. watts}$$

- iii) The slip at which maximum torque occurs varies with the variation of rotor resistance.

iv) Maximum torque is developed at starting when rotor resistance is equal to the standstill rotor reactance i.e. when $R_2 = X_2$

v) Addition of rotor resistance reduces the starting current

$$I_{st} \propto \frac{E_2}{\sqrt{R_2^2 + X_2^2}}$$

vi) Addition of rotor resistance improves the power factor at starting.

$$\text{Power factor at starting} = \frac{r_2}{\sqrt{R_2^2 + X_2^2}} \propto \frac{R_2}{X_2}$$

Relation between starting torque, maximum torque and full load torque

$$\text{Let } \frac{R_2}{X_2} = a$$

$$T = \frac{sE_2^2 R_2}{R_2^2 + (sX_2)^2} \text{ sync. watts}$$

Divide numerator and denominator by X_2^2

$$T = \frac{E_2^2 \cdot s \cdot \frac{R_2}{X_2} \cdot \frac{1}{X_2}}{\left(\frac{R_2}{X_2}\right)^2 + s^2}$$

$$T = \frac{E_2^2}{X_2^2} \cdot \frac{s \cdot a}{a^2 + s^2}$$

$$\text{or } T \propto \frac{s \cdot a}{a^2 + s^2}$$

$$T = k \frac{s \cdot a}{a^2 + s^2}$$

$$T_{st} = k \frac{a}{a^2 + 1}$$

$$T_{\max} = \frac{k}{2} \text{ since } s = a \text{ at maximum torque}$$

$$T_{fl} = k \frac{s_f a}{a^2 + s_f^2}$$

$$\frac{T_{fl}}{T_{\max}} = \frac{2s_f a}{a^2 + s_f^2}$$

$$\frac{T_{st}}{T_{\max}} = \frac{2a}{a^2 + 1}$$

Qn.1 A 746KW, 3 phase, 50Hz, 16 pole induction motor has a rotor impedance of $(0.02 + j0.15 \Omega)$ at standstill. Full load torque is obtained at 360rpm. Calculate i) the ratio of maximum to full load torque ii) the speed for maximum torque and iii) the rotor resistance to be added to get maximum starting torque.

Ans:-

$$N_s = \frac{120f}{p} = \frac{120 \times 50}{16} = 375 \text{ rpm}$$

$$\text{Slip at full load, } s_f = \frac{N_s - N}{N_s} = \frac{375 - 360}{375} = 0.04$$

$$a = \frac{R_2}{X_2} = \frac{0.02}{0.15} = 0.133$$

$$\frac{T_{fl}}{T_{\max}} = \frac{2s_f a}{a^2 + s_f^2} = \frac{2 \times 0.04 \times 0.133}{0.133^2 + 0.04^2} = 0.55$$

$$\frac{T_{\max}}{T_{fl}} = 1.817$$

$$\text{Slip at maximum torque, } s_m = a = 0.133$$

$$\text{Speed at maximum torque, } N = (1 - s_m)N_s = 325.1 \text{ rpm}$$

To get maximum torque at starting, $R + R_2 = X_2$.

$$\text{External rotor resistance, } R = 0.15 - 0.02 = 0.13 \Omega$$

Qn.2 A 3 phase induction motor has starting torque of 100% and a maximum torque of 200% of the full load torque. Find slip at maximum torque.

Ans:-

$$\frac{T_{\max}}{T_{fl}} = 2 \text{ and } \frac{T_{st}}{T_{fl}} = 1$$

$$\frac{T_{st}}{T_{\max}} = \frac{1}{2} = \frac{2a}{a^2 + 1}$$

$$a^2 - 4a + 1 = 0$$

$$a = 0.268 \text{ or } 3.732 \text{ (for induction motor, maximum value of slip = 1)}$$

$$a = 0.268$$

$$\text{Slip at maximum torque, } s_m = 26.8\%$$

Qn.3 The per phase values of rotor resistance and standstill reactance of a 3 phase 50Hz 6 pole induction motor are 0.02Ω and 0.15Ω respectively. Calculate the ratio of maximum to full load torque if full load occurs at 4% slip. What value of resistance inserted in rotor circuit can result in starting torque, half of the maximum value?

Ans:-

$$a = \frac{R_2}{X_2} = \frac{0.02}{0.15} = 0.1333$$

$$\frac{T_{fl}}{T_{\max}} = \frac{2s_f a}{a^2 + s_f^2} = \frac{2 \times 0.04 \times 0.1333}{0.04^2 + 0.1333^2} = 0.55$$

$$\frac{T_{\max}}{T_{fl}} = 1.817$$

$$\frac{T_{st}}{T_{\max}} = \frac{2a}{a^2 + 1} = \frac{1}{2}$$

$$a^2 - 4a + 1 = 0$$

$$a = 0.268 = \frac{R + R_2}{X_2}$$

$$R + R_2 = 0.268 \times X_2 = 0.04$$

External resistance, $R = 0.02\Omega$

Qn.4:- A 3-phase, 400V, 50Hz, 4-pole induction motor has star connected stator winding. The rotor resistance and reactance are 0.1Ω and 1Ω respectively. The full load speed is 1440 rpm. Calculate the torque developed on full load by the motor. Assume stator to rotor turns ratio as 2:1.

Ans:-

$$N_s = \frac{120f}{p} = \frac{120 \times 50}{4} = 1500 \text{ rpm}$$

$$\text{Slip at full load, } s_f = \frac{N_s - N}{N_s} = \frac{1500 - 1440}{1500} = 0.04$$

$$E_2 = \frac{E_1}{2} = \frac{230.94}{2} = 115.47 \text{ V}$$

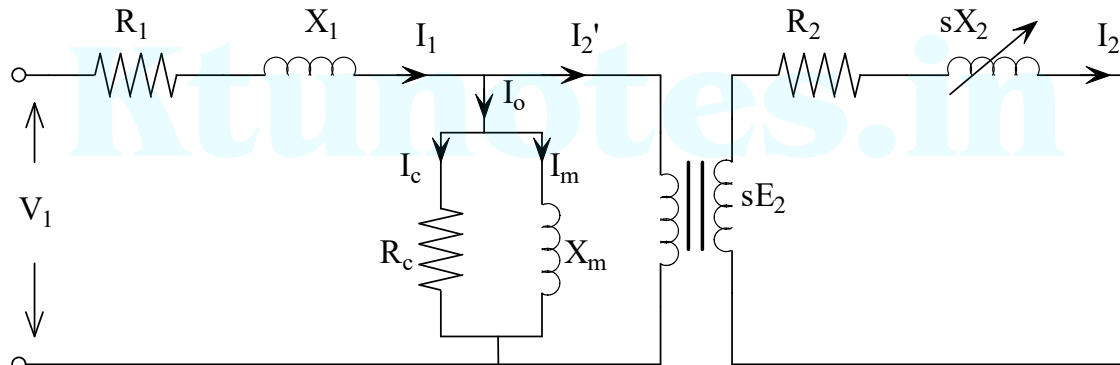
$$T = \frac{3}{2\pi \frac{N_s}{60}} \cdot \frac{sE_2^2 R_2}{R_2^2 + (sX_2)^2} = \frac{3}{2\pi \frac{1500}{60}} \cdot \frac{0.04 \times 115.47^2 \times 0.1}{0.1^2 + (0.04 \times 1)^2} = 87.81 \text{ N-m}$$

EQUIVALENT CIRCUIT

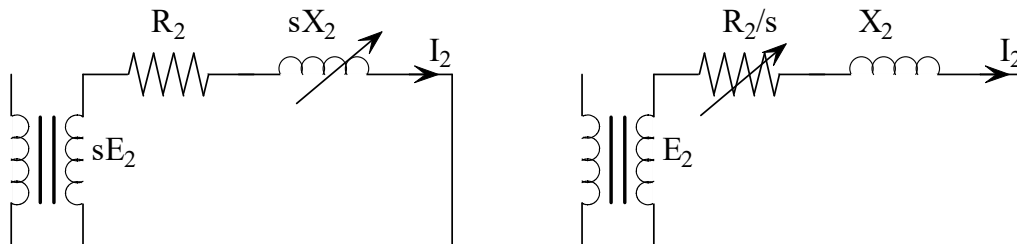
The behavior of an induction motor is similar to that of a transformer.

The mutual flux Φ_m induces emfs in stator and rotor. These emfs, under running conditions are

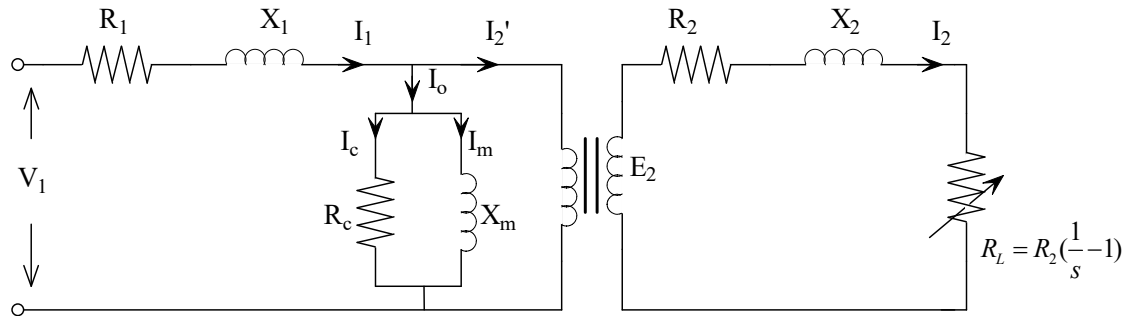
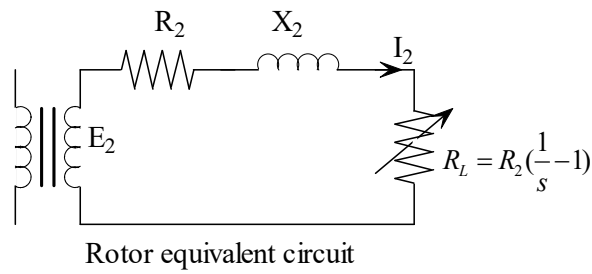
E_1 and sE_2 respectively.



Rotor side:



Rotor equivalent circuit



$$I_2 = \frac{sE_2}{\sqrt{R_2^2 + (sX_2)^2}} = \frac{E_2}{\sqrt{\left(\frac{R_2}{s}\right)^2 + X_2^2}}$$

Here, $\frac{R_2}{s} = R_2 + R_2\left(\frac{1}{s} - 1\right)$ where

R_2 = actual rotor winding resistance

$R_L = R_2\left(\frac{1}{s} - 1\right)$ = Load resistance (Electrical equivalent of mechanical load on the motor)

Also, $3I_2^2 R_2$ = rotor copper loss

$3I_2^2 R_L$ = Mechanical power developed

$$\frac{E_2}{E_1} = \frac{K_{w2}T_2}{K_{w1}T_1} = K$$

Shift all the elements from rotor side to stator side

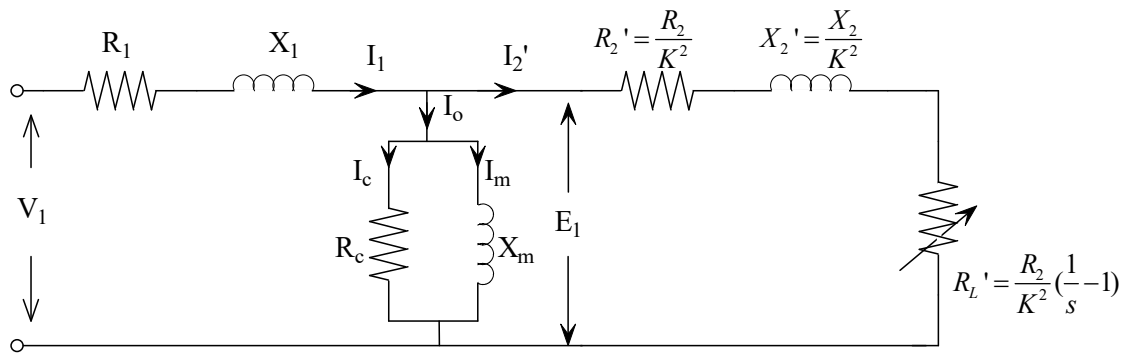
$$E_2' = \frac{E_2}{K} = E_1$$

$$R_2' = \frac{R_2}{K^2}$$

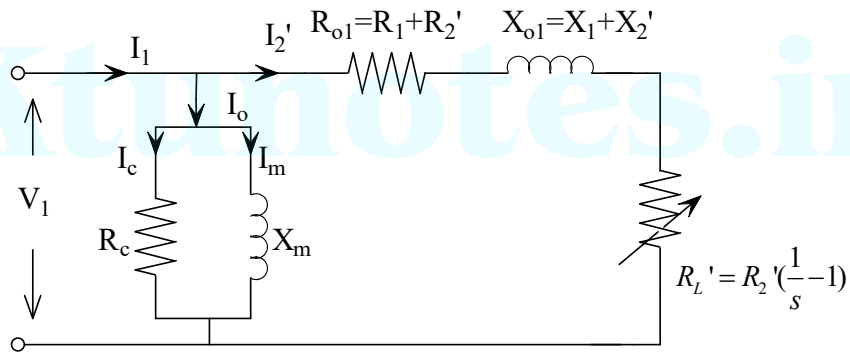
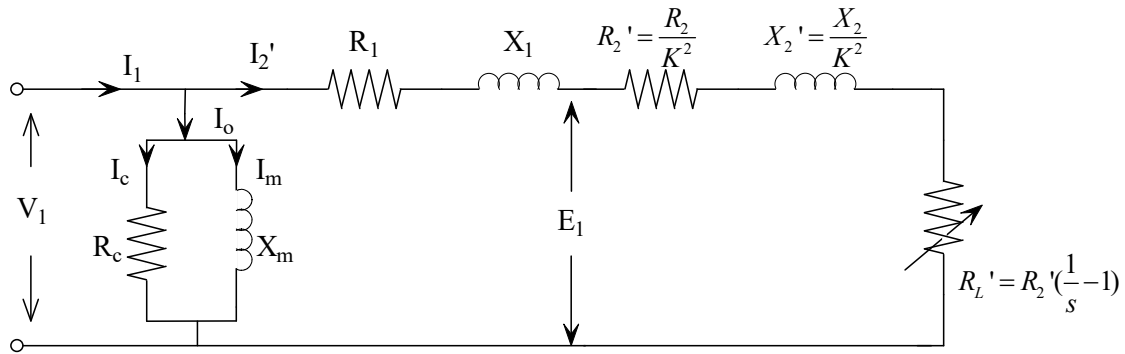
$$X_2' = \frac{X_2}{K^2}$$

$$I_2' = KI_2$$

Exact equivalent circuit is given below



Approximate equivalent circuit is given below



From equivalent circuit,

$$\text{Rotor copper loss } P_{cu} = 3I_2'^2 R_2'$$

$$\text{Rotor output} = 3I_2'^2 R_L' = 3I_2'^2 R_2' \left(\frac{1}{s} - 1\right) = \text{Mechanical power developed } (P_m)$$

$$\text{Rotor input } P_2 = \text{Rotor copper loss} + \text{Rotor output}$$

$$= P_{cu} + P_m = \frac{3I_2'^2 R_2'}{s}$$

$$\text{Rotor input} = P_2$$

$$\text{Rotor copper loss} = sP_2$$

$$\text{Mechanical power developed} = (1-s)P_2 = P_m$$

$$\text{Shaft output } P_{out} = P_m - \text{friction \& windage loss}$$

TORQUE EQUATION

$$\text{Torque developed} = \frac{\text{mechanical power developed}}{2\pi N/60}$$

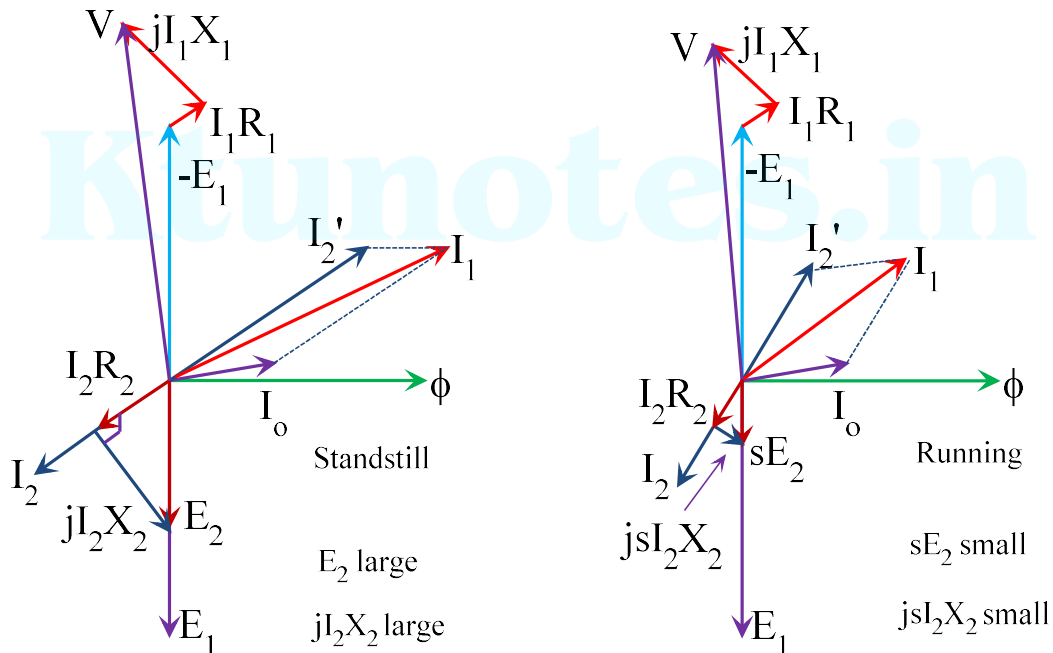
$$T_g = \frac{\left(\frac{3I_2^2 R_2(1-s)}{s}\right)}{2\pi N/60} \quad \text{But, } N = (1-s)N_s$$

$$T_g = \frac{\left(\frac{3I_2^2 R_2}{s}\right)}{2\pi N_s/60} = \frac{3}{2\pi N_s/60} \times \frac{I_2^2 R_2}{s} \quad \text{But, } I_2 = \frac{sE_2}{\sqrt{R_2^2 + (sX_2)^2}}$$

$$T_g = \frac{3}{2\pi N_s/60} \times \frac{sE_2^2 R_2}{R_2^2 + (sX_2)^2} \text{ N-m}$$

$$T_g = \frac{sE_2^2 R_2}{R_2^2 + (sX_2)^2} \text{ sync. watts}$$

PHASOR DIAGRAM



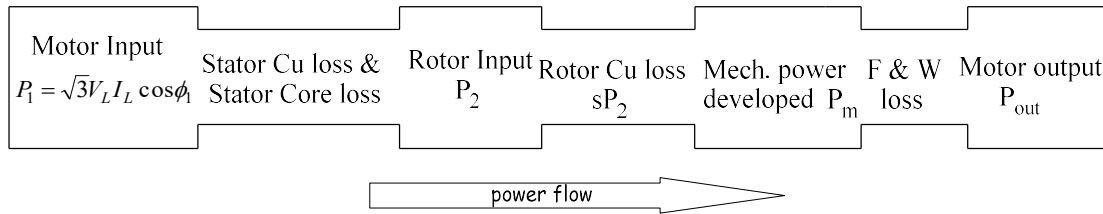
LOSSES AND EFFICIENCY

An induction motor stator is supplied with electrical power. Power is transferred to the rotor through the air gap and is made available at the shaft to do mechanical work. The whole of the power input to the stator is not available at the shaft as output. Some power is lost in the stator and rotor windings and on the core as heat. In addition there is friction and windage loss due to the rotation of the rotor.

Various losses are

- $I^2 R$ loss in the stator winding
- Core loss (hysteresis and eddy current loss) in the stator core
- $I^2 R$ loss in the rotor winding
- core loss in the rotor core (negligible because of frequency of rotor currents under normal running condition is always small)
- Friction losses in the bearings, brushes and slip rings

vi) Windage loss due to the rotation of rotor in air



$$\text{Efficiency } \eta = \frac{\text{Motor Output } (P_{\text{out}})}{\text{Motor Input } (P_1)} \times 100\%$$

Qn.1:- A 3-phase, 6-pole, 50Hz, star connected induction motor delivers useful power 25kW while running at a speed of 950 rpm. It is connected to a supply of 400V and takes a current of 60A. Its stator resistance per phase is 0.14Ω . Mechanical losses are 900W. Calculate i) shaft torque ii) gross torque developed iii) rotor copper loss iv) stator copper loss and v) overall efficiency. The power factor of the motor is 0.75 (lagging).

Ans:-

$$N_s = \frac{120f}{p} = \frac{120 \times 50}{6} = 1000 \text{ rpm}$$

$$\text{Slip, } s = \frac{N_s - N}{N_s} = \frac{1000 - 950}{1000} = 0.05$$

$$P_m = P_{\text{out}} + \text{mechanical loss} = 25000 + 900 = 25900 \text{ W}$$

$$\text{Rotor input, } P_2 = \frac{P_m}{(1-s)} = \frac{25900}{(1-0.05)} = 27263.16 \text{ W}$$

$$\text{Rotor copper loss} = sP_2 = 0.05 \times 27263.16 = 1363.16 \text{ W}$$

$$\text{Stator copper loss} = 3I_1^2 R_1 = 3 \times 60^2 \times 0.14 = 1512 \text{ W}$$

$$\text{Motor input} = P_1 = \sqrt{3} V_L I_L \cos \phi = \sqrt{3} \times 400 \times 60 \times 0.75 = 31176.9 \text{ W}$$

$$\text{Efficiency} = \frac{25000}{31176.9} \times 100 = 80.19\%$$

$$\text{Shaft torque, } T_{\text{sh}} = \frac{P_{\text{out}}}{2\pi \times \frac{N}{60}} = \frac{25000}{2\pi \times \frac{950}{60}} = 251.3 \text{ N-m}$$

$$\text{Gross torque, } T_g = \frac{P_m}{2\pi \times \frac{N}{60}} = \frac{25900}{2\pi \times \frac{950}{60}} = 260.34 \text{ N-m}$$

Qn.2:- An induction motor has an efficiency of 90% when delivering an output of 37kW. At this load, the stator copper loss and rotor copper loss each equals the stator iron loss. The mechanical losses are one-third of the no-load losses. Determine the slip. (Calicut University June 2009)

Ans:-

$$P_{\text{in}} = \frac{P_{\text{out}}}{\eta} = \frac{37000}{0.9} = 41111.1 \text{ W}$$

$$\text{Total losses} = P_{\text{in}} - P_{\text{out}} = 41111.1 - 37000 = 4111.1 \text{ W}$$

$$\text{Total losses} = \text{stator Cu loss} + \text{stator core loss} + \text{rotor Cu loss} + \text{mechanical loss}$$

$$= 3 \times \text{stator core loss} + \text{mechanical loss} = 4111.1 \text{ W}$$

$$\text{Stator core loss} = \frac{4111.1 - m}{3} \quad \text{----- (i) where } m = \text{mechanical loss}$$

But,

No load loss = stator core loss + mechanical loss (since copper loss on no-load neglected)

$$\text{mechanical loss} = \frac{1}{3} \times \text{No-load loss} = \frac{1}{3} \times (\text{stator core loss} + \text{mechanical loss})$$

$$m = \frac{1}{3} \times \left(\frac{4111.1 - m}{3} + m \right) \quad \text{--- (ii)}$$

$$3m = \frac{4111.1 - m}{3} + m$$

$$7m = 4111.1$$

$$m = 587.3W$$

$$\text{Mechanical loss} = 587.3W$$

$$\text{Stator core loss} = \frac{4111.1 - m}{3} = \frac{4111.1 - 587.3}{3} = 1174.6W$$

$$\text{Rotor copper loss} = sP_2 = 1174.6W$$

$$\begin{aligned} \text{Rotor input } P_2 &= P_{\text{out}} + \text{mechanical loss} + \text{Rotor copper loss} = \\ &= 37000 + 587.3 + 1174.6 = 38761.9W \end{aligned}$$

$$\text{Slip, } s = \frac{\text{Rotor Cu loss}}{\text{Rotor input}} = \frac{1174.6}{38761.9} = 0.0303$$

Qn 3:- A 6-pole, 50Hz, 3-phase induction motor running on full load develops a useful torque of 160Nm when the rotor emf makes 120 complete cycles per minute. Calculate the shaft power output. If the mechanical torque lost in friction and that for core loss is 10Nm, compute (a) the copper loss in the rotor windings (b) the input to the motor and (c) the efficiency. The total stator loss is given to be 800W.

Ans:-

$$f_2 = sf = \frac{120}{60} = 2Hz$$

$$s = \frac{2}{50} = 0.04$$

$$N_s = \frac{120f}{p} = \frac{120 \times 50}{6} = 1000rpm$$

$$N = (1 - s)N_s = (1 - 0.04) \times 1000 = 960rpm$$

$$\text{Shaft power output, } P_{\text{out}} = 2\pi \times \frac{N}{60} T_{sh} = \frac{2\pi \times 960 \times 160}{60} = 16085W$$

$$\text{Mechanical power developed, } P_m = 2\pi \times \frac{N}{60} T_g = \frac{2\pi \times 960 \times (160 + 10)}{60} = 17090.3W$$

$$\text{Rotor input, } P_2 = \frac{P_m}{(1 - s)} = \frac{17090.3}{(1 - 0.04)} = 17802.36W$$

$$\text{Rotor copper loss, } sP_2 = 0.04 \times 17802.36 = 712W$$

$$\text{Motor input, } P_{\text{in}} = \text{Rotor input} + \text{stator losses} = 17802.36 + 800 = 18602.36W$$

$$\text{Efficiency, } \eta = \frac{P_{\text{out}}}{P_{\text{in}}} = \frac{16085}{18602.36} = 86.47\%$$

Qn.3 :- The Power input to a 500V, 50Hz, 6 pole, 3 phase induction motor running at 976 rpm is 40KW. The stator losses are 1 KW and the friction and windage losses total 2 KW. Calculate i) the slip ii) the rotor copper loss iii) shaft power and iv) the efficiency.

(Ans: i)0.024, ii)0.936KW, iii)36.064KW, iv) 90.16%)

Qn.4:- An 18.65KW, 6pole, 50Hz, 3 phase slip ring induction motor runs at 960 rpm on full load with a rotor current per phase of 35A. Allowing 1 KW for mechanical losses, find the resistance/phase of 3 phase rotor winding. (Ans : 0.223Ω/phase)

Qn.5:- A 100KW 3300V 50Hz 3 phase star connected induction motor has a synchronous speed of 500 rpm. The full load slip is 1.8% and full load power factor 0.85. Stator copper loss = 1200W. Iron loss = 3500W. Rotational losses = 1200W. Calculate i) the rotor copper loss ii) the line current and iii) full load efficiency. (Ans: i) 1.855KW ii) 22.4A, iii)91.7%)

Effect of change in supply voltage on torque and slip

$$T_g = \frac{sE_2^2 R_2}{R_2^2 + (sX_2)^2} \text{ sync. watts}$$

But, E_2 = rotor induced emf at standstill

$$E_2 \propto \phi \propto V$$

Hence,

$$T_g \propto \frac{sV^2 R_2}{R_2^2 + (sX_2)^2}$$

Since, during running condition, $s^2 X_2^2 \ll R_2^2$

$$T_g \propto \frac{sV^2 R_2}{R_2^2} \propto \frac{sV^2}{R_2}$$

$$T_g \propto sV^2$$

Any change in supply voltage changes the value of torque.

For constant torque, if supply voltage is decreased, slip increases.

Qn 1:- A 6-pole, 50 Hz, 3-phase induction motor has a rotor resistance of 0.25Ω/phase and a maximum torque of 10 Nm at 875 rpm. Calculate (a) the torque when the slip is 5%, and (b) the resistance to be added to the rotor circuit to obtain 60% of the maximum torque at starting. The stator impedance is assumed to be negligible.

Ans:-

$$T_{\max} = \frac{E_2^2}{2X_2} \text{ sync. watts}$$

$$T_{\max} \propto \frac{V^2}{2X_2}$$

$$N_s = \frac{120f}{p} = \frac{120 \times 50}{6} = 1000 \text{ rpm}$$

$$s_m = \frac{N_s - N}{N_s} = \frac{1000 - 875}{1000} = 0.125$$

$$s_m = \frac{R_2}{X_2}$$

$$X_2 = \frac{R_2}{s_m} = \frac{0.25}{0.125} = 2\Omega$$

$$T_{\max} \propto \frac{V^2}{2X_2} = \frac{V^2}{2 \times 2} = 0.25V^2$$

$$T = \frac{sE_2^2 R_2}{R_2^2 + (sX_2)^2} \text{ sync. watts}$$

$$T \propto \frac{sV^2 R_2}{R_2^2 + (sX_2)^2} = \frac{0.05 \times V^2 \times 0.25}{0.25^2 + (0.05 \times 2)^2} = \frac{0.0125V^2}{0.0725} = 0.1724V^2$$

$$\frac{T}{T_{\max}} = \frac{0.1724}{0.25} = 0.6896$$

$$T = T_{\max} \times 0.6896 = 10 \times 0.6896 = 6.896 Nm$$

$$\frac{T_{st}}{T_{\max}} = \frac{2a}{a^2 + 1} = 0.6$$

$$0.6a^2 - 2a + 0.6 = 0$$

$$a = 3 \text{ or } \frac{1}{3}$$

But, a should be less than 1, otherwise $T_{\max}$ occurs in the region $s > 1$.

$$a = \frac{1}{3} = \frac{R + R_2}{X_2}$$

$$R + R_2 = \frac{1}{3} \times X_2 = 0.667$$

External resistance, $R = 0.667 - 0.25 = 0.417 \Omega$

Qn 2:- A 60kW, 400V, 3-phase, 50Hz wound rotor induction motor has a full slip of 0.04 when operating at rated voltage and frequency with rotor winding shorted at the slip rings. The slip at the maximum torque is 0.2. Neglecting the rotational and stator resistance losses, find the maximum torque and full-load rotor resistance losses. (Calicut University, December 2008)

Ans:- $(1-s)P_2 = 60000W$

$$P_2 = \frac{60000}{(1-0.04)} = 62500W$$

$$T_f = 62500 \text{ sync.watts}$$

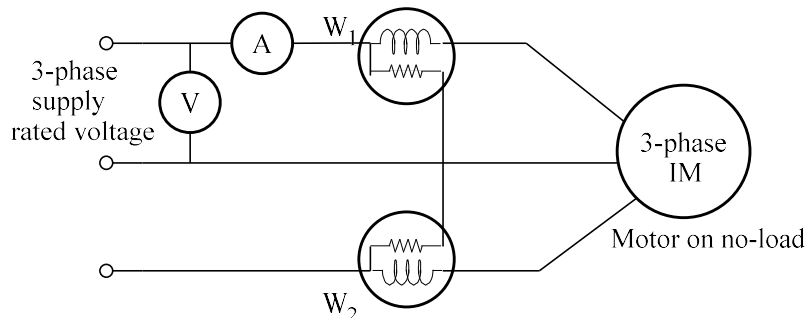
$$\frac{T_m}{T_f} = \frac{s_m^2 + s_f^2}{2s_f s_m} = \frac{0.2^2 + 0.04^2}{2 \times 0.04 \times 0.2} = 2.6$$

$$T_m = 162500 \text{ sync.watts}$$

$$\text{Full-load rotor copper loss} = sP_2 = 2500W$$

NO LOAD AND BLOCKED ROTOR TESTS

Equivalent circuit parameters can be obtained by means of non-loading tests as in the case of the transformer – no-load test (corresponding to the OC test on transformer) and the blocked rotor test (corresponding to the SC test on transformer).



No load test : (Similar to OC test in transformer)

(To determine the rotational losses (= Core loss + F&W loss) and constants R_c and X_m)

In this test, motor is run on no-load (rotor is allowed to rotate freely without any external mechanical load on the shaft) at rated voltage and frequency. Input power is measured by 2 wattmeters.

Let V_{oc} and I_{oc} be the voltmeter and ammeter readings (Line values).

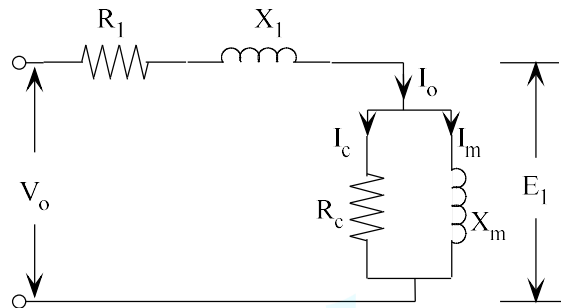
Total power, $W_{oc} = W_1 + W_2$

Let V_o and I_o be the per phase values.

(If stator windings are star connected, $V_o = \frac{V_{oc}}{\sqrt{3}}$, $I_o = I_{oc}$)

If stator windings are delta connected, $V_o = V_{oc}$, $I_o = \frac{I_{oc}}{\sqrt{3}}$)

Equivalent circuit on no-load is given below.



Here, $\frac{R_2'}{s}$ is very high since slip s at no load is very small (of the order of 0.1%). Hence rotor circuit is considered to be open.

No load input W_0 = Stator Core loss + F&W loss + Stator Cu loss

(Rotor Cu loss is negligible since slip is very small)

No-load current, $I_0 = 30$ to 40% of rated current; hence stator copper loss at no load needs to be accounted for.

(In a transformer, I_0 is around 5% of rated current since the flux path consists only iron of low reluctance. In an induction motor, flux path consists of air which has large reluctance. Hence, for same flux (flux = mmf/reluctance) large mmf and hence I_0 is required in case of induction motor. Also, the no load current I_0 in an induction motor consists of F&W loss component in addition to magnetizing and core loss components.)

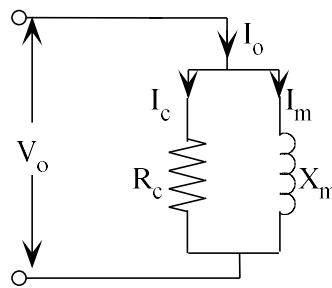
On no load, stator copper loss = $3I_0^2 R_1$ (R_1 found from stator resistance test)

Rotational losses $P_i = W_{OC} - 3I_0^2 R_1$

(Rotational losses = core loss + F&W loss)

Note : No load power factor is low (I_0 is mainly utilized for magnetizing the core which is inductive in nature; $I_m > I_c$); one of the wattmeter may read negative; therefore it is usually necessary to reverse the current coil (or potential coil) connections of the wattmeter.

Approximate equivalent circuit on no-load is given below.



To determine the equivalent circuit parameters R_o and X_o :-

$$\text{No load power factor, } \cos \Phi_0 = \frac{W_{oc}}{3V_o I_o}$$

$$\text{Core loss current, } I_c = I_o \cos \Phi_0$$

$$\text{Magnetizing current, } I_m = I_o \sin \Phi_0 = \sqrt{I_o^2 - I_c^2}$$

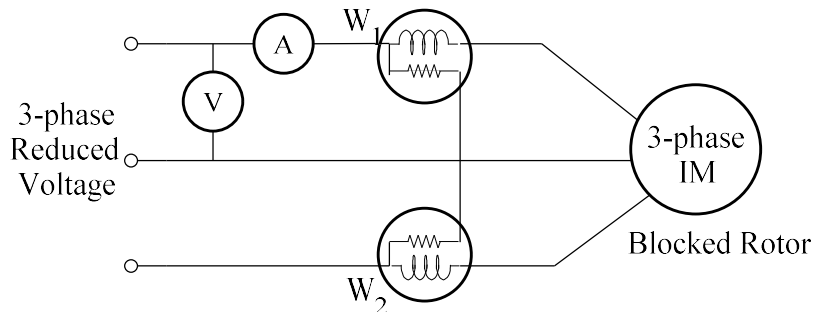
$$R_c = \frac{V_o}{I_c} = \frac{V_o}{I_o \cos \Phi_0}$$

$$X_m = \frac{V_o}{I_m} = \frac{V_o}{I_o \sin \Phi_0}$$

$$3I_c^2 R_c = \text{Core loss} + \text{F\&W loss} = W_{oc}$$

Blocked rotor test: (Similar to SC test in transformer)

Aim: To determine the winding resistances R_1 & R_2' and leakage reactances X_1 & X_2' .



Rotor shaft is blocked or held stationary by belt-pulley arrangement or by hand. A reduced voltage at rated frequency is applied to the stator winding so that rated current flows.

Let V_{sc} and I_{sc} be the voltmeter and ammeter readings (Line values).

$$\text{Total power} = W_{sc} = W_1 + W_2$$

Let V_s , I_s and W_s be the per phase values.

$$(\text{If stator windings are star connected, } V_s = \frac{V_{sc}}{\sqrt{3}}, \quad I_s = I_{sc})$$

$$(\text{If stator windings are delta connected, } V_s = V_{sc}, \quad I_s = \frac{I_{sc}}{\sqrt{3}})$$

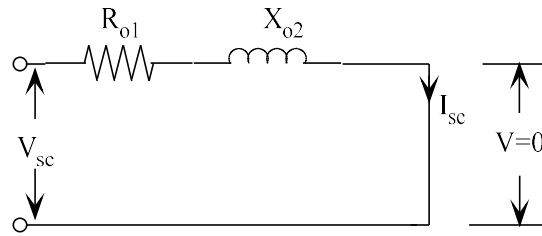
Power input under short circuit = stator copper loss + rotor copper loss

Since stator voltage during blocked rotor test is small, flux and hence I_o is very small and neglected.

Since flux is small, core loss is neglected.

Also, F&W is absent since motor is at standstill.

Equivalent circuit during blocked rotor circuit is given below.



$$R_{01} = \frac{W_{sc}}{3I_s^2}$$

$$Z_{01} = \frac{V_s}{I_s}$$

$$X_{01} = \sqrt{Z_{01}^2 - R_{01}^2}$$

$$R_2' = R_{01} - R_1$$

R_1 = stator resistance per phase (determined from stator resistance test)

There is no practical method of separating X_1 and X_2' . Usually, it is assumed that $X_1 = X_2' = \frac{X_{01}}{2}$.

(For precise calculation, it is a good practice to consult the manufacturer's data sheet)

Note : Under running conditions, the frequency of the stator currents is line frequency while that of the rotor currents is sf, only a few herz. In the blocked rotor test, however, the slip is unity and the rotor and stator currents are at the same frequency. Since both rotor and stator quantities are to be evaluated, the frequency of the blocked rotor test voltage is usually a compromise between the normal stator and rotor frequencies. IEEE test code recommends that the blocked rotor test be made using 25% rated frequency with the test voltage adjusted to obtain approximately rated current.

Qn.1:- A 10KW 400V 4pole delta connected squirrel cage induction motor gave the following test results: No load test : 400V 8A 250W

Blocked rotor test : 90V 35A 1350W

The stator resistance is 0.72Ω/ph. Calculate the rotational losses and the equivalent circuit parameters.

Ans:

$$V_o = V_{oc} = 400V, \quad I_o = \frac{I_{oc}}{\sqrt{3}} = \frac{8}{\sqrt{3}} = 4.619A$$

$$V_s = V_{sc} = 90V, \quad I_s = \frac{I_{sc}}{\sqrt{3}} = \frac{35}{\sqrt{3}} = 20.207A$$

$$\text{Rotational losses } P_i = W_{oc} - 3I_o^2 R_1 = 250 - 3 \times 4.619^2 \times 0.72 = 203.92W$$

$$\cos \Phi_0 = \frac{W_{oc}}{3V_o I_o} = \frac{250}{3 \times 400 \times 4.619} = 0.0451$$

$$\sin \Phi_0 = 0.999$$

$$R_c = \frac{V_0}{I_c} = \frac{V_0}{I_o \cos \Phi_0} = \frac{400}{4.619 \times 0.0451} = 1920.15\Omega$$

$$X_m = \frac{V_0}{I_m} = \frac{V_0}{I_o \sin \Phi_0} = \frac{400}{4.619 \times 0.999} = 86.69\Omega$$

$$R_{01} = \frac{W_{sc}}{3I_s^2} = \frac{1350}{3 \times 20.207^2} = 1.102\Omega$$

$$Z_{01} = \frac{V_s}{I_s} = \frac{90}{20.207} = 4.454\Omega$$

$$X_{01} = \sqrt{Z_{01}^2 - R_{01}^2} = \sqrt{4.454^2 - 1.102^2} = 4.315\Omega$$

$$R_2' = R_{01} - R_1 = 1.102 - 0.72 = 0.382\Omega$$

$$X_1 = X_2' = \frac{X_{01}}{2} = \frac{4.315}{2} = 2.16\Omega$$

Qn.2:- Determine the parameters of the equivalent circuit of a 3 phase, star connected, 400V, 50Hz, 4-pole squirrel cage induction motor with the following test data:-

No load : 400V 3.2A 250W

Blocked rotor : 100V 7A 500W

The stator resistance is 1Ω/phase.

Ans:-

$$V_o = \frac{V_{oc}}{\sqrt{3}} = \frac{400}{\sqrt{3}} = 230.94V, \quad I_o = I_{oc} = 3.2A$$

$$V_s = \frac{V_{sc}}{\sqrt{3}} = \frac{100}{\sqrt{3}} = 57.735V, \quad I_s = I_{sc} = 7A$$

$$\cos \Phi_0 = \frac{W_{oc}}{3V_o I_o} = \frac{250}{3 \times 230.94 \times 3.2} = 0.1127$$

$$\sin \Phi_0 = 0.9936$$

$$R_c = \frac{V_o}{I_c} = \frac{V_o}{I_o \cos \Phi_0} = \frac{230.94}{3.2 \times 0.1127} = 640.36\Omega$$

$$X_m = \frac{V_o}{I_m} = \frac{V_o}{I_o \sin \Phi_0} = \frac{230.94}{3.2 \times 0.9936} = 72.63\Omega$$

$$R_{01} = \frac{W_{sc}}{3I_s^2} = \frac{500}{3 \times 7^2} = 3.4\Omega$$

$$Z_{01} = \frac{V_s}{I_s} = \frac{57.735}{7} = 8.248\Omega$$

$$X_{01} = \sqrt{Z_{01}^2 - R_{01}^2} = \sqrt{8.248^2 - 3.4^2} = 7.514\Omega$$

$$R_2' = R_{01} - R_1 = 3.4 - 1 = 2.4\Omega$$

Qn.3:- A 11KW 415V 4pole delta connected squirrel cage induction motor gave the following test results:

No load test : 415V 5.8A 488W

Blocked rotor test : 40V 18.4A 510W

The stator resistance is 0.7Ω/ph. For full load condition, find i) line current, ii) power factor, iii) input power, iv) slip and v) efficiency.

Ans:-

$$V_o = V_{oc} = 415V, \quad I_o = \frac{I_{oc}}{\sqrt{3}} = \frac{5.8}{\sqrt{3}} = 3.35A$$

$$V_s = V_{sc} = 40V, \quad I_s = \frac{I_{sc}}{\sqrt{3}} = \frac{18.4}{\sqrt{3}} = 10.62A$$

$$\text{Rotational losses } P_i = W_{OC} - 3I_o^2 R_1 = 488 - 3 \times 3.35^2 \times 0.7 = 464.4W$$

$$\cos \Phi_0 = \frac{W_{oc}}{3V_o I_o} = \frac{488}{3 \times 415 \times 3.35} = 0.117$$

$$\sin \Phi_0 = 0.993$$

$$R_c = \frac{V_0}{I_c} = \frac{V_0}{I_0 \cos \Phi_0} = \frac{415}{3.35 \times 0.117} = 1058.8 \Omega$$

$$X_m = \frac{V_0}{I_m} = \frac{V_0}{I_0 \sin \Phi_0} = \frac{415}{3.35 \times 0.993} = 124.75 \Omega$$

$$R_{01} = \frac{W_{sc}}{3I_s^2} = \frac{510}{3 \times 10.62^2} = 1.5 \Omega$$

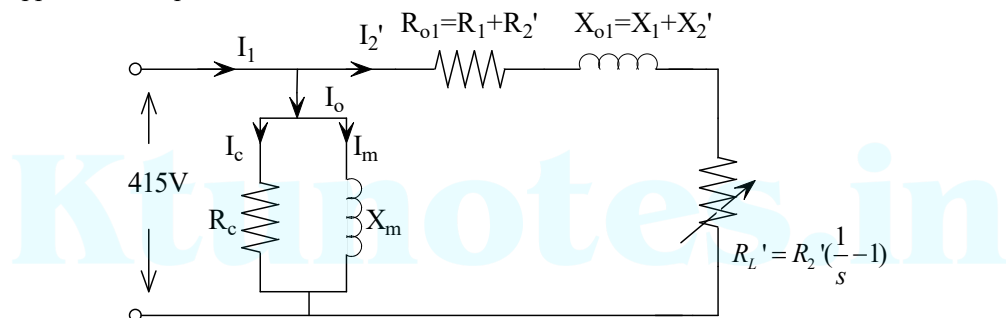
$$Z_{01} = \frac{V_s}{I_s} = \frac{40}{10.62} = 3.77 \Omega$$

$$X_{01} = \sqrt{Z_{01}^2 - R_{01}^2} = \sqrt{3.77^2 - 1.5^2} = 3.46 \Omega$$

$$R_2' = R_{01} - R_1 = 1.5 - 0.7 = 0.8 \Omega$$

$$X_1 = X_2' = \frac{X_{01}}{2} = \frac{3.46}{2} = 1.73 \Omega$$

From approximate equivalent circuit,



$$I_o = 3.35 \angle -83.28^\circ A$$

$$I_2' = \frac{V}{\sqrt{(R_{01} + R_L')^2 + X_{01}^2}}$$

$$\text{Power output} = 3I_2'^2 R_L' = \frac{3V^2 R_L'}{(R_{01} + R_L')^2 + X_{01}^2} = 11000$$

$$\frac{3 \times 415^2 \times R_L'}{(1.5 + R_L')^2 + 3.46^2} = 11000$$

$$11000R_L'^2 - 483675R_L' + 156420 = 0$$

$$R_L' = 43.64 \Omega$$

$$\text{Rotor current, } I_2' = \frac{V \angle 0^\circ}{(R_{01} + R_L') + jX_{01}} = \frac{415 \angle 0^\circ}{(1.5 + 43.64) + j3.46} = 9.17 \angle -4.38^\circ A$$

$$I_1 = I_o + I_2' = 3.35 \angle -83.28^\circ + 9.17 \angle -4.38^\circ A = 10.35 \angle -22.9^\circ$$

$$\text{Line current} = \sqrt{3} \times 10.35 = 17.93 A$$

$$\text{Power factor} = \cos(-22.9) = 0.92 \text{ lag}$$

$$\text{Input power} = 3 \times 415 \times 10.35 \times \cos(22.9) = 11870 W$$

$$\text{Efficiency} = \frac{11000}{11870} \times 100 = 92.67\%$$

$$\text{Slip} = \frac{R_2'}{R_2' + R_L'} = \frac{0.8}{0.8 + 43.64} = 0.018 = 1.8\%$$

Module 4

Performance of three phase Induction motors using Circle diagram, Numerical Problems. Cogging and Crawling in cage motors, Double cage Induction motor-Torque-Slip Characteristics.

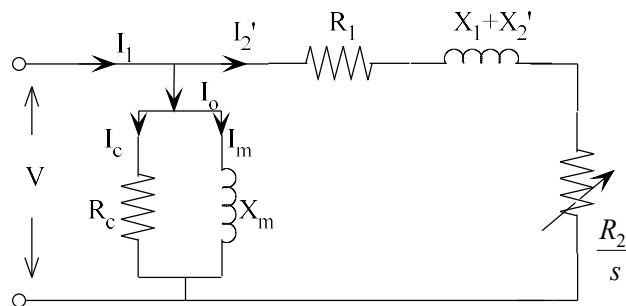
Starting of Induction motors – Types of Starters – DOL starter, Autotransformer Starter, Star Delta starter, Rotor Resistance Starter-Numerical Problems.

Braking of Induction motors – Plugging, Dynamic braking, Regenerative braking

Speed control – Stator Voltage control, V/f control, Rotor Resistance Control.

CIRCLE DIAGRAM

The performance characteristics of induction motors can be determined approximately by graphical method such as circle diagram. This is applicable both for the squirrel cage and slip ring induction motors.

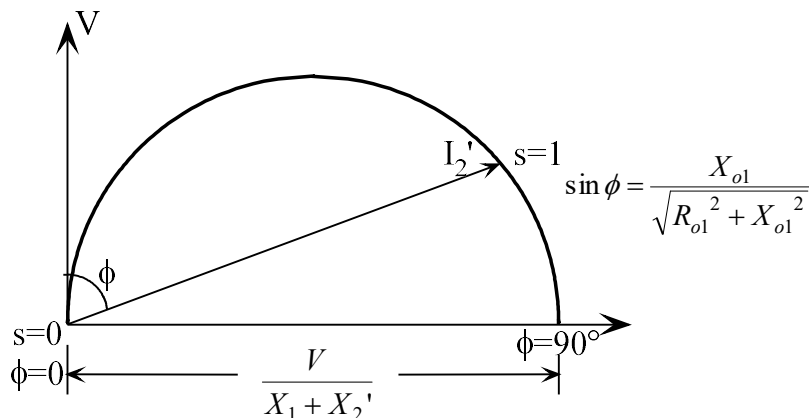


From the approximate equivalent circuit,

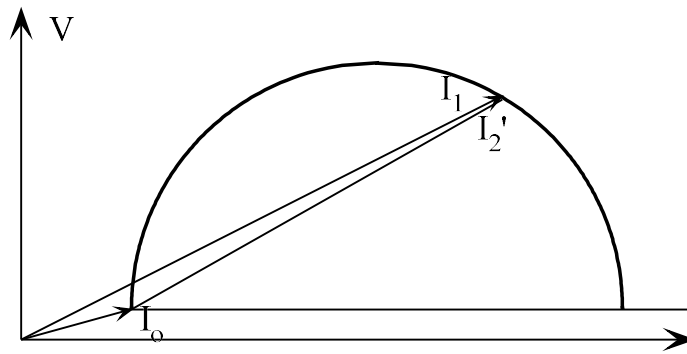
$$I_2' = \frac{V}{\sqrt{(R_1 + \frac{R_2'}{s})^2 + (X_1 + X_2')^2}} = \frac{V}{X_1 + X_2'} \sin \Phi \quad \text{where } \sin \Phi = \frac{X_1 + X_2'}{\sqrt{(R_1 + \frac{R_2'}{s})^2 + (X_1 + X_2')^2}}$$

If the leakage reactances X_1 and X_2' are assumed to remain constant regardless of load, and the applied voltage V is constant, the above equation represents the polar equation of a circle with diameter $\frac{V}{X_1 + X_2'}$. When load changes, slip s varies and hence Φ and I_2' is varied. The locus of the

current, however, lies on a circle. Thus in the case of induction motors, the locus of the current due to load lies on a circle and the diagram is known as a circle diagram.



If no load current taken by the motor is also to be accounted for to obtain the stator current, the diagram can then be as shown below. The stator current I_1 is then the phasor sum of I_2' and I_0 .



Procedure to draw the circle diagram:

1. Draw the lines by taking the current (I) in X-axis, voltage (V) in Y-axis. (V & I are line values). (Circle diagram can be drawn with phase values as well as line values).
2. From the No-load test find out the current I_0 and draw the OA vector with the magnitude of I_0 from the origin by suitable current scale, which lags the voltage (Y-axis) V by an angle Φ_0 where $\Phi_0 = \cos^{-1}\left(\frac{W_{oc}}{\sqrt{3}V_{oc}I_{oc}}\right)$.
3. From the current I_{sc} find out the I_{SN} (short circuit current corresponding to the normal voltage) through the formula $I_{SN} = I_{sc}\left(\frac{V_{rated}}{V_{sc}}\right)$, draw the OB vector with the magnitude of I_{SN} from the origin by the same current scale, which lags the voltage (Y-axis) V by an angle Φ_{SC} where $\Phi_{SC} = \cos^{-1}\left(\frac{W_{sc}}{\sqrt{3}V_{sc}I_{sc}}\right)$.
4. Join the points B and A to get the output line.
5. Draw the parallel line for the X-axis from point A and for the Y-axis from point B upto the X-axis (point E), let both the lines intersect at point D.
6. Then draw the bisector for the output line and extend it to the line AD let the point of intersection be C.
7. By keeping the point C as center draw a semi circle with radius CA.
8. Let EB be the line of total loss [EB = ED + DB where ED = constant loss and DB = variable loss]
9. In the line DB locate the point G to separate the stator and rotor copper losses by using the formula (rotor copper loss/stator copper loss) = $\frac{\text{Rotor Copper loss}}{\text{Stator Copper loss}} = \frac{I_2'^2 R_2'}{I_2'^2 R_1} = \frac{R_2'}{R_1}$ where R_1 = stator resistance per phase and R_2 = rotor resistance per phase.
Or, $\frac{BG}{BD} = \frac{R_2'}{R_{01}} = \frac{\text{Rotor Copper loss}}{\text{Stator Copper loss} + \text{Rotor Copper loss}}$.
10. To get the torque line, join the points A and G.
11. To find the full load quantities, draw line BK (=Full load output/power scale). Now, draw line PK parallel to output line meeting the circle at point P.
12. Draw line PT parallel to Y-axis meeting output line at Q, torque line at R, constant loss line at S and X-axis at T.

Note: Choose the current scale such that the circle diagram will be as large as possible. The larger the circle diagram more will be the accuracy. Select power scale = $\sqrt{3} \times V_{rated} \times \text{current scale}$.

Qn 1:- Draw the circle diagram for a 3-phase, 4-pole, 50 Hz, 3.7 kW, 400V, 7.5A delta connected induction motor from the following data (line values)

No Load test : 400V, 3A, 300W

Blocked Rotor test : 90V, 7.5A, 500W

Stator resistance per phase is 3Ω /phase.

From the circle diagram, determine:

- i) Power factor, slip, efficiency, speed and torque at full load,
- ii) Starting torque,
- iii) Maximum power output,
- iv) Maximum power input,
- v) Maximum torque developed,
- vi) Slip for maximum torque,
- vii) Maximum power factor

Ans: -

$$V_{oc} = 400V, I_{oc} = 3A, W_o = 300W$$

$$V_{sc} = 90V, I_{sc} = 7.5A, W_{sc} = 500W$$

$$I_s = \frac{I_{sc}}{\sqrt{3}} = \frac{7.5}{\sqrt{3}} = 4.33A$$

$$R_1 = 3\Omega$$

$$R_{o1} = \frac{W_{sc}}{3I_s^2} = \frac{500}{3 \times 4.33^2} = 8.89\Omega$$

$$R_2' = R_{o1} - R_1 = 8.89 - 3 = 5.89\Omega$$

$$\frac{BG}{BD} = \frac{R_2'}{R_{o1}} = \frac{5.89}{8.89} = 0.66$$

$$BG = 0.66 \times BD$$

Selection of current and power scale

$$\text{Current scale} = 1\text{cm} = 1.5A$$

$$I_o = 3A \text{ (=2cm)}$$

$$I_{SN} = \left(\frac{V_{rated}}{V_{sc}}\right) I_{sc} = \left(\frac{400}{90}\right) \times 7.5 = 33.33A \text{ (=22.2cm)}$$

$$\Phi_o = \cos^{-1}\left(\frac{W_{oc}}{\sqrt{3}V_{oc}I_{oc}}\right) = \cos^{-1}\left(\frac{300}{\sqrt{3} \times 400 \times 3}\right) = 81.7^\circ$$

$$\Phi_{sc} = \cos^{-1}\left(\frac{W_{sc}}{\sqrt{3}V_{sc}I_{sc}}\right) = \cos^{-1}\left(\frac{500}{\sqrt{3} \times 90 \times 7.5}\right) = 64.68^\circ$$

$$\text{Power Scale} = \sqrt{3} \times V_{rated} \times \text{current scale} = \sqrt{3} \times 400 \times 1.5 = 1039.2W = 1\text{cm}$$

PERFORMANCE AT FULL LOAD FROM CIRCLE DIAGRAM

$$\text{Full load output} = 3700W = PQ = 3.56\text{cm}$$

$$\text{Full load current} = OP \times \text{current scale} = 5 \times 1.5 = 7.5A$$

$$\text{Full load power factor} = \frac{PT}{OP} \times 100\% = \frac{4.3}{5} \times 100\% = 0.86\text{lag}$$

$$\text{Rotor copper loss at full load} = QR \times \text{power scale} = 0.2 \times 1039.2 = 207.84W$$

$$\text{Stator copper loss at full load} = RS \times \text{power scale} = 0.1 \times 1039.2 = 103.92W$$

$$\text{Constant loss} = \text{ST} \times \text{power scale} = 0.3 \times 1039.2 = 311.76\text{W}$$

$$\text{Rotor input at full load} = \text{PR} \times \text{power scale} = 3.8 \times 1039.2 = 3948.96\text{W}$$

$$\text{Torque developed at full load} = \text{PR} \times \text{power scale (sync.watts)}$$

$$= \text{PR} \times \text{power scale} \times \frac{60}{2\pi N_s} \text{ N-m}$$

$$= 3.8 \times 1039.2 \times \frac{60}{2\pi \times 1500} = 25.14\text{N-m}$$

$$\text{Stator input at full load} = \text{PT} \times \text{power scale} = 3.9 \times 1039.2 = 4052.88\text{W}$$

$$\text{Efficiency at full load} = \frac{PQ}{PT} \times 100\% = \frac{3.56}{3.9} \times 100\% = 91.28\%$$

$$\text{Slip at full load } s = \frac{QR}{PR} \times 100\% = \frac{0.2}{3.8} \times 100\% = 5.26\%$$

$$\text{Speed at full load} = (1-s) \times N_s = (1-0.0526) \times 1500 = 1421 \text{ rpm}$$

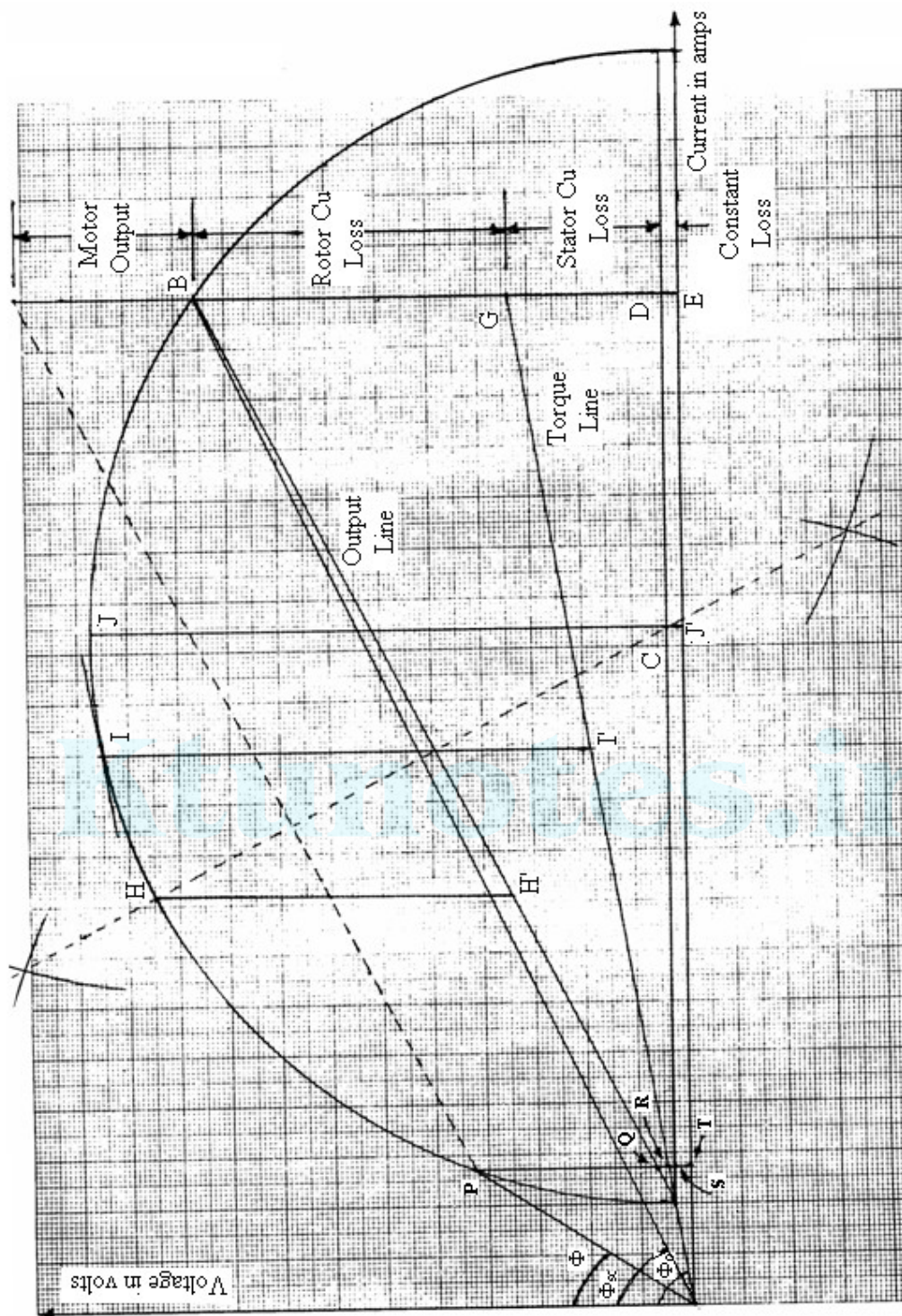
$$\begin{aligned} \text{Starting torque developed} &= \text{BG} \times \text{power scale} \times \frac{60}{2\pi N_s} \text{ N-m} = 6.2 \times 1039.2 \times \frac{60}{2\pi \times 1500} = \\ &= 41.02\text{N-m} \end{aligned}$$

$$\text{Maximum torque developed} = \text{I I}' = 9.6 \times 1039.2 \times \frac{60}{2\pi \times 1500} = 63.5\text{N-m}$$

$$\text{Maximum output} = \text{HH}' = 7.1 \times 1039.2 = 7378.32\text{W}$$

$$\text{Maximum input} = \text{JJ}' = 11.7 \times 1039.2 = 12158.6\text{W}$$

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Qn 2:- Draw the circle diagram of a 7.46kW, 200V, 50Hz, 3-phase slip ring induction motor with a star connected stator and rotor, a winding ratio of unity, a stator resistance of $0.38\Omega/\text{phase}$ and rotor resistance of $0.24\Omega/\text{phase}$. Following are test readings:

No Load : 200V, 7.7A, $\cos\phi_o = 0.195$; Short Circuit : 100V, 47.6A, $\cos\phi_{sc} = 0.454$. Find a) starting torque, b) maximum output, c) Maximum torque in synchronous watts, d) maximum power factor and e) slip for maximum torque. (Calicut University, June 2009)

Ans:-

$$I_o = 7.7A; \phi_o = 78.76^\circ$$

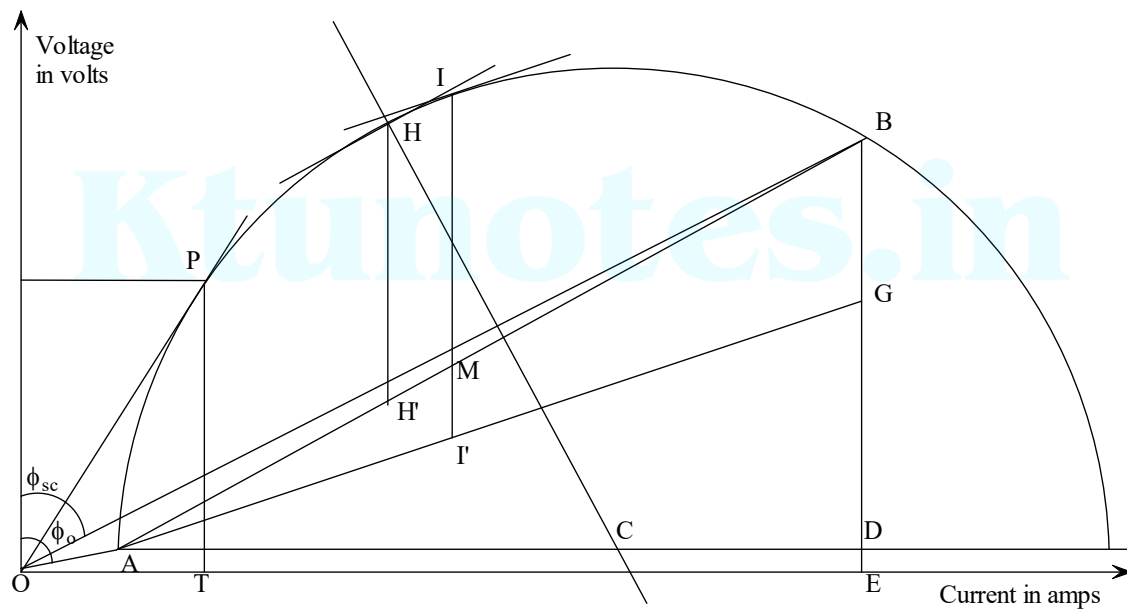
$$I_{sc} = 47.6A; \phi_{sc} = 63^\circ$$

Power scale, $1\text{ cm} = \sqrt{3} \times 200 \times 5.5 = 1905.3\text{ W}$

$$I_{sN} = \frac{V_{rated}}{V_{sc}} \times I_{sc} = \frac{200}{100} \times 47.6 = 95.2 A$$

$$R_2' = \left(\frac{N_1}{N_2}\right)^2 \times R_2 = 0.24\Omega$$

$$\frac{R_2'}{R_{o1}} = 0.387 = \frac{BG}{BD}$$



Slip at maximum torque, $\frac{MI'}{II'} = \frac{1.4}{6.4} = 0.219$

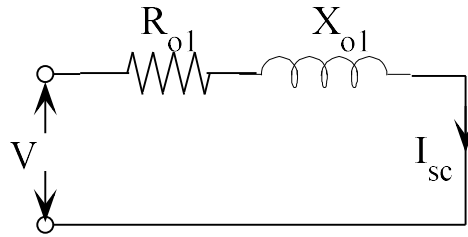
STARTING OF INDUCTION MOTORS

An induction motor at standstill is like a transformer with short circuited secondary. (At starting since the relative speed between rotor conductors and rotating magnetic field is maximum, the rotor induced emf will be maximum, and hence large current will flow through the rotor. Proportionately, heavy currents flow in stator also). It takes a high current (about 5 to 8 times full load current) if started at full voltage. This heavy current may not be dangerous to the motor since this current flows only for a short duration of time; but cause a large drop in line voltage supplying the motor. Hence, the machines connected to the same supply may be adversely affected.

It is therefore recommended that large three phase squirrel cage induction motors be started with reduced voltage applied across the stator terminals at starting and then increasing the voltage to the full-line value, when the motor picks up speed.

Small motors upto 5 hp ratings may however be started direct-on-line (D.O.L).

Equivalent circuit of the motor at standstill condition is shown below (neglecting the no load current).



$$I_{st} = \frac{V}{R_{o1} + jX_{o1}} = \frac{V}{Z_{o1}} = I_{sc}$$

where I_{sc} = short circuited stator current (at rated voltage)

$$I_{sc} = \frac{\text{Rated Voltage per phase}}{Z_{01}}$$

By applying reduced voltage ($= xV$ where x less than 1) at starting, the starting current can be reduced.

$$I_{st} = xI_{sc}$$

$$\text{Torque, } T = \frac{\text{Rotor copper loss}}{s} = \frac{3I_2^2 R_2}{s}$$

$$\text{But } I_1 \propto I_2$$

$$\text{Hence, } T \propto \frac{I_1^2}{s}$$

$$\text{At starting, } T_{st} \propto I_{st}^2$$

$$\text{At full load, } T_{fl} \propto \frac{I_f^2}{s_f}$$

$$\frac{T_{st}}{T_{fl}} = \left(\frac{I_{st}}{I_f} \right)^2 \times s_f = x^2 \left(\frac{I_{sc}}{I_f} \right)^2 \times s_f$$

Hence, with reduced voltage, starting current is reduced; but the starting torque is also decreased.

Various methods of starting 3 phase *squirrel cage induction motors* are as follows:

1. Direct-on-Line (DOL) Starting

This method means switching the motor directly on to the supply without any device for reducing the starting current. This method is restricted to small motors upto 5 HP. Here, the starting period lasts only for a few seconds.

If stator is star connected,

$$I_{st} = \frac{V/\sqrt{3}}{Z_{01}} = I_{sc}$$

Starting current drawn from the mains =

$$I_{stline} = I_{st}$$

$$\frac{T_{st}}{T_{fl}} = \left(\frac{I_{st}}{I_f}\right)^2 \times s_f = \left(\frac{I_{sc}}{I_f}\right)^2 \times s_f$$

If stator is delta connected,

$$I_{st} = \frac{V}{Z_{01}} = I_{sc}$$

Starting current drawn from the mains =

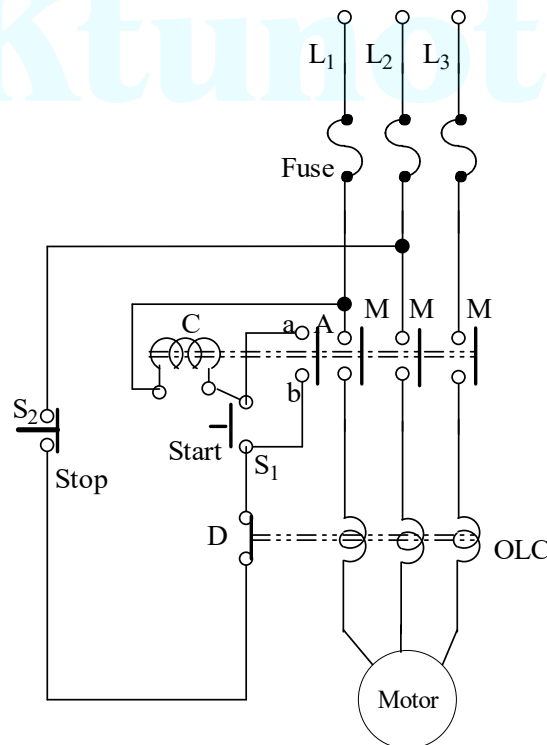
$$I_{stline} = \sqrt{3}I_{st} = \sqrt{3}I_{sc}$$

$$\frac{T_{st}}{T_{fl}} = \left(\frac{I_{st}}{I_f}\right)^2 \times s_f = \left(\frac{I_{sc}}{I_f}\right)^2 \times s_f$$

With DOL starting, inrush starting current is permissible.

A DOL starter is shown in figure below.

On pressing the START push button S_1 , the contactor coil C is energized from two line conductors L_1 and L_2 . The three main contacts M and the auxiliary contact A close and the terminals a and b are short-circuited. The motor is thus connected to the supply. When the pressure on S_1 is released, it moves back under spring action. Even then the coil C remains energized through ab . Thus, the main contacts M remain closed and the motor continues to get supply. When the STOP push button S_2 is pressed, the supply through the contactor coil is disconnected. Since the coil C is de-energised, the main contacts M and auxiliary contact A are opened. The supply to motor is disconnected and the motor stops.



S_1, A, M - normally open

S_2, D - normally closed

Undervoltage protection : When the voltage falls below a certain value, or in the event of failure of supply during motor operation, the coil C is de-energised. The motor is then disconnected from the supply.

Overload protection: In case of an overload, one or all the overload coils (OLC) are energized. The normally closed contact D is opened and the contactor coil C is de-energised to disconnect the supply to the motor.

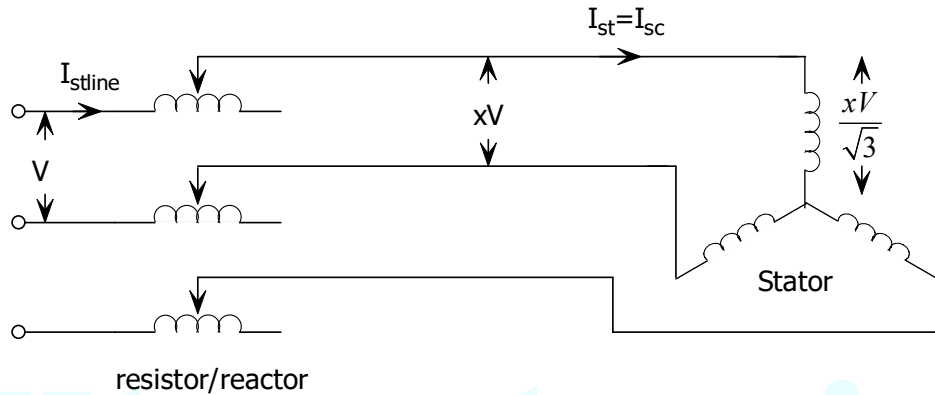
2. Stator impedance Starting

In this method, a resistor or reactor is inserted in series with the stator winding. At the time of starting, some voltage drop occurs across the starting impedance and a reduced voltage ($=xV$ where x is less than 1) appears across the stator terminals. As the motor speeds up, the impedance is cut out in steps and finally short circuited when the motor speed is near to its operating speed.

$$I_{st} = \frac{xV / \sqrt{3}}{Z_{01}} = xI_{sc}$$

$$\text{Starting current drawn from the mains} = I_{stline} = I_{st}$$

$$\frac{T_{st}}{T_{fl}} = \left(\frac{I_{st}}{I_f}\right)^2 \times s_f = x^2 \left(\frac{I_{sc}}{I_f}\right)^2 \times s_f$$



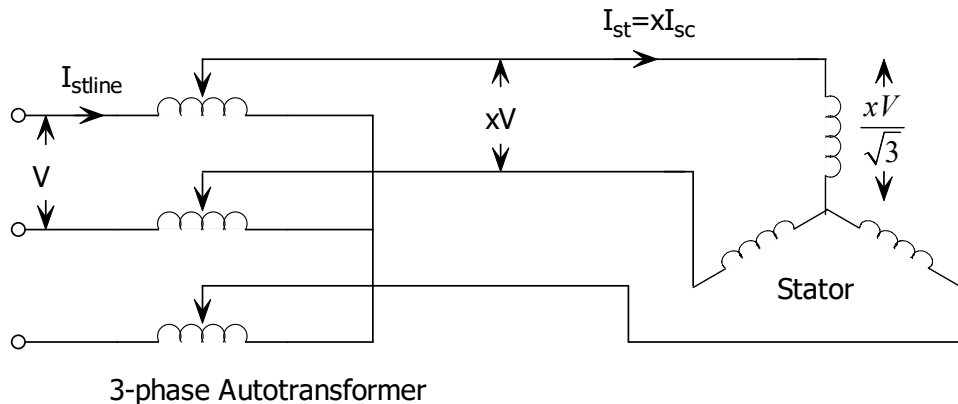
Conclusions :

- 1) T_{st} with stator impedance starter $= x^2 \times T_{st}$ with direct starting
- 2) I_{stline} with stator impedance starter $= x \times I_{stline}$ with direct starting

3. Auto Transformer Starting

Reduced voltage (xV) is applied to the stator terminals at starting using a 3 phase autotransformer. As the motor picks up speed upto around 80% of its normal speed, the autotransformer is cut out of the circuit and full rated voltage gets applied across the motor terminals.

$$I_{st} = \frac{xV / \sqrt{3}}{Z_{01}} = xI_{sc}$$



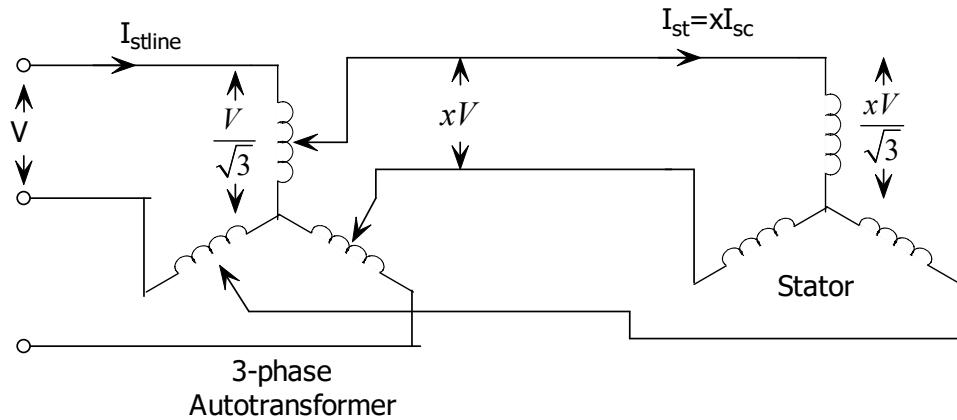
For an autotransformer, input VA = output VA (neglecting losses).

Hence

$$\frac{V}{\sqrt{3}} I_{stline} = \frac{xV}{\sqrt{3}} I_{st} = x^2 \frac{V}{\sqrt{3}} I_{sc}$$

$$\text{or } I_{stline} = x^2 I_{sc}$$

$$\frac{T_{st}}{T_{fl}} = \left(\frac{I_{st}}{I_f}\right)^2 \times s_f = x^2 \left(\frac{I_{sc}}{I_f}\right)^2 \times s_f$$



Conclusions :

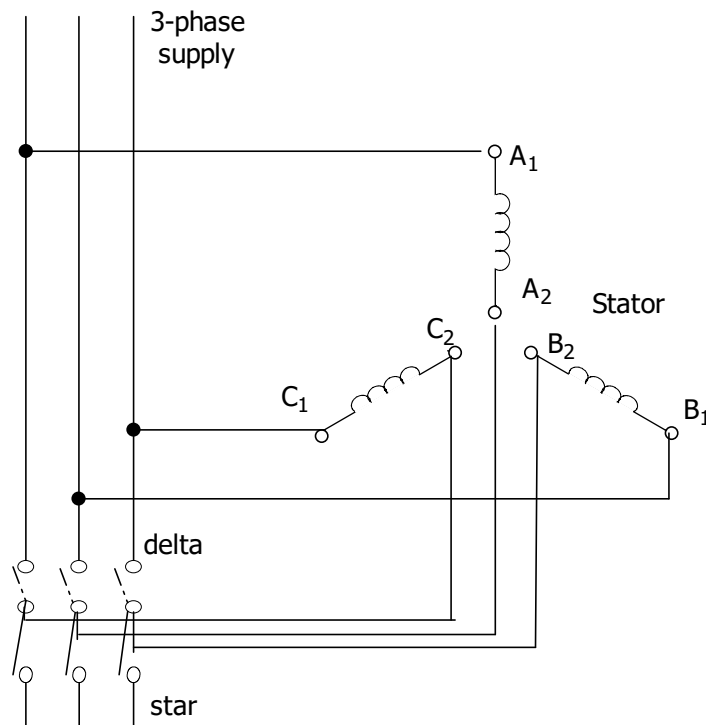
$$3) T_{st} \text{ with autotransformer starter} = x^2 \times T_{st} \text{ with direct starting}$$

$$4) I_{stline} \text{ with autotransformer starter} = x^2 \times I_{stline} \text{ with direct starting}$$

4. Star-Delta Starter

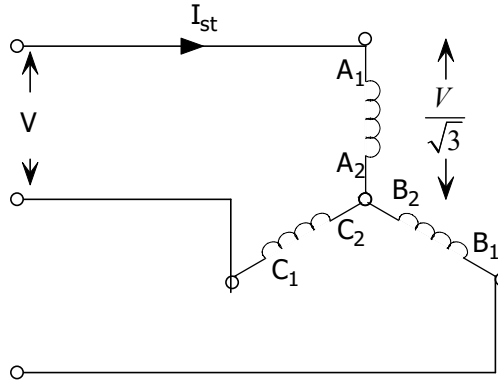
Star-Delta starter is used if the motor is delta connected for normal operation and all the 6 stator terminals are available.

At starting, the stator phases are connected in star by means of a TPDT switch (position 1). When the motor attains around 80% of normal speed, TPDT switch is moved to position 2 thereby connecting the stator winding in delta.



When star connected, the applied voltage per phase is reduced by a factor $\frac{1}{\sqrt{3}}$ than the voltage when directly connected in delta.

If started in star,

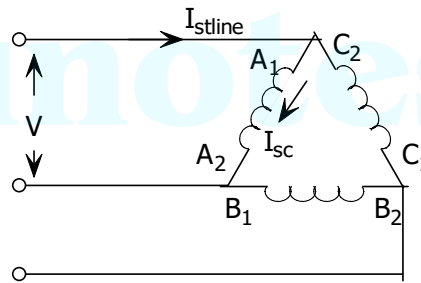


$$I_{st} = \frac{V/\sqrt{3}}{z_{o1}} = \frac{I_{sc}}{\sqrt{3}}$$

$$I_{stline} = I_{st} = \frac{I_{sc}}{\sqrt{3}}$$

$$\frac{T_{st}}{T_{fl}} = \left(\frac{I_{st}}{I_f}\right)^2 \times s_f = \frac{1}{3} \left(\frac{I_{sc}}{I_f}\right)^2 \times s_f$$

If started in delta (ie. Direct starting),



$$I_{st} = \frac{V}{z_{o1}} = I_{sc}$$

$$I_{stline} = \sqrt{3}I_{st} = \sqrt{3}I_{sc}$$

$$\frac{T_{st}}{T_{fl}} = \left(\frac{I_{st}}{I_f}\right)^2 \times s_f = \left(\frac{I_{sc}}{I_f}\right)^2 \times s_f$$

$$\frac{I_{stline} \text{ with Y-}\Delta \text{ starter}}{I_{stline} \text{ with direct starting}} = \frac{I_{sc}/\sqrt{3}}{\sqrt{3}I_{sc}} = \frac{1}{3}$$

$$\frac{T_{st} \text{ with Y-}\Delta \text{ starter}}{T_{st} \text{ with direct starting}} = \frac{1}{3}$$

Note : A star-delta starter is equivalent to autotransformer starter with $x = \frac{1}{\sqrt{3}} = 0.578$

Star-delta starter is cheap as compared to autotransformer starter and is, therefore, used extensively. For applications requiring starting torque not exceeding about 50% of rated torque, star-delta starting is the most suitable method.

Qn.1:- A 3-phase squirrel cage induction motor has a short circuit current equal to 5 times its full load current. Calculate the starting torque as a percentage of full-load torque if the motor is started by a)

direct switching to the supply b) a star delta starter c) an autotransformer and d) a resistance in the stator circuit. The starting line current in (c) and (d) is limited to 2.5 times the full-load current and the full-load slip is 2.5%.

Ans:-

$$I_{sc} = 5I_f$$

a) With direct-on-line starter,

$$\frac{T_{st}}{T_{fl}} = \left(\frac{I_{sc}}{I_f}\right)^2 \times s_f = 5^2 \times 0.025 = 0.625$$

b) With star delta starter

$$\frac{T_{st}}{T_{fl}} = \frac{1}{3} \times \left(\frac{I_{sc}}{I_f}\right)^2 \times s_f = \frac{1}{3} \times 5^2 \times 0.025 = 0.208$$

c) With autotransformer starter

$$x^2 I_{sc} = 2.5 \times I_f$$

$$x^2 = 2.5 \times \frac{I_f}{I_{sc}} = 2.5 \times \frac{1}{5} = 0.5$$

$$x = 0.707$$

$$\frac{T_{st}}{T_{fl}} = x^2 \times \left(\frac{I_{sc}}{I_f}\right)^2 \times s_f = 0.5 \times 5^2 \times 0.025 = 0.3125$$

d) With stator resistor starter

$$\frac{T_{st}}{T_{fl}} = \left(\frac{I_{st}}{I_f}\right)^2 \times s_f = 2.5^2 \times 0.025 = 0.15625$$

Qn.2:- A 3-phase squirrel cage induction motor has maximum torque equal to twice the full load torque. Determine the ratio of motor starting torque to its full load torque, if it is started by a) direct-on-line starter, b) star-delta starter and c) autotransformer starter with 70% tapping. The per phase rotor resistance and per phase standstill reactance referred to stator are 0.2Ω and 2Ω respectively. Neglect stator impedance.

Ans:-

$$a = \frac{R_2}{X_2} = \frac{0.2}{2} = 0.1$$

With direct-on-line starter,

$$\frac{T_{st}}{T_{max}} = \frac{2a}{a^2 + 1} = \frac{2 \times 0.1}{0.1^2 + 1} = 0.198$$

$$\frac{T_{st}}{T_{fl}} = \frac{T_{st}}{T_{max}} \times \frac{T_{max}}{T_{fl}} = 0.198 \times 2 = 0.396$$

With star-delta starter,

$$T_{st} \text{ with start-delta} = \frac{1}{3} \times T_{st} \text{ with direct-on-line} = \frac{1}{3} \times 0.396 = 0.132$$

With autotransformer starter,

$$T_{st} \text{ with autotransformer} = x^2 \times T_{st} \text{ with direct-on-line} = 0.7^2 \times 0.396 = 0.194$$

Qn 3:- A 3-phase cage induction motor starts with a current at a value 5 times that of normal. Full load slip is 4%. What autotransformer ratio would enable the motor to be started with not more than 1.5 times the full load current drawn from supply? Calculate the starting torque under these conditions and compare it with that obtained by using a stator resistance starter under the same limitations of line current.

Ans:-

$$I_{sc} = 5I_f$$

a) Autotransformer starter

$$I_{st} = x^2 I_{sc} = 1.5 \times I_f$$

$$x^2 = 1.5 \times \frac{I_f}{I_{sc}} = 1.5 \times \frac{1}{5} = 0.3$$

$$x = 0.5477$$

$$\frac{T_{st}}{T_{fl}} = x^2 \times \left(\frac{I_{sc}}{I_f}\right)^2 \times s_f = 0.3 \times 5^2 \times 0.04 = 0.3$$

b) Stator resistance starter

$$I_{st} = x I_{sc} = 1.5 I_f$$

$$x = 1.5 \times \frac{I_f}{I_{sc}} = 1.5 \times \frac{1}{5} = 0.3$$

$$\frac{T_{st}}{T_{fl}} = x^2 \times \left(\frac{I_{sc}}{I_f}\right)^2 \times s_f = 0.3^2 \times 5^2 \times 0.04 = 0.09$$

$$\frac{T_{st} \text{ with autotransformer}}{T_{st} \text{ with stator resistor}} = \frac{0.3}{0.09} = 3.333$$

Qn 4:- A 415V, 11kW, 3-phase induction motor has an efficiency of 88% and power factor of 0.86 at full load. At standstill, the motor draws 7 times the full load current and develops 1.96 times the full load torque when rated voltage is applied. An autotransformer starter is to be installed to reduce the starting current and to develop full-load torque at starting. Neglecting magnetizing current, find i) the voltage to be applied to the motor ii) the full load line current iii) the line current likely to be drawn by the motor at starting and iv) the line current likely to be drawn from the supply.

Ans:-

$$I_{sc} = 7I_f$$

$$\frac{T_{st}}{T_{fl}} = \left(\frac{I_{sc}}{I_f}\right)^2 \times s_f = 7^2 \times s_f = 1.96$$

$$s_f = \frac{1.96}{7^2} = 0.04$$

With Autotransformer starter

$$\frac{T_{st}}{T_{fl}} = x^2 \times \left(\frac{I_{sc}}{I_f}\right)^2 \times s_f = x^2 \times 7^2 \times 0.04 = 1$$

$$x^2 = 0.51$$

$$x = 0.714$$

i) Applied voltage = $xV = 0.714 \times 415 = 296.31V$

ii) Full load current, $I_f = \frac{11 \times 1000}{\sqrt{3} \times 415 \times 0.88 \times 0.86} = 20.22A$

iii) Starting current without starter, $I_{st} = I_{sc} = 7 \times I_f = 7 \times 20.22 = 141.54A$

iv) Starting current with autotransformer,

$$I_{st} = x^2 I_{sc} = x^2 \times (7 \times I_f) = 0.51 \times (7 \times 20.22) = 72.185A$$

Qn 5:- An induction motor is started directly from the mains. If the starting torque = full load torque, find the starting current in terms of full load current if the slip of motor on full load is 4%.

(Ans:- $5 I_f$)

Qn 6:- Find a suitable autotransformer ratio for starting a squirrel cage induction motor with a supply current not to exceed 2.5 times the full load current. The motor has the following data : Full load slip

= 3%, short circuit current = 5 times Full load current. Find the starting torque in terms of full load torque.

(Ans:- 0.707 and 0.375)

Qn 7:- Calculate the % tapplings required on an autotransformer starter for a squirrel cage induction motor to start the motor against 40% of full load torque. Short circuit current under normal full voltage = 4 times the full load current. Full load slip is 4%.

(Ans:- 79%)

Qn 8:- A 5KW, 3 phase, 415V, 50Hz induction motor is to be started by a star/delta starter. Full load efficiency of the motor is 85% and the power factor is 0.8 lagging, short circuit current is 5 times the full load current. Neglecting magnetizing current, find the line current at start from the mains.

(Ans:- 17.05A)

Qn 9:- The short circuit current of a certain induction motor is 5 times full load current. Calculate the starting torque as % of full load torque if the motor is started by i) direct switching to supply ii) star-delta starter and iii) an auto transformer starter which limits starting current to 3 times full load current. Assume full load slip = 4%.

(Ans:- i) 1 ii) 0.33 iii) 0.36)

Qn 10:- Determine the approximate starting torque of an induction motor in terms of the full-load torque when started by means of:

- i) Star-delta starter and ii) autotransformer with 50% tapping. Ignore magnetizing current. The short circuit current of the motor at normal voltage is 5 times the full load current, and the full-load slip is 5%. (*Calicut University, December 08*)

Ans:- i) $T_{st} = 41.67\% I_f$ and ii) $T_{st} = 31.25\% I_f$

Starting of Slip Ring Induction Motors – Rotor Resistance Starting

A wound rotor induction motor is started by adding external resistance in the rotor circuit with full voltage applied across the stator terminals. At the time of starting, addition of rotor resistance i) decreases its starting current ii) increases its starting torque and iii) improves its starting power factor. The external resistance is kept at such a value as to give maximum or near to maximum torque at starting. As the motor accelerates, the external resistance is gradually reduced to zero.

Starting current, $I_{st} = \frac{V}{(R_{o1} + R) + jX_{o1}}$ where R=external resistance to be added at starting.

Let

R_m = rotor resistance/phase

V = Line voltage

E = rotor induced emf at standstill

X = rotor standstill reactance/phase

n = number of resistance elements or sections

I_{1max} = upper current limit during starting

I_{1min} = lower current limit during starting

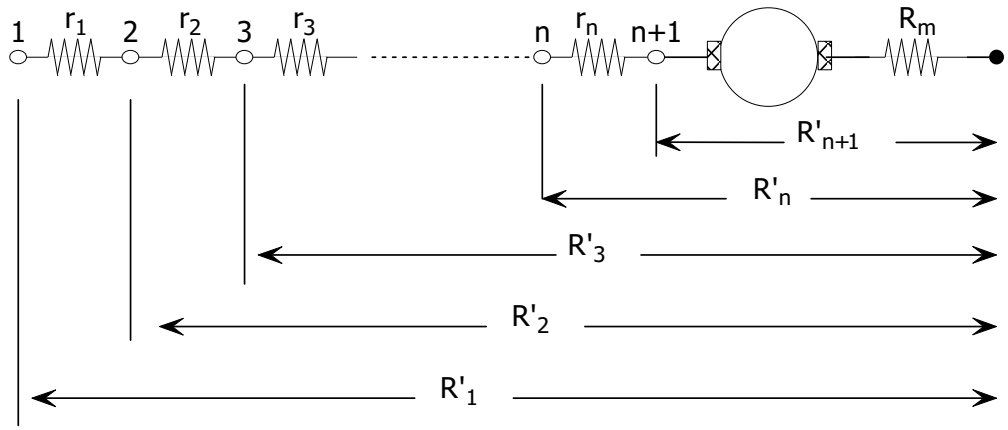
Consider one phase of the rotor of a wound-rotor induction motor with resistance R_m and standstill reactance X.

$r_1, r_2, r_3, \dots, r_n$ = resistances of n resistance elements (or sections)

$R'_1, R'_2, R'_3, \dots, R'_{n+1}$ = total resistance on stud 1, 2, 3 etc.

Assumptions:-

- During starting time, constant load torque is assumed
- No load current neglected
- Stator current is fluctuating between fixed limits I_{1max} and I_{1min} .



At the time of start, movable handle is at stud No. 1. Rotor resistance is R'_1 . When supply is switched on, the input current shoots to $I_{1\max}$.

$$I_{1\max} = \frac{V}{\sqrt{\left(\frac{R'_1}{s_1}\right)^2 + X_2^2}} = \frac{V}{\sqrt{R_1'^2 + X_2^2}} \quad (s_1=1, \text{ at starting})$$

On Stud 1, R'_1 remains in circuit until the motor has started and the current has fallen from $I_{1\max}$ to $I_{1\min}$. At the same time, slip falls from s_1 to s_2 .

$$I_{1\min} = \frac{V}{\sqrt{\left(\frac{R'_1}{s_2}\right)^2 + X_2^2}}$$

As soon as $I_{1\min}$ is reached at stud 1, resistance r_1 is cut out by moving the handle from stud 1 to stud 2. Since speed does not change instantaneously, slip remains same but current becomes $I_{1\max}$.

$$I_{1\max} = \frac{V}{\sqrt{\left(\frac{R'_2}{s_2}\right)^2 + X_2^2}}$$

At stud 2, speed rises so that slip falls to s_3 and current decreases to $I_{1\min}$.

$$I_{1\min} = \frac{V}{\sqrt{\left(\frac{R'_2}{s_3}\right)^2 + X_2^2}}.$$

This process continues.

$$I_{1\max} = \frac{V}{\sqrt{\left(\frac{R'_1}{s_1}\right)^2 + X_2^2}} = \frac{V}{\sqrt{\left(\frac{R'_2}{s_2}\right)^2 + X_2^2}} = \frac{V}{\sqrt{\left(\frac{R'_3}{s_3}\right)^2 + X_2^2}} = \dots\dots\dots$$

$$\frac{R'_1}{s_1} = \frac{R'_2}{s_2} = \frac{R'_3}{s_3} = \dots\dots\dots = \frac{R'_n}{s_n} = \frac{R'_{n+1}}{s_{n+1}} = \frac{R_m}{s_m}$$

where s_m = slip under normal operating conditions (when all external resistance is reduced to zero)

$$\frac{s_2}{s_1} = \frac{s_3}{s_2} = \frac{s_4}{s_3} = \dots\dots\dots \frac{s_m}{s_n} = \frac{R'_2}{R'_1} = \frac{R'_3}{R'_2} = \frac{R'_4}{R'_3} \dots\dots\dots = \frac{R'_{n+1}}{R'_n} = \frac{R_m}{R'_n} = \alpha$$

$$R'_1 = s_1 \frac{R_m}{s_m} = \frac{R_m}{s_m}$$

$$R'_2 = \alpha R'_1$$

$$R'_3 = \alpha R'_2 = \alpha^2 R'_1$$

.

.

.

$$R'_{n+1} = \alpha^n R'_1$$

$$R_m = \alpha^n R'_1$$

$$\text{Hence, } R_m = \alpha^n R'_1 = \alpha^n \frac{R_m}{s_m}$$

$$\alpha^n = s_m \quad \text{OR} \quad \alpha = s_m^{\frac{1}{n}}$$

Resistance of the sections are

$$r_1 = R'_1 - R'_2 = R'_1(1 - \alpha)$$

$$r_2 = R'_2 - R'_3 = R'_2(1 - \alpha) = \alpha R'_1(1 - \alpha) = \alpha r_1$$

$$r_3 = R'_3 - R'_4 = R'_3(1 - \alpha) = \alpha R'_2(1 - \alpha) = \alpha^2 r_1$$

$$r_4 = \alpha^3 r_1$$

.

.

$$r_n = \alpha^{n-1} r_1$$

If $I_{1\max} = I_{fl}$, $s_m = s_f$.

Qn 1:- Calculate the values of resistance elements of a 4-step starter for a 3-phase, 400V, wound rotor induction motor. The full load slip is 3% and the maximum starting current is limited to its full-load value. Rotor resistance per phase is 0.015Ω .

Ans:-

$$n=4$$

$$s_m = s_f = 0.03$$

$$\alpha = s_m^{\frac{1}{n}} = 0.03^{\frac{1}{4}} = 0.416$$

$$R_m = 0.015\Omega$$

$$R'_1 = \frac{R_m}{s_m} = \frac{0.015}{0.03} = 0.5\Omega$$

$$r_1 = R'_1(1 - \alpha) = 0.5(1 - 0.416) = 0.292\Omega$$

$$r_2 = \alpha r_1 = 0.416 \times 0.292 = 0.121\Omega$$

$$r_3 = \alpha^2 r_1 = 0.416^2 \times 0.292 = 0.051\Omega$$

$$r_4 = \alpha^3 r_1 = 0.416^3 \times 0.292 = 0.021\Omega$$

Qn 2:- Calculate the steps in a 4-step starter for a 3-phase slip ring induction motor, from the following data:- Maximum starting current = full load current; full-load slip = 2.5%; Rotor resistance per phase = 0.02Ω .

$$\text{Ans:- } \alpha = s_m^{\frac{1}{n}} = 0.3976$$

$$R'_1 = \frac{R_m}{s_m} = 0.8\Omega$$

$$r_1 = R'_1(1 - \alpha) = 0.482\Omega$$

$$r_2 = \alpha r_1 = 0.19\Omega$$

$$r_3 = \alpha^2 r_1 = 0.0765\Omega$$

$$r_4 = \alpha^3 r_1 = 0.03\Omega$$

Qn 3:- Design the 5 sections of a 6-stud starter for a 3-phase slip-ring induction motor. The full load slip is 2% and the maximum starting current is limited to twice full-load current. Rotor resistance per phase is 0.03Ω .

Ans:-

$$\text{Rotor current, } I_2 = \frac{sE_2}{\sqrt{R_2 + (sX_2)^2}} \approx \frac{sE_2}{R_2}$$

$$I_2 \propto s$$

$$s_m = 2 \times 0.02 = 0.04$$

$$n=5$$

$$\alpha = s_m^{\frac{1}{n}} = 0.04^{\frac{1}{5}} = 0.525$$

$$R_m = 0.03\Omega$$

$$R'_1 = \frac{R_m}{s_m} = \frac{0.03}{0.04} = 0.75\Omega$$

$$r_1 = R'_1 (1 - \alpha) = 0.75(1 - 0.525) = 0.356\Omega$$

$$r_2 = \alpha r_1 = 0.525 \times 0.356 = 0.187\Omega$$

$$r_3 = \alpha^2 r_1 = 0.525^2 \times 0.356 = 0.098\Omega$$

$$r_4 = \alpha^3 r_1 = 0.525^3 \times 0.356 = 0.052\Omega$$

$$r_5 = \alpha^4 r_1 = 0.525^4 \times 0.356 = 0.027\Omega$$

HIGH TORQUE CAGE ROTORS

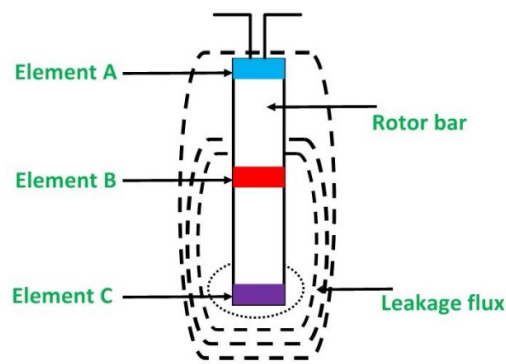
Conventional squirrel-cage motors suffers from the disadvantage of low starting torque because of low rotor resistance. The starting torque can be increased by using bar material of higher resistivity. A higher rotor resistance gives a higher starting torque and lower starting line current at a higher power factor. However, higher rotor resistance reduces the full-load speed, increases rotor ohmic losses and lower efficiency. A low rotor resistance is required for normal operation while running, so that the slip is low and efficiency is high. Therefore, for good starting performances, the rotor resistance should be high, and under normal operating speeds, the rotor resistance should be low. In wound-rotor induction motors these conditions are fulfilled by connecting external resistances in the rotor circuit at the time of starting. As the motor speeds up, the external resistance is cut out and the rotor windings are short circuited through slip rings.

In order to obtain high rotor resistance at starting and low rotor resistance at running, two types of rotor connections are used in cage motors:

1. Deep bar rotor
2. Double-cage rotor

DEEP-BAR CAGE MOTORS

In this type of construction bars of narrow width are laid down in deep semi-enclosed slots as shown in figure. The magnetic leakage flux pattern set up by the bar current is indicated in dotted lines.



A bar may be assumed to be made up of a number of narrow layers connected in parallel. The above figure shows three layers A, B and C. The topmost layer element A is linked with the minimum leakage flux. Its leakage inductance is minimum. On the other hand, the bottom layer C links with maximum leakage flux and its leakage inductance is maximum.

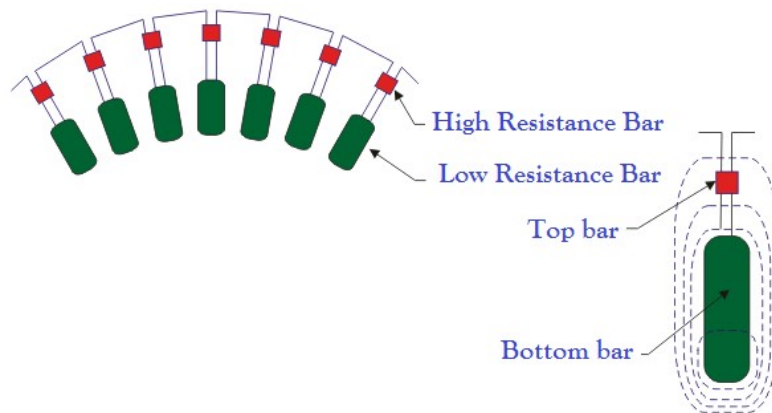
At starting, rotor frequency is equal to supply frequency. Bottom layer element C offers more impedance to the flow of current than the top layer element A. Hence, maximum current flows through the top layer and minimum current flows through the bottom layer.

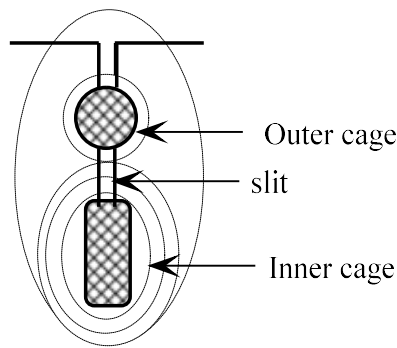
Non-uniform current distribution causes the effective resistance more than its DC resistance. More torque and less current are achieved at starting due to more rotor resistance at starting.

As the motor speeds up to a value close to synchronous speed, rotor frequency is very low. Reactances of various elementary strips will be almost equal, current density over the conductor cross section becomes nearly uniform so that it offers a resistance almost equal to its dc value. Low resistance causes reduced copper losses and improved efficiency.

DOUBLE-CAGE INDUCTION MOTORS

An induction motor with two windings or cages is used for obtaining high starting torque at low starting current. The stator of a double-cage rotor is similar to that of an ordinary induction motor. In the double-cage rotor there are two layers of bars as shown in figure.





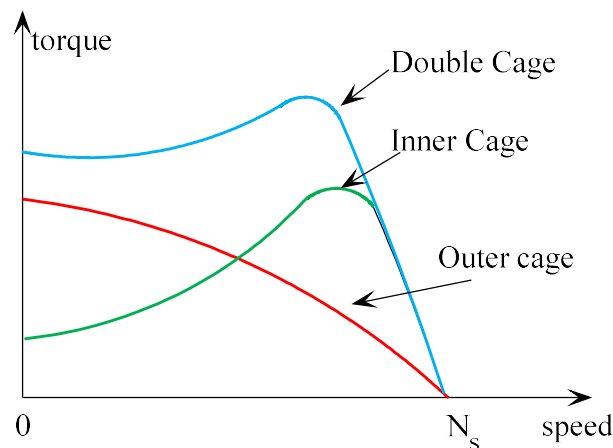
In double cage rotor, there are two layer of the bars. Each layer is short circuited by the end rings. The outer cage bars have a smaller cross-sectional area than the inner bars and are made of high resistivity materials like brass, aluminium, bronze etc. The bars of the inner cage are made of low resistance copper. Thus the resistance of the outer cage is greater than the resistance of the inner cage. There is a slit between the top and bottom slots. The slit increases the permeance for leakage flux around the inner-cage bars. Thus, the leakage flux linking the inner-cage winding is much larger than that of the outer-cage winding and the inner winding, therefore, has a greater self inductance.

At starting, rotor frequency is same as the supply frequency ($f_2=f_1$). Hence, the leakage reactance of the inner-cage winding ($=2\pi fL$) is much larger than that of the outer-cage winding. Therefore, most of the starting current is flowing in the outer-cage winding which offers low-impedance to the flow of current. The high resistance outer cage winding, therefore, develops a high starting torque.

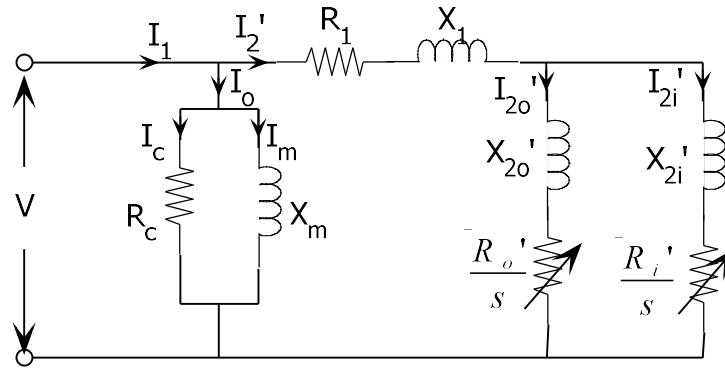
As the motor speed increases, the frequency of the rotor emf ($f_2=sf$) decreases. At normal operating speed, the leakage reactances ($=2\pi sfL$) of both the windings become negligibly small. The rotor current division between the two cages is governed mainly by their resistances. Since the resistance of the outer cage is about 5 to 6 times that of the inner cage, most of the rotor current flows through the inner cage. Hence under normal operating speed, the torque is developed mainly by the low-resistance inner cage.

It is to be noted that for low-starting torque requirements an ordinary cage motor is generally used. For higher torque requirements a deep-bar cage motor is used. A double-cage motor is used for still higher torques. For large-size motors with very large starting torques and exceptionally long starting periods, slip-ring construction is used.

Torque slip characteristics of double cage induction motor



Equivalent Circuit of a double cage induction motor



Let R_1 = stator resistance per phase

X_1 = stator reactance per phase

R_{2o}' = resistance of outer cage/phase referred to stator

X_{2o}' = standstill leakage reactance of outer cage/phase referred to stator

R_{2i}' = resistance of inner cage/phase referred to stator

X_{2i}' = standstill leakage reactance of inner cage/phase referred to stator

At slip s , outer cage impedance = $Z_{2o}' = \frac{R_{2o}'}{s} + jX_{2o}'$

At slip s , inner cage impedance = $Z_{2i}' = \frac{R_{2i}'}{s} + jX_{2i}'$

Stator impedance, $Z_1 = R_1 + jX_1$

Torque developed by outer cage, $T_{2o} \propto I_{2o}^2 \frac{R_{2o}'}{s}$

Torque developed by inner cage, $T_{2i} \propto I_{2i}^2 \frac{R_{2i}'}{s}$

Qn 1:- The standstill impedance of the outer cage of a double cage induction motor is $(0.4+j0.3)\Omega$ and that of the inner cage is $(0.1+j1.6)\Omega$. Compare the relative currents and torques of the two cages (a) at standstill, (b) at a slip of 4%. Neglect stator impedance.

Ans:-

(a) At standstill,

$$Z_{2o}' = \frac{R_{2o}'}{s} + jX_{2o}' = R_{2o}' + jX_{2o}' = 0.4 + j0.3 = 0.5 \angle 36.87^\circ \Omega$$

$$Z_{2i}' = \frac{R_{2i}'}{s} + jX_{2i}' = R_{2i}' + jX_{2i}' = 0.1 + j1.6 = 1.603 \angle 86.42^\circ \Omega$$

$$I_{2o}' = \frac{V \angle 0^\circ}{Z_{2o}'}$$

$$I_{2i}' = \frac{V \angle 0^\circ}{Z_{2i}'}$$

$$\text{Hence, } \frac{I_{2o}'}{I_{2i}'} = \frac{Z_{2i}'}{Z_{2o}'} = \frac{1.603}{0.5} = 3.206$$

$$\frac{T_{2o}}{T_{2i}} = \frac{I_{2o}'^2 R_{2o}'}{I_{2i}'^2 R_{2i}'} = 3.206^2 \times \frac{0.4}{0.1} = 41.11$$

(b) During running condition with $s = 0.04$,

$$Z_{2o}' = \frac{R_{2o}'}{s} + jX_{2o}' = \frac{0.4}{0.04} + j0.3 = 10 + j0.3 = 10.0045\Omega$$

$$Z_{2i}' = \frac{R_{2i}'}{s} + jX_{2i}' = \frac{0.1}{0.04} + j1.6 = 2.5 + j1.6 = 2.968\angle 32.62^\circ\Omega$$

$$I_{2o}' = \frac{V\angle 0^\circ}{Z_{2o}'}$$

$$I_{2i}' = \frac{V\angle 0^\circ}{Z_{2i}'}$$

$$\text{Hence, } \frac{I_{2o}'}{I_{2i}'} = \frac{Z_{2i}'}{Z_{2o}'} = \frac{2.968}{10.0045} = 0.297$$

$$\frac{T_{2o}'}{T_{2i}'} = \frac{I_{2o}'^2 \frac{R_{2o}'}{s}}{I_{2i}'^2 \frac{R_{2i}'}{s}} = 0.297^2 \times \frac{0.4}{0.1} = 0.353$$

Qn 2:- The standstill impedance of the outer cage of a double cage induction motor is $(3+j1)\Omega$ and that of the inner cage is $(0.6+j4)\Omega$. Determine the slip at which the two cages develop equal torques. (Ans:- 30.2%)

BRAKING OF INDUCTION MOTOR

Methods employed for braking of an induction motor are i) regenerative braking ii) plugging and iii) dynamic braking

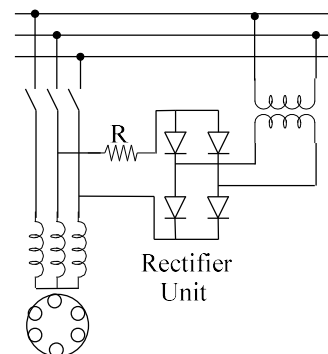
REGENERATIVE BRAKING

When the rotor of an induction motor runs faster than the stator field, the slip becomes negative and the machine generates power. Therefore, whenever the motor has a tendency to run faster than the rotating field, regenerative braking occurs and the kinetic energy of the rotating parts is returned to the mains. The speed of the motor decreases. The braking torque makes the motor run at a constant speed. In hoists and cranes, the drive motor has a tendency to run faster than the synchronous speed. This situation occurs when the hoist is raising an empty cage or lowering a loaded cage.

When fed from a source of fixed frequency, regenerative braking is possible only for speeds greater than synchronous speed. With a variable frequency source, it can also be obtained for speeds below synchronous speed.

DYNAMIC BRAKING

In DC dynamic braking, the stator running at a speed is connected to a dc supply. The dc flowing through the stator sets up a stationary field. This induces rotor currents which produce a torque to bring the rotor to rest quickly. The torque developed and the retardation during braking can be controlled by the amount of dc power. Additional resistances are used in stator and rotor circuit to control the dc excitation and braking torques respectively. The dc supply can be either separate dc source or a rectified dc supply derived from the ac source through a diode rectifier.



In AC dynamic braking, the stator is switched to a capacitor bank. The machine runs as a self-excited induction generator. All the mechanical energy is dissipated as electrical energy in the rotor resistance. This method is uneconomical due to the cost of the capacitors.

PLUGGING

By changing the phase sequence of the input to an induction motor, the direction of the stator field can be reversed. In practice, this is done by interchanging the supply to any two terminals of the motor. A braking torque is developed and the motor comes to rest very fast. The motor must be switched off from the mains when zero speed is reached. Otherwise, the torque developed accelerates the motor in the reverse direction.

$$\text{Slip during plugging, } s_n = \frac{-N_s - N}{-N_s} = \frac{2N_s - (N_s - N)}{N_s} = 2 - s$$

At the instant of plugging, s_n is nearly 2 and the relative speed of stator field with respect to rotor is twice the relative speed at standstill. The rotor induced emf can be twice of its value at zero speed. Consequently, the motor current is large. In case of wound-rotor motors, a resistance of twice the starter resistance is inserted in the rotor to limit the braking current to starting value. This increases the braking torque also.

EFFECTS OF HARMONICS

a) **Time harmonics of the supply voltage:** If voltage applied to the induction motor contains harmonics, air gap flux may have harmonic contents, but of little importance since the torque developed by such harmonics are usually very small.

b) **Space harmonics of the air-gap flux:** Air gap flux contains harmonics developed by the windings, slotting, magnetic saturation, air gap length irregularity etc. These fields induce emfs and circulate harmonic currents in the rotor winding. These harmonic currents in the rotor interact with the harmonic fields and develops harmonic torques ($T \propto \Phi I_2$).

Effects of space harmonics are

- Crawling (motor may exhibit a tendency to run stably at low speed eg:- $1/7^{\text{th}}$ normal speed)
- Cogging (motor may refuse to start at all)
- Excessive vibration may be set up at speeds below normal speed
- Noise is produced

In a 3-phase machine, when sinusoidally varying currents (i.e. no time harmonics) flow through the windings, the space harmonic waves rotate at $1/n$ times the speed of the fundamental wave. The space harmonic waves rotate in the same direction as the fundamental wave if $n=6m+1$ (eg:- 7^{th} and 13^{th}) and in the opposite direction if $n=6m-1$ (eg:- 5^{th} and 11^{th}).

Amplitude of n^{th} harmonic = $1/n^2$ * amplitude of fundamental

No. of poles of n^{th} harmonic = n * no. of poles of fundamental

Synchronous speed of n^{th} harmonic = $1/n$ * synchronous speed of fundamental

Synchronous speeds for space harmonics of a 3-phase, 4-pole, 50Hz induction motor					
Space Harmonic	1	5	7	11	13
Synchronous speed (rpm)	1500	-300	214.3	-136.4	115.4

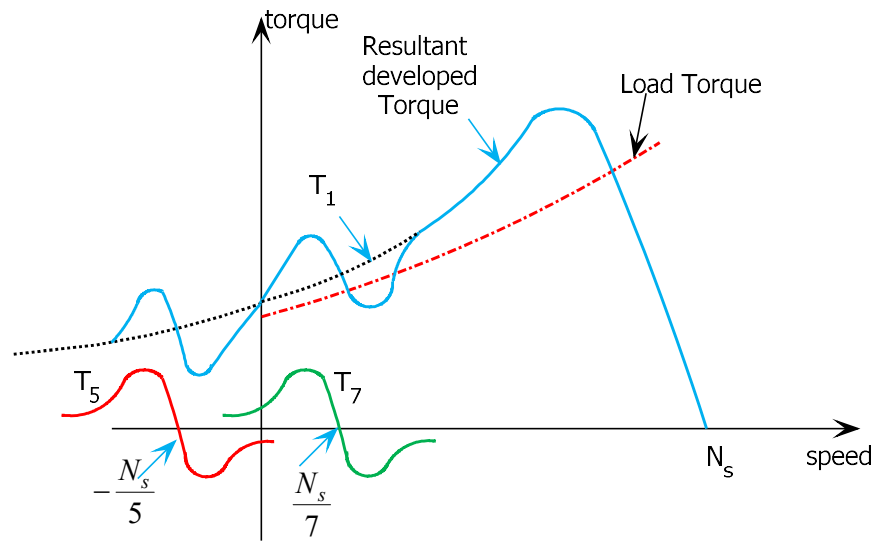
CRAWLING

Consider a 3 phase winding carrying sinusoidal currents produces harmonics of the order $n=6m\pm1$ where m is an integer.

When $m=1$, 3-phase winding produces a forward rotating 7^{th} harmonic and a backward rotating 5^{th} harmonic. These harmonic fluxes generate rotor emfs, currents and torques of the same T-s shape as that of the fundamental but with synchronous speed $1/5^{\text{th}}$ (backward) and $1/7^{\text{th}}$ (forward) of the synchronous speed of fundamental.

The 7^{th} harmonic torque reaches its maximum just before $1/7^{\text{th}}$ synchronous speed, but beyond this speed, the 7^{th} harmonic torque becomes negative, since slip in the harmonic field is negative. The resultant T-s curve combined with the fundamental shows dip. With certain slot ratio, the dip due to

7th harmonic may become very pronounced. If the torque developed by the motor falls below the load torque, the motor can not accelerate up to its full speed but continues to run at a speed a little lower than $1/7^{\text{th}}$ synchronous speed. This is called crawling.



COGGING

A squirrel-cage rotor may exhibit a peculiar behavior in starting for certain relationships between the number of poles and the stator and rotor slots.

Let No. of stator slots S_s = No. of rotor slots S_r = 24. Stator slotting produces forward rotating 13th harmonic and reverse rotating 11th harmonic. Similarly, rotor slotting also produces forward rotating 13th and reverse rotating 11th harmonics. Each pair of harmonics of same order together produce a synchronizing torque (similar to that in a synchronous motor) so that the rotor will remain at rest unless the fundamental torque is large enough to move the machine from rest which is unlikely. A similar condition will occur when S_r is an integral multiple of S_s . The phenomenon by which the machine refuse to start is called cogging.

Effects of harmonics can be reduced by

- Chording
- using fractional slot windings
- skewing (making non-parallel) either stator or rotor slots
- increasing the air gap length
- by suitable selection of the number of rotor and stator slots

SPEED CONTROL OF 3-PHASE INDUCTION MOTOR

The speed of an induction motor is given by

$$N = (1-s)N_s \text{ and } N_s = \frac{120f}{P}$$

Basic methods of speed control of induction motor are

- By changing the number of poles
- By varying the supply frequency
- By varying the stator voltage
- By varying the rotor resistance
- By injecting emf of slip frequency into the rotor circuit

Method (i) is applicable only for squirrel cage induction motors and (iv) and (v) only for slip ring induction motors.

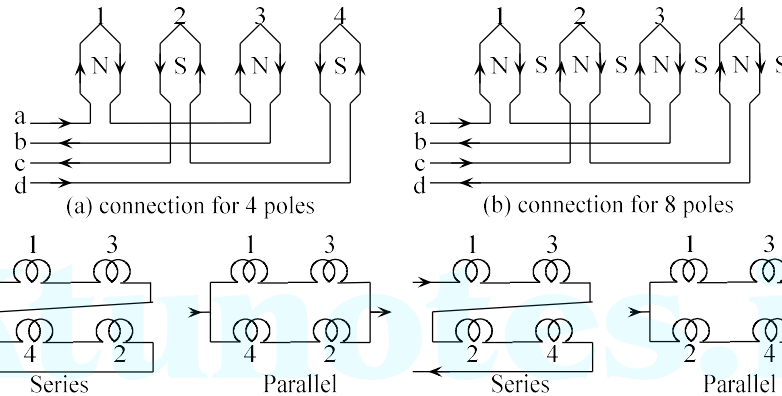
POLE CHANGING

For a given frequency, the synchronous speed is inversely proportional to the number of poles. Synchronous speed and hence the motor speed can be changed by changing the number of poles.

Since there is no winding in rotor circuit of a squirrel cage induction motor, the rotor can adjust to any number of stator poles. In slip ring induction motor, since the rotor is wound for same number of stator poles, change in number of stator poles requires change in number of rotor poles which complicates the machine. Hence, this method of speed control is only used for squirrel cage induction motors.

If an induction motor is to run at different speeds, one way is to have different stator windings for the motor so that it will have different synchronous speeds and the running speeds. But this method is expensive.

Another method is to use one winding but with suitable connections for a change-over to double/halve the number of poles. The stator winding per phase is divided into two coil groups. By reversing the current in one coil group, change in number of poles by a factor 2 can be achieved. The two coil groups can be connected in series or parallel.



The three phases of the machine can be connected in star or delta. By choosing a suitable combination of series or parallel connections between coil groups of each phase and star or delta connection between the phases, speed change can be obtained with constant torque operation, constant power operation or variable torque operation.

STATOR VOLTAGE CONTROL

By reducing stator voltage, speed of a high-slip induction motor can be reduced by an amount which is sufficient for the speed control of fan and pump drives (Load torque, $T_L = N^2$).

$$\text{Torque developed (Motor torque), } T = \frac{sE_2^2 R_2}{R_2^2 + (sX_2)^2} \text{ sync. watts}$$

But, E_2 = rotor induced emf at standstill

$$E_2 \propto \phi \propto V$$

Hence,

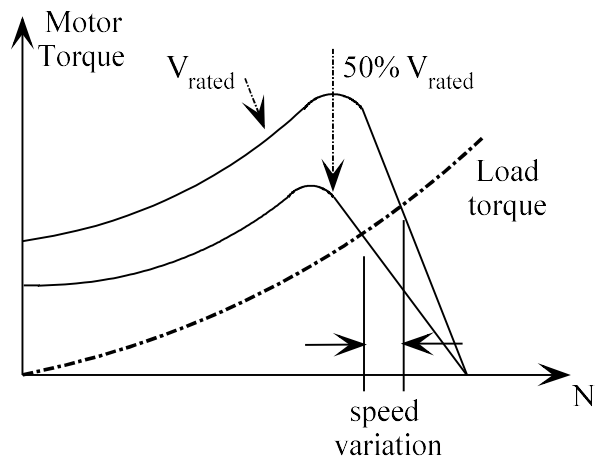
$$T \propto \frac{sV^2 R_2}{R_2^2 + (sX_2)^2}$$

Since, during running condition, $s^2 X_2^2 \ll R_2^2$

$$T \propto \frac{sV^2 R_2}{R_2^2} \propto \frac{sV^2}{R_2}$$

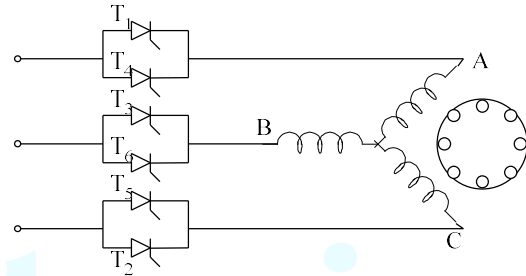
$$T \propto sV^2$$

Here, the speed variation is very much limited.



But, when V is reduced, current drawn by the motor increases and motor may get overheated.

The variation in applied voltage is achieved by means of a 3-phase AC voltage controller. The voltage is varied by means of phase angle control of the anti-parallel thyristors.



ROTOR RESISTANCE CONTROL (only for slip ring induction motor)

With the addition of external resistance in the rotor circuit, maximum torque remains the same but the slip or speed at which the maximum torque occurs varies. For the same torque, the speed falls with an increase in rotor resistance. Because of low cost and high torque capability at low speeds, rotor resistance control is used in cranes. Main disadvantage is the low efficiency due to additional Cu loss in external rotor resistance.

$$\text{Torque developed, } T = \frac{sE_2^2 R_2}{R_2^2 + (sX_2)^2} \text{ sync. watts}$$

$$T \propto \frac{sV^2 R_2}{R_2^2 + (sX_2)^2}$$

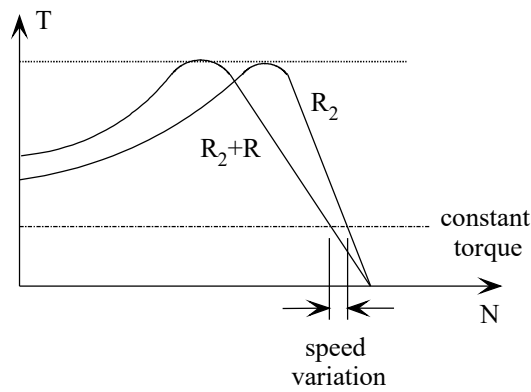
Since, during running condition, $s^2 X_2^2 \ll R_2^2$

$$T \propto \frac{sV^2 R_2}{R_2^2} \propto \frac{sV^2}{R_2}$$

If supply voltage is kept constant,

$$T \propto \frac{s}{R_2}$$

For a given torque, addition of rotor resistance increases the slip and hence speed is decreased.



Starting torque increases proportional to rotor resistance.

Qn 1:- A 4-pole 50 Hz 3-phase induction motor has rotor resistance of 0.2Ω /phase and rotor standstill reactance of 1Ω /phase. On full load it is running with a slip of 4%. Calculate the extra resistance required in the rotor circuit per phase to reduce the speed to 1260 rpm, on the same load condition.

Ans:-

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{4} = 1500 \text{ rpm}$$

$$s_2 = \frac{N_s - N_2}{N_s} = \frac{1500 - 1260}{1500} = 0.16$$

$$T = \frac{sE_2^2 R_2}{R_2^2 + (sX_2)^2}$$

For $s_1 = 0.04$,

$$T_1 = \frac{0.04 \times E_2^2 \times 0.2}{0.2^2 + (0.04 \times 1)^2} = 0.1923 E_2^2$$

For $s_2 = 0.16$,

$$T_2 = \frac{0.16 \times E_2^2 \times R}{R^2 + (0.16 \times 1)^2} = \frac{0.16 \times R}{R^2 + 0.0256} E_2^2$$

For same load condition, $T_1 = T_2$.

$$\frac{0.16 \times R}{R^2 + 0.0256} = 0.1923$$

$$0.16 \times R = 0.1923 \times (R^2 + 0.0256) = 0.1923 R^2 + 0.004923$$

$$0.1923 R^2 - 0.16 R + 0.004923 = 0$$

$$R = 0.8\Omega \text{ or } 0.032\Omega$$

$$R = 0.8\Omega$$

External rotor resistance = $0.8 - 0.2 = 0.6\Omega$ /phase.

Qn 2:- A 3-phase 6-pole 50Hz induction motor when fully loaded with a slip of 3%. Find the value of the resistance necessary in series per phase of the rotor to reduce the speed by 10%. Assume rotor resistance/phase is 0.2Ω .

Ans:-

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{6} = 1000 \text{ rpm}$$

$$N_1 = (1 - s)N_s = (1 - 0.03) \times 1000 = 970 \text{ rpm}$$

$$N_2 = 0.9 \times 970 = 873 \text{ rpm}$$

$$s_2 = \frac{N_s - N_2}{N_s} = \frac{1000 - 873}{1000} = 0.127$$

$$T = \frac{sE_2^2 R_2}{R_2^2 + (sX_2)^2} \approx \frac{sE_2^2}{R_2} \propto \frac{s}{R_2}$$

$$T_1 \propto \frac{0.03}{0.2} = 0.15$$

$$T_2 \propto \frac{0.127}{(0.2 + R)}$$

But, $T_1 = T_2$.

$$\frac{0.127}{(0.2 + R)} = 0.15$$

$$0.127 = (0.2 + R) \times 0.15$$

$$R = 0.647 \Omega$$

STATOR FREQUENCY CONTROL

$$\text{Synchronous speed, } N_s = \frac{120f}{P}$$

By varying frequency the supply frequency, N_s (hence N) can be controlled over a wide range.

The torque and speed of induction motor can be controlled by changing the supply frequency. If the voltage is maintained fixed at its rated value while reducing the frequency below its rated value, the flux increases. This will cause saturation of the air-gap flux, and the motor parameters would not be valid in determining the torque speed characteristics. Magnetizing current also increases. At low frequency, the reactances decrease and the motor current may be too high. Core loss also increases while reducing frequency keeping voltage constant.

The synchronous speed corresponding to the rated frequency is called base speed. For speed control below base speed, while frequency is reduced the voltage is also reduced in proportion so that V/f ratio is constant. In constant V/f control, flux remains constant at its rated value. With reduction in frequency, magnetizing current remains constant while core loss decreases. Maximum torque remains constant.

(Note:-For speed control below base speed, frequency is decreased keeping V/f ratio constant.

If frequency f is reduced keeping supply voltage V constant,

$$E_1 = 4K_f K_c K_d f \phi T$$

$$E_1 \propto f \phi$$

$$V \propto f \phi$$

$$\frac{V}{f} \propto \phi$$

$$\frac{V}{f} \propto B_m$$

When f is reduced keeping V constant, air gap flux ϕ is increased and it may cause saturation of stator and rotor cores resulting in very large increase in magnetizing current.

$$\text{Also, hysteresis loss } W_h \propto f B_m^{1.6} \propto f \left(\frac{V}{f}\right)^{1.6} = \frac{V^{1.6}}{f^{0.6}}$$

$$W_h \propto \frac{V^{1.6}}{f^{0.6}} \propto \frac{1}{f^{0.6}} \text{ (if } f \text{ is reduced by 50\%, } W_h \text{ is increased by 52\%)}$$

$$\text{Eddy current loss } W_e \propto f^2 B_m^2 \propto f^2 \frac{V^2}{f^2} = V^2$$

$$W_e \propto V^2$$

Hence, if supply frequency f is decreased keeping supply voltage constant, hysteresis loss increases but eddy current loss remains constant. Since core loss increases, it causes overheating and decreased efficiency.

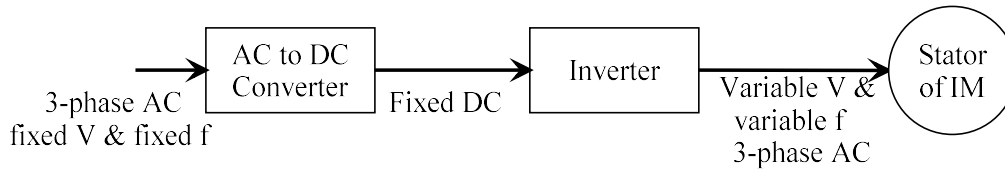
For speed control above base speed, frequency f is increased keeping supply voltage V constant at its rated value.

Flux decreases, magnetizing current decreases, core loss decreases, stator current decreases since reactance increases, copper loss decreases.

But, $\phi \propto \frac{V}{f}$

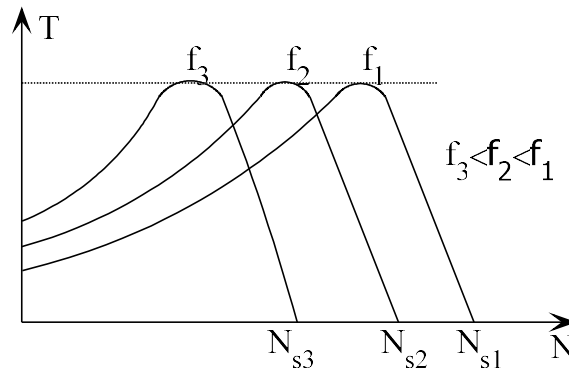
If f is increased, flux will decrease thereby decreasing the torque developed ($T \propto \phi$).

Frequency can be varied by using a static inverter set (figure given below) or by using motor alternator set.

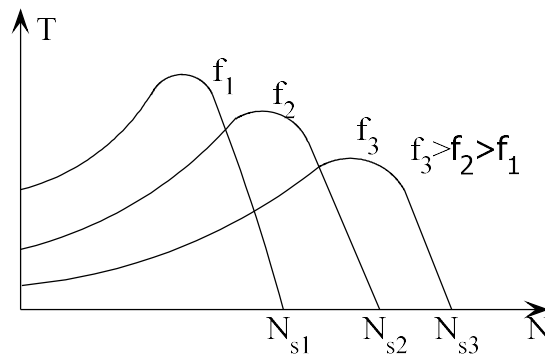


Normal supply available is constant voltage, constant frequency 3-phase ac supply. The converter converts this fixed ac into fixed dc supply. Inverter converts this dc supply into variable voltage variable frequency ac supply keeping V/f ratio constant. By selecting the proper frequency and maintaining V/f constant, smooth speed control of the induction motor is possible.

Torque slip characteristics for different frequencies for speed control below base speed (V/f is kept constant) are given below. Note that the maximum torque remains constant.



Torque slip characteristics for different frequencies for speed control above base speed (V is kept constant at rated value) are given below. Note that the maximum torque decreases as frequency increases.



Applications of induction motors

a) Squirrel-cage motor

- i) General purpose with normal torque and normal starting current

Applications: Fans, blowers, centrifugal pumps, line shafting etc.

- ii) High torque, low starting current

Applications: Conveyors, compressors, crushers, agitators, reciprocating pumps etc.

- iii) High torque, medium and high slip

Applications: Used for high inertia loads, such as sheets, punch presses die stamping etc.

- iv) Low starting torque, normal starting current

Applications: For loads of low inertia and low starting torque requirement, such as fans and centrifugal pumps

b) Wound-rotor motor

These motors are used in situation where high starting torque is required and where the starting current should be low such as in fans, pumps, conveyors, hoists, compressors etc.

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Module 5

Induction generator – Principle of operation, Grid Connected and Self Excited Operation of Induction Generators, Torque-Slip Characteristics of an Induction machine.

Single phase Induction motors - Double field revolving theory, Equivalent Circuit, Torque-Slip Characteristics, Types of Single-phase induction motor, Applications.

Selection of AC motors for different applications.

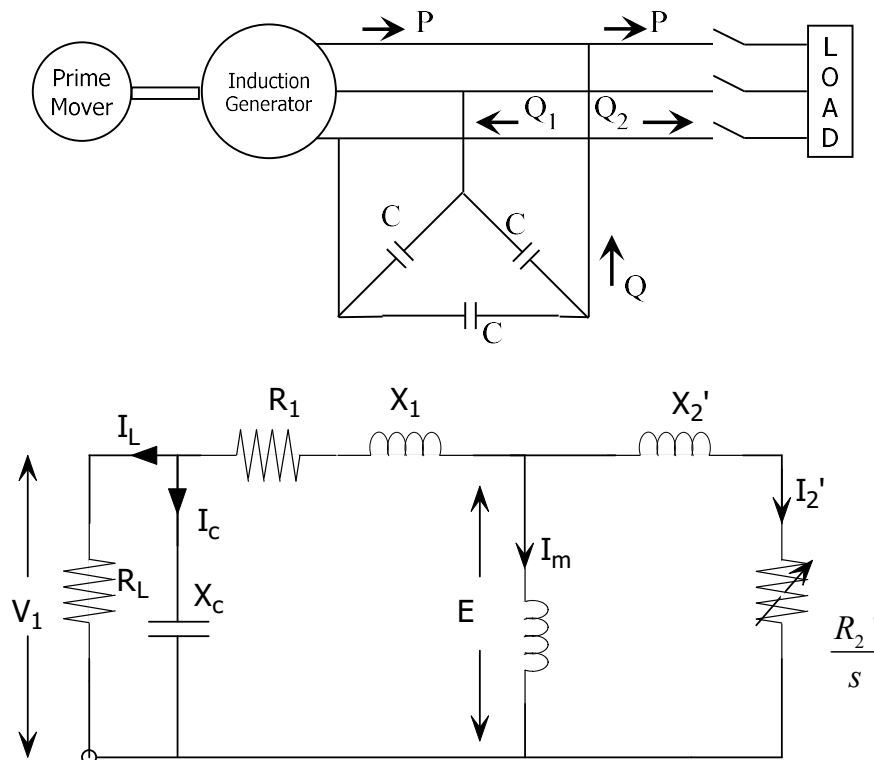
INDUCTION GENERATOR

An induction generator is asynchronous in nature because of which it is commonly used as windmill generator since a windmill runs at non-fixed speed. These are used in remote areas to supplement power received from weak transmission links.

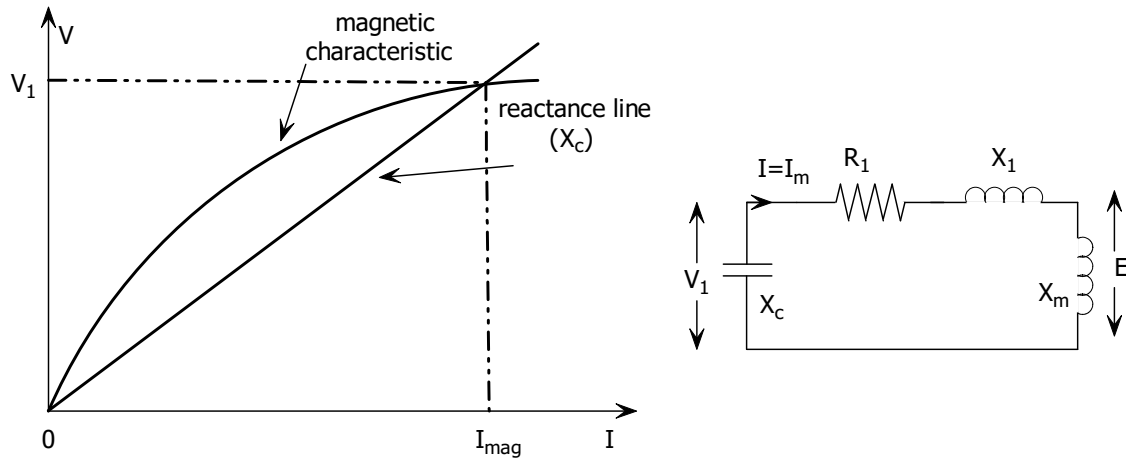
If the induction machine is driven at a speed greater than synchronous speed by a prime mover, the direction of induced torque reverses and it acts as an induction generator. The rotating magnetic field is set up by the magnetizing current drawn from the mains. Based on the way with which the generator gets the required lagging reactive-power, they are classified into i) Line excited induction generator and ii) self-excited induction generator.

Line excited (grid connected) induction generator : Induction machine connected to supply mains and driven at super-synchronous speed by its prime mover is called a line excited induction generator. The generator draws the required lagging reactive power from the mains.

Self excited (stand-alone) induction generator : In this generators, the necessary lagging reactive power for its excitation is obtained by a capacitor bank connected across the generator terminals. If the rotor of the machine is driven by its prime-mover, the presence of residual flux (present in the rotor core) causes a small emf to get induced in the stator windings at a frequency proportional to the rotor speed. This voltage appears across the 3-phase capacitor and gives rise to leading current drawn by the capacitor which is equivalent to lagging current supplied to the generator. The flux set up by this current assists the initial residual flux. Hence the net flux will increase causing the voltage to build up further. The steady state voltage induced on no load is given by the intersection of magnetization characteristic of the machine and capacitance V-A characteristic.

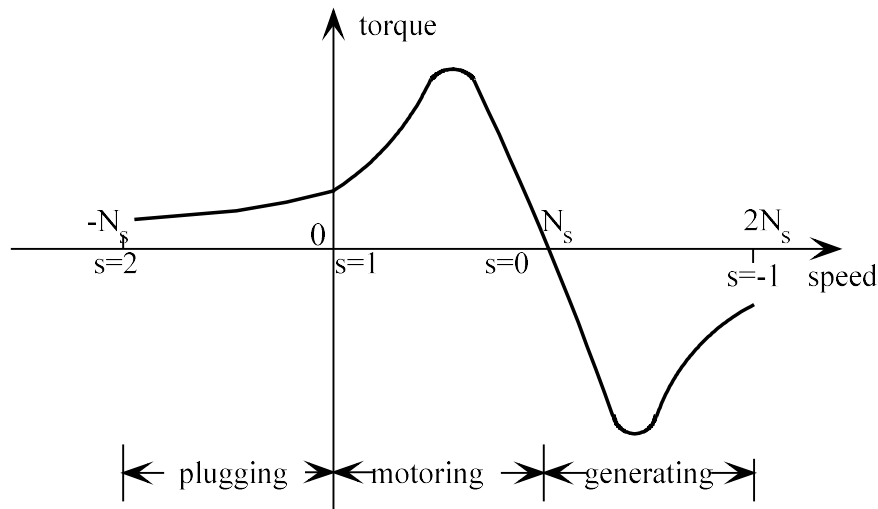


Consider that the generator is run at synchronous speed at no-load (switch off) with delta-connected capacitors connected across its terminals. The rotor circuit behaves as open and so the circuit model is given below. The current fed into the motor is $I_1 = I_m$, the magnetizing current. If the voltage drop in stator impedance is ignored, $V_1 = E$ (induced emf) $= X_m I_m = X_c I_m$. The intersection of the magnetization and reactance line gives the no-load voltage. This is the case at rated frequency. V_1 will vary slightly as the actual stator frequency changes from the rated value. This variation may be ignored. Voltage build up depends upon the value of capacitor. Higher the value of the capacitance, greater is the voltage build up.



The voltage and frequency of line excited induction generators cannot be controlled. Only the active power can be controlled through the speed control of prime-mover driving the 3-phase induction generator. Advantages of line-excited induction generators are i) these generators do not hunt ii) short circuit current of the generators is limited. The disadvantages are i) they require considerable amount of reactive power from the existing supply system ii) efficiency is relatively poor and iii) it can work at leading power factor only.

Torque-slip characteristics of induction machine



PLUGGING

This mode of operation is sometimes utilized in drive applications where the drive system is required to stop very quickly. Suppose an induction motor is running at a steady-state speed. If its terminal phase sequence is changed suddenly, the stator rotating field will rotate opposite to the rotation of the rotor, producing the plugging operation. The motor will come to zero speed rapidly and will accelerate in the opposite direction, unless the supply is disconnected at zero speed.

COMPARISON OF INDUCTION GENERATOR & SYNCHRONOUS GENERATOR

In a synchronous generator, the waveform of generated voltage is synchronized with (directly corresponds to) the rotor speed. The frequency of output can be given as $f = \frac{PN}{120}$ Hz where N is speed of the rotor in rpm and P is number of poles. In case of induction generators, the output voltage frequency is regulated by the power system to which the induction generator is connected. If induction generator is supplying a standalone load, the output frequency will be slightly lower (by 2 or 3%) than calculated from the formula $f = \frac{PN}{120}$.

Separate DC excitation system is required in an alternator (synchronous generator). Induction generator takes reactive power from the power system for field excitation. If an induction generator is meant to supply a standalone load, a capacitor bank needs to be connected to supply reactive power.

Construction of induction generator is less complicated as it does not require brushes and slip ring arrangement. Brushes are required in synchronous generator to supply DC voltage to the rotor for excitation.

SINGLE PHASE MOTORS

Single-phase motors are small motors, mostly built in the fractional horse-power range.

Single-phase motors are relatively simple in construction. However, they are not always easy to analyze.

Single-phase motors are of 3 types:

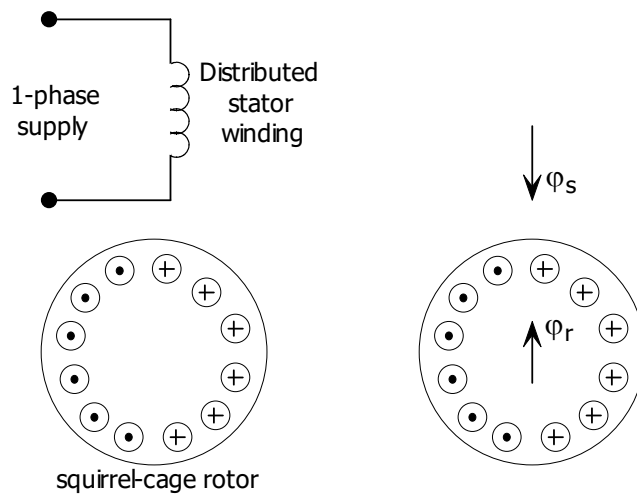
1. Single-phase induction motors: - Majority of fractional horse-power motors are of the induction type. They are classified according to the methods used to start them. Common types are resistance-start (split-phase), capacitor-start, capacitor-run, and shaded pole.
2. Single-phase synchronous motors :- Motors of synchronous type run at constant speed. Two types of single-phase synchronous motors are in common use: the reluctance type and the hysteresis type.
3. Single-phase series (or universal) motor:- Motors of the series type can be used with either a dc supply or a single-phase ac supply. These motors provide high starting torque and can operate at high speed.

SINGLE-PHASE INDUCTION MOTORS

Motors of the induction type have cage rotors and a single-phase distributed stator winding. Such a motor inherently does not develop any starting torque and therefore will not start to rotate if the stator winding is connected to an ac supply. However, if the rotor is given a spin or started by auxiliary means, it will continue to run.

Single phase induction motors are not self starting. Why?

Consider that the rotor is stationary and the stator winding is connected to single-phase supply. A pulsating mmf and hence a pulsating flux ϕ_s is established in the machine along the axis of the stator winding. This pulsating stator flux induces current by transformer action in the rotor circuit, which in turn produces a pulsating flux ϕ_r acting along the same axis as the stator flux ϕ_s . By Lenz's law, these two fluxes tend to oppose each other. As the angle between these fluxes is zero, no starting torque is produced.



Now, assume that the rotor is running. This can be done either by spinning the rotor or by using auxiliary circuits. The single-phase induction motor can develop torque when it is in the running condition, which can be explained by using double revolving field theory.

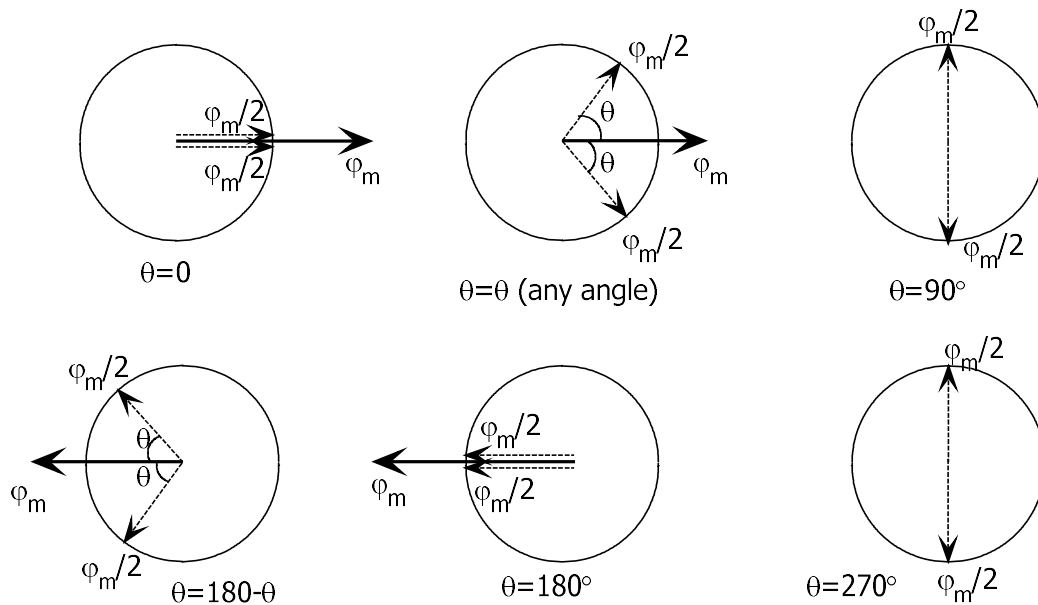
DOUBLE REVOLVING FIELD THEORY

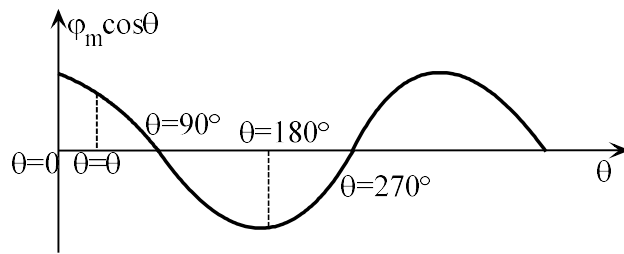
A pulsating flux can be resolved into two rotating fluxes of half the magnitude but rotating at the same synchronous speed in opposite directions.

This can be proved either graphically.

Let ϕ_f ($=\phi_m/2$) be forward component rotating in anticlockwise direction while ϕ_b ($=\phi_m/2$) be backward component rotating in clockwise direction. At start, both the components are in same direction. The resultant flux is the algebraic sum of the magnitudes of two components ($\Phi_R = \phi_m$).

After 90° , the two components are opposite to each other and the resultant flux is zero. Hence continuous rotation of the two components gives the original alternating stator flux.





Both of these rotating mmf's produce induction motor torque, although in opposite directions. At standstill, these two torques, forward and backward, are equal in magnitude and therefore, the resultant starting torque is zero. At any other speed, the two torques are unequal and the resultant torque keeps the motor rotating in the direction of rotation.

SLIP

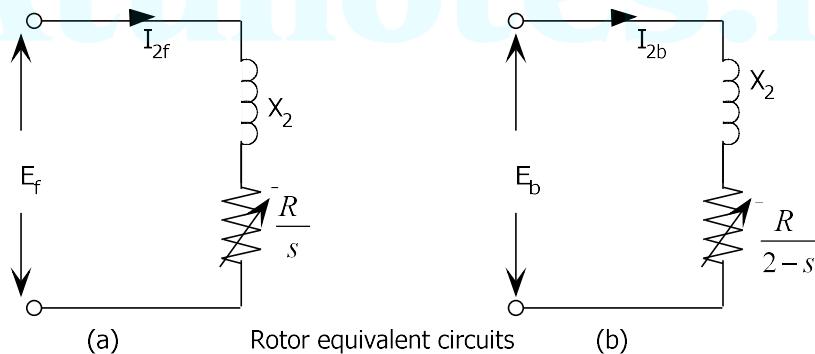
Assume that the rotor is rotating in the direction of the forward rotating field at a speed N rpm and the synchronous speed is N_s rpm.

The slip with respect to the forward field is $s_f = \frac{N_s - N}{N_s} = s$

The rotor rotates opposite to the rotation of the backward field. Therefore, the slip with respect to the backward field is

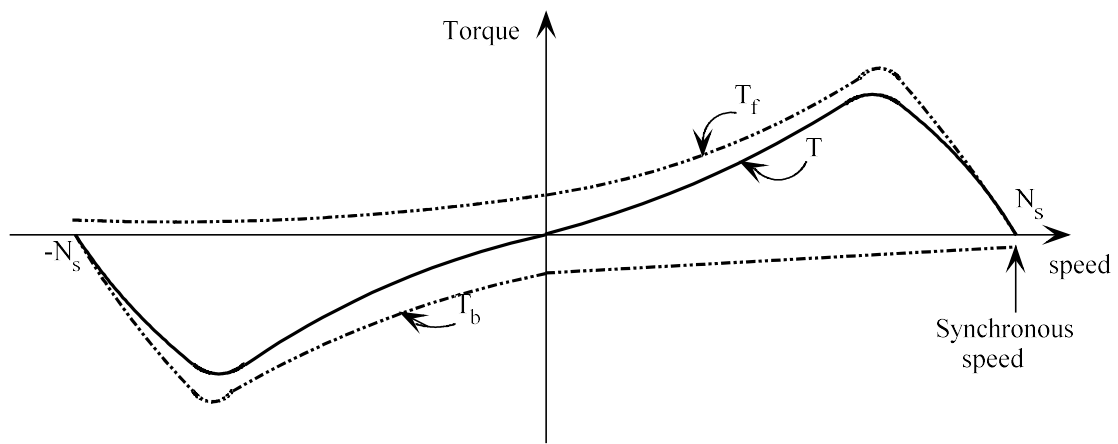
$$s_b = \frac{-N_s - N}{-N_s} = \frac{2N_s - (N_s - N)}{N_s} = 2 - s$$

The rotor circuits for the forward and backward rotating fluxes are shown below.



At standstill, the impedances are equal and so are the currents ($I_{2f} = I_{2b}$). The rotating forward and backward fluxes in the air gap are equal in magnitude.

However, when the rotor rotates, the impedances of the rotor circuits are unequal and the rotor current I_{2b} is higher than the rotor current I_{2f} . Their mmf's which oppose the stator mmf's, will result in a reduction of the backward-rotating flux. Consequently, as the speed increases, the forward flux remains essentially constant to induce voltage in the stator winding, which is almost the same as the applied voltage (if the voltage drops across the stator winding resistance and the leakage reactance are neglected). Hence, with the rotor in motion, the forward torque increases and the backward torque decreases.



EQUIVALENT CIRCUIT

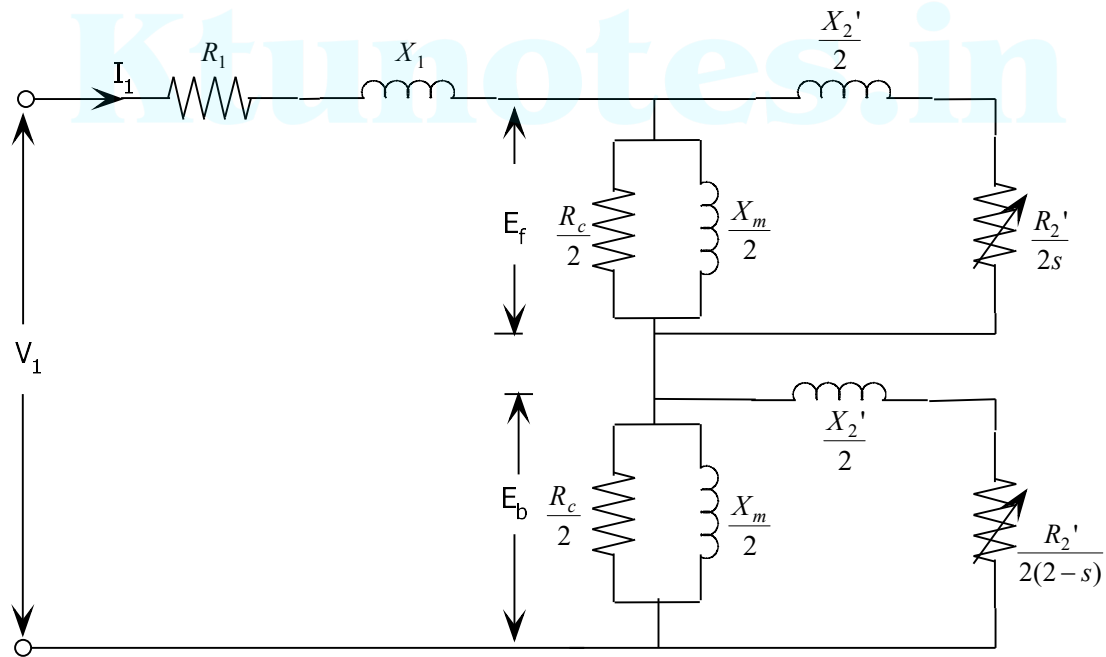
According to the double revolving field theory, the equivalent circuit can be split into two halves, representing the effects of forward and backward fields.

$$E_f = 4.44 f T \phi_f$$

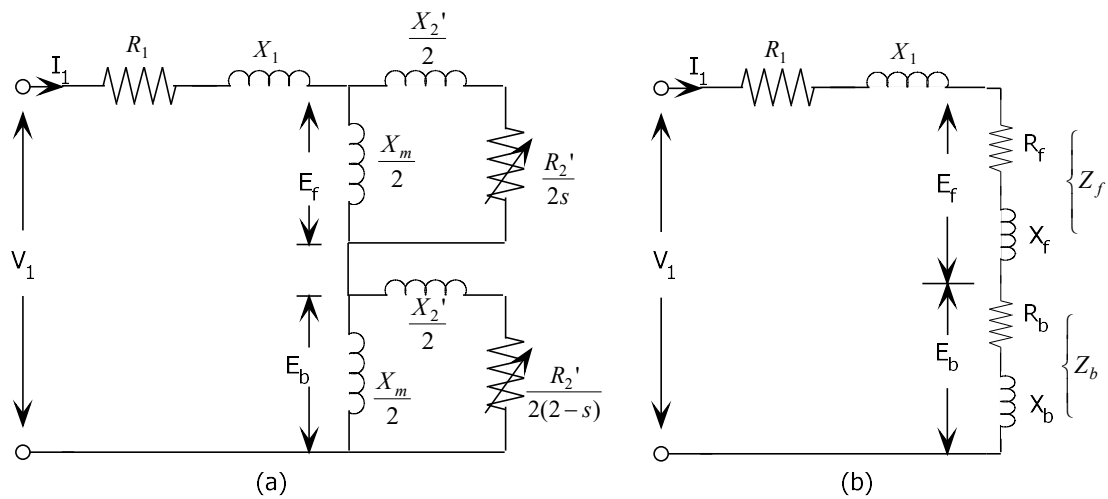
$$E_b = 4.44 f T \phi_b$$

At standstill, as $\phi_f = \phi_b$ (these being the revolving air gap fluxes), $E_f = E_b$.

Exact equivalent circuit is shown below.



Approximate equivalent of 1-phase induction motor is shown below.



The air gap powers due to the forward field and backward fields are

$$P_{2f} = I_1^2 R_f$$

$$P_{2b} = I_1^2 R_b$$

The corresponding torques are

$$T_f = \frac{P_{2f}}{\left(\frac{2\pi N_s}{60}\right)}$$

$$T_b = \frac{P_{2b}}{\left(\frac{2\pi N_s}{60}\right)}$$

The resultant torque is

$$T = T_f - T_b = \frac{I_1^2}{\left(\frac{2\pi N_s}{60}\right)} (R_f - R_b)$$

The mechanical power developed is

$$P_{mech} = \frac{2\pi N}{60} \times T = \frac{2\pi N_s(1-s)}{60} \times T = I_1^2 (R_f - R_b)(1-s)$$

The power output is

$$P_{out} = P_{mech} - \text{Rotational loss}$$

where rotational loss = Core loss + F&W loss

Rotor copper loss produced by the forward field = sP_{2f}

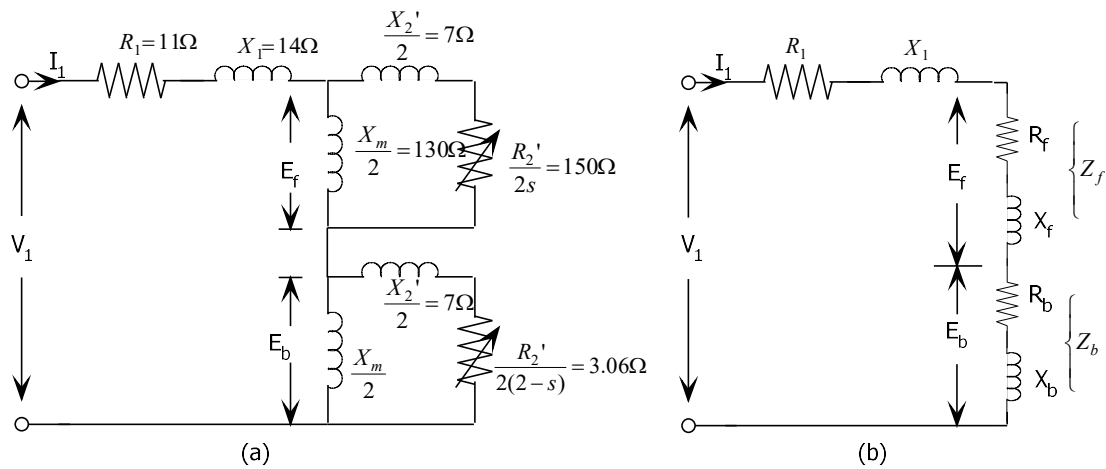
Rotor copper loss produced by the backward field = $(2-s)P_{2b}$

Total rotor copper loss = $sP_{2f} + (2-s)P_{2b}$

Pr. A 230V, 50Hz, 4-pole, single-phase induction motor runs at 1440rpm. Its equivalent circuit parameters are : $R_1=11\Omega$, $X_1=14\Omega$, $R_2'=12\Omega$, $X_2'=14\Omega$, $X_m=260\Omega$, core loss = 23W, and mechanical losses = 13W. Determine the torque developed, mechanical power developed and efficiency.

$$Z_f = \frac{j130 \times (150 + j7)}{j130 + (150 + j7)} = 61.43 + j73.9\Omega$$

$$Z_b = \frac{j130 \times (3.06 + j7)}{j130 + (3.06 + j7)} = 2.75 + j6.7\Omega$$



$$I_1 = \frac{V \angle 0^\circ}{R_1 + R_f + R_b + j(X_1 + X_f + X_b)} = \frac{230 \angle 0^\circ}{75.18 + j94.6} = 1.9 \angle -51.5^\circ A$$

$$P_{2f} = I_1^2 R_f = 1.9^2 \times 61.43 = 221.76 W$$

$$P_{2b} = I_1^2 R_b = 1.9^2 \times 2.75 = 9.9 W$$

$$T_f = \frac{P_{2f}}{\left(\frac{2\pi N_s}{60}\right)} = \frac{221.76}{50\pi} = 1.4 Nm$$

$$T_b = \frac{P_{2b}}{\left(\frac{2\pi N_s}{60}\right)} = \frac{9.9}{50\pi} = 0.06 Nm$$

$$T = T_f - T_b = 1.4 - 0.06 = 1.34 Nm$$

$$P_{mech} = \frac{2\pi N}{60} \times T = \frac{2\pi \times 1440}{60} \times 1.34 = 202 W$$

$$\text{Output Power, } P_{out} = P_{mech} - \text{Rotational loss} = 202 - (23 + 13) = 166 W$$

$$\text{Motor input, } P_{in} = V_1 I_1 \cos \phi_1 = 230 \times 1.9 \times \cos 51.5^\circ = 272 W$$

$$\text{Efficiency} = 61\%$$

STARTING OF SINGLE-PHASE INDUCTION MOTORS

A single-phase induction motor with one stator winding inherently does not produce any starting torque. In order to make the motor start rotating, some arrangement is required so that the motor produces a starting torque. In the running condition, the motor will produce torque with only one stator winding.

The simplest method of starting a single-phase induction motor is to provide an auxiliary winding on the stator in addition to the main winding and start the motor as a two-phase machine. The two windings are placed in the stator with their axes displaced 90 electrical degrees in space. The impedances of the two circuits are such that the currents in the main and the auxiliary windings are phase-shifted from each other. The motor is equivalent to an unbalanced two-phase motor. However, the result is a rotating stator field that can produce the starting torque. The two windings can be properly designed to make the motor behave as a balanced two-phase motor.

In the running condition, a single-phase induction motor can develop a torque with only the main winding. Therefore, as the motor speeds up, the auxiliary winding can be taken out of the circuit. In most motors, this is done by connecting a centrifugal switch in the auxiliary circuit. At about 75% of

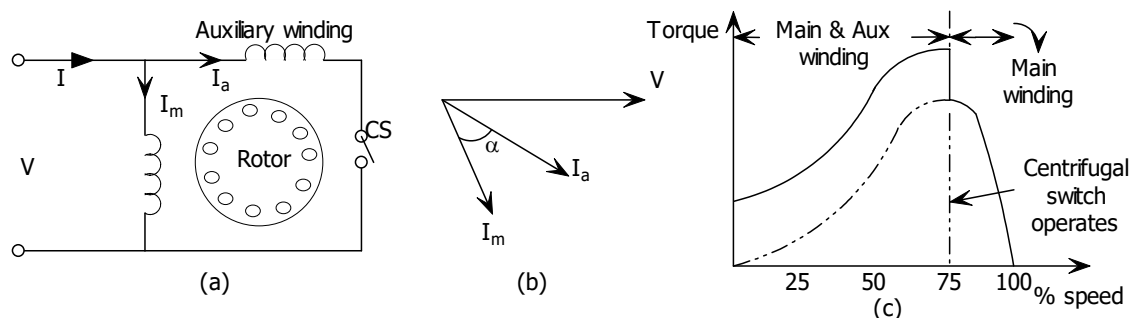
the synchronous speed, the centrifugal switch operates and disconnects the auxiliary winding from the supply.

TYPES OF SINGLE-PHASE INDUCTION MOTORS

Single phase induction motors are known by various names. The names are descriptive of the methods used to produce the phase difference between the currents in the main and auxiliary windings. Some of the commonly used types of single-phase induction motors are described here.

SPLIT-PHASE (Resistance-start) MOTOR

The auxiliary winding has a higher resistance to reactance ratio than the main winding, so the two currents are out of phase as shown in the phasor diagram.



The high resistance-to-reactance ratio is usually obtained by using finer wire for the auxiliary winding. This is permissible because the auxiliary winding is in the circuit only during the starting period. The centrifugal switch cuts it out at about 75% of the synchronous speed.

This motor has low to moderate starting torque, which depends on the two currents and the phase angle between them. The starting torque can be increased by inserting a series resistance in the auxiliary winding.

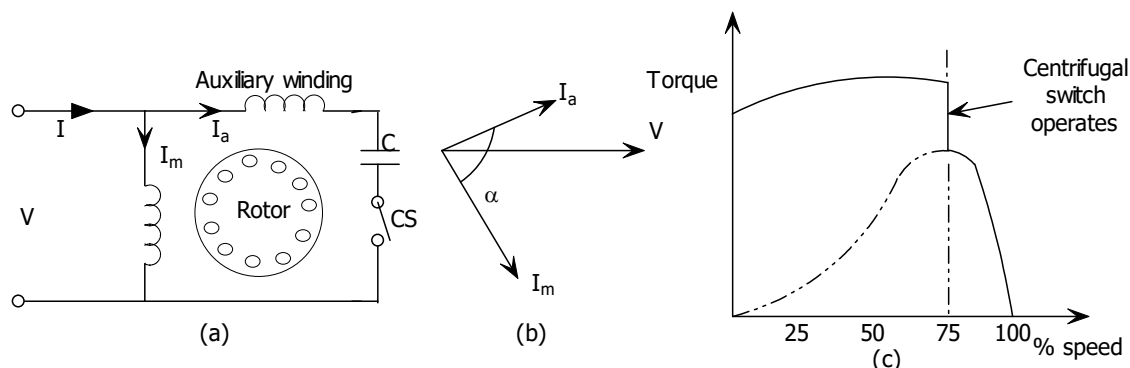
Applications : Fans, blowers, centrifugal pumps

Available range is 1/20 to 1/2 kW

CAPACITOR-START MOTORS

Higher starting torque can be obtained if a capacitor is connected in series with the auxiliary winding. This increases the phase angle between the winding currents as shown in phasor diagram. Because the capacitor is in the circuit only during the starting period, it can be an inexpensive ac electrolytic type. High starting torque is the outstanding feature of this arrangement.

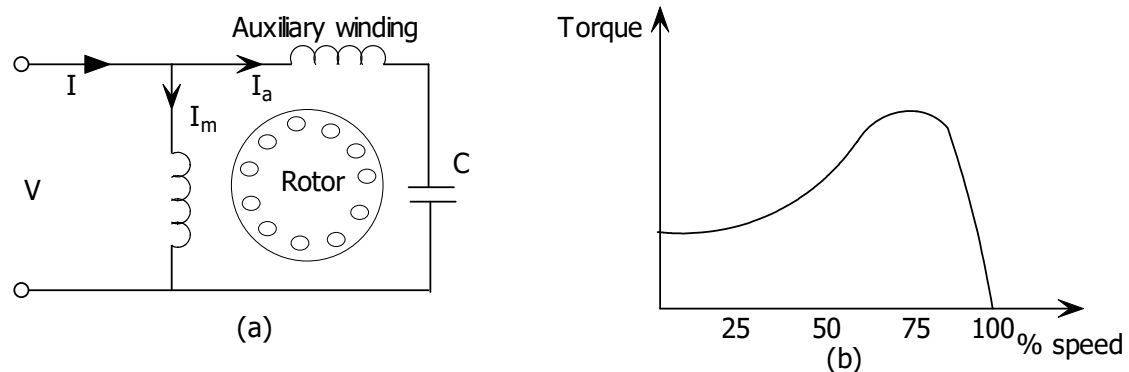
As in split phase motor, the starting winding will be disconnected from the supply when the speed reaches about 70% of its synchronous speed.



Applications : Domestic water pump, compressor, large fans, high inertia loads

CAPACITOR START & RUN MOTOR

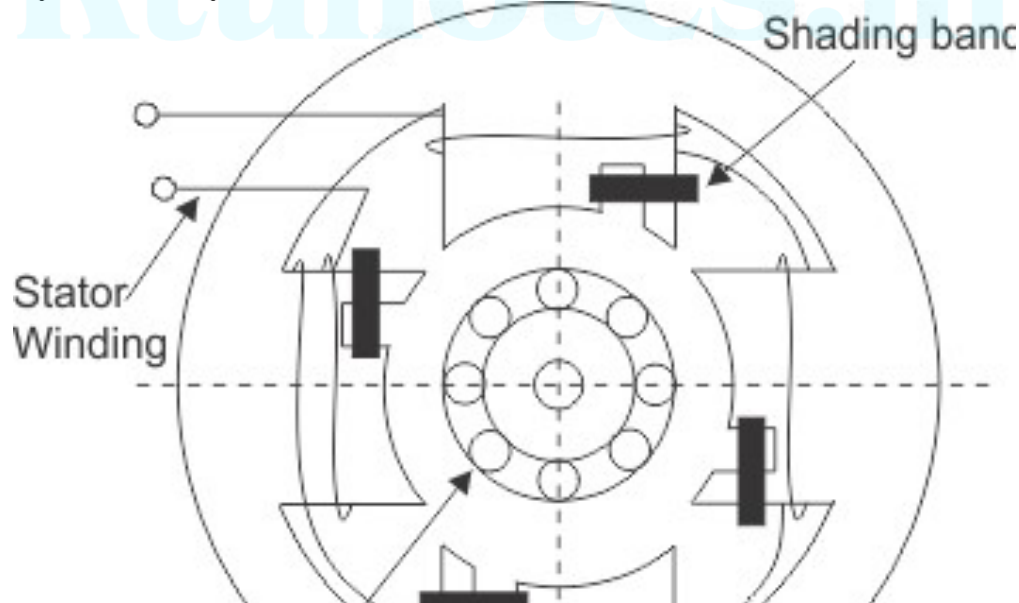
In this motor, the capacitor that is connected in series with the auxiliary winding is not cut out after starting and is left in the circuit all the time. This simplifies the construction and decreases the cost because the centrifugal switch is not needed. The power factor, torque pulsation and efficiency are also improved because of the motor runs as a two-phase motor. The motor will run more quietly. The capacitor value is of the order of $20\text{-}50\mu\text{F}$ and because it operates continuously, it is an ac paper oil type. The capacitor is a compromise between the best starting and running values and therefore starting torque is sacrificed.



Applications : Ceiling fans, washing machine, blowers

SHADED POLE INDUCTION MOTOR

The stator of the shaded pole single phase induction motor has salient or projected poles. These poles are shaded by copper band or ring which is inductive in nature. The poles are divided into two unequal halves. The smaller portion carries the copper band and is called as shaded portion of the pole.



When a single phase supply is given to the stator of shaded pole induction motor an alternating flux is produced. This change of flux induces emf in the shaded coil. Since this shaded portion is short circuited, the current is produced in it in such a direction to oppose the main flux. The flux in shaded pole lags behind the flux in the unshaded pole. The phase difference between these two fluxes produces resultant rotating flux.

Stator winding current is alternating in nature and so is the flux produced by the stator current. In order to clearly understand the working of shaded pole induction motor consider three regions-

- i) When the flux changes its value from zero to nearly maximum positive value.
- ii) When the flux remains almost constant at its maximum value.
- iii) When the flux decreases from maximum positive value to zero.

REGION 1: *When the flux changes its value from zero to nearly maximum positive value*

In this region the rate of rise of flux and hence current is very high. According to Faraday's law whenever there is change in flux, emf gets induced. Since the copper band is short circuit the current starts flowing in the copper band due to this induced emf. This current in copper band produces its own flux. Now according to Lenz's law the direction of this current in copper band is such that it opposes its own cause i.e rise in current. So the shaded ring flux opposes the main flux, which leads to the crowding of flux in non shaded part of stator and the flux weaken in shaded part. This non uniform distribution of flux causes magnetic axis to shift in the middle of the non shaded part.

REGION 2: *When the flux remains almost constant at its maximum value*

In this region the rate of rise of current and hence flux remains almost constant. Hence there is very little induced emf in the shaded portion. The flux produced by this induced emf has no effect on the main flux and hence distribution of flux remains uniform and the magnetic axis lies at the center of the pole.

REGION 3: *When the flux decreases from maximum positive value to zero*

In this region the rate of decrease in the flux and hence current is very high. According to Faraday's law whenever there is change in flux emf gets induced. Since the copper band is short circuit the current starts flowing in the copper band due to this induced emf. This current in copper band produces its own flux. Now according to Lenz's law the direction of the current in copper band is such that it opposes its own cause i.e decrease in current. So the shaded ring flux aids the main flux, which leads to the crowding of flux in shaded part of stator and the flux weaken in non shaded part. This non uniform distribution of flux causes magnetic axis to shift in the middle of the shaded part of the pole. This shifting of magnetic axis continues for negative cycle also and leads to the production of rotating magnetic field. The direction of this field is from non shaded part of the pole to the shaded part of the pole.

The advantages of shaded pole induction motor are

Very economical and reliable.

Construction is simple and robust because there is no centrifugal switch.

The disadvantages of shaded pole induction motor are

- i) Low power factor
- ii) The starting torque is very poor
- iii) The efficiency is very low as, the copper losses are high due to presence of copper band.
- iv) The speed reversal is also difficult and expensive as it requires another set of copper rings.

Applications of shaded pole motor :-

Due to their low starting torques and reasonable cost these motors are mostly employed in small instruments, hair dryers, toys, record players, small fans, electric clocks etc. These motors are usually available in a range of 1/300 to 1/20 KW.

SELECTION OF AC MOTORS FOR DIFFERENT APPLICATIONS:

S No	Application	Requirement	Motor
1	Textile mill - Ginning	Constant speed from 200 to 1500rpm	3-phase squirrel cage IM
	Textile mill - Cording	High starting torque, high overload capacity, 600 or 700rpm constant operating speed	3-phase slip ring IM
	Textile mill - Looms	Frequent starts and stops, constant speed operation, high starting torque	3-phase squirrel cage IM
2	Cranes	Accurate control of acceleration and retardation; wide speed control	3-phase slip ring IM
3	Steel mills	High starting torque, high overload capacity, frequent starts and stops, wide range of speed control	3-phase slip-ring iM
4	Sugar mills	500 to 1500 rpm; high starting torque	Two speed pole changing 3-phase squirrel cage motor
5	Compressors	Controlled acceleration, 700 to 1000rpm	3-phase IM or synchronous motor with frequency control
6	Pumps and blowers	Speed range 1:5	3-phase squirrel cage or slip-ring IM
7	Cement mills	Small starting current, high overload capacity, frequent starts and stops	3-phase slip ring IM with rotor resistance starting or synchronous motor or squirrel cage IM with frequency control
8	Domestic fans and pumps		1-phase induction motors

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

FIFTH SEMESTER B.TECH DEGREE EXAMINATION

MODEL QUESTION PAPER

Course Code: EET307

Course Name: SYNCHRONOUS & INDUCTION MACHINES

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks.

Marks

- | | | |
|----|---|-----|
| 1 | List the advantages of stationary armature type alternators over rotating armature types | (3) |
| 2 | Define coil pitch factor and distribution factor of an alternator | (3) |
| 3 | State and explain Blondel's Two Reaction Theory | (3) |
| 4 | What is meant by synchronisation? List the conditions to be met while synchronising an alternator to the common bus bars. | (3) |
| 5 | With the help of neat figures, explain why a synchronous motor is not self-starting. | (3) |
| 6 | Differentiate between slip ring and squirrel cage induction motors. | (3) |
| 7 | Explain the phenomenon of crawling and cogging in induction motors. | (3) |
| 8 | Explain any two braking techniques of induction motors. | (3) |
| 9 | Differentiate between synchronous and induction generators. | (3) |
| 10 | What is double field revolving theory? | (3) |

PART B

Answer any one full question from each module. Each question carries 14 marks.

Module 1

- | | | |
|----|--|------|
| 11 | a) List the causes of harmonics in alternators and suggest ways to mitigate them. | (5) |
| | b) A 3- Φ , 10 pole alternator has 2 slots/ pole/ phase on its stator with 10 conductors per slot. The air gap flux is sinusoidally distributed and equals 0.05 Wb. The stator has a double layer winding with a coil span of 150°. If the alternator is running at 600 rpm, calculate the emf generated /phase at no load. | (9) |
| 12 | With the help of neat diagrams, explain the effects of armature reaction in alternators under lagging, leading and unity power factors. | (14) |

Module 2

- | | | |
|----|---|------|
| 13 | A 220V, 6 pole, 50 Hz, star connected alternator gave the following test results: | (14) |
|----|---|------|

I_f A	0.2	0.4	0.6	0.8	1	1.2	1.4	1.8	2.2	2.6	3	3.4
V_{oc} (line) V	29	58	87	116	146	172	194	232	261	284	300	310
V_{ZPF} (line) V						0	29	88	140	177	208	230
I_{sc} A	6.6	13.2	20	26.5	32.4	40	46.3	59				

Find % voltage regulation at full load current of 40A at power factor 0.8 lag by (i) m.m.f method (ii) ZPF method. $R_a = 0.06 \Omega$ /phase.

- | | | |
|----|--|------|
| 14 | a) Two 3 Φ , 6.6 kV star connected alternators supply a load of 3000kW at 0.8 p.f lag. The synchronous impedance/phase of machine A is $0.5 + j 10 \Omega$ and that of machine B is $0.4 + j 12 \Omega$. The excitation of machine A is adjusted so that it delivers 150 A at a lagging power factor and the governors are so set that the load is equally shared between the machines. Determine the current, power | (10) |
|----|--|------|

factor and induced emf of each machine.

- b) With the help of a neat circuit diagram, explain how an alternator is synchronised to the bus bars by bright lamp method. (4)

Module 3

- 15 a) With the help of a neat circuit diagram, explain how V and inverted V curves are obtained. (6)
- b) A 2000V, 3-phase, 4 pole star connected synchronous motor runs at 1500 rpm. The excitation is constant and corresponds to an open circuit voltage of 2000V. The resistance is negligible compared to synchronous reactance of 3Ω per phase. Determine power input, power factor, torque developed for an armature current of 200A. (8)
- 16 a) In rice/flour mills driven by squirrel cage induction motors, the hopper is loaded with the grains only after starting the motor. Similarly, the delivery valve of centrifugal pumps driven by squirrel cage induction motor is opened only after starting the motor. What is the reason behind this? Justify your answer with a relevant performance curve of squirrel cage induction motor. (4)
- b) A 6-pole, 50 Hz, 3- Φ induction motor running on full load develops a useful torque of 150 Nm at a rotor frequency of 1.5 Hz. Calculate the shaft power output. If the mechanical torque lost in friction is 10 Nm, determine a) rotor copper loss b) input to the motor c) the efficiency. The total stator loss is 700W. (10)

Module 4

- 17 For the following test data, calculate (i) line current (ii) power factor (iii) rotor copper loss (iv) slip (v) efficiency (vi) maximum output power (vii) maximum torque and (viii) starting torque: (14)
- Induction Motor Details: 3.73kW, 200V, 50Hz, 4pole, 3 ϕ star connected
- No Load Test: 200V, 350W, 5A
- Blocked Rotor Test: 100V, 26A, 1700W
- Rotor Copper Loss at standstill is 60% of the total copper loss.
- 18 Explain the methods of speed control in three phase induction motors. (14)

Module 5

- 19 a) Explain the working principle and modes of operation of an Induction Generator. (8)
- b) With the help of a neat figure, explain the torque-slip characteristics of an induction machine (6)
- 20 Explain the working of split phase and capacitor start single phase induction motors with the help of neat circuit diagrams and phasor diagrams. Also mention the applications of each. (14)

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
FOURTH SEMESTER B.TECH DEGREE EXAMINATION, JULY 2017

Course Code: EE202

Course Name: SYNCHRONOUS & INDUCTION MACHINES

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 5 marks.

Marks

- | | | |
|---|--|-----|
| 1 | Enumerate various methods for minimizing harmonics in alternators. | (5) |
| 2 | What are the different methods of finding the voltage regulation of an alternator. | (5) |
| 3 | Describe the slip test method for finding the X_d and X_q of synchronous machines. | (5) |
| 4 | Explain the constructional details of a synchronous motor. | (5) |
| 5 | Differentiate between the phenomenon cogging and crawling of an induction motor. | (5) |
| 6 | List the various methods adopted for braking of an induction motor. | (5) |
| 7 | Compare induction generator with synchronous generator. | (5) |
| 8 | With a sketch describe the principle of working of a shaded pole motor. | (5) |

PART B

Answer any two full questions, each carries 10 marks.

- | | | |
|----|--|------|
| 9 | a) Explain various types of armature winding of an alternator. | (5) |
| | b) Find the distribution and pitch factor of a 3-phase, 4-pole, 24 slots alternator having its armature coils short pitched by one slot. | (5) |
| 10 | A 0.5MVA, 1.1kV, 50Hz, 3-phase, star connected alternator has R_a and X_s per phase as 0.1Ω and 1.5Ω respectively. Find its voltage regulation at different power factor of (i) unity, (ii) 0.9 lag and (iii) 0.8 lead at full load. | (10) |
| 11 | A 220V, 6 pole, 50Hz star connected alternator gave the following test results:- | (10) |

Field current in A	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.8	2.2	2.6	3	3.4
OC Voltage in volts	29	58	87	116	146	172	194	232	261	284	300	310
ZPF test line voltage in volts						0	29	88	140	177	208	230
SCC in A	6.6	13.2	20	26.5	32.4	40	46.3	59				

Find percentage voltage regulation at full load current of 40A at power factor 0.8 lag by (i) mmf method (ii) ZPF method. $R_a=0.06\Omega/\text{phase}$.

PART C

Answer any two full questions, each carries 10 marks.

- | | | |
|----|---|------|
| 12 | Explain the two reaction theory of salient pole alternator. | (10) |
| 13 | a) Describe the constructional features of 3-phase slip ring induction motor. | (5) |
| | b) A 6-pole, 50Hz, 3-phase, slip ring induction motor, the rotor resistance and the reactance at standstill per phase are 0.3 and 1.5Ω respectively. The emf between the slip rings on open circuit is 175V. Calculate (i) slip (ii) rotor emf/phase (iii) rotor frequency and reactance when the motor runs at a speed of 950rpm. | (5) |
| 14 | a) What are the various methods of synchronization of alternators. | (5) |

- b) Explain the effect of excitation on armature current and power factor of a synchronous motor and hence deduce the V and inverted V curves. (5)

PART D

Answer any two full questions, each carries 10 marks.

- 15 a) What are the different types of starters used for starting a 3-phase induction motor? (5)
- b) A 3-phase induction motor has a short circuit current 5 times of full load current at 5% slip. Determine the starting torque and starting current if the impressed voltage is reduced to 60% of the normal voltage by using starting resistance starter. The full load current and torque are 10A and 10Nm respectively. (5)
- 16 a) Explain the principle of operation of an induction generator. (5)
- b) Describe the double field revolving theory of a 1-phase induction motor. (5)
- 17 a) Draw the circle diagram of a 3-phase, 20hp, 400V, 50Hz star connected induction motor with the following test data:- (5)
- No load test 400V 9A $\cos\Phi=0.2$
- Block rotor test 200V 50A $\cos\Phi=0.4$
- Stator and rotor copper losses are divided equally in the block rotor test.
- b) From the above circle diagram obtain (a) line current (b) power factor (c) slip (d) efficiency at full load. (5)

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
FOURTH SEMESTER B.TECH DEGREE EXAMINATION, APRIL 2018

Course Code: EE202

Course Name: SYNCHRONOUS & INDUCTION MACHINES

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 5 marks.

Marks

- | | | |
|---|--|-----|
| 1 | Two alternators run on the same shaft (ie, runs at same speed). One generates voltage at 50Hz and the other at 40Hz. Find the maximum possible speed. Also find another possible speed. | (5) |
| 2 | Draw the phasor diagram of cylindrical rotor alternator supplying leading current and develop an expression for induced emf E. | (5) |
| 3 | The excitation of an alternator is running on an infinite bus is steadily increased from a small value over a wide range keeping the power output constant. By drawing phasor diagram show the magnitude armature current decreases to a minimum value and then increases. | (5) |
| 4 | In rice/flour mills driven by squirrel cage induction motor, the hopper is loaded with the grains only after starting the motor. Similarly, the delivery valve of centrifugal pumps driven by squirrel cage induction motor is opened only after starting the motor. What is the reason behind this? Justify your answer with a relevant performance curve of squirrel cage induction motor. | (5) |
| 5 | No load and blocked rotor tests on induction motor are effectively open circuit and short circuit tests. Justify. | (5) |
| 6 | Name the induction motor used to drive lifts. What property of this makes it suitable to drive lifts? How this property is achieved. | (5) |
| 7 | Using equivalent circuit justify that induction machine becomes a generator when the machine is driven above synchronous speed. | (5) |
| 8 | Show the construction of one pole of a shaded pole induction motor and explain how it produces a sweeping flux when an alternator current flows through the winding. | (5) |

PART B

Answer any two full questions, each carries 10 marks.

- | | | |
|----|---|------|
| 9 | The armature of 60Hz 11kV 450 rpm star connected alternator has 8 conductors per slot. The winding is short chorded by 3 slots to eliminate 5 th harmonics completely. Find the flux per pole to generate rated voltage on open circuit. | (10) |
| 10 | a) Prove that even harmonics will not be present in a full pitched winding. | (5) |
| | b) Give that procedure for experimental determination of synchronous impedance. | (5) |
| 11 | In an alternator a field current of I amp was required to drive rated current on short circuit and a field current of 2.5 times I was required to develop rated voltage on open circuit. Using emf method, find the voltage regulation of the alternator when delivering rated current at 0.8 pf lag. Assume armature resistance = 20% of synchronous impedance. (You can assume any value for rated voltage and rated current. The final answer will be the same.) | (10) |

PART C

Answer any two full questions, each carries 10 marks.

- 12 Two 100MW alternators operate in parallel. At no load both machines operate at 50Hz. The maximum load that can be shared without overloading either of the machines is 180MW and this happens at 48Hz. Find how will they share a total load of 160MW. (10)
- 13 a) Draw the phasor diagram of a salient pole alternator working at lagging power factor and derive an expression for internal power factor angle ψ . (5)
- b) From rotor equivalent circuit of an induction motor derive the expression for torque in synchronous watts. (5)
- 14 While running at 1440rpm a 3-phase induction motor draws 50kW from the mains. The stator iron and copper losses amount to 2kW. Find the rotor copper loss, torque in synchronous watts as well as in Newton Metre. If there is a mechanical loss of 1.08kW, find the overall efficiency of the motor. (10)

PART D

Answer any two full questions, each carries 10 marks.

- 15 A 400V 50Hz 7.5A 10HP delta connected induction motor was drawing 3A and 540W on no load test. In blocked rotor test rated current was driven with 100V and the power consumed was 450W. Draw the circle diagram and locate the maximum output point on the diagram. Find out slip efficiency and torque in synchronous watts at this point. (10)
- 16 a) Draw the schematic diagram of a star-delta starter. (5)
- b) Explain the working of synchronous induction motor. (5)
- 17 a) Show that the magnetic field produced by a single-phase armature can be modeled as double revolving field. By drawing torque speed characteristics prove that a single-phase induction motor is not self-starting. (10)

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
FOURTH SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2018

Course Code: EE202

Course Name: SYNCHRONOUS & INDUCTION MACHINES

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 5 marks.

Marks

- 1 Compare salient pole and cylindrical rotor type of alternators. (5)
- 2 What is armature reaction? Explain the effects of armature reaction in an alternator. (5)
- 3 Draw the phasor diagram of a salient pole alternator working at a lagging power factor and derive an expression for the regulation. (5)
- 4 Sketch and explain the V and inverted V-curves of a synchronous motor. (5)
- 5 What is the need for starter in a 3-phase induction motor? Explain the principle of operation of a star-delta starter. (5)
- 6 Sketch the equivalent circuit of a double cage induction motor and explain the parameters. (5)
- 7 Explain the principle of operation of synchronous induction motor. What are its advantages over synchronous motors? (5)
- 8 Why is a 1-phase induction motor not self-starting? How is it made self starting? (5)

PART B

Answer any two full questions, each carries 10 marks.

- 9 a) Define pitch factor and distribution factor as related to an ac winding. Derive expressions for both. (5)
- b) A 3-phase, 10-pole alternator has 2 slots/pole/phase on its stator with 10 conductors per slot. The air gap flux is sinusoidally distributed and equals 0.05Wb. The stator has a double layer winding with a coil span of 150° . If the alternator is running at 600rpm, calculate the emf generated/phase at no load. (5)
- 10 The following data pertains to a 5000kVA, 6600V, 3-phase, 50 Hz star connected alternator:- (10)

Field current	32	50	75	100	140
OC Voltage	3100	4900	6600	7500	8300
ZPF voltage (line) (full load)	0	1850	4250	5800	7000

Determine the regulation by ZPF method at full load unity power factor. Neglect armature resistance. Draw its phasor diagram also.

- 11 a) How do harmonics affect the emf generated in an alternator? What are the techniques used to minimize these harmonic voltages in the induced emf? (5)
- b) A 3-phase, 10kVA, 400V, 50Hz, star connected alternator supplies the rated load at 0.8 p.f. If the armature resistance is 0.5Ω and synchronous reactance is 10Ω , find the load angle and voltage regulation. (5)

PART C

Answer any two full questions, each carries 10 marks.

- 12 a) Explain the procedure for conduct of slip test using a neat circuit diagram. (5)
- b) The efficiency of a 3-phase 400V star connected synchronous motor is 95% and (5)

- it takes 24A at full load. What will be the induced emf and mechanical power developed at full load, 0.9 p.f. lead. The synchronous reactance is $0.2+j2\Omega$.
- 13 a) Describe with the help of a neat circuit diagram, the two bright and one dark lamp method of synchronizing an alternator to the AC mains. (5)
- b) A 6-pole, 50Hz, 3-phase induction motor running on full load develops a useful torque of 150Nm at a rotor frequency of 1.5Hz. Calculate the shaft power output. If the mechanical torque lost in friction is 10Nm, determine a) rotor copper loss b) input to the motor c) the efficiency. The total stator loss is 700W. (5)
- 14 a) Describe a set of torque-slip characteristics of a 3-phase induction motor. Explain the effect of change in rotor resistance on the characteristics. (5)
- b) Two 3-phase 6.6kV star connected alternators supply a load of 3000kW at 0.8pf lag. The synchronous impedance/phase of machine A is $0.5+j10\Omega$ and that of machine B is $0.4+j12\Omega$. The excitation of machine A is adjusted so that it delivers 150A at a lagging power factor and the governors are so set that the load is equally shared between the machines. Determine the current, power factor and induced emf of each machine. (5)

PART D

Answer any two full questions, each carries 10 marks.

- 15 a) What is “crawling” in induction motors? What are its causes and how can it be eliminated? (5)
- b) Draw the equivalent circuit of a 1-phase induction motor and explain how it is used to predetermine the performance of the machine. (5)
- 16 a) A 3-phase, 400V, 14.91 kW induction motor gave the following test results:- (5)
- | | | | |
|------------------|------|-------|-----|
| No load test | 400V | 1250W | 9A |
| Block rotor test | 150V | 4000W | 38A |
- Draw the circle diagram and determine the full load current, power factor, slip and efficiency.
- b) With a neat diagram, explain the capacitor start and run type of induction motor. (5)
- 17 a) Describe the principle of operation of an induction generator. Compare grid connected and self excited type of induction generators. (5)
- b) Describe any two methods of speed control of a 3-phase induction motor. (5)

**APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
FOURTH SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2018**

Course Code: EE202

Course Name: SYNCHRONOUS & INDUCTION MACHINES

ANSWERS

- 9 b) A 3-phase, 10-pole alternator has 2 slots/pole/phase on its stator with 10 conductors (6) per slot. The air gap flux is sinusoidally distributed and equals 0.05Wb. The stator has a double layer winding with a coil span of 150° . If the alternator is running at 600rpm, calculate the emf generated/phase at no load.

Ans:-

$m = 2$; $\alpha = 30^\circ$; No. of slots = $2 \times 10 \times 3 = 60$; No. of slots/pole = $6 = 180^\circ$;

$$f = \frac{PN}{120} = \frac{10 \times 600}{120} = 50 \text{ Hz}$$

$$\beta = \frac{180^\circ}{6} = 30^\circ$$

$$K_d = \frac{\sin(m\beta/2)}{m \sin(\beta/2)} = \frac{\sin 30^\circ}{2 \times \sin 15^\circ} = 0.966 \quad (1 \text{ mark})$$

$$K_c = \cos \frac{\alpha}{2} = \cos 15^\circ = 0.966 \quad (1 \text{ mark})$$

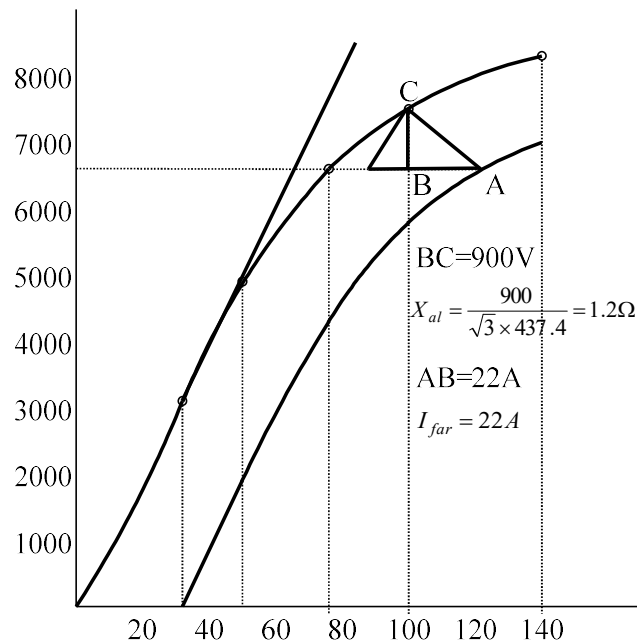
$$T = \frac{60 \times 10}{3 \times 2} = 100 \quad (1 \text{ mark})$$

$$E_{ph} = 4K_f K_c K_d f \phi T = 4 \times 1.11 \times 0.966 \times 0.966 \times 50 \times 0.05 \times 100 = 1035.8 \text{ V} \quad (2 \text{ marks})$$

- 10 The following data pertains to a 5000kVA, 6600V, 3-phase, 50 Hz star connected (10) alternator:-

Field current	32	50	75	100	140
OC Voltage	3100	4900	6600	7500	8300
ZPF voltage (line) (full load)	0	1850	4250	5800	7000

Determine the regulation by ZPF method at full load unity power factor. Neglect armature resistance. Draw its phasor diagram also.



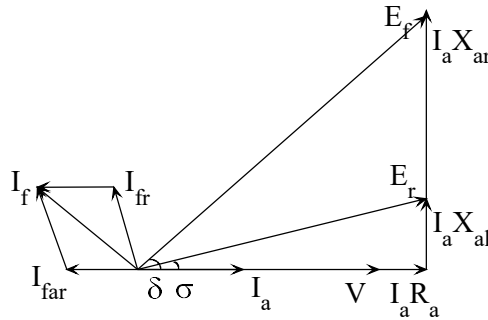
$$\text{Full load current, } I_a = \frac{5000 \times 10^3}{\sqrt{3} \times 6600} = 437.4 A$$

$$V = \frac{6600}{\sqrt{3}} = 3810.5 V$$

Plot of OCC & ZPFC (3 marks)

$X_{al} = 1.2 \Omega$, $I_{far} = 22 A$ from Potier triangle (3 marks)

Phasor diagram (2 marks)



$$\bar{E}_r = V \angle 0^\circ + I_a \angle 0^\circ \times (R_a + jX_{al}) = 3810.5 \angle 0^\circ + 437.4 \angle 0^\circ \times j1.2 = 3846.5 \angle 7.8^\circ V = E_r \angle \sigma^\circ$$

$$E_r(\text{line}) = 6662 V$$

$I_{fr} = 80 A$ (from OCC corresponding to 6662V)

$$\bar{I}_{fr} = I_{fr} \angle 90^\circ + \sigma^\circ = 80 \angle 97.8^\circ A$$

$$I_{far} = AB = 22 A$$

$$\bar{I}_{far} = I_{far} \angle 180^\circ - \Phi^\circ = 22 \angle 180^\circ A$$

$$\bar{I}_f = \bar{I}_{fr} + \bar{I}_{far} = 80 \angle 97.8^\circ + 22 \angle 180^\circ = 85.8 \angle 112.5^\circ = I_f \angle 90^\circ + \delta^\circ$$

$E_f(\text{line}) = 7100 V$ (corresponding to $I_f = 85.8 A$)

$$\bar{E}_f = 4099 \angle 22.5^\circ V$$

$$\text{Voltage regulation} = \frac{4099 - 3810.5}{3810.5} \times 100\% = 7.6\% \quad (2 \text{ marks})$$

- 11 b) A 3-phase, 10kVA, 400V, 50Hz, star connected alternator supplies the rated load at 0.8 p.f. lag. If the armature resistance is 0.5Ω and synchronous reactance is 10Ω , find the load angle and voltage regulation. (6)

$$\text{Full load current, } I_a = \frac{10 \times 10^3}{\sqrt{3} \times 400} = 14.43 A \quad (1 \text{ mark})$$

$$V = \frac{400}{\sqrt{3}} = 230.94 V$$

$$\bar{E}_f = V \angle 0^\circ + I_a \angle -36.87^\circ \times (R_a + jX_s) = 230.94 \angle 0^\circ + 14.43 \angle -36.87^\circ \times (0.5 + j10) = 341.85 \angle 18.97^\circ V$$

(3 marks)

$$\text{Voltage regulation} = \frac{341.85 - 230.94}{230.94} \times 100 = 48\% \quad (2 \text{ marks})$$

- 12 b) The efficiency of a 3-phase 400V star connected synchronous motor is 95% and it takes 24A at full load. What will be the induced emf and mechanical power developed at full load, 0.9 p.f. lead. The synchronous reactance is $0.2 + j2 \Omega$. (6)

$$I_{a1} \cos \phi_1 = I_{a2} \cos \phi_2 \quad I_{a2} = \frac{24 \times 1}{0.9} = 26.67 A \quad (1 \text{ mark})$$

$$\bar{E}_f = \bar{V} - \bar{I}_a(R_a + jX_s) = 230.94\angle 0^\circ - 26.7\angle +25.84^\circ \times (0.2 + j2) = 254.4\angle -11.4^\circ V \quad (2 \text{ marks})$$

$$\text{Power input} = \sqrt{3} \times V_L \times I_a \times \cos \phi = \sqrt{3} \times 400 \times 26.7 \times 0.9 = 16648W$$

$$\text{Armature Cu loss} = 3 \times I_a^2 \times R_a = 3 \times 26.7^2 \times 0.2 = 427.8W$$

$$\text{Mechanical power developed} = P_m = 16648 - 427.8 = 16220W \quad (3 \text{ marks})$$

- 13 b) A 6-pole, 50Hz, 3-phase induction motor running on full load develops a useful torque of 150Nm at a rotor frequency of 1.5Hz. Calculate the shaft power output. If the mechanical torque lost in friction is 10Nm, determine a) rotor copper loss b) input to the motor c) the efficiency. The total stator loss is 700W. (6)

$$s = \frac{1.5}{50} = 0.03$$

$$N_s = \frac{120f}{p} = \frac{120 \times 50}{6} = 1000rpm$$

$$N = (1-s)N_s = (1-0.03) \times 1000 = 970rpm$$

$$\text{Shaft power output, } P_{out} = 2\pi \times \frac{N}{60} T_{sh} = \frac{2\pi \times 970 \times 150}{60} = 15236.7W \quad (1 \frac{1}{2} \text{ marks})$$

$$\text{Mechanical power developed, } P_m = 2\pi \times \frac{N}{60} T_g = \frac{2\pi \times 970 \times (150 + 10)}{60} = 16252.5W$$

$$\text{Rotor input, } P_2 = \frac{P_m}{(1-s)} = \frac{16252.5}{(1-0.03)} = 16755W \quad (1 \frac{1}{2} \text{ marks})$$

$$\text{Rotor copper loss, } sP_2 = 0.03 \times 16755 = 502.7W \quad (1 \frac{1}{2} \text{ marks})$$

$$\text{Motor input, } P_{in} = \text{Rotor input} + \text{stator losses} = 16755 + 502.7 = 17257.7W$$

$$\text{Efficiency, } \eta = \frac{P_{out}}{P_{in}} = \frac{15236.7}{17257.7} = 88.3\% \quad (1 \frac{1}{2} \text{ marks})$$

- 14 b) Two 3-phase 6.6kV star connected alternators supply a load of 3000kW at 0.8pf lag. The synchronous impedance/phase of machine A is $0.5 + j10\Omega$ and that of machine B is $0.4 + j12\Omega$. The excitation of machine A is adjusted so that it delivers 150A at a lagging power factor and the governors are so set that the load is equally shared between the machines. Determine the current, power factor and induced emf of each machine. (6)

$$\sqrt{3}V_{I_1} \cos \phi_1 = \sqrt{3}V_{I_2} \cos \phi_2 = 1500 \times 10^3$$

$$\cos \phi_1 = \sqrt{3}V_{I_2} \cos \phi_2 = \frac{1500 \times 10^3}{\sqrt{3} \times 6600 \times 150} = 0.875lag \quad (1 \frac{1}{2} \text{ marks})$$

$$\bar{I}_1 = 150\angle -29^\circ A$$

$$I = \frac{3000 \times 10^3}{\sqrt{3} \times 6600 \times 0.8} = 328A \quad \bar{I} = 328\angle -36.87^\circ A$$

$$\bar{I}_2 = \bar{I} - \bar{I}_1 = 328\angle -36.87^\circ - 150\angle -29^\circ = 180.6\angle -43.4^\circ \quad (1 \text{ mark})$$

$$\cos \phi_2 = 0.727lag \quad (\frac{1}{2} \text{ mark})$$

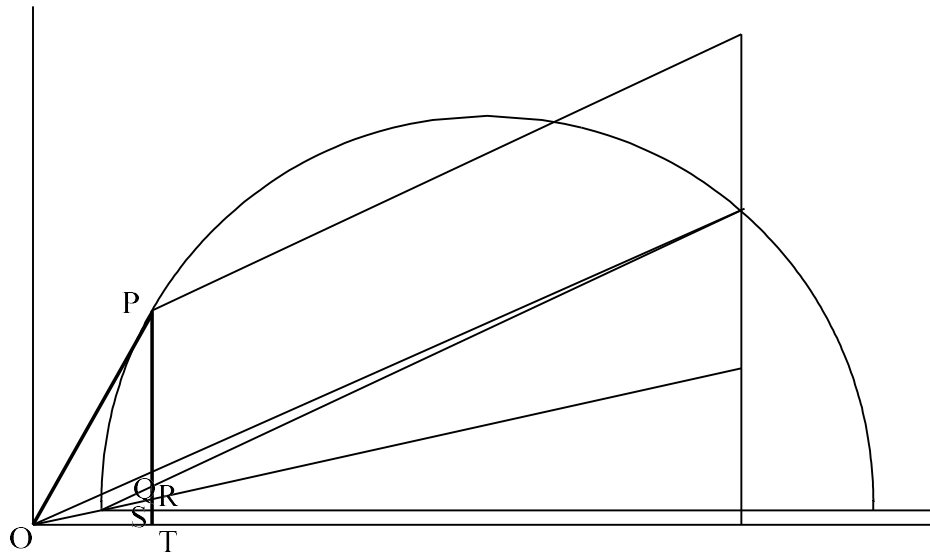
$$E_1 = V + I_1 Z_1 = 3810.5\angle 0^\circ + 150\angle -29^\circ \times (0.5 + j10) = 4776.8\angle 15.5^\circ V \quad (1 \frac{1}{2} \text{ marks})$$

$$E_2 = V + I_2 Z_2 = 3810.5\angle 0^\circ + 180.6\angle -43.4^\circ \times (0.4 + j12) = 5565\angle 15.9^\circ V \quad (1 \frac{1}{2} \text{ marks})$$

- 16 a) A 3-phase, 400V, 14.91 kW induction motor gave the following test results:- (6)
No load test 400V 1250W 9A

Block rotor test 150V 4000W 38A

Draw the circle diagram and determine the full load current, power factor, slip and efficiency.



Circle diagram (3 marks)

Full load current = 32A Power factor 0.7 lag (1 mark)

$$\text{Slip} = \frac{QR}{PR} = \frac{0.2}{2.6} = 7.7\% \text{ (1 mark)}$$

$$\text{Efficiency} = \frac{PQ}{PT} = \frac{2.4}{3} = 80\% \text{ (1 mark)}$$

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
FOURTH SEMESTER B.TECH DEGREE EXAMINATION, MAY 2019

Course Code: EE202

Course Name: SYNCHRONOUS AND INDUCTION MACHINES (EE)

Max. Marks: 100

Graph sheets shall be provided.

Duration: 3 Hours

PART A

Answer all questions, each carries 5 marks.

Mark
s

- 1 Derive the emf equation of an alternator. (5)
- 2 Draw the phasor diagram of a cylindrical rotor type alternator with a) unity power factor load and (b) leading power factor load (5)
- 3 Write the necessary conditions for synchronization of alternators. (5)
- 4 Synchronous motor is not self starting. Why? (5)
- 5 What is crawling in induction motor? How can it be eliminated? (5)
- 6 Explain V/f speed control method in induction motor. (5)
- 7 Explain the working principle of synchronous induction motor. (5)
- 8 Why single phase induction motor is not self starting. How it is made self starting? (5)

PART B

Answer any two full questions, each carries 10 marks.

- 9 a) Compare salient pole alternator with smooth cylindrical alternator? (4)
b) A 3-phase, 8 pole, 750rpm, star connected alternator has 72 slots on armature. Each slot has 12 conductors and winding is short pitched by two slots. Find the induced emf between the lines, given flux per pole 0.06 Wb. (6)
- 10 Following test results are obtained on a 6600V alternator (10)

Open circuit voltage in volts	3100	4900	6600	7500	8300
Field currents in Amperes	16	25	37.5	50	70

A field current of 20A is found to circulate full load current on armature with short circuited. Calculate full load regulation at 0.8 pf lag by using mmf method. Neglect armature resistance.
- 11 a) Explain the causes of harmonics in alternator. How can it be eliminated? (5)
b) A 3-phase star connected alternator supplies a load of 1000kW at a pf of 0.8 lagging with a terminal voltage of 11kV. Its armature resistance is 0.4Ω per phase while synchronous reactance is 3Ω per phase. Calculate the line value of emf induced and full load regulation. (5)

PART C

Answer any two full questions, each carries 10 marks.

- 12 a) Explain the procedure to conduct slip test with a neat circuit diagram (5)

- b) Explain synchronisation of alternators using dark lamp method. (5)
- 13 a) Explain any one method of starting of synchronous motor. (4)
- b) A 2000V, 3-phase, 4 pole star connected synchronous motor runs at 1500 rpm. (6)
The excitation is constant and corresponds to an open circuit voltage of 2000V.
The resistance is negligible compared to synchronous reactance of 3Ω per phase.
Determine power input, power factor, torque developed for an armature current of 200A
- 14 a) Explain the effect of change in excitation of an alternator. (5)
- b) A 400V, 4-pole, 3-phase, 50Hz, star connected induction motor has rotor (5)
resistance and reactance per phase 0.01Ω and 0.1Ω respectively. Determine a)
starting torque b) slip at maximum torque and c) maximum torque. Given rotor to
stator turns is 0.25.

PART D

Answer any two full questions, each carries 10 marks.

- 15 Draw the circle diagram of a 20 Hp, 400 V, 50 Hz, 3-phase, star-connected (10)
induction motor from the following test data (line values)
No load Test : 400V, 9 A , 0.2 pf
Blocked Rotor Test : 200V, 50 A, 0.4 pf
From the circle diagram, find the line current and power factor at full load.
The stator and rotor copper losses are divided equally in the blocked rotor test.
- 16 a) Explain with neat diagram, star-delta starter in 3-phase induction motor. (5)
- b) Compare induction generator with synchronous generator. (5)
- 17 a) Explain types of single phase induction motors with relevant figures. (8)
- b) Draw the equivalent circuit of single phase induction motor (2)

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
FOURTH SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2019

Course Code: EE202

Course Name: SYNCHRONOUS AND INDUCTION MACHINES (EE)

Max. Marks: 100

Graphs sheets shall be supplied.

Duration: 3 Hours

PART A

Answer all questions, each carries 5 marks.

Marks

- | | | |
|---|---|-----|
| 1 | Explain the term pitch factor in an alternator and derive an expression for it. Also discuss the effect of short pitching on harmonics. | (5) |
| 2 | Define armature reaction. Explain the effect of armature reaction on the terminal voltage of an alternator at zero leading power factor load. | (5) |
| 3 | Explain why X_d and X_q are different for salient pole alternators whereas they are the same for the smooth rotor machines. | (5) |
| 4 | Explain the working of a synchronous condenser. | (5) |
| 5 | How would you rate the performance of an autotransformer starter with a DOL starter used in an induction motor? | (5) |
| 6 | Equivalent circuit parameters of a 3 phase induction motor can be determined from no load and blocked rotor tests. Justify. | (5) |
| 7 | Explain the principle of shaded pole induction motor with suitable diagram. | (5) |
| 8 | Using equivalent circuit prove that the induction machine becomes a generator when the machine is driven above synchronous speed. | (5) |

PART B

Answer any two full questions, each carries 10 marks.

- | | | | | | | | | | | | | | | | | | | | | | | |
|----------------------|--|-----------|------|------|------|------|------|------|------|-----|-----|----------------------|------|------|------|------|------|------|------|------|------|------|
| 9 | a) Differentiate between salient pole type and cylindrical type alternator with sketches. | (5) | | | | | | | | | | | | | | | | | | | | |
| | b) Derive the EMF equation of an alternator. | (5) | | | | | | | | | | | | | | | | | | | | |
| 10 | The open-circuit test data for a 3-phase, 3.5 MVA, 4.16 kV, 50 Hz star connected synchronous generator are given below.
Open-circuit characteristic:
<table border="0" style="width: 100%;"><tr><td>I_f(A)</td><td>50</td><td>100</td><td>150</td><td>200</td><td>250</td><td>300</td><td>350</td><td>400</td><td>450</td></tr><tr><td>V_{OC} (line) (kV)</td><td>1.62</td><td>3.15</td><td>4.16</td><td>4.75</td><td>5.13</td><td>5.37</td><td>5.55</td><td>5.65</td><td>5.75</td></tr></table> <p>A field current of 200A is found necessary to circulate full load current on short circuit of the alternator. The machine supplies full-load at a p.f. of 0.8 lagging. Determine its voltage regulation by (i) EMF method (ii) MMF method. Neglect resistance.</p> | I_f (A) | 50 | 100 | 150 | 200 | 250 | 300 | 350 | 400 | 450 | V_{OC} (line) (kV) | 1.62 | 3.15 | 4.16 | 4.75 | 5.13 | 5.37 | 5.55 | 5.65 | 5.75 | (10) |
| I_f (A) | 50 | 100 | 150 | 200 | 250 | 300 | 350 | 400 | 450 | | | | | | | | | | | | | |
| V_{OC} (line) (kV) | 1.62 | 3.15 | 4.16 | 4.75 | 5.13 | 5.37 | 5.55 | 5.65 | 5.75 | | | | | | | | | | | | | |
| 11 | a) Show that the output emf wave of an alternator do not contain even harmonics. | (5) | | | | | | | | | | | | | | | | | | | | |

- b) Find the distribution factor and pitch factor of a 3 phase 4 pole 24 slots alternator having its armature coils short pitched by one slot. (5)

PART C

Answer any two full questions, each carries 10 marks.

- 12 a) Derive the expression for mechanical power developed in a cylindrical rotor type synchronous motor. (6)
- b) A 2.3kV, 3 phase star connected synchronous motor has $Z_s = (0.2 + j 2.2)$ ohms per phase. The motor is operating at 0.5pf leading with a line current of 200A. Determine the induced emf per phase. (4)
- 13 a) Derive an expression for developed torque in a 3-phase induction motor and find the condition for maximum torque. Also sketch the torque-slip curve. (6)
- b) A 746KW, 3-phase, 50Hz, 16 pole induction motor has a rotor impedance of $(0.02 + j0.15)$ ohm at stand still. Full load torque is obtained at 360 rpm. Calculate (i) the ratio of maximum to full load torque and (ii) the speed corresponding to max torque. (4)
- 14 a) Explain the effect of change in excitation in parallel operation of alternators. (6)
- b) The power input to a 500V, 50Hz, 6-pole, 3-phase induction motor running at 975 rpm is 40KW. The stator losses are 1KW and the friction and windage losses total 2KW. Calculate: (i) the slip (ii) the rotor copper loss (iii) shaft power and (iv) the efficiency. (4)

PART D

Answer any two full questions, each carries 10 marks.

- 15 Draw the circle diagram for a 3.73kW, 200V, 50Hz, 4 pole 3 phase star connected induction motor from the following data : (10)
- No load Test : 200V, 5A, 350W
- Blocked Rotor Test: 100V, 26A, 1700W
- Rotor copper loss at stand still = half of the total copper loss
- Construct the circle diagram and estimate: (i) full load current, (ii) power factor at full-load and (iii) maximum torque in terms of full-load torque.
- 16 a) Explain the phenomenon crawling as applied to induction motor. (5)
- b) Explain how pulling into step is achieved in synchronous induction motor. (5)
- 17 a) Differentiate between plugging and regenerative braking as applied to induction motors. (4)
- b) If the standstill impedance of the outer cage of a double cage induction motor is $(2 + j0.4)$ ohm and of the inner cage is $(0.4 + j2)$ ohm, compare the relative torques of two cages (i) at standstill (ii) at a slip of 5%. (6)